



Cambridge International Examinations
Cambridge International General Certificate of Secondary Education (9–1)

PHYSICS

Paper 2 Multiple Choice (Extended)

0972/21

May/June 2018

45 minutes

Additional Materials: Multiple Choice Answer Sheet
Soft clean eraser
Soft pencil (type B or HB recommended)

READ THESE INSTRUCTIONS FIRST

Write in soft pencil.

Do not use staples, paper clips, glue or correction fluid.

Write your name, Centre number and candidate number on the Answer Sheet in the spaces provided unless this has been done for you.

DO NOT WRITE IN ANY BARCODES.

There are **forty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A, B, C** and **D**.

Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

Read the instructions on the Answer Sheet very carefully.

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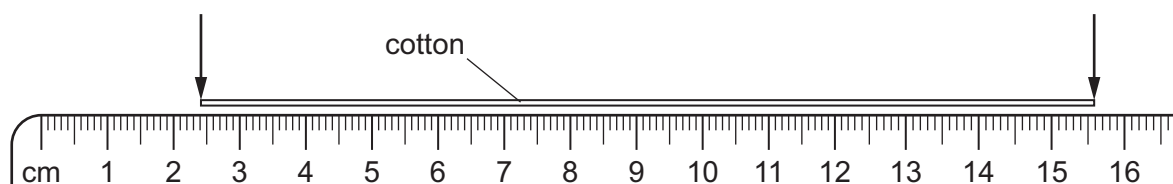
Electronic calculators may be used.

Take the weight of 1.0 kg to be 10 N (acceleration of free fall = 10 m/s^2).

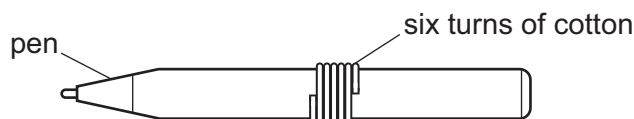
This document consists of **15** printed pages and **1** blank page.



- 1 A length of cotton is measured between two points on a ruler.



When the length of cotton is wound closely around a pen, it goes round six times.

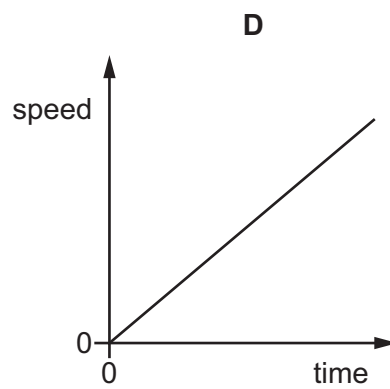
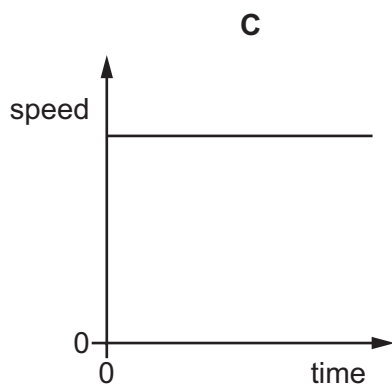
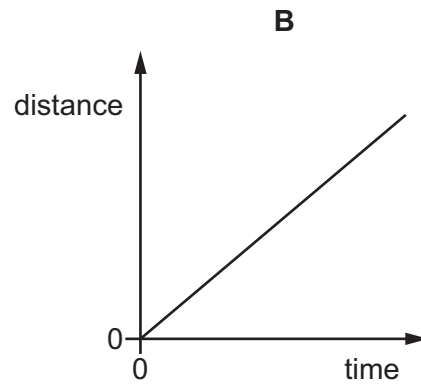
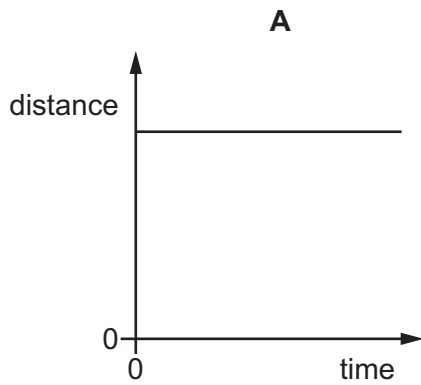


What is the distance once round the pen?

- A** 2.2 cm **B** 2.6 cm **C** 13.2 cm **D** 15.6 cm
- 2 When does an object falling vertically through the air reach terminal velocity?
- A** when the acceleration of the object becomes negative
B when the acceleration of the object is equal to g
C when the air resistance equals the weight of the object
D when the air resistance is greater than the weight of the object

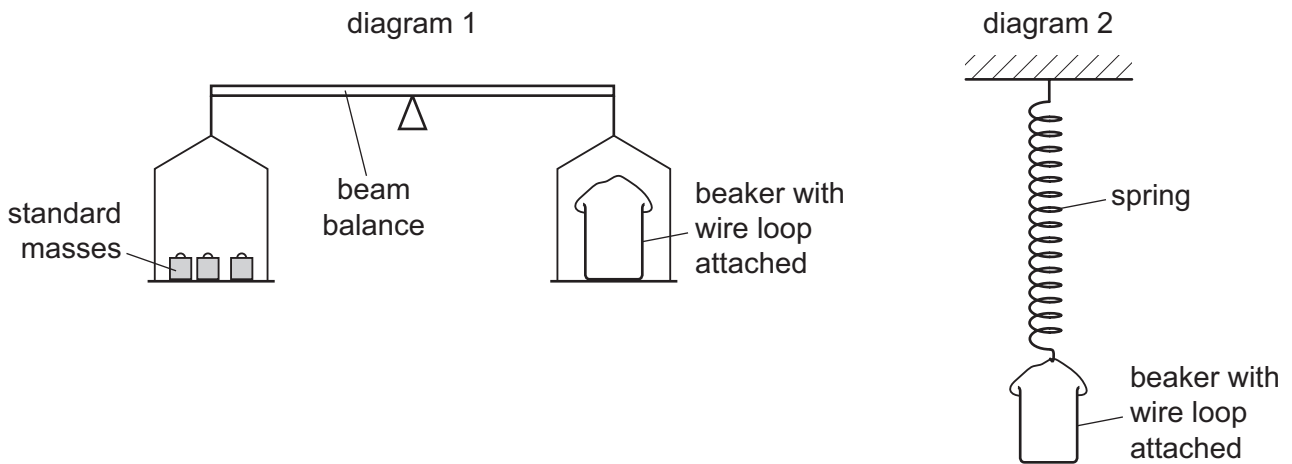
- 3 A car is moving along a straight, level road, with a constant acceleration.

Which graph shows the motion of the car?



- 4 Diagram 1 shows a beam balance. A beaker with a wire loop balances the standard masses.

The beaker is then removed and hung from a spring. The spring extends by 5.0 cm, as in diagram 2.



The experiment is repeated with the same apparatus on the Moon, where the acceleration of free fall is less than on Earth.

Which statement describes what happens on the Moon?

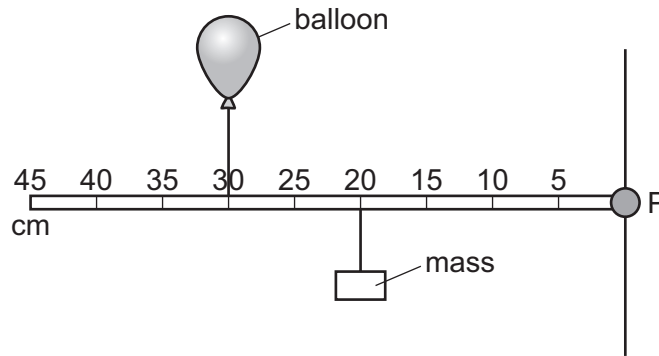
- A The beam balance is balanced and the spring extends by 5.0 cm.
- B The beam balance is balanced and the spring extends by less than 5.0 cm.
- C The right-hand balance pan is higher and the spring extends by 5.0 cm.
- D The right-hand balance pan is higher and the spring extends by less than 5.0 cm.

- 5 An object always has mass but does not always have weight.

What must be present and acting on the mass for it to have weight?

- A a gravitational field
 - B a set of scales
 - C displaced water
 - D friction due to air resistance
- 6 A force acting on a moving ball causes its motion to change. This force stays constant.
- What makes the force produce a greater change in the motion of the ball?
- A decreasing the total mass of the ball
 - B increasing the temperature of the ball
 - C using a ball with a hollow centre but the same mass
 - D using a different material for the ball so that it has a lower density but the same mass

- 7 A balloon and a mass are attached to a rod that is pivoted at P.



The balloon is filled with helium, a gas less dense than air, so that it applies an upward force on the rod.

The rod is horizontal and stationary.

Which action causes the rod to rotate clockwise?

- A Move both the balloon and mass 10 cm to the left.
 - B Move both the balloon and mass 10 cm to the right.
 - C Move both the balloon and mass to the 25 cm mark.
 - D Move the balloon to the 20 cm mark and the mass to the 30 cm mark.
- 8 A car is moving in a straight line on a level road. Its engine provides a forward force on the car. A second force of equal size acts on the car due to resistive forces.

Which statement describes what happens?

- A The car changes direction.
 - B The car moves at a constant speed.
 - C The car slows down.
 - D The car speeds up.
- 9 Which expression gives the momentum of an object?
- A mass $\times$ acceleration
 - B mass $\times$ gravitational field strength
 - C mass $\times$ velocity
 - D $\frac{1}{2} \times \text{mass} \times (\text{velocity})^2$

- 10** As energy is transferred into different forms, it eventually becomes dissipated.

What does this mean?

- A** All the energy disappears.
- B** The energy finally changes into every possible form of energy.
- C** The energy spreads out among the objects and their surroundings.
- D** The total amount of energy becomes less.

- 11** A ball of mass 1.2 kg is dropped from a height of 30 m. As it falls, 25% of its initial gravitational potential energy is transferred to thermal energy.

What is the kinetic energy of the ball just before it hits the ground?

- A** 27 J
- B** 90 J
- C** 270 J
- D** 360 J

- 12** A girl hangs by her hands from a bar in the gymnasium. She pulls herself up until her chin is level with the bar.

The mass of the girl is 48 kg.

She pulls herself up through a distance of 0.25 m.

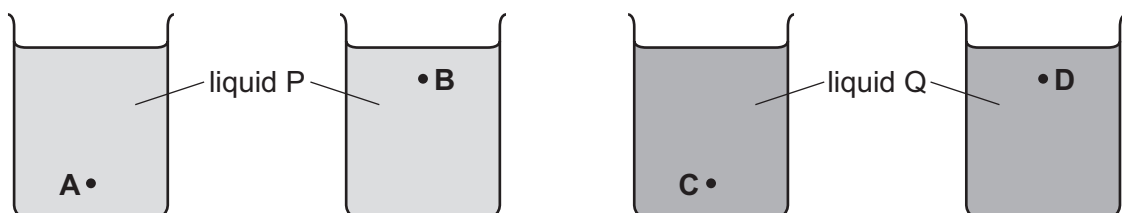
She does this in 2.0 s.

What is the useful power she uses to pull herself up?

- A** 6.0 W
- B** 24 W
- C** 60 W
- D** 240 W

- 13** Four identical beakers are filled with equal volumes of liquids P or Q, as shown. Liquid P is more dense than liquid Q.

At which point is the pressure the least?



- 14** An oil tank has a base of area 2.5 m^2 and is filled with oil to a depth of 1.2 m.

The density of the oil is 800 kg/m^3 .

What is the force exerted on the base of the tank due to the oil?

- A** 960 N
- B** 2400 N
- C** 9600 N
- D** 24 000 N

- 15** When molecules of a gas rebound from a wall of a container, the wall experiences a pressure.

What is the cause of this pressure?

- A** the change in energy of the molecules
- B** the change in momentum of the molecules
- C** the change in power of the molecules
- D** the change in speed of the molecules

- 16** A student wishes to calibrate a mercury-in-glass thermometer with a °C scale.

Which values should she use for the lower fixed point and for the upper fixed point?

	lower fixed point	upper fixed point
A	melting point of ice	boiling point of mercury
B	melting point of ice	boiling point of water
C	melting point of mercury	boiling point of mercury
D	melting point of mercury	boiling point of water

- 17** Which statements about boiling and about evaporation are both correct?

	boiling	evaporation
A	takes place only at the surface	takes place only at the surface
B	takes place only at the surface	takes place throughout the liquid
C	takes place throughout the liquid	takes place only at the surface
D	takes place throughout the liquid	takes place throughout the liquid

- 18** On a cold day, a metal front-door knob X and a similar plastic knob Y are at the same temperature.

Why does X feel cooler to the touch than Y?

- A** X convects thermal energy better than Y.
- B** X is a better thermal conductor than Y.
- C** X is a better insulator than Y.
- D** X is a better radiator of thermal energy than Y.

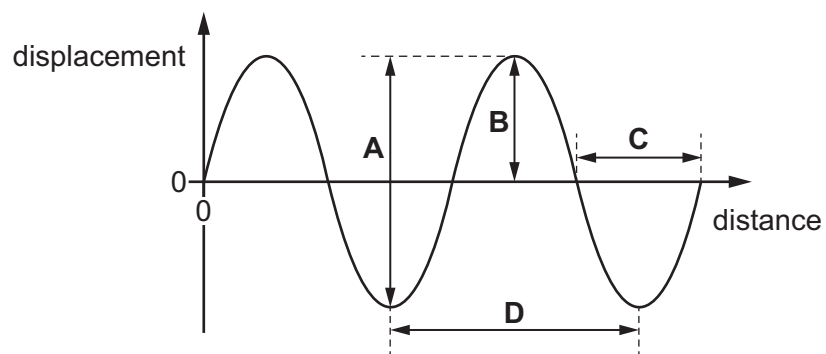
- 19 Light travels at a speed of $2.0 \times 10^8 \text{ m/s}$ in a glass block.

In the glass, the wavelength of the light is $4.0 \times 10^{-7} \text{ m}$.

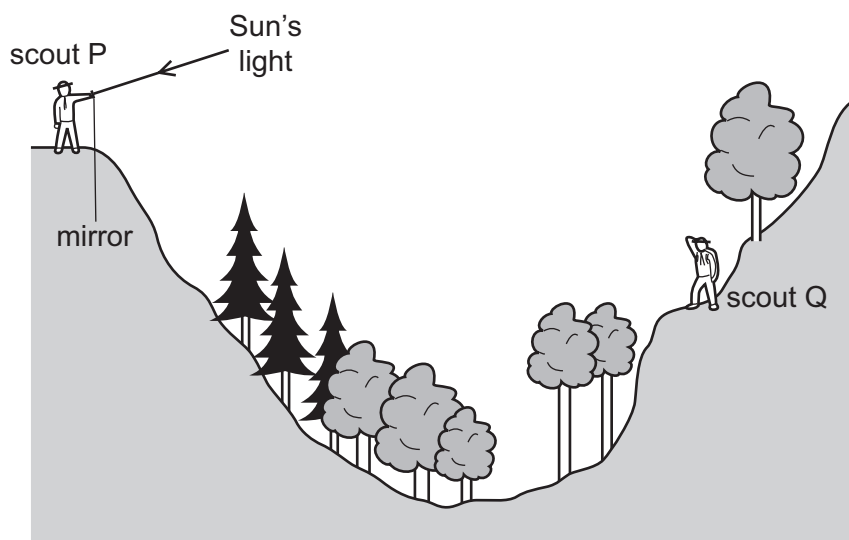
What is the frequency of the light?

- A $2.0 \times 10^{-15} \text{ Hz}$
- B $1.3 \times 10^{-2} \text{ Hz}$
- C 80 Hz
- D $5.0 \times 10^{14} \text{ Hz}$

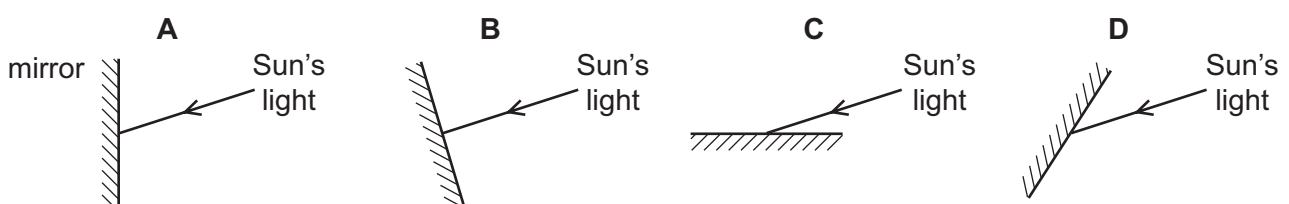
- 20 Which arrow on the graph shows the amplitude of the wave?



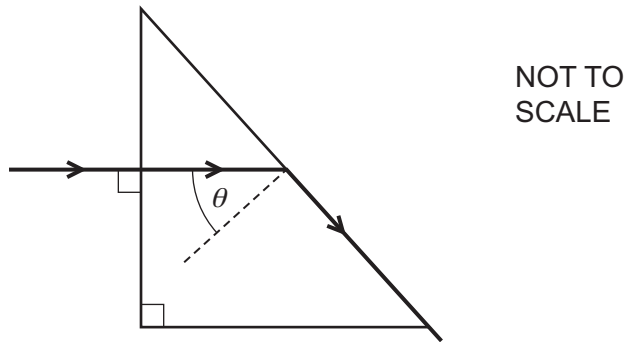
- 21 Scout P signals to scout Q on the other side of a valley by using a mirror to reflect the Sun's light.



Which mirror position allows the Sun's light to be reflected to scout Q?



- 22** A prism is made from transparent plastic. In this plastic, light travels at $0.80c$, where c is its speed in air. Light enters one face of the prism at right-angles as shown.



The light just escapes from the sloping face of the prism.

What is angle θ ?

- A** 37° **B** 39° **C** 51° **D** 53°
- 23** The Sun emits infra-red radiation and light.

Light from the Sun reaches the Earth in 8 minutes.

Which row gives correct information about the infra-red radiation?

	wavelength of infra-red radiation	time taken for infra-red radiation to reach Earth
A	longer than wavelength of light	8 minutes
B	longer than wavelength of light	much less than 8 minutes
C	shorter than wavelength of light	8 minutes
D	shorter than wavelength of light	much more than 8 minutes

- 24** A dolphin has a range of audible frequencies of 150 Hz – 150 kHz .

Which range of frequencies can be heard both by humans with good hearing and by dolphins?

- A** 20 Hz – 150 Hz
B 20 Hz – 150 kHz
C 20 kHz – 150 kHz
D 150 Hz – 20 kHz

- 25** A permanent magnet is placed close to a bar of soft iron.



What are the polarities of end P and of end Q?

	end P	end Q
A	N	N
B	N	S
C	S	N
D	S	S

- 26** A steel magnet is placed inside a coil of wire.

Which method is used to demagnetise the magnet?

- A** connect the coil to an a.c. power supply
 - B** connect the coil to an a.c. power supply and slowly remove the magnet from the coil
 - C** connect the coil to a d.c. power supply
 - D** connect the coil to a d.c. power supply and slowly remove the magnet from the coil
- 27** The electromotive force (e.m.f.) of a rechargeable battery is 6.0 V.
- What does this mean?
- A** 6.0 J is the maximum energy the battery can provide in 1.0 s.
 - B** 6.0 J is the total energy the battery can provide before it has to be recharged.
 - C** 6.0 J of energy is provided by the battery to drive a charge of 1.0 C around a complete circuit.
 - D** 6.0 J of energy is provided by the battery to drive a current of 1.0 A around a complete circuit.
- 28** A student measures the potential difference across a device and the current in the device.

Which calculation gives the resistance of the device?

- A** current + potential difference
- B** current ÷ potential difference
- C** potential difference ÷ current
- D** potential difference × current

- 29** A water heater is connected to a 230 V supply and there is a current of 26 A in the heater. It takes 20 minutes to heat the water to the required temperature.

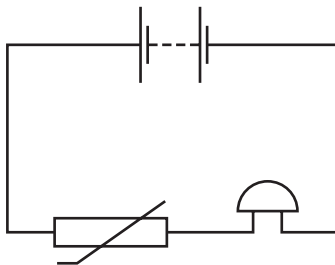
How much energy is supplied by the heater?

- A** $6.0 \times 10^3 \text{ J}$ **B** $1.0 \times 10^4 \text{ J}$ **C** $1.2 \times 10^5 \text{ J}$ **D** $7.2 \times 10^6 \text{ J}$

- 30** Which electrical symbol represents a diode?



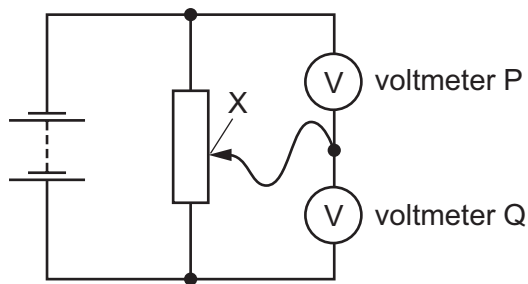
- 31** A student sets up this circuit.



What is the purpose of the circuit?

- A** to allow a lamp to be made dimmer or brighter as required
B to amplify the sound of a voice
C to light a lamp in the dark
D to sound a bell when the temperature rises

32 The diagram shows two voltmeters P and Q connected to a potential divider.

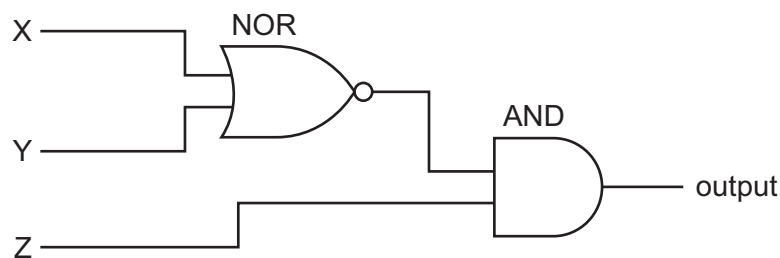


The sliding connection at point X is moved towards the top of the diagram.

What happens to the reading on P and to the reading on Q?

	reading on P	reading on Q
A	decreases	decreases
B	decreases	increases
C	increases	decreases
D	increases	increases

33 The diagram represents a digital circuit using a NOR gate and an AND gate.



What is the truth table for this circuit?

A

X	Y	Z	output
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	0
1	1	1	1

B

X	Y	Z	output
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	0
1	0	1	0
1	1	0	0
1	1	1	0

C

X	Y	Z	output
0	0	0	1
0	0	1	1
0	1	0	1
0	1	1	0
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	0

D

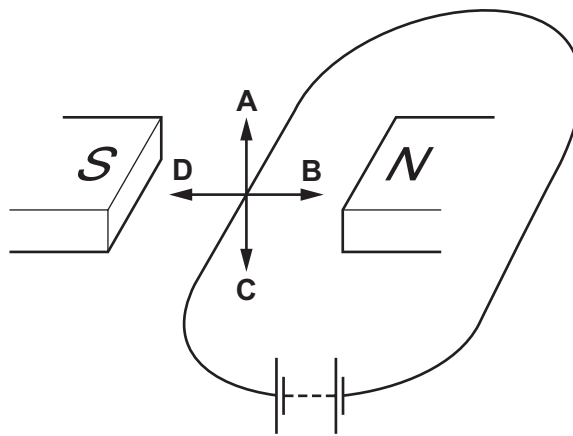
X	Y	Z	output
0	0	0	1
0	0	1	0
0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	1

34 Which statement about electromagnetic induction is correct?

- A** A strong magnet that is held stationary near a stationary conductor causes a greater effect than a weak magnet.
- B** The effect occurs when a magnet and a conductor are both moved with the same speed and in the same direction.
- C** The effect occurs when a magnet is moved away from a nearby conductor.
- D** The effect only occurs when a magnet is moved towards a conductor.

35 The diagram shows a current-carrying conductor in a magnetic field.

Which arrow shows the direction of the force acting on the conductor?

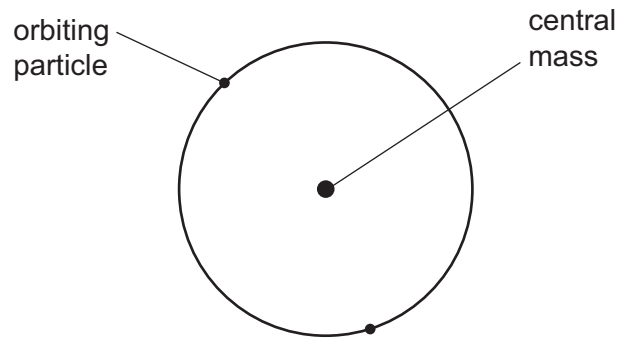


36 Power losses in transmission cables are reduced by increasing the transmission voltage.

What is the explanation for this reduction?

- A** The current decreases, reducing thermal energy losses.
- B** The current increases, increasing the flow of charge.
- C** The resistance of the cable increases, reducing the current.
- D** The resistance of the cable decreases.

37 In the atomic model, an atom consists of a central mass, orbited by much smaller particles.



What is the name of the central mass and of the orbiting particles?

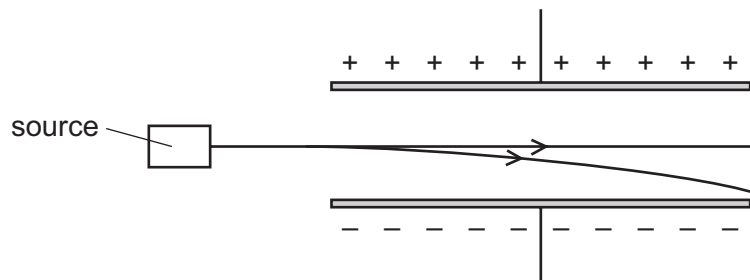
	central mass	orbiting particles
A	neutron	α -particles
B	neutron	electrons
C	nucleus	α -particles
D	nucleus	electrons

38 Nuclear fusion is a reaction that takes place in stars.

Which row describes this reaction?

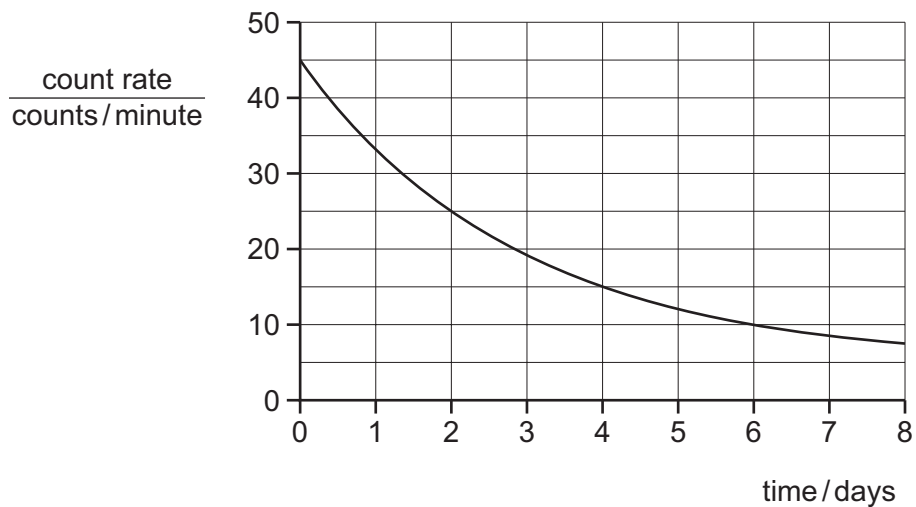
	action of atomic nuclei	energy
A	an atomic nucleus splits into two or more smaller nuclei	absorbed
B	an atomic nucleus splits into two or more smaller nuclei	released
C	atomic nuclei join together to form a larger nucleus	absorbed
D	atomic nuclei join together to form a larger nucleus	released

- 39 The diagram shows emissions from a source passing into the electric field between two charged plates.



What is emitted by this source?

- A** neutrons and γ -rays only
B α -particles and β -particles only
C α -particles and γ -rays only
D β -particles and γ -rays only
- 40 The graph shows how the count rate registered by a counter near to a sample of a radioactive isotope changes over a period of a few days. The background count rate is 5 counts per minute.



What is the half-life of the isotope?

- A** 2.0 days **B** 2.5 days **C** 3.0 days **D** 4.0 days

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Cambridge Assessment International Education
Cambridge International General Certificate of Secondary Education (9–1)

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May/June 2018

MARK SCHEME

Maximum Mark: 40

Published

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Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

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Question	Answer	Marks
1	A	1
2	C	1
3	D	1
4	B	1
5	A	1
6	A	1
7	B	1
8	B	1
9	C	1
10	C	1
11	C	1
12	C	1
13	D	1
14	D	1
15	B	1
16	B	1
17	C	1
18	B	1
19	D	1
20	B	1
21	A	1
22	D	1
23	A	1
24	D	1
25	C	1
26	B	1
27	C	1
28	C	1

Question	Answer	Marks
29	D	1
30	B	1
31	D	1
32	B	1
33	B	1
34	C	1
35	A	1
36	A	1
37	D	1
38	D	1
39	C	1
40	A	1



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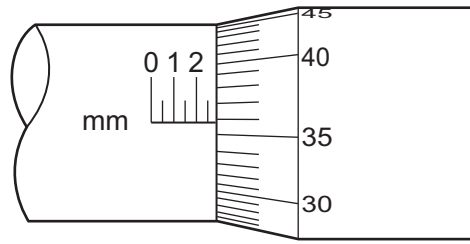
Electronic calculators may be used.

Take the weight of 1.0 kg to be 10 N (acceleration of free fall = 10 m/s^2).

This document consists of **16** printed pages.



- 1 The diagram shows part of a micrometer screw gauge.

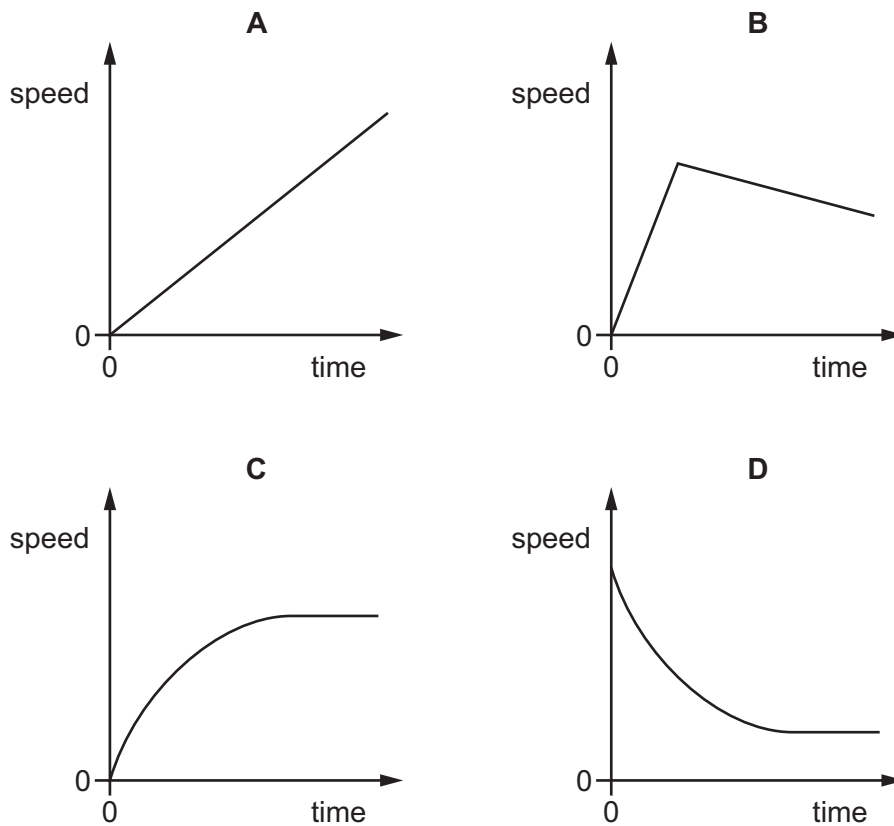


What is the smallest reading that can be achieved using this micrometer screw gauge?

- A** 0.0001 mm **B** 0.01 mm **C** 0.1 mm **D** 1 mm

- 2 A small, light ball is dropped from the top of a tall building.

Which graph shows how the speed of the ball changes with time?

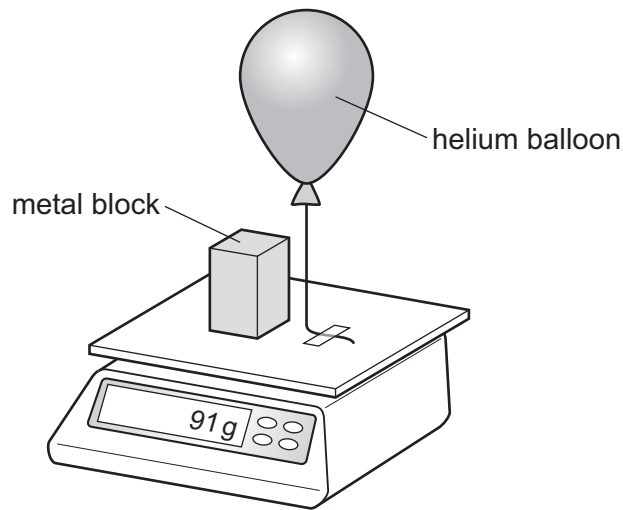


- 3 A runner runs 300 m at an average speed of 3.0 m/s. She then runs another 300 m at an average speed of 6.0 m/s.

What is her average speed for the total distance of 600 m?

- A** 2.0 m/s **B** 4.0 m/s **C** 4.5 m/s **D** 8.0 m/s

- 4 A helium balloon is tied to a top-pan balance. A metal block of mass 100 g is placed on the balance. The reading on the balance is 91 g.



Which statement can be deduced from this experiment?

- A The balloon exerts a downward force of 0.09 N on the top-pan balance.
- B The helium has a mass of -9 g .
- C The helium has a mass of $+9\text{ g}$.
- D The resultant downward force on the top-pan balance is 0.91 N.

- 5 A liquid has a volume of 0.040 m^3 and a mass of 30 000 g.

What is the density of the liquid?

- A 0.075 kg/m^3
- B 7.5 kg/m^3
- C 750 kg/m^3
- D 7500 kg/m^3

- 6 A resultant force of 4.0 N acts on an object of mass 0.50 kg for 3.0 seconds.

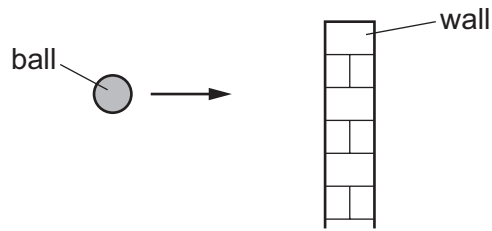
What is the change in velocity caused by this force?

- A 4.0 m/s
- B 6.0 m/s
- C 12 m/s
- D 24 m/s

- 7 Which quantities are both vectors?

- A acceleration and force
- B acceleration and pressure
- C density and force
- D density and pressure

- 8 A moving ball with a momentum of 25 kg m/s collides head-on with a wall.



It rebounds from the wall with the same speed but in the opposite direction. The time of collision is 50 ms .

What is the average force exerted on the wall by the ball during the collision?

- A** 0.50 N **B** 1.00 N **C** 500 N **D** 1000 N
- 9 Which device is designed to convert chemical energy into kinetic energy?
- A** an a.c. generator
B a battery-powered torch
C a car engine
D a wind-up mechanical clock
- 10 An object, initially at rest, is dropped from a height of 12.0 m . The change in gravitational potential energy when it falls to the ground is 565 J .

The frictional forces are negligible.

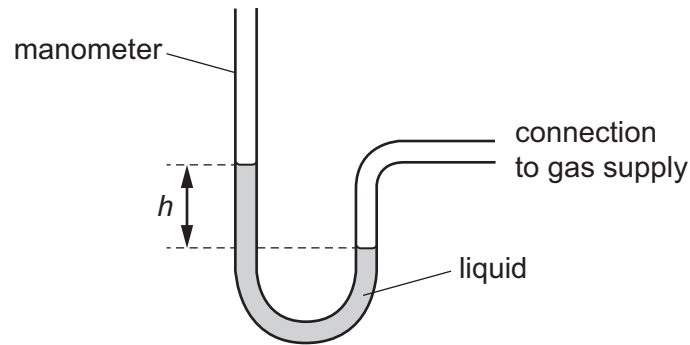
What is its speed when it hits the ground?

- A** 4.71 m/s **B** 15.5 m/s **C** 47.1 m/s **D** 240 m/s
- 11 A man climbs a ladder.

Which two quantities can be used to calculate the useful power of the man?

- A** the weight of the man and the time taken only
B the weight of the man and the vertical distance moved only
C the work done by the man and the time taken only
D the work done by the man and the vertical distance moved only

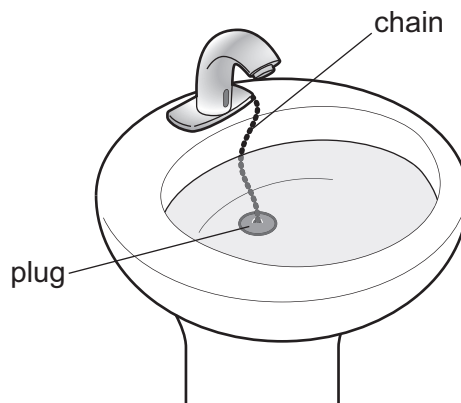
- 12 A manometer is used to measure the pressure of a gas supply.



Which change gives a greater value of height h ?

- A using a less dense liquid
 - B using a more dense liquid
 - C using a narrower tube
 - D using a wider tube
- 13 A washbasin has an exit pipe covered with a plug of area 12 cm^2 . A chain is attached to the centre of the plug to assist in pulling the plug away from the exit hole. The washbasin contains water to a depth of 0.080 m .

The density of the water is 1000 kg/m^3 .

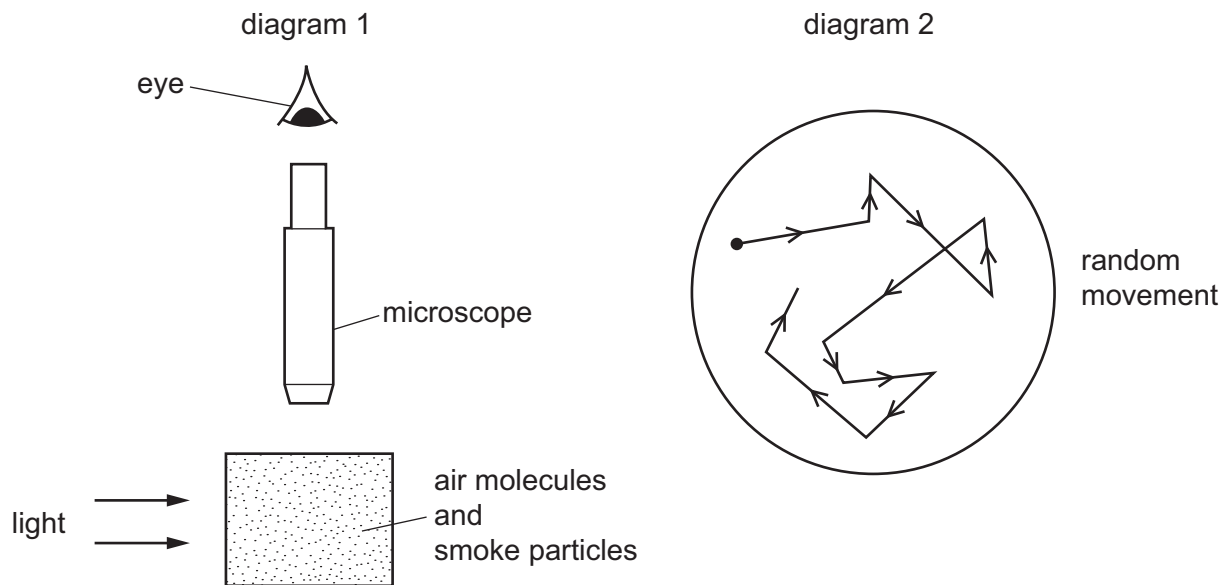


What is the force acting on the plug due to the water?

- A 0.96 N
- B 800 N
- C 9600 N
- D 80000 N

- 14** Diagram 1 shows apparatus being used to observe smoke particles.

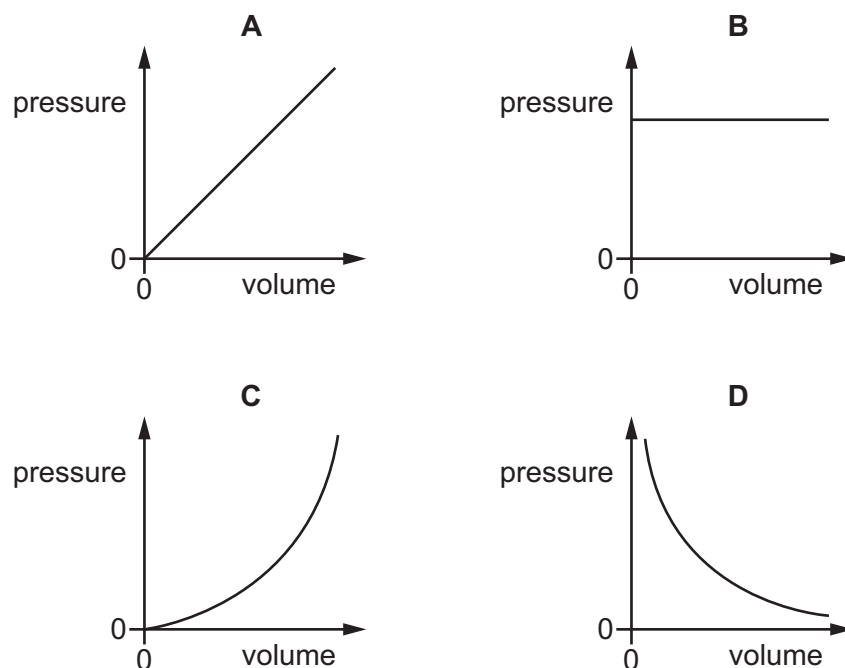
Diagram 2 shows how a smoke particle moves randomly.



Why do the smoke particles move randomly?

- A** They are hit by air molecules.
 - B** They are less dense than air.
 - C** They are moved by convection currents.
 - D** They gain energy from the light.
- 15** The gas in a sealed container is compressed at constant temperature.

Which graph shows how the pressure of the gas changes with its volume?



- 16 The distance between two electricity pylons is 60 m. An engineer fits a cable of length 62 m between the pylons.

Why does the engineer choose a cable that is longer than the distance between the two pylons?

- A to allow for contraction of the cable in cold weather
- B to create a slope in the cable for electrons to flow down
- C to keep the current low and the voltage high
- D to reduce magnetic fields around the cable

- 17 The diagram shows a liquid-in-glass thermometer.



Which change increases the sensitivity of the thermometer?

- A a narrower capillary tube
 - B a wider capillary tube
 - C thicker glass around the bulb
 - D thinner glass around the bulb
- 18 An object of mass 800 g and specific heat capacity $250 \text{ J/(kg } ^\circ\text{C)}$ is heated. It absorbs 5300 J of energy.

What is the increase in temperature of the object?

- A 0.027°C
 - B 17°C
 - C 27°C
 - D $17\,000^\circ\text{C}$
- 19 On a cold day, a shiny metal rod feels colder to the touch than a black plastic rod.

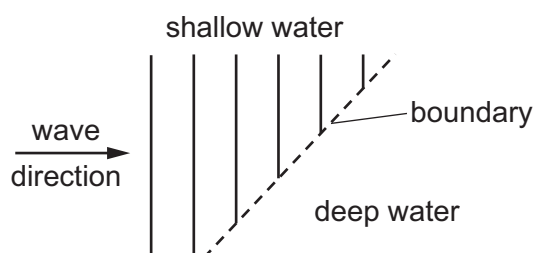
Which statement explains this observation?

- A The metal rod is a better absorber of infra-red radiation than the plastic rod.
- B The metal rod is a better thermal conductor than the plastic rod.
- C The metal rod is a worse absorber of infra-red radiation than the plastic rod.
- D The metal rod is a worse thermal conductor than the plastic rod.

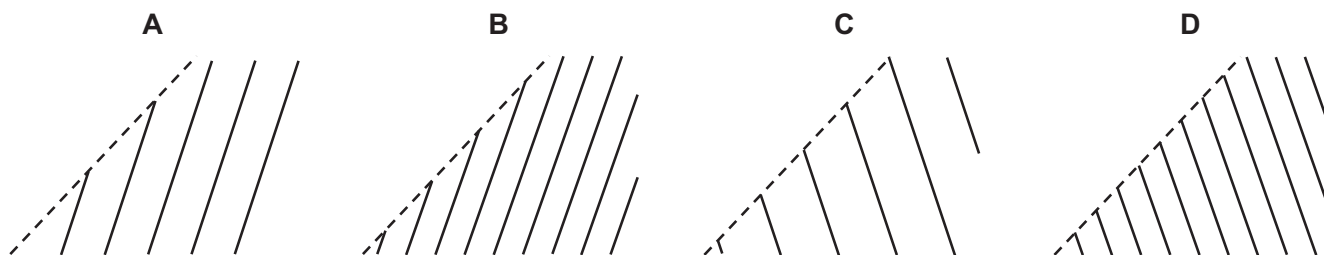
- 20** One end of a copper bar is heated to a high temperature.

Which mechanism is responsible for the transfer of thermal energy to the other end of the copper bar?

- A** the lattice vibrations of copper ions only
 - B** the lattice vibrations of copper ions and the movement of high energy electrons along the bar
 - C** the movement of high energy copper ions along the bar
 - D** the movement of high energy electrons along the bar only
- 21** Plane water waves travel from a shallow region into a deeper region. They travel more quickly in the deeper water.



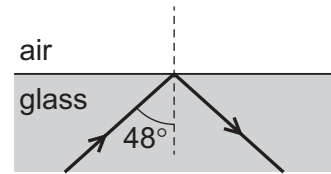
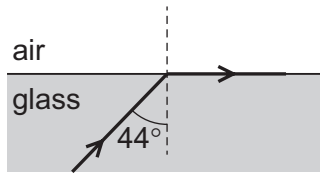
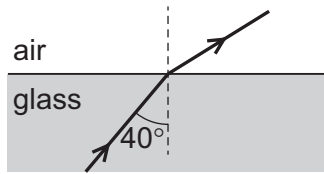
Which diagram shows the wave pattern in the deeper water?



- 22** Which statement about the image formed by a plane mirror is correct?

- A** The image is larger than the object.
- B** The image is smaller than the object.
- C** The image is twice as far from the mirror as the object.
- D** The image is virtual.

- 23** A ray of light is incident on a glass-air surface. The diagrams show the ray of light at different angles of incidence in the glass.



What is the refractive index of the glass?

- A** 1.35 **B** 1.44 **C** 1.50 **D** 1.55

- 24** Different parts of the electromagnetic spectrum are used for different purposes. Below are four statements about parts of the spectrum.

statement 1: Infra-red waves are used in television remote controllers.

statement 2: Radio waves are used to transmit television pictures from satellites to Earth.

statement 3: Ultraviolet waves are used for intruder alarms.

statement 4: X-rays are used for security checks.

Which statements are correct?

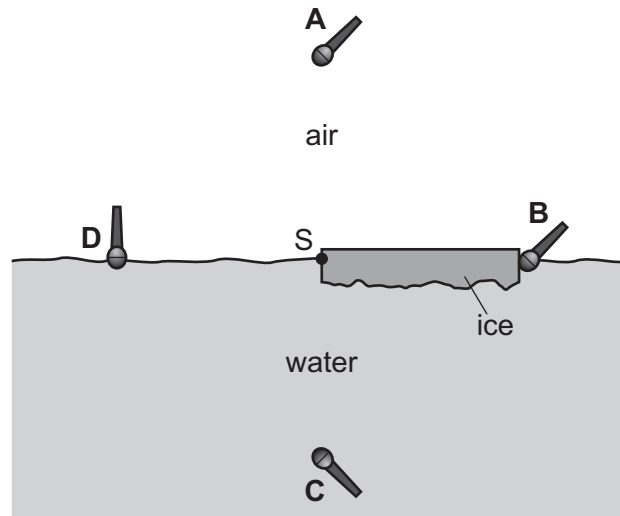
- A** 1 and 2 **B** 1 and 4 **C** 2 and 3 **D** 3 and 4

- 25** Iron is used for the core of a transformer and steel is used to make a bar magnet.

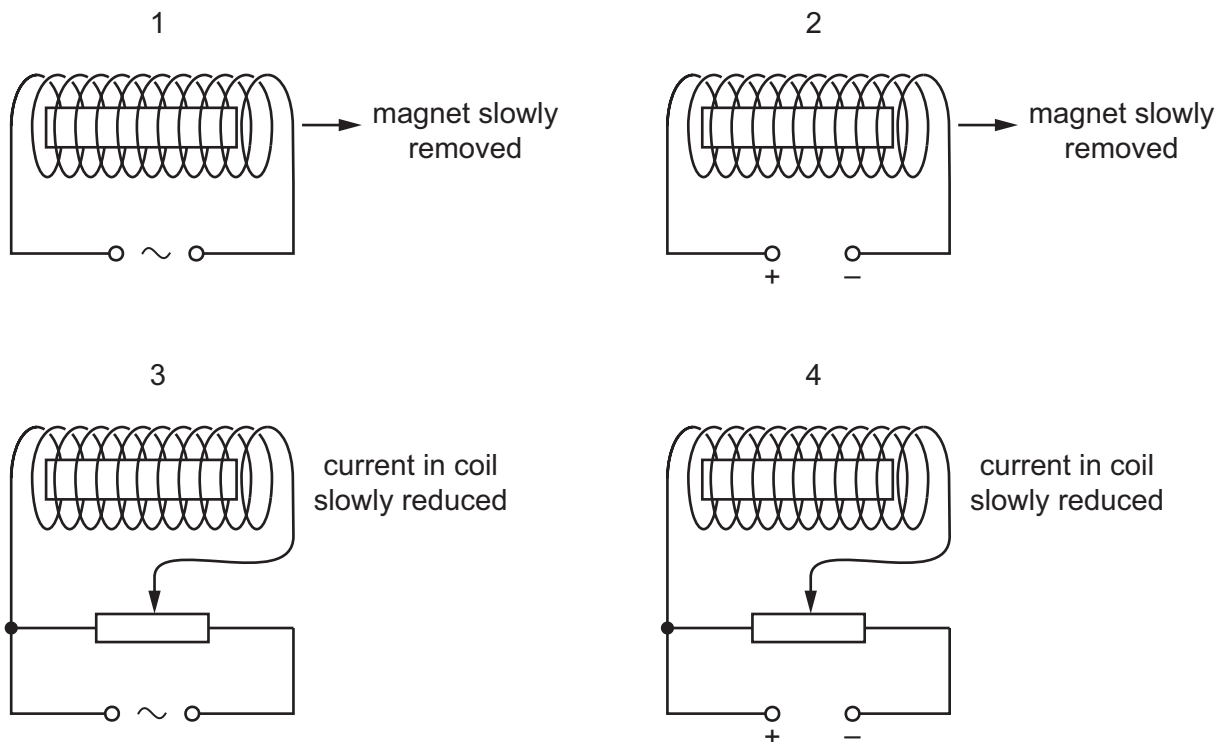
Which statement explains these uses of iron and of steel?

- A** Iron is a magnetic material and steel is a non-magnetic material.
B Iron is a permanent magnetic material and steel is a temporary magnetic material.
C Iron is a temporary magnetic material and steel is a permanent magnetic material.
D Iron is a non-magnetic material and steel is a magnetic material.

- 26** A sheet of ice floats on water. A source of sound S is positioned at the edge of the ice sheet. Four microphones are placed equal distances from S.
- Four microphones are placed equal distances from S.
- Which microphone detects the sound from S first?



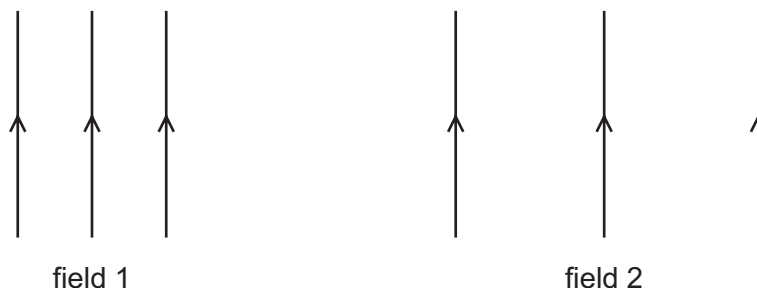
- 27** A bar magnet is placed inside a current-carrying coil. The diagram shows four different experiments.



In which experiments is the magnet demagnetised?

- A** 1 and 2 **B** 1 and 3 **C** 2 and 4 **D** 3 and 4

28 The diagrams represent two different electric fields.



A single electron is located in each of the fields.

Which row gives the correct direction of the force on the electron and the field in which there is a larger force on it?

	direction of the force	larger force
A	↓	field 1
B	↓	field 2
C	↑	field 1
D	↑	field 2

29 The resistance of a component in a circuit is found using an ammeter and a voltmeter.

How are the ammeter and the voltmeter connected?

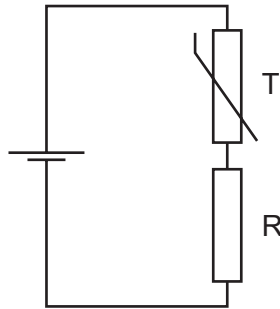
- A** the voltmeter and ammeter in parallel with the component
- B** the voltmeter and ammeter in series with the component
- C** the voltmeter in parallel with the component and the ammeter in series with the component
- D** the voltmeter in series with the component and the ammeter in parallel with the component

30 The table gives data for four different electrical devices.

Which device develops the greatest power?

	device	voltage	current
A	car headlight	12 V	3.0 A
B	cooling fan	110 V	0.40 A
C	electric spark generator	400 kV	0.10 mA
D	mains lamp	240 V	0.20 A

- 31** The circuit diagram shows a fixed resistor R and a thermistor T used in a potential divider circuit.



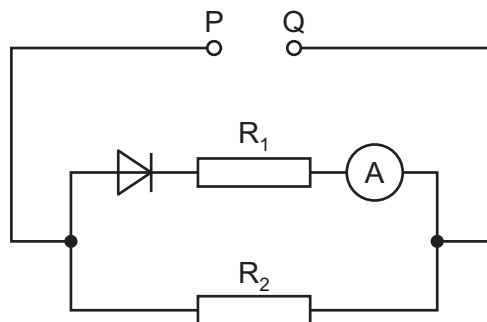
V_R and V_T are the potential differences across R and T respectively.

What happens to V_R and to V_T as the temperature of the thermistor decreases?

	V_R	V_T
A	decreases	increases
B	increases	decreases
C	stays the same	decreases
D	stays the same	increases

- 32** The circuit diagram shows a power supply connected to some circuit components.

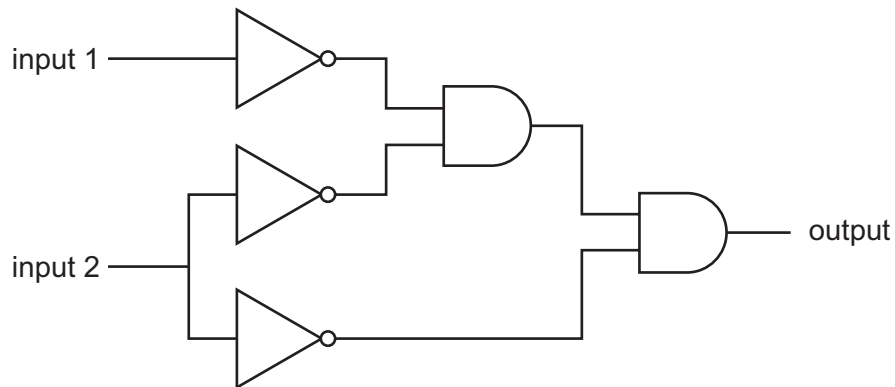
In the diagram, P and Q are the terminals of the d.c. power supply.



Under which circumstances does the ammeter show a reading other than zero?

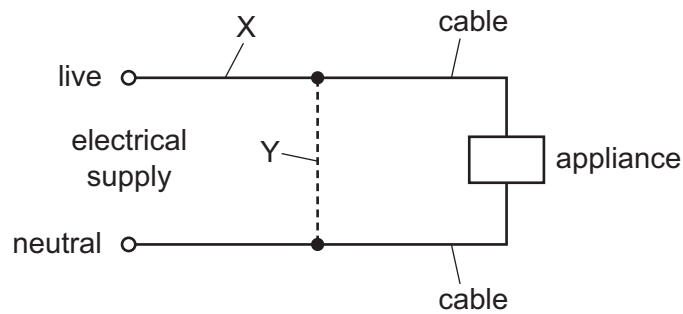
- A** when P is positive or negative
- B** the ammeter always shows a zero reading
- C** only when P is negative
- D** only when P is positive

- 33 The diagram shows a digital circuit with two inputs and one output.



What describes the action of the circuit?

- A** AND **B** NAND **C** NOR **D** OR
- 34 Either a fuse or a circuit-breaker can be used to protect electrical cables from large currents that could cause overheating.



When a fuse is used, where should it be connected, and when a circuit-breaker is used, where should it be connected?

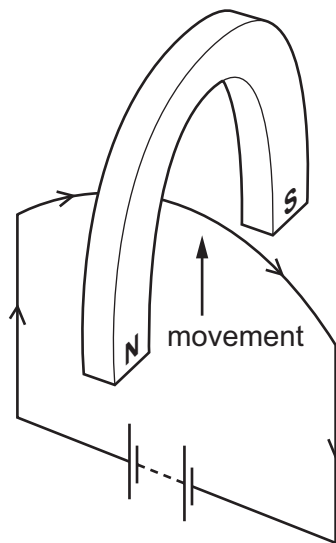
	position of fuse	position of circuit-breaker
A	X	X
B	X	Y
C	Y	X
D	Y	Y

- 35** There is a current of 3.0 A in a conductor.

The current is changed to 6.0 A in the opposite direction.

Which effect does this have on the magnetic field around the conductor?

- A** It decreases and it acts in the opposite direction.
 - B** It decreases and it acts in the same direction.
 - C** It increases and it acts in the opposite direction.
 - D** It increases and it acts in the same direction.
- 36** A wire is placed in a strong magnetic field. When a current is passed through the wire it moves upwards, as shown.

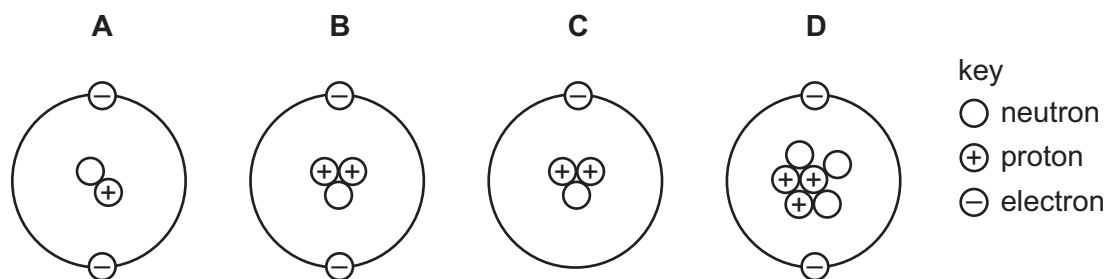


The current is reversed.

In which direction does the wire move?

- A** downwards
- B** towards the north pole
- C** towards the south pole
- D** upwards

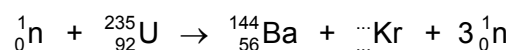
37 Which diagram represents the structure of a neutral atom?



38 When a uranium-235 nucleus absorbs a neutron, it becomes unstable and undergoes fission.

The fission process produces a barium (Ba) nucleus, a krypton (Kr) nucleus and 3 neutrons.

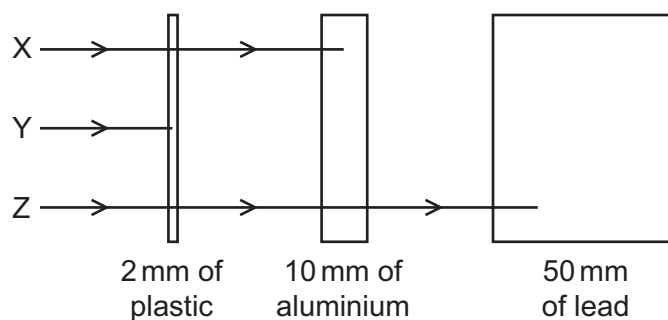
The fission process is represented by the nuclear equation shown.



Which symbol represents the resulting krypton nucleus?

- A** ${}_{36}^{89}\text{Kr}$ **B** ${}_{34}^{91}\text{Kr}$ **C** ${}_{35}^{91}\text{Kr}$ **D** ${}_{36}^{91}\text{Kr}$

39 The diagram shows the paths of three different types of radiation X, Y and Z.



Which row correctly identifies X, Y and Z?

	X	Y	Z
A	α-particles	β-particles	γ-rays
B	β-particles	α-particles	γ-rays
C	β-particles	γ-rays	α-particles
D	γ-rays	α-particles	β-particles

- 40** A scientist measures the count rate of a radioactive sample in a laboratory over a period of 12 weeks.

The background radiation count rate in the laboratory remains constant at 20 counts per minute.

The table shows the scientist's results before the background radiation count rate is taken into account.

time /weeks	count rate /counts per minute
0	100
2	80
4	65
6	54
8	45
10	39
12	34

In which range does the half-life of the radioactive isotope lie?

- A** between 4 and 6 weeks
- B** between 6 and 8 weeks
- C** between 8 and 10 weeks
- D** more than 12 weeks

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Cambridge Assessment International Education
Cambridge International General Certificate of Secondary Education (9–1)

PHYSICS

0972/21

Paper 2 Multiple Choice

October/November 2018

MARK SCHEME

Maximum Mark: 40

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

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This document consists of **3** printed pages.

Question	Answer	Marks
1	B	1
2	C	1
3	B	1
4	D	1
5	C	1
6	D	1
7	A	1
8	D	1
9	C	1
10	B	1
11	C	1
12	A	1
13	A	1
14	A	1
15	D	1
16	A	1
17	A	1
18	C	1
19	B	1
20	B	1
21	C	1
22	D	1
23	B	1
24	B	1
25	C	1
26	B	1
27	B	1
28	A	1

Question	Answer	Marks
29	C	1
30	D	1
31	A	1
32	D	1
33	C	1
34	A	1
35	C	1
36	A	1
37	B	1
38	A	1
39	B	1
40	A	1



Cambridge Assessment International Education
Cambridge International General Certificate of Secondary Education (9–1)

PHYSICS

Paper 2 Multiple Choice (Extended)

0972/21

May/June 2019

45 minutes

Additional Materials: Multiple Choice Answer Sheet
Soft clean eraser
Soft pencil (type B or HB recommended)

READ THESE INSTRUCTIONS FIRST

Write in soft pencil.

Do not use staples, paper clips, glue or correction fluid.

Write your name, centre number and candidate number on the Answer Sheet in the spaces provided unless this has been done for you.

DO NOT WRITE IN ANY BARCODES.

There are **forty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A, B, C** and **D**.

Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

Read the instructions on the Answer Sheet very carefully.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.

Any rough working should be done in this booklet.

Electronic calculators may be used.

Take the weight of 1.0 kg to be 10 N (acceleration of free fall = 10 m/s^2).

This document consists of **16** printed pages.

1 Which quantity can be measured directly using a micrometer screw gauge?

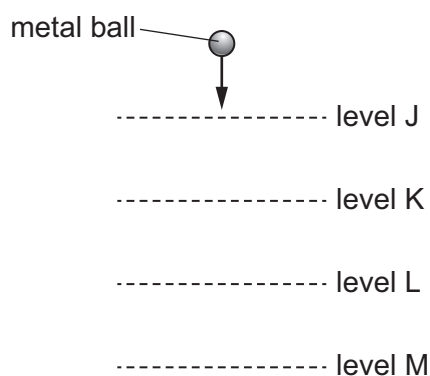
- A the area of a sheet of paper
- B the mass of a sheet of paper
- C the thickness of a sheet of paper
- D the volume of a sheet of paper

2 The velocity of an object increases from 30 m/s to 50 m/s in 5.0 seconds.

What is the average acceleration of the object?

- A 0.10 m/s^2 B 0.25 m/s^2 C 4.0 m/s^2 D 10 m/s^2

3 A heavy metal ball falls vertically downwards through air past four equally spaced levels J, K, L and M.



The times taken to fall from one level to the next are measured.

Where is the speed of the ball greatest and which time is shortest?

	speed is greatest between	time is shortest between
A	J and K	J and K
B	J and K	L and M
C	L and M	J and K
D	L and M	L and M

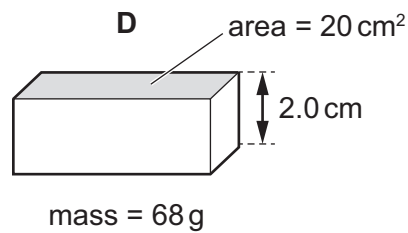
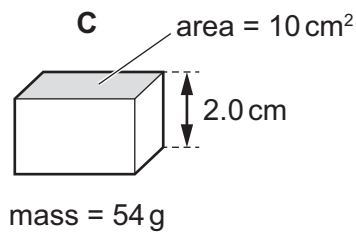
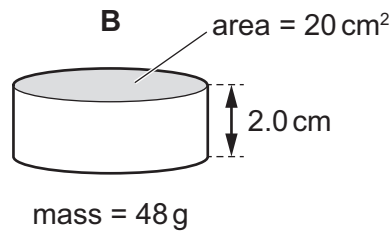
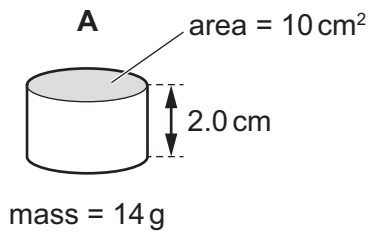
- 4 A body is moved from place X to place Y where the gravitational field strength is different.

What happens to its mass and to its weight due to the move?

	mass	weight
A	changes	changes
B	changes	stays the same
C	stays the same	changes
D	stays the same	stays the same

- 5 The diagrams show four solid blocks with their dimensions and masses.

Which block has the greatest density?



- 6 A student wishes to determine the spring constant of a spring where it obeys Hooke's law.

Different loads are hung from the spring and its length is measured for each different load.

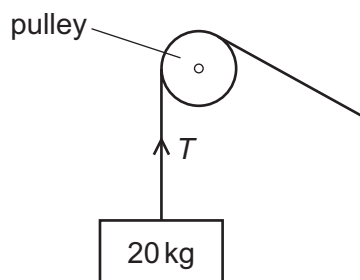
The table shows the results of the experiment.

weight of load / N	0	2.0	4.0	5.0
length of spring / cm	12	20	28	38

What is the value of the spring constant of the spring?

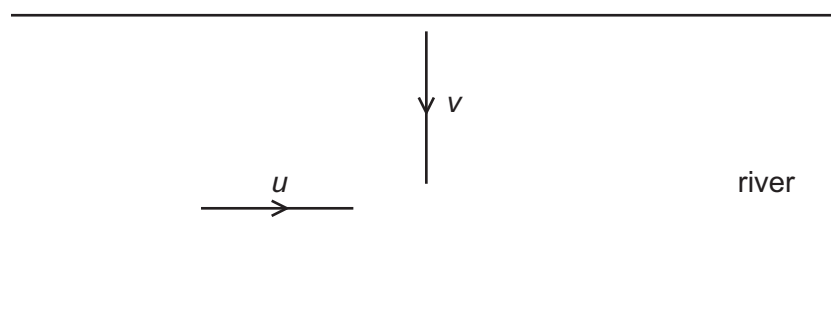
- A** 0.13 N/cm **B** 0.14 N/cm **C** 0.19 N/cm **D** 0.25 N/cm

- 7 A mass of 20 kg is held stationary by a rope passing over a frictionless pulley.

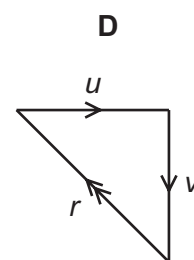
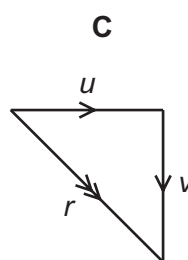
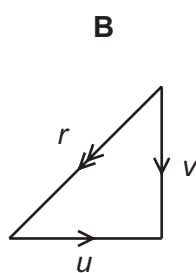
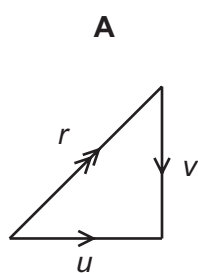


What is the tension T in the rope?

- A** 10 kg **B** 20 kg **C** 100 N **D** 200 N
- 8 A boat starts moving across a river at velocity v perpendicular to the river bank. The boat encounters a current along the river of velocity u , as shown.

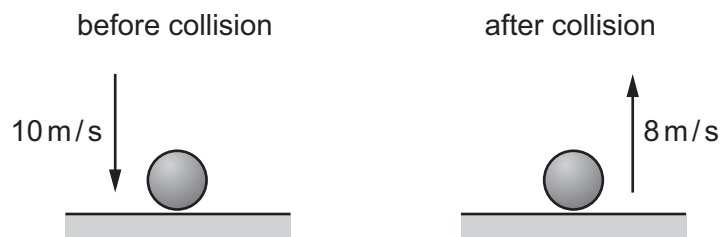


Which vector diagram shows the resultant velocity r of the boat?



- 9 A ball of mass 0.50 kg falls and hits the floor at 10 m/s.

It rebounds at speed 8.0 m/s, as shown.



The collision between the ball and the floor lasts for 0.50 s.

What is the average force acting on the ball during the collision?

- A 2.0 N upwards
 - B 2.0 N downwards
 - C 18 N upwards
 - D 18 N downwards
- 10 An object falls from the top of a building that is 25 m high. Air resistance is negligible.

What is the speed of the object when it hits the ground?

- A 10 m/s
- B 22 m/s
- C 25 m/s
- D 625 m/s

- 11 A machine is very efficient.

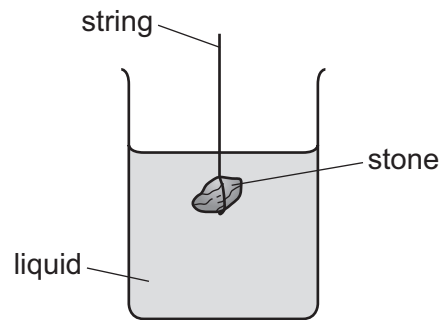
What does this mean?

- A It produces a large amount of power.
 - B It uses very little energy.
 - C It wastes very little energy.
 - D It works very quickly.
- 12 A crane takes 2.0 minutes to lift a load to the top of a building. The change in gravitational potential energy of the load is 360 kJ.

What is the useful power output of the crane?

- A 3.0 kW
- B 180 kW
- C 720 kW
- D 43 200 kW

- 13** The diagram shows a stone suspended on a string under the surface of a liquid. The stone experiences a pressure caused by the liquid.



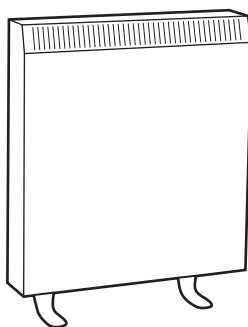
What would increase the pressure on the stone?

- A** decreasing the surface area of the stone
 - B** increasing the mass of the stone
 - C** lowering the stone deeper into the liquid
 - D** using a liquid with a lower density
- 14** A stationary smoke particle is hit by a fast-moving nitrogen molecule.

Which row describes the motion of the smoke particle and of the nitrogen molecule after the collision?

	smoke particle	nitrogen molecule
A	moves	rebounds
B	moves	stops
C	remains stationary	rebounds
D	remains stationary	stops

- 15 A night storage heater contains a large block of material that is heated electrically during the night. During the day the block cools down, releasing thermal energy into the room.



Which thermal capacity and which night-time temperature increase will cause the most energy to be stored by the block?

	thermal capacity of block	night-time temperature increase
A	large	large
B	large	small
C	small	large
D	small	small

- 16 100 g of water at 25 °C is poured into an insulating cup. 50 g of ice at 0 °C is added to the water. The water is stirred until the temperature of the water has fallen to 0 °C.

18 g of ice remains unmelted.

The specific heat capacity of water is 4.2 J/g °C.

Which value does this experiment give for the specific latent heat of fusion of ice?

- A** 210 J/g **B** 330 J/g **C** 580 J/g **D** 770 J/g

- 17 In which does thermal conduction **not** occur?

- A** a gas
B a liquid
C a solid
D a vacuum

- 18** The metal surface of a kettle is hot.

What happens to the cool air outside the kettle when it comes into contact with the hot kettle?

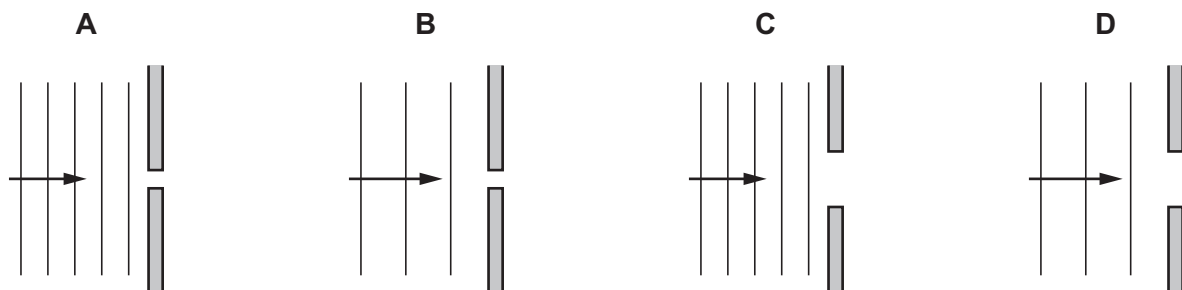
- A** The density of the air decreases and the air falls.
 - B** The density of the air decreases and the air rises.
 - C** The density of the air increases and the air falls.
 - D** The density of the air increases and the air rises.
- 19** Some hot water is sealed inside a metal can. The can is in a vacuum in outer space. The hot water slowly cools down.

How does the thermal energy escape into space?

- A** by conduction then convection
 - B** by conduction then radiation
 - C** by evaporation then convection
 - D** by evaporation then radiation
- 20** When water waves pass through a gap they diffract.

The diagrams show wavefronts approaching a narrow gap.

In which diagram will the diffraction be least?

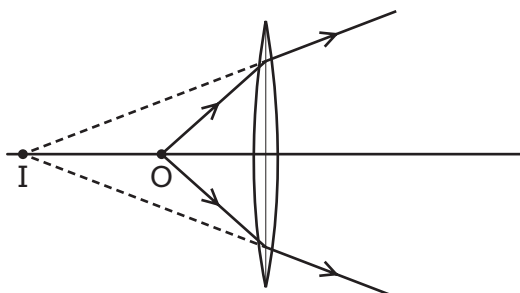


- 21** An object is placed 30 cm in front of a plane mirror.

Which statement describes the image of the object?

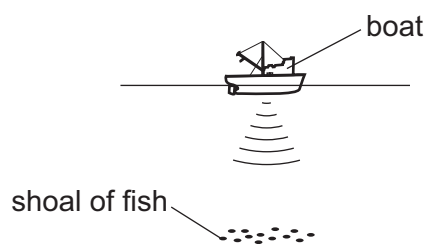
- A** The image is the same size and 30 cm from the object.
- B** The image is the same size and 60 cm from the object.
- C** The image is smaller and 30 cm from the object.
- D** The image is smaller and 60 cm from the object.

- 22** A small object O is placed near a converging lens, as shown. The lens forms an image I.



Which statement is correct?

- A** The image I is diminished.
 - B** The image I is inverted.
 - C** The image I is real.
 - D** The object O is closer to the lens than its principal focus.
- 23** An eclipse of the Sun happens when the Moon comes between the Earth and the Sun.
- Which statement is correct?
- A** Infra-red radiation from the Sun disappears before visible light and ultra-violet radiation.
 - B** Ultra-violet radiation from the Sun disappears before visible light and infra-red radiation.
 - C** Visible light from the Sun disappears before ultra-violet radiation and infra-red radiation.
 - D** Infra-red radiation, ultra-violet radiation and visible light from the Sun all disappear at the same moment.
- 24** A pulse of sound is produced at the bottom of a boat. The sound travels through the water and is reflected from a shoal of fish. The sound reaches the boat again after 1.2 s. The speed of sound in the water is 1500 m/s.



How far below the bottom of the boat is the shoal of fish?

- A** 450 m
- B** 900 m
- C** 1800 m
- D** 3600 m

25 What is the approximate value of the speed of sound in air at normal temperature?

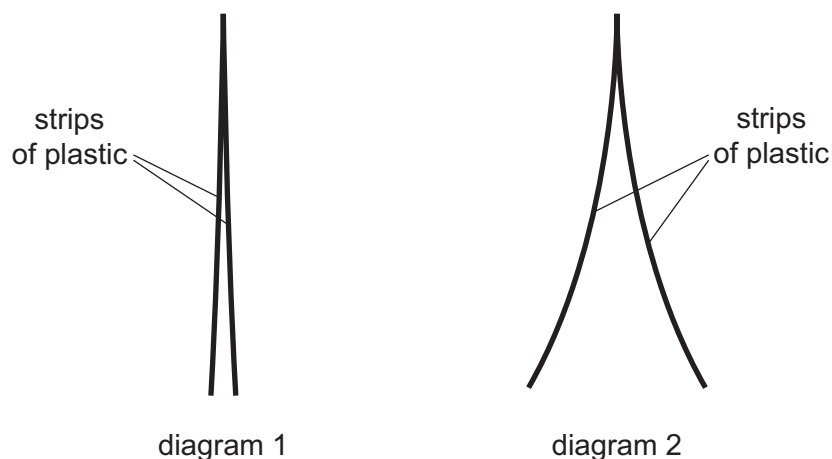
- A** 340 m/s **B** 34 000 m/s **C** 340 km/s **D** 3.0×10^8 m/s

26 Why is soft iron used for the core of an electromagnet?

- A** Soft iron easily becomes a permanent magnet.
B Soft iron is a good electrical conductor.
C Soft iron is a poor thermal conductor.
D Soft iron loses its magnetism when the current in the coil is switched off.

27 Diagram 1 shows two thin, uncharged strips of plastic.

Diagram 2 shows the same strips after they have been rubbed with a dry cloth.



Which row describes the charge on the strips after rubbing, and the force between the strips after rubbing?

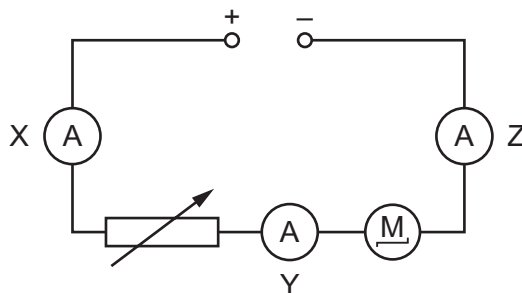
	charge on strips	force between strips
A	opposite	attraction
B	opposite	repulsion
C	the same	attraction
D	the same	repulsion

- 28 A cell has an electromotive force (e.m.f.) of 1.5 V.

What does this statement mean?

- A The cell converts 1.0 J of energy when driving 1.5 C of charge round a complete circuit.
 - B The cell converts 1.5 J of energy when driving 1.0 C of charge round a complete circuit.
 - C The cell converts 1.5 J of energy per second when driving 1.0 C of charge round a complete circuit.
 - D The cell converts 1.5 W of power when driving 1.0 C of charge round a complete circuit.
- 29 Which two changes to a metal wire both increase resistance?
- A decreasing its length and increasing its temperature
 - B increasing its length and decreasing its temperature
 - C decreasing its thickness and increasing its temperature
 - D increasing its thickness and decreasing its temperature
- 30 The diagram shows a circuit containing a d.c. power supply, a motor and a variable resistor.

Three ammeters X, Y and Z show the current in different parts of the circuit.



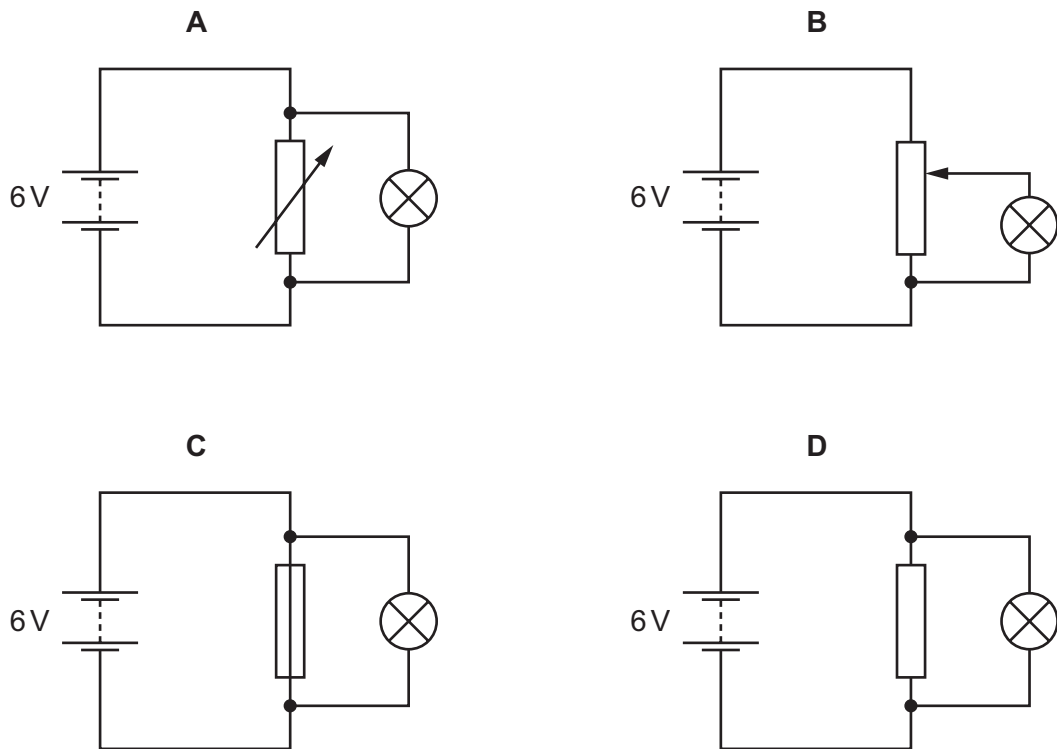
The reading on X is 4.0 A.

Which statement is correct?

- A The readings on Y and Z are both less than 4.0 A.
- B The readings on Y and Z are both equal to 4.0 A.
- C The readings on Y and Z are both greater than 4.0 A.
- D The reading on Z is zero.

- 31** A lamp is to be connected in a circuit so that the potential difference (p.d.) across it can be varied from 0 to 6 V.

Which circuit would be most suitable?

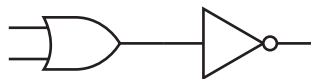


- 32** Which logic gate is represented by the symbol shown?



- A** AND **B** NAND **C** NOR **D** OR

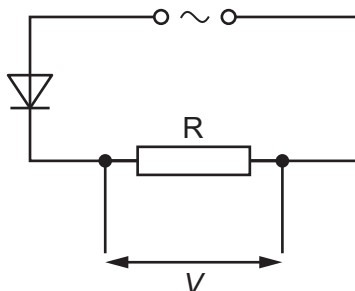
- 33** The diagram shows a combination of logic gates.



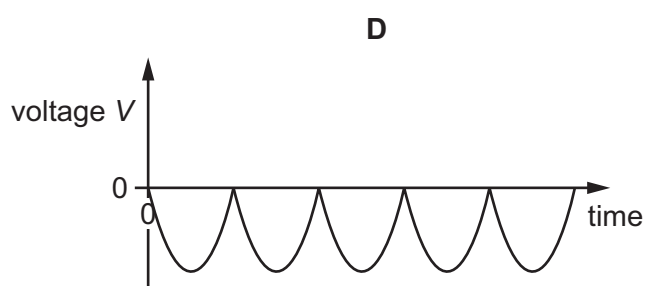
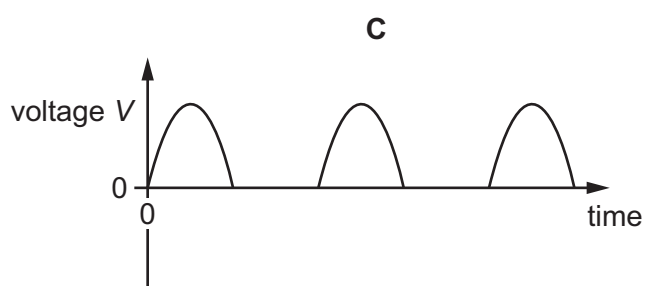
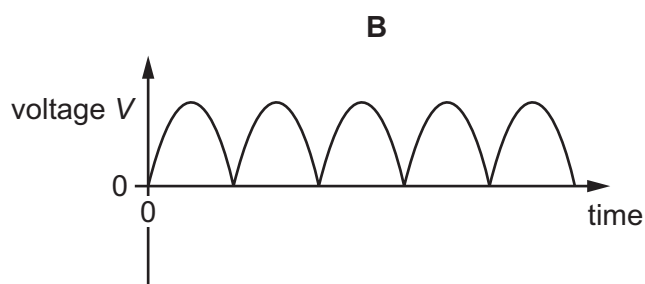
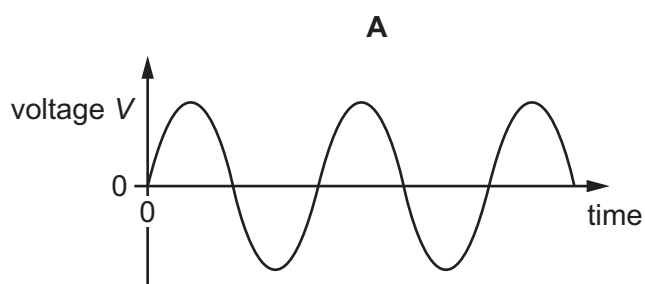
Which single logic gate is equivalent to this combination?

- A** AND **B** NOR **C** NOT **D** OR

- 34** An alternating current (a.c.) power supply is connected in series with a resistor R and a diode.



Which graph shows how the voltage V across the resistor R varies with time?



- 35** An electric heater is plugged into the mains supply using a fused plug.

The current in the heater is 10 A.

The cable attached to the heater is rated at 15 A.

The fuses available are rated at 1 A, 3 A, 5 A and 13 A.

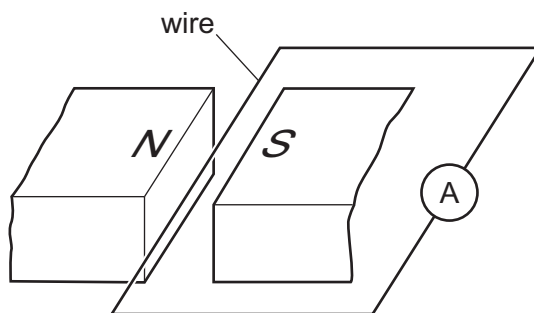
Which fuse should be used?

- A** 1 A **B** 3 A **C** 5 A **D** 13 A

- 36** What is the purpose of the commutator in a d.c. electric motor?

- A** to control the speed at which the motor rotates
B to reverse the direction of the voltage across the power supply every half-turn
C to reverse the direction of the current in the motor coil every half-turn
D to reverse the direction of spin of the motor coil every half-turn

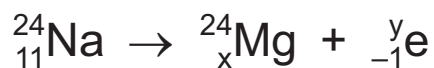
- 37 The diagram shows a wire between two magnetic poles. The wire is connected in a circuit with an ammeter.



The wire is moved downwards, towards the bottom of the page. A current is induced in the wire.

In which direction is the force on the wire caused by this current?

- A towards the bottom of the page
 - B towards the left of the page
 - C towards the right of the page
 - D towards the top of the page
- 38 The chemical symbol for sodium is Na. The equation represents the radioactive decay of sodium-24.



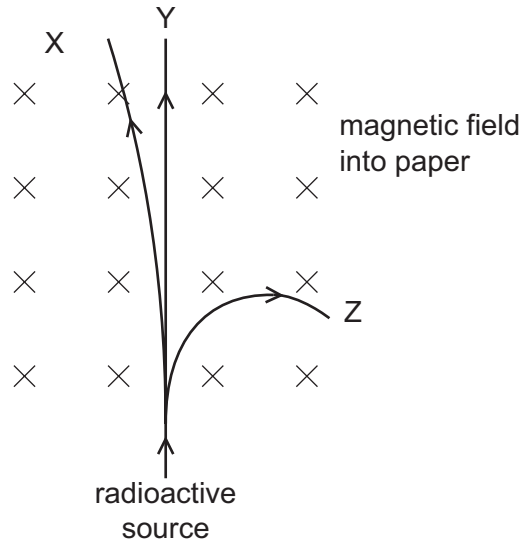
What are the numbers x and y?

	x	y
A	10	0
B	10	1
C	12	0
D	12	1

- 39 A radioactive source emits α -particles, β -particles and γ -rays into a vacuum where there is a magnetic field.

The magnetic field acts perpendicularly into the plane of the paper.

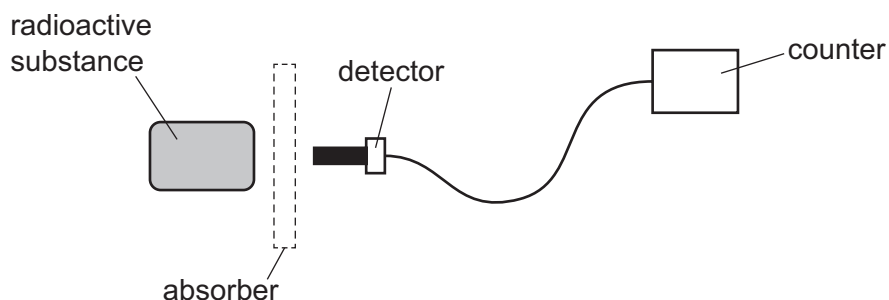
The paths X, Y and Z of the three types of radiation through the magnetic field are shown.



Which radiation follows path X, path Y and path Z?

	X	Y	Z
A	α -particles	β -particles	γ -rays
B	α -particles	γ -rays	β -particles
C	β -particles	α -particles	γ -rays
D	β -particles	γ -rays	α -particles

- 40 A student measures the level of radiation emitted from a radioactive substance. He places a detector very close to the substance. He puts different absorbers between the radioactive substance and the detector.



The student's results are shown. These results are corrected for background radiation.

absorber	counter reading counts per minute
none	95
thin paper	52
few mm of aluminium	52
several cm of lead	12

Which types of radiation are being emitted by the substance?

- A α -particles and β -particles only
- B α -particles and γ -rays only
- C β -particles and γ -rays only
- D α -particles, β -particles and γ -rays

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Cambridge Assessment International Education
Cambridge International General Certificate of Secondary Education (9–1)

PHYSICS (9–1)

0972/21

Paper 2 Multiple Choice (Extended)

May/June 2019

MARK SCHEME

Maximum Mark: 40

Published

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Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

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This document consists of **3** printed pages.

Question	Answer	Marks
1	C	1
2	C	1
3	D	1
4	C	1
5	C	1
6	D	1
7	D	1
8	C	1
9	C	1
10	B	1
11	C	1
12	A	1
13	C	1
14	A	1
15	A	1
16	B	1
17	D	1
18	B	1
19	B	1
20	C	1
21	B	1
22	D	1
23	D	1
24	B	1
25	A	1
26	D	1
27	D	1
28	B	1

Question	Answer	Marks
29	C	1
30	B	1
31	B	1
32	A	1
33	B	1
34	C	1
35	D	1
36	C	1
37	D	1
38	C	1
39	B	1
40	B	1



Cambridge Assessment International Education
Cambridge International General Certificate of Secondary Education (9–1)

PHYSICS

0972/22

Paper 2 Multiple Choice (Extended)

May/June 2019

45 minutes

Additional Materials: Multiple Choice Answer Sheet
Soft clean eraser
Soft pencil (type B or HB recommended)

READ THESE INSTRUCTIONS FIRST

Write in soft pencil.

Do not use staples, paper clips, glue or correction fluid.

Write your name, centre number and candidate number on the Answer Sheet in the spaces provided unless this has been done for you.

DO NOT WRITE IN ANY BARCODES.

There are **forty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A, B, C** and **D**.

Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

Read the instructions on the Answer Sheet very carefully.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.

Any rough working should be done in this booklet.

Electronic calculators may be used.

Take the weight of 1.0 kg to be 10 N (acceleration of free fall = 10 m/s^2).

This document consists of **16** printed pages.

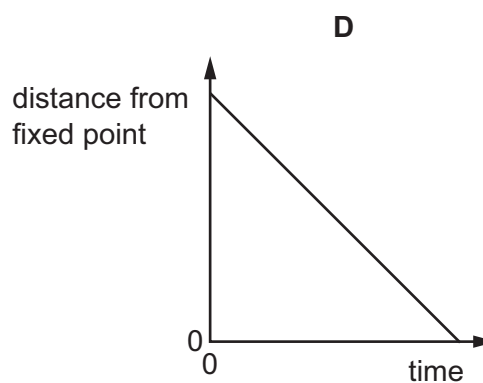
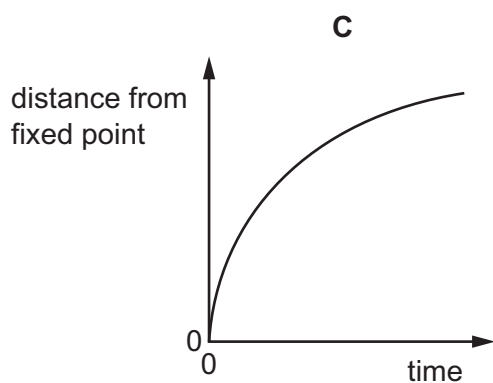
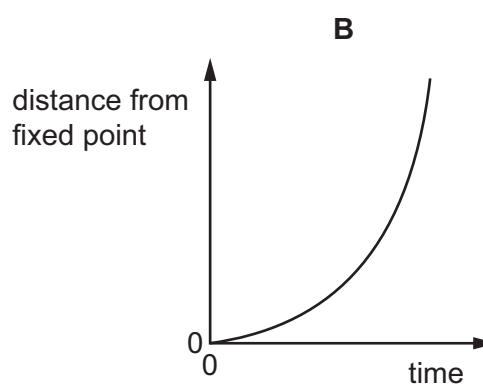
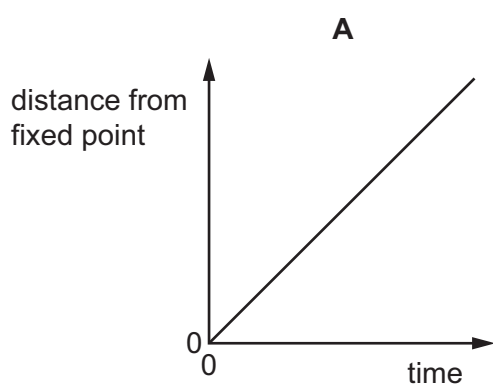
1 Which quantity can be measured directly using a micrometer screw gauge?

- A the area of a sheet of paper
- B the mass of a sheet of paper
- C the thickness of a sheet of paper
- D the volume of a sheet of paper

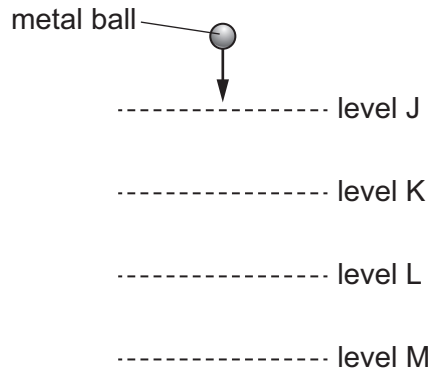
2 Four objects are moving along a straight line.

The distance of an object from a fixed point on the line is plotted against time for each object.

Which object is decelerating?



- 3 A heavy metal ball falls vertically downwards through air past four equally spaced levels J, K, L and M.



The times taken to fall from one level to the next are measured.

Where is the speed of the ball greatest and which time is shortest?

	speed is greatest between	time is shortest between
A	J and K	J and K
B	J and K	L and M
C	L and M	J and K
D	L and M	L and M

- 4 Four students make statements about the mass of an object.

Which statement is correct?

- A** The mass of an object depends on the gravitational field which acts on the object.
- B** The mass of an object divided by its weight is equal to the acceleration with which it falls freely.
- C** The mass of an object increases when the temperature of the object increases.
- D** The mass of an object resists change in motion of the object.
- 5 A metal has a density of 8.0 g/cm^3 . A solid cube of mass 1.0 kg is made from this metal.

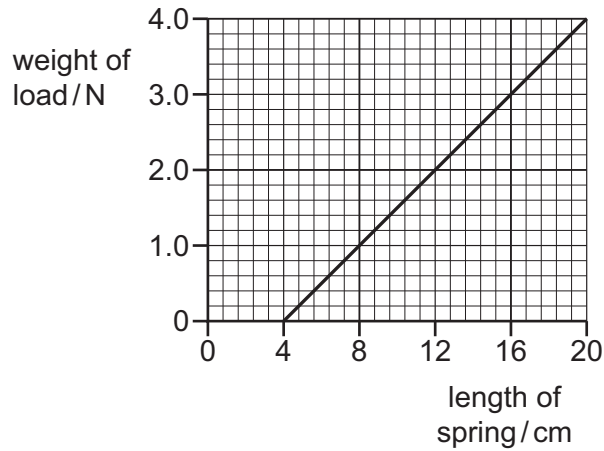
How long is each side of the cube?

- A** 0.50 cm **B** 2.0 cm **C** 5.0 cm **D** 42 cm

- 6 An experiment is carried out to determine the spring constant for a spring.

Different loads are hung from the spring and its length is measured for each different load.

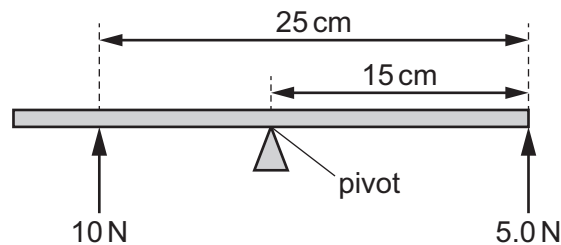
The graph shows how its length varies with load.



What is the value of the spring constant?

- A** 0.20 N/cm **B** 0.25 N/cm **C** 4.0 N/cm **D** 5.0 N/cm
- 7 A beam is pivoted at its centre of mass.

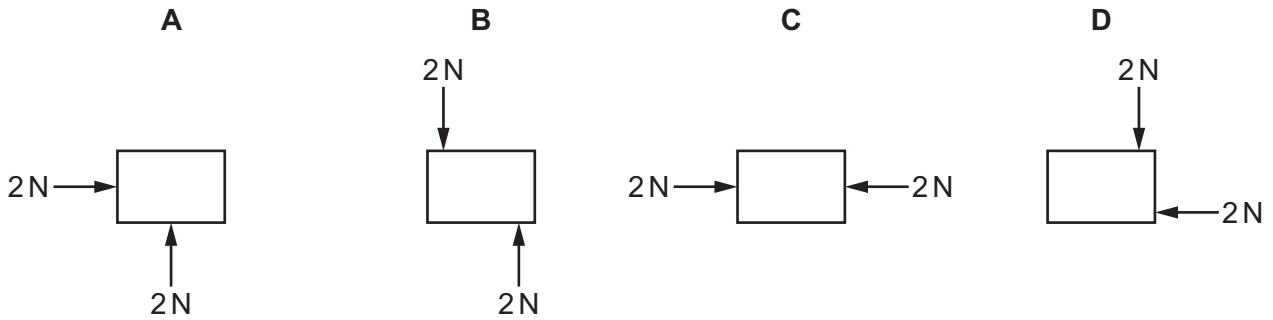
It is acted upon by two forces, 10 N and 5.0 N, as shown.



What is the resultant moment about the pivot?

- A** 25 N cm anticlockwise
B 25 N cm clockwise
C 175 N cm anticlockwise
D 175 N cm clockwise

8 Which object is in equilibrium?



9 An object of mass 4.0 kg is moving with a velocity of 3.0 m/s in a straight line.

What is the momentum of the object?

- A** 0.75 kg m/s **B** 1.3 kg m/s **C** 12 kg m/s **D** 24 kg m/s

10 A machine is very efficient.

What does this mean?

- A** It produces a large amount of power.
B It uses very little energy.
C It wastes very little energy.
D It works very quickly.

11 A force F acts on a body and the body moves a distance d in the direction of the force.

Which expression for the work done W is correct?

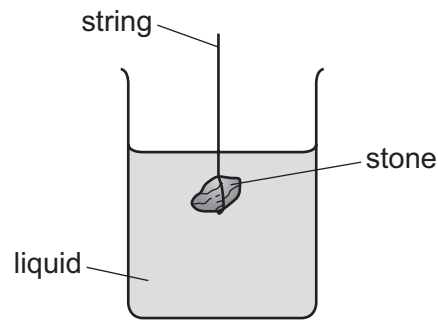
- A** $W = Fd$ **B** $W = Fd^2$ **C** $W = \frac{1}{2}Fd$ **D** $W = \frac{1}{2}Fd^2$

12 A boy produces an average power output of 60 W as he rides his bicycle for 2.0 minutes.

How much useful energy does he expend?

- A** 0.50 J **B** 30 J **C** 120 J **D** 7200 J

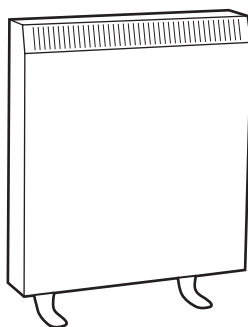
- 13** The diagram shows a stone suspended on a string under the surface of a liquid. The stone experiences a pressure caused by the liquid.



What would increase the pressure on the stone?

- A** decreasing the surface area of the stone
 - B** increasing the mass of the stone
 - C** lowering the stone deeper into the liquid
 - D** using a liquid with a lower density
- 14** At room temperature, iron is difficult to compress.
- At the same temperature, oxygen is much easier to compress.
- Which comparison of the structures of iron and oxygen explains this?
- A** The iron particles are closer together.
 - B** The iron particles have a greater mass.
 - C** The iron particles can be magnetised.
 - D** The iron particles have less average kinetic energy.

- 15** A night storage heater contains a large block of material that is heated electrically during the night. During the day the block cools down, releasing thermal energy into the room.



Which thermal capacity and which night-time temperature increase will cause the most energy to be stored by the block?

	thermal capacity of block	night-time temperature increase
A	large	large
B	large	small
C	small	large
D	small	small

- 16** 100 g of water at 25 °C is poured into an insulated cup. 50 g of ice at 0 °C is added to the water. The water is stirred until the temperature of the water has fallen to 0 °C.

18 g of ice remains unmelted.

The specific heat capacity of water is 4.2 J/g °C.

Which value does this experiment give for the specific latent heat of fusion of ice?

- A** 210 J/g **B** 330 J/g **C** 580 J/g **D** 770 J/g

- 17** The handle of a metal saucepan is made of plastic. As the saucepan heats up, the handle gets warmer.

Which statement explains this?

- A** Molecules of the plastic radiate their energy to other molecules.
B Molecules of the plastic vibrate more and pass on their energy to nearby molecules.
C The free electrons in the plastic transfer the thermal energy along the handle.
D The heated molecules very slowly move along the plastic handle.

- 18** The metal surface of a kettle is hot.

What happens to the cool air outside the kettle when it comes into contact with the hot kettle?

- A** The density of the air decreases and the air falls.
- B** The density of the air decreases and the air rises.
- C** The density of the air increases and the air falls.
- D** The density of the air increases and the air rises.

- 19** Vacuum flasks usually have silvered walls that help to keep the contents of the flask hot.

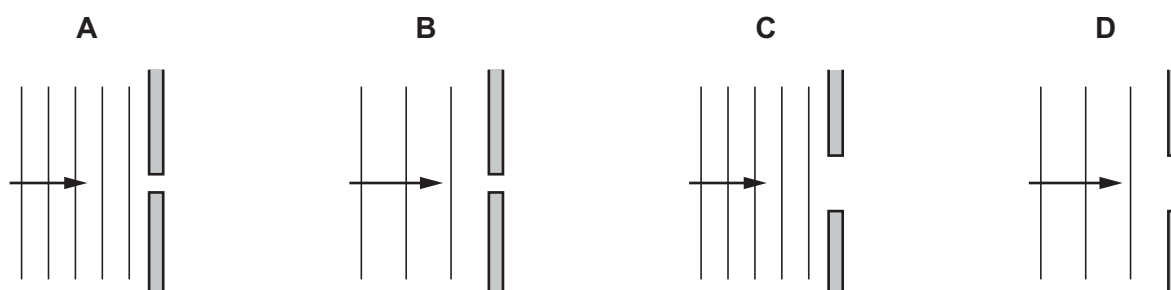
Why are the walls silvered?

- A** to absorb thermal energy from the air around the flask
- B** to increase the rate of convection inside the flask
- C** to reduce energy loss to the surroundings by conduction
- D** to reflect thermal radiation back into the flask

- 20** When water waves pass through a gap they diffract.

The diagrams show wavefronts approaching a narrow gap.

In which diagram will the diffraction be least?



- 21** Which conditions are necessary for light to be totally internally reflected?

	the incident light is in	angle of incidence
A	the less dense medium	less than the critical angle
B	the less dense medium	greater than the critical angle
C	the more dense medium	less than the critical angle
D	the more dense medium	greater than the critical angle

- 22** Light is travelling through air. It strikes a glass block at an angle of incidence of 45° . The glass has a refractive index of 1.4.

What is the angle of refraction of the light as it enters the glass?

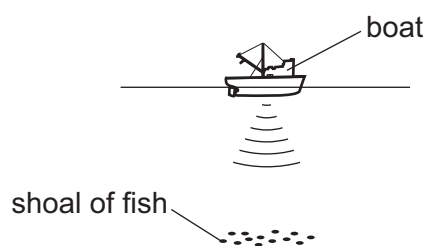
- A** 29° **B** 30° **C** 32° **D** 82°

- 23** An eclipse of the Sun happens when the Moon comes between the Earth and the Sun.

Which statement is correct?

- A** Infra-red radiation from the Sun disappears before visible light and ultra-violet radiation.
B Ultra-violet radiation from the Sun disappears before visible light and infra-red radiation.
C Visible light from the Sun disappears before ultra-violet radiation and infra-red radiation.
D Infra-red radiation, ultra-violet radiation and visible light from the Sun all disappear at the same moment.

- 24** A pulse of sound is produced at the bottom of a boat. The sound travels through the water and is reflected from a shoal of fish. The sound reaches the boat again after 1.2 s. The speed of sound in the water is 1500 m/s.



How far below the bottom of the boat is the shoal of fish?

- A** 450 m **B** 900 m **C** 1800 m **D** 3600 m

- 25** An observer stands at the finish line of a 100 m race. He wants to time the winner's run. He starts his stop-watch as soon as he sees the smoke from the starting gun instead of when he hears the bang.

What is the reason for doing this?

- A** Light travels much faster than sound.
B There is a risk he might respond to an echo from a wall.
C Humans react slower to sound than to light.
D Humans react more quickly to sound than to light.

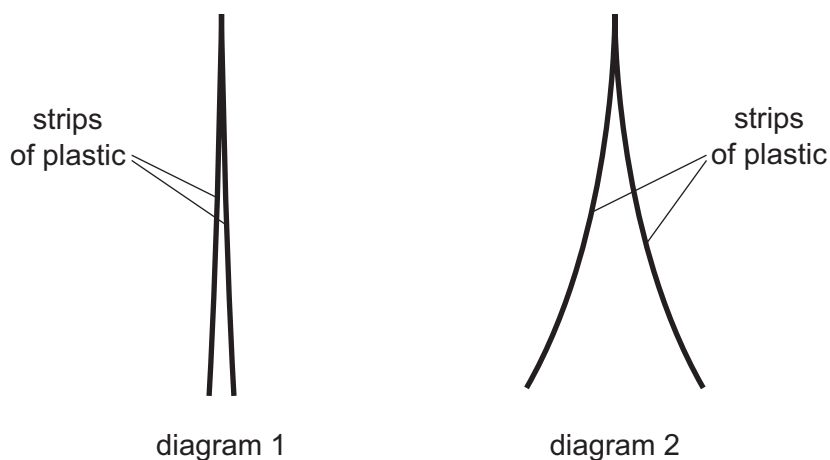
- 26** A soft iron bar is a long way from any magnetic field.

How can the material of the bar be described?

- A** It is magnetic and strongly magnetised.
- B** It is magnetic and unmagnetised.
- C** It is non-magnetic and strongly magnetised.
- D** It is non-magnetic and unmagnetised.

- 27** Diagram 1 shows two thin, uncharged strips of plastic.

Diagram 2 shows the same strips after they have been rubbed with a dry cloth.



Which row describes the charge on the strips after rubbing, and the force between the strips after rubbing?

	charge on strips	force between strips
A	opposite	attraction
B	opposite	repulsion
C	the same	attraction
D	the same	repulsion

- 28** A cell has an electromotive force (e.m.f.) of 1.5 V.

What does this statement mean?

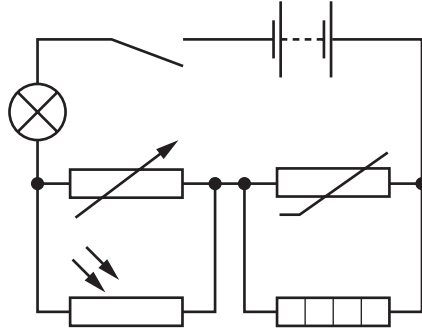
- A** The cell converts 1.0 J of energy when driving 1.5 C of charge round a complete circuit.
- B** The cell converts 1.5 J of energy when driving 1.0 C of charge round a complete circuit.
- C** The cell converts 1.5 J of energy per second when driving 1.0 C of charge round a complete circuit.
- D** The cell converts 1.5 W of power when driving 1.0 C of charge round a complete circuit.

- 29** A metal wire of length 100 cm and cross-sectional area 0.20 mm^2 has a resistance of 8.0Ω .

What is the resistance of a wire of the same metal of length 50 cm and cross-sectional area of 0.40 mm^2 ?

- A** 2.0Ω **B** 8.0Ω **C** 16Ω **D** 32Ω

- 30** The diagram shows a circuit.

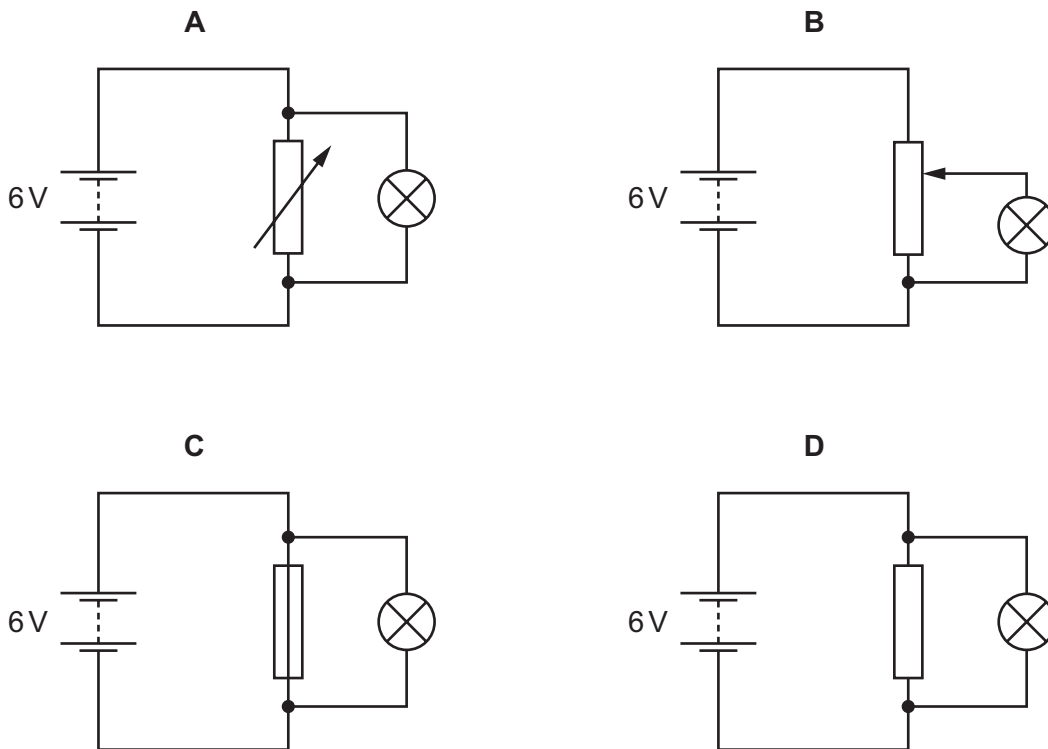


What is connected in parallel with the thermistor?

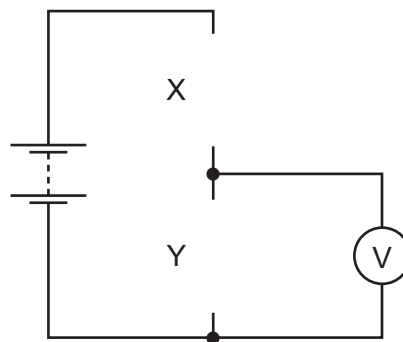
- A** heater
B lamp
C light-dependent resistor
D variable resistor

- 31** A lamp is to be connected in a circuit so that the potential difference (p.d.) across it can be varied from 0 to 6 V.

Which circuit would be most suitable?



- 32** The circuit below can be completed by inserting components at X and at Y. The completed circuit is a potential divider in which the potential difference across component Y increases when the temperature increases.



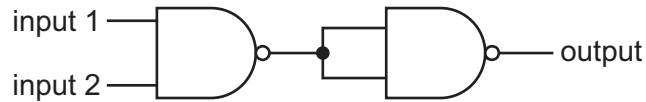
Which row shows the components X and Y?

	X	Y
A	light-dependent resistor	resistor
B	resistor	light-dependent resistor
C	resistor	thermistor
D	thermistor	resistor

33 Which two logic gates each have a high output (1) when both of their inputs are low (0)?

- A** AND and OR
- B** AND and NOR
- C** NAND and NOR
- D** NAND and OR

34 Two NAND gates are joined together as shown.



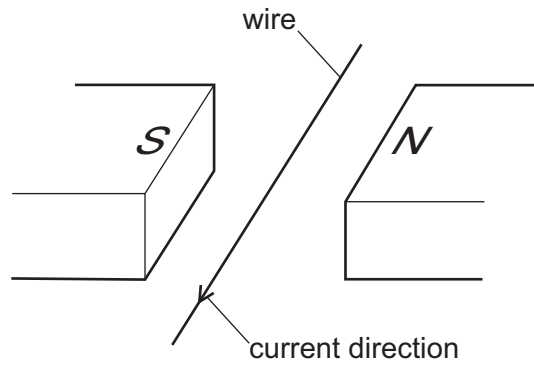
Which single logic gate is equivalent to this combination?

- A** AND
- B** NAND
- C** NOR
- D** OR

35 Which components are designed to improve the safe working of a mains electrical supply?

	circuit breaker	earth wire	fuse
A	✓	✓	x
B	✓	x	✓
C	x	✓	✓
D	✓	✓	✓

- 36 The diagram shows a current-carrying wire placed between two magnetic poles. The current is in the direction shown.



What is the direction of the force on the wire?

- A towards the bottom of the page
- B towards the top of the page
- C towards the left-hand side of the page
- D towards the right-hand side of the page

37 Diagram 1 shows an a.c. generator. The coil is turning as shown.

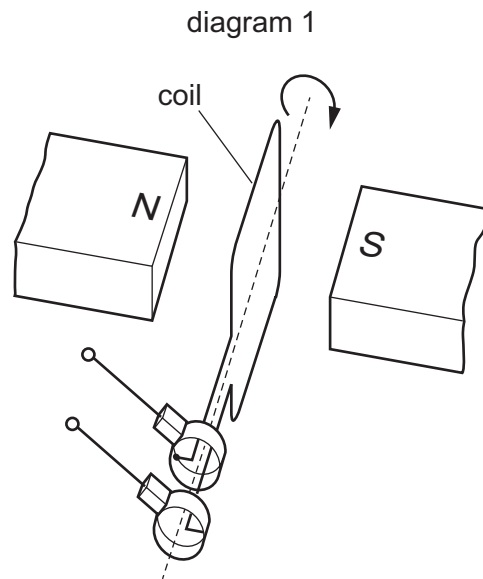
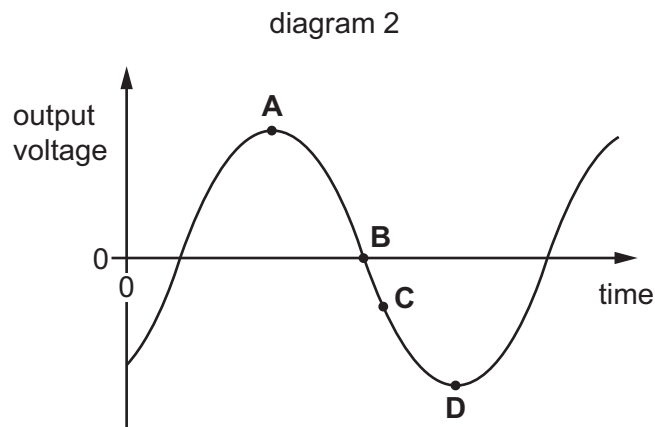


Diagram 2 shows the output voltage produced by the generator as the coil turns.

Which point on diagram 2 shows the voltage induced when the coil is moving through the position shown in diagram 1?



38 Which observation provides evidence for the nuclear atom?

- A attraction of opposite charges
- B emission of γ -rays during the decay of a radioactive nuclide
- C scattering of α -particles by thin metal foils
- D scattering of γ -rays by a thin metal foil

- 39 The chemical symbol for uranium is U. The equation represents the radioactive decay of uranium-235.

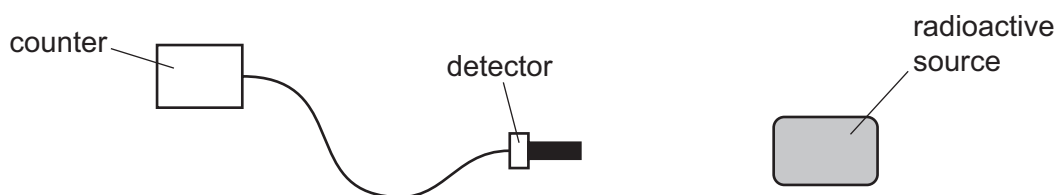


What are the numbers x and y?

	x	y
A	231	94
B	231	90
C	239	94
D	239	90

- 40 An experiment is carried out to measure the radiation from a radioactive source that has a half-life of 10 minutes.

The source is placed close to a detector that is connected to a counter, as shown.



The average background count-rate is 20 counts/minute.

At the start of the experiment, the count-rate recorded by the counter is 1000 counts/minute.

What is the count-rate 10 minutes later?

- A** 490 counts/minute
- B** 500 counts/minute
- C** 510 counts/minute
- D** 530 counts/minute

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Cambridge Assessment International Education
Cambridge International General Certificate of Secondary Education (9–1)

PHYSICS (9–1)

0972/22

Paper 2 Multiple Choice (Extended)

May/June 2019

MARK SCHEME

Maximum Mark: 40

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the May/June 2019 series for most Cambridge IGCSE™, Cambridge International A and AS Level and Cambridge Pre-U components, and some Cambridge O Level components.

This document consists of **3** printed pages.

Question	Answer	Marks
1	C	1
2	C	1
3	D	1
4	D	1
5	C	1
6	B	1
7	B	1
8	C	1
9	C	1
10	C	1
11	A	1
12	D	1
13	C	1
14	A	1
15	A	1
16	B	1
17	B	1
18	B	1
19	D	1
20	C	1
21	D	1
22	B	1
23	D	1
24	B	1
25	A	1
26	B	1
27	D	1
28	B	1

Question	Answer	Marks
29	A	1
30	A	1
31	B	1
32	D	1
33	C	1
34	A	1
35	D	1
36	A	1
37	B	1
38	C	1
39	B	1
40	C	1



Cambridge Assessment International Education
Cambridge International General Certificate of Secondary Education (9–1)

PHYSICS

0972/21

Paper 2 Multiple Choice (Extended)

October/November 2019

45 minutes

Additional Materials: Multiple Choice Answer Sheet
Soft clean eraser
Soft pencil (type B or HB recommended)



READ THESE INSTRUCTIONS FIRST

Write in soft pencil.

Do not use staples, paper clips, glue or correction fluid.

Write your name, centre number and candidate number on the Answer Sheet in the spaces provided unless this has been done for you.

DO NOT WRITE IN ANY BARCODES.

There are **forty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A, B, C** and **D**.

Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

Read the instructions on the Answer Sheet very carefully.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.

Any rough working should be done in this booklet.

Electronic calculators may be used.

Take the weight of 1.0 kg to be 10 N (acceleration of free fall = 10 m/s^2).

This document consists of **15** printed pages and **1** blank page.

- 1 A student measures the diameter of a pencil.

Which measuring instrument will give the most precise reading?

- A a measuring tape
- B a metre rule
- C a micrometer screw gauge
- D a ruler

- 2 A light object is dropped from rest. It falls a large distance vertically through air.

How can the motion of the object be described?

- A constant acceleration
- B increasing acceleration
- C decreasing acceleration and then moving at terminal velocity
- D increasing acceleration and then moving at terminal velocity

- 3 A car travels at an average speed of 60 km/h for 15 minutes.

How far does the car travel in 15 minutes?

- A 4.0 km
- B 15 km
- C 240 km
- D 900 km

- 4 Which quantity is a force due to a gravitational field?

- A density
- B mass
- C weight
- D volume

- 5 The density of air is 1.2 kg/m^3 .

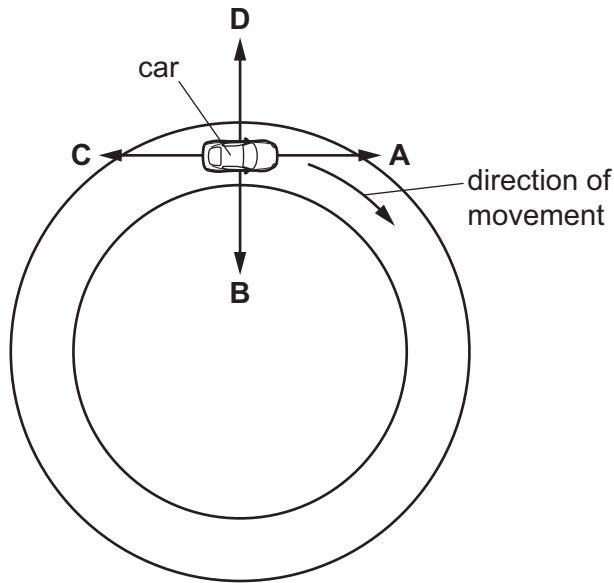
A room has dimensions $5.0 \text{ m} \times 4.0 \text{ m} \times 3.0 \text{ m}$.

What is the mass of the air in the room?

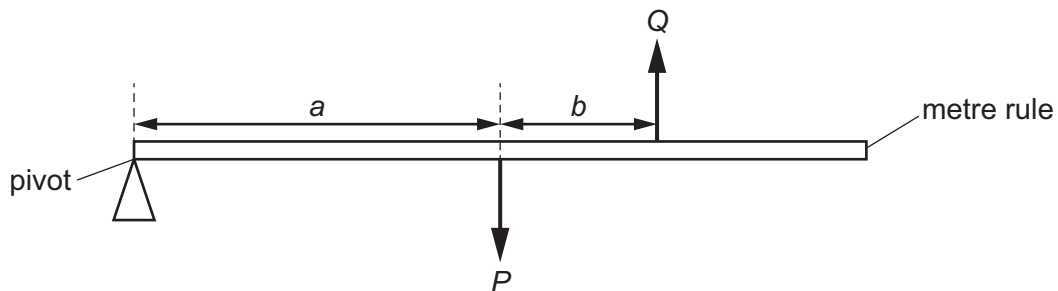
- A 0.02 kg
- B 0.10 kg
- C 50 kg
- D 72 kg

- 6 A car is travelling around a circular track at a constant speed, as shown.

In which direction is the resultant force on the car?



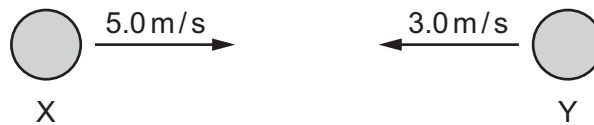
- 7 Two forces P and Q act on a metre rule as shown. The metre rule is pivoted at one end. The rule starts to rotate in a clockwise direction.



Which statement is correct?

- A P equals Q
 - B P is less than Q
 - C $(P \times a)$ is equal to $(Q \times b)$
 - D $(P \times a)$ is greater than $(Q \times (a + b))$
- 8 Which statement gives a complete description of any object that is in equilibrium?
- A There are no forces acting.
 - B There is no resultant force.
 - C There is no resultant force and no resultant turning effect.
 - D There is no resultant turning effect.

- 9 Two objects X and Y move directly towards each other. The objects have the same mass. Object X has a velocity of 5.0 m/s to the right. Object Y has a velocity of 3.0 m/s to the left.

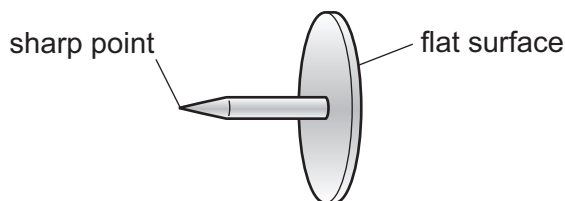


Object X and object Y collide and stick together.

What is their velocity after colliding?

- A** 1.0 m/s to the left
B 1.0 m/s to the right
C 4.0 m/s to the left
D 4.0 m/s to the right
- 10 Brakes are used to slow down a moving car.
- Into which form of energy is most of the kinetic energy converted as the car slows down?
- A** chemical
B elastic
C thermal
D sound
- 11 A man carries 20 tiles from the ground to the roof of a house. Each tile has a mass of 1.2 kg . The roof of the house is 15 m above the ground.
- How much work does the man do against gravity on the tiles in carrying them to the roof?
- A** 36 J **B** 180 J **C** 360 J **D** 3600 J
- 12 A car is moving along a straight horizontal road. The car has 1.6 MJ of kinetic energy. The car accelerates for 20 s until the kinetic energy of the car increases to 2.5 MJ .
- What is the minimum average power developed by the car engine for this acceleration?
- A** 45 W **B** 205 W **C** 45 kW **D** 205 kW

- 13** A drawing pin (thumb tack) has a sharp point at one end and a flat surface at the other end.



The pin is pushed into a wooden board.

How do the pressure and the force at the sharp point compare with the pressure and the force on the flat surface?

	force at the sharp point	pressure at the sharp point
A	greater than on the flat surface	greater than on the flat surface
B	greater than on the flat surface	less than on the flat surface
C	the same as on the flat surface	greater than on the flat surface
D	the same as on the flat surface	less than on the flat surface

- 14** An object is 20 cm below the surface of a liquid. The density of the liquid is 1200 kg/m^3 .

What is the pressure on the object due to the liquid?

- A** 600 Pa **B** 2400 Pa **C** 60 000 Pa **D** 240 000 Pa

- 15** Which statement about the evaporation of a liquid is correct?

- A** The least energetic molecules escape from the surface and the temperature of the liquid decreases.
- B** The least energetic molecules escape from the surface and the temperature of the liquid increases.
- C** The most energetic molecules escape from the surface and the temperature of the liquid decreases.
- D** The most energetic molecules escape from the surface and the temperature of the liquid increases.

- 16** A bubble of gas is formed deep under water. The bubble has a volume of 40 cm^3 and the pressure inside the bubble is P .

The bubble rises up through the water. The volume of the bubble increases to 56 cm^3 and the pressure becomes 100 kPa. The temperature of the gas does not change.

What is the initial pressure P ?

- A** 71 Pa **B** 71 kPa **C** 140 Pa **D** 140 kPa

17 Which change in the design of a liquid-in-glass thermometer makes it more sensitive?

- A a larger liquid reservoir
- B a longer tube
- C a smaller liquid reservoir
- D a wider tube

18 A liquid turns into a gas. This occurs only at one particular temperature, and the change happens throughout the liquid.

What is this process called?

- A boiling
- B condensation
- C evaporation
- D fusion

19 One end of a rod of copper is placed in hot water. Thermal energy travels along the rod to make the other end warmer.

What is the behaviour of the copper at an atomic level that accounts for most of the transfer of thermal energy from one end to the other?

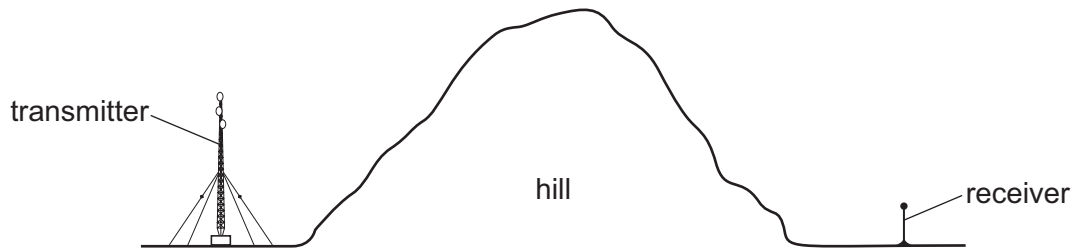
- A Atoms at the hot end gain kinetic energy and move towards the other end.
- B Atoms at the hot end expand, colliding with other atoms and transferring energy.
- C Free electrons at the hot end gain energy and move towards the other end, colliding with atoms along the rod.
- D Free electrons at the hot end gain energy from the hot water and move directly to the other end.

20 A surface is made so that it is a good source of infrared radiation.

Which surface is **not** suitable?

- A a surface that is painted matt black
- B a surface that is painted white
- C a surface that is heated to a high temperature
- D a surface that has a large surface area

- 21** A large hill blocks the direct path between a transmitter of radio waves and a receiver, as shown.

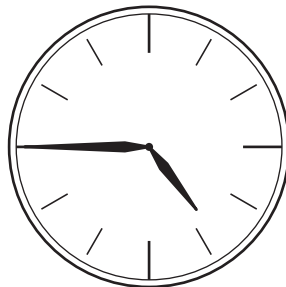


The receiver picks up the signal from the transmitter even though the radio waves do not travel through the hill.

Which row is correct?

	A possible way for this to happen is	A stronger signal is received using
A	diffraction round the hill.	longer wavelengths.
B	diffraction round the hill.	shorter wavelengths.
C	refraction round the hill.	longer wavelengths.
D	refraction round the hill.	shorter wavelengths.

- 22** The diagram shows the image of a clock in a plane mirror.

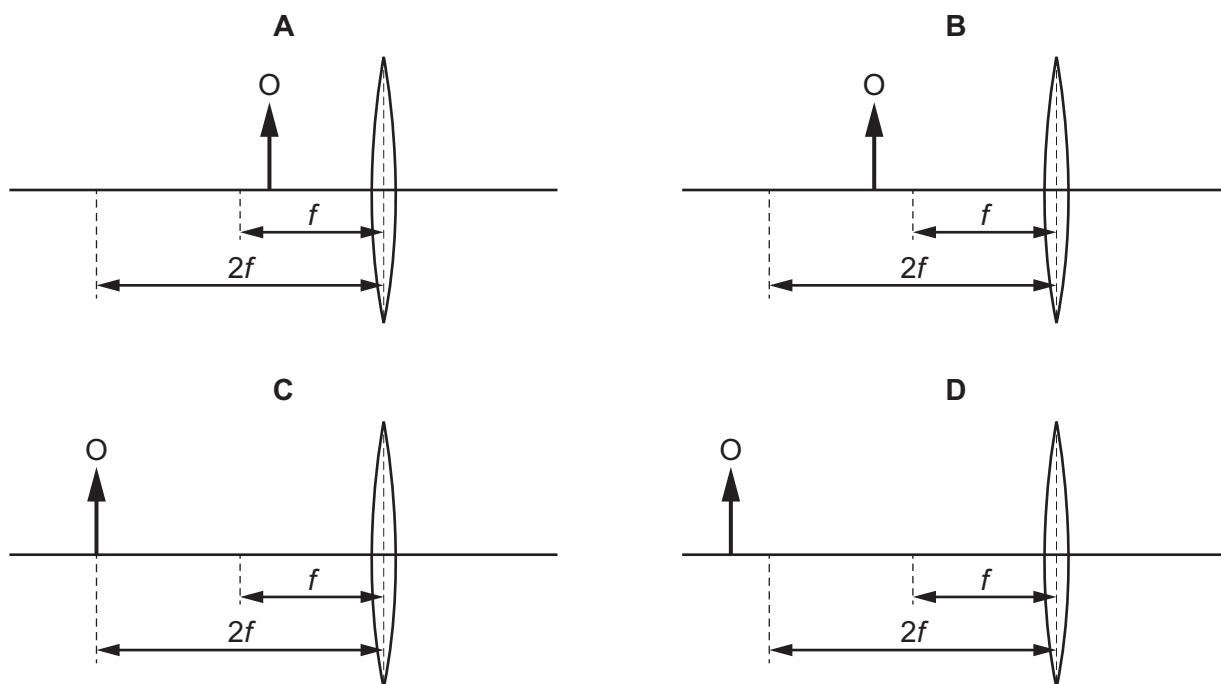


Which is the actual time?

- A** 04:15 **B** 04:45 **C** 07:15 **D** 07:45

23 A converging lens produces an image of an object O. The focal length of the lens is f .

Which position of the object produces a virtual image?



24 The diagram shows the electromagnetic spectrum.

γ -rays	X-rays	ultraviolet	visible light	infrared	microwaves	radio waves
----------------	--------	-------------	---------------	----------	------------	-------------

increasing 

A word is missing from the label below the spectrum.

Which word is missing?

- A** amplitude
- B** frequency
- C** speed
- D** wavelength

25 Which row gives a possible set of values for the speed of sound in ice, in water and in steam?

	speed of sound in ice m/s	speed of sound in water m/s	speed of sound in steam m/s
A	500	1500	4000
B	1500	4000	500
C	4000	500	1500
D	4000	1500	500

26 A steel bar is placed in an East-West direction for it to be demagnetised. No other magnet is nearby.

Which method is **not** suitable?

- A** Hammering the bar.
- B** Heating the bar to a very high temperature.
- C** Slowly taking the bar out of a coil that carries an alternating current.
- D** Slowly taking the bar out of a coil that carries a direct current.

27 A plastic rod is rubbed with a dry cloth. The rod becomes positively charged.

Why has the rod become positively charged?

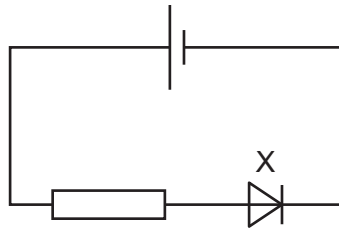
- A** It has gained electrons.
- B** It has gained neutrons.
- C** It has lost electrons.
- D** It has lost neutrons.

28 A circuit contains a cell of electromotive force (e.m.f.) of 2.0 V. The current in the circuit is 2.0 A.

How much energy is converted by the cell in 2.0 minutes?

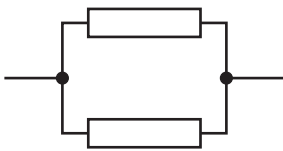
- A** 2.0 J
- B** 4.0 J
- C** 8.0 J
- D** 480 J

- 29 The circuit diagram shows a cell connected in series to a resistor and a component X.



What is component X?

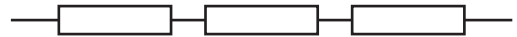
- A bell
 - B diode
 - C heater
 - D thermistor
- 30 Identical resistors are connected together to form arrangements X, Y and Z.



X



Y



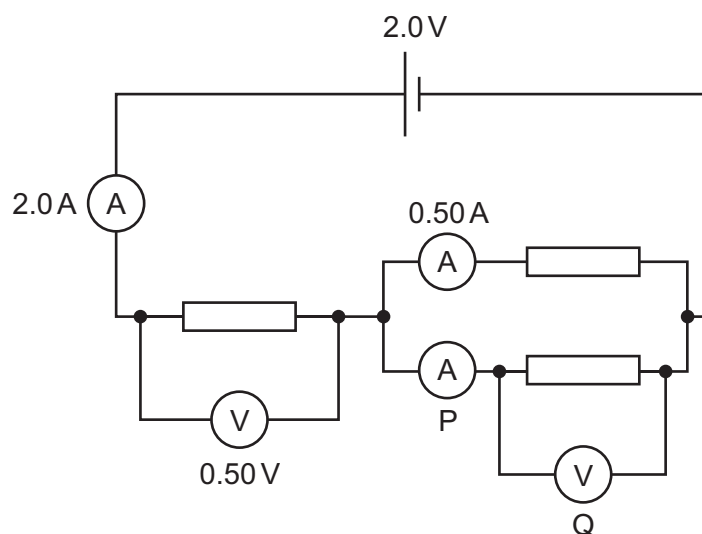
Z

What is the correct order of the resistances of the arrangements from the largest to the smallest?

- A $X \rightarrow Y \rightarrow Z$
- B $Y \rightarrow X \rightarrow Z$
- C $Z \rightarrow X \rightarrow Y$
- D $Z \rightarrow Y \rightarrow X$

- 31 A circuit contains a cell of electromotive force (e.m.f.) 2.0 V, three resistors, three ammeters and two voltmeters. One ammeter is labelled P and one voltmeter is labelled Q.

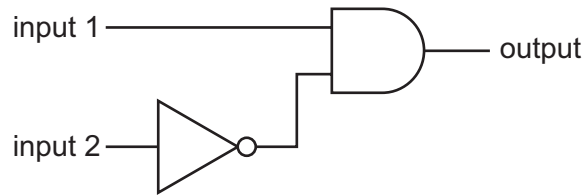
The readings on the other two ammeters and on the other voltmeter are shown.



What is the reading on ammeter P and what is the reading on voltmeter Q?

	reading on P / A	reading on Q / V
A	1.5	1.5
B	1.5	2.5
C	2.5	1.5
D	2.5	2.5

32 There are two inputs and one output for the combination of logic gates shown.



Which truth table represents the operation of this combination of logic gates?

A

input 1	input 2	output
0	0	0
0	1	0
1	0	1
1	1	1

B

input 1	input 2	output
0	0	0
0	1	0
1	0	1
1	1	0

C

input 1	input 2	output
0	0	1
0	1	1
1	0	0
1	1	0

D

input 1	input 2	output
0	0	1
0	1	0
1	0	1
1	1	1

33 Where must a fuse be connected in a mains electric circuit?

- A** the earth wire only
- B** the live wire only
- C** the neutral wire only
- D** the live wire and the earth wire

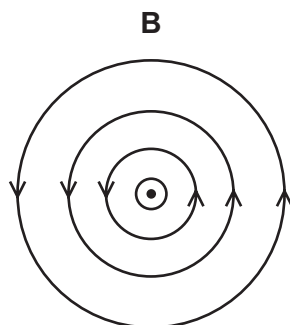
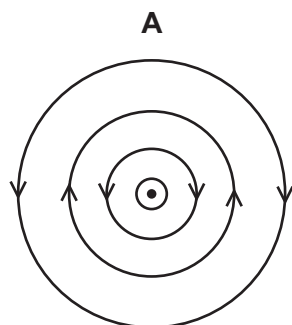
34 A student investigates the output voltage induced across a coil of wire by a bar magnet.

When will the induced voltage have the greatest value?

- A** The student slowly moves the bar magnet into the coil of wire.
- B** The student leaves the bar magnet stationary in the coil of wire.
- C** The student quickly removes the bar magnet from the coil of wire.
- D** The student places the bar magnet at rest outside the coil of wire.

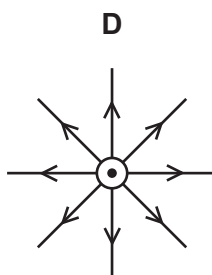
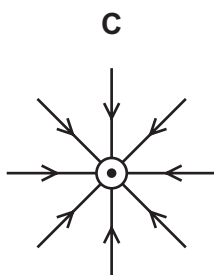
35 There is a current in a wire. The direction of the current is out of the page.

Which diagram shows the magnetic field pattern produced?



key

⊙ wire carrying a current
out of the page



- 36** Diagram 1 shows a coil of wire P between the poles of a magnet. The ends of coil P are connected to a battery by slip rings.

Diagram 2 shows a coil of wire Q between the poles of a different magnet. The ends of coil Q are connected to a battery by a split-ring commutator.

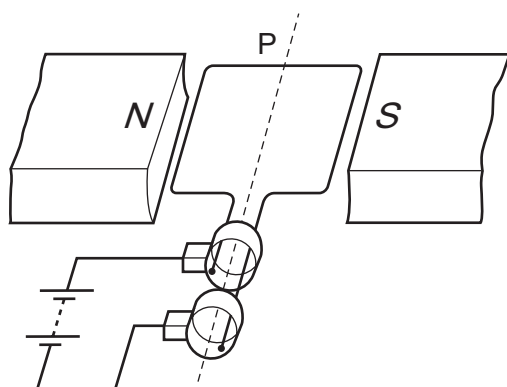


diagram 1

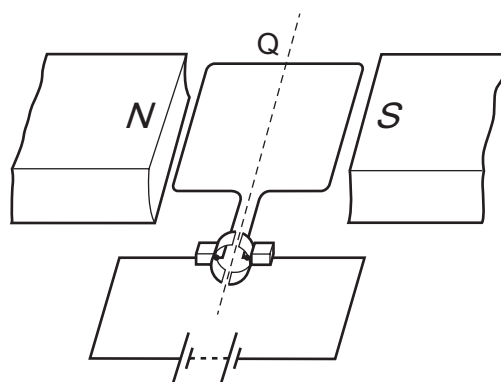
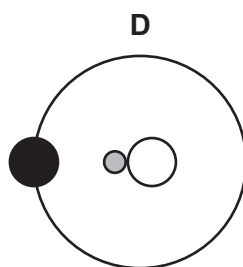
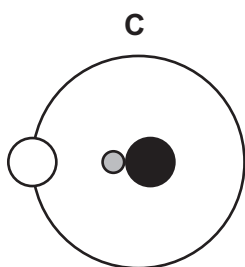
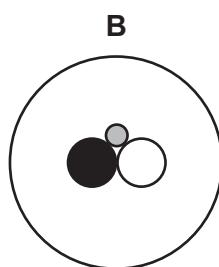
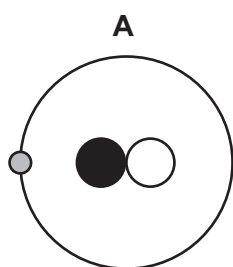


diagram 2

What happens to coils P and Q?

	coil P	coil Q
A	continuously turns anticlockwise	makes one quarter turn anticlockwise then stops
B	continuously turns clockwise	makes one quarter turn clockwise then stops
C	makes one quarter turn anticlockwise then stops	continuously turns anticlockwise
D	makes one quarter turn clockwise then stops	continuously turns clockwise

- 37** Which diagram shows a possible structure of a neutral atom?



key

● proton

○ neutron

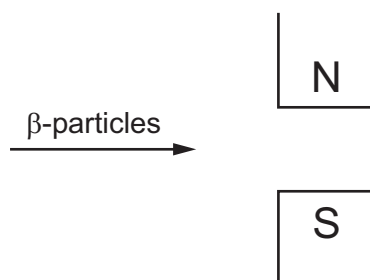
● electron

- 38 The scattering of particles by a thin gold foil provided scientists with evidence for the nuclear atom.

Which particles were scattered by the gold nuclei in the thin foil?

- A α -particles
- B β -particles
- C neutrons
- D protons

- 39 The diagram shows β -particles being directed between the poles of a magnet.



In which direction will the particles be deflected?

- A into the page
 - B out of the page
 - C towards the bottom of the page
 - D towards the top of the page
- 40 Why are some radioactive sources stored in boxes made from lead?
- A Lead absorbs emissions from the radioactive sources.
 - B Lead decreases the half-life of radioactive sources.
 - C Lead increases the half-life of radioactive sources.
 - D Lead repels emissions from the radioactive sources.

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Cambridge Assessment International Education
Cambridge International General Certificate of Secondary Education (9–1)

PHYSICS

0972/21

Paper 2 Multiple Choice (Extended)

October/November 2019

MARK SCHEME

Maximum Mark: 40

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the October/November 2019 series for most Cambridge IGCSE™, Cambridge International A and AS Level components and some Cambridge O Level components.

This document consists of **3** printed pages.

Question	Answer	Marks
1	C	1
2	C	1
3	B	1
4	C	1
5	D	1
6	B	1
7	D	1
8	C	1
9	B	1
10	C	1
11	D	1
12	C	1
13	C	1
14	B	1
15	C	1
16	D	1
17	A	1
18	A	1
19	C	1
20	B	1
21	A	1
22	C	1
23	A	1
24	D	1
25	D	1
26	D	1
27	C	1
28	D	1

Question	Answer	Marks
29	B	1
30	D	1
31	A	1
32	B	1
33	B	1
34	C	1
35	B	1
36	C	1
37	A	1
38	A	1
39	B	1
40	A	1



Cambridge IGCSE™ (9–1)

PHYSICS

0972/21

Paper 2 Multiple Choice (Extended)

May/June 2020

45 minutes

You must answer on the multiple choice answer sheet.

You will need: Multiple choice answer sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)

INSTRUCTIONS

- There are **forty** questions on this paper. Answer **all** questions.
- For each question there are four possible answers **A**, **B**, **C** and **D**. Choose the **one** you consider correct and record your choice in soft pencil on the multiple choice answer sheet.
- Follow the instructions on the multiple choice answer sheet.
- Write in soft pencil.
- Write your name, centre number and candidate number on the multiple choice answer sheet in the spaces provided unless this has been done for you.
- Do **not** use correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.
- Take the weight of 1.0 kg to be 10 N (acceleration of free fall = 10 m/s^2).

INFORMATION

- The total mark for this paper is 40.
- Each correct answer will score one mark. A mark will not be deducted for a wrong answer.
- Any rough working should be done on this question paper.

This document has **16** pages. Blank pages are indicated.



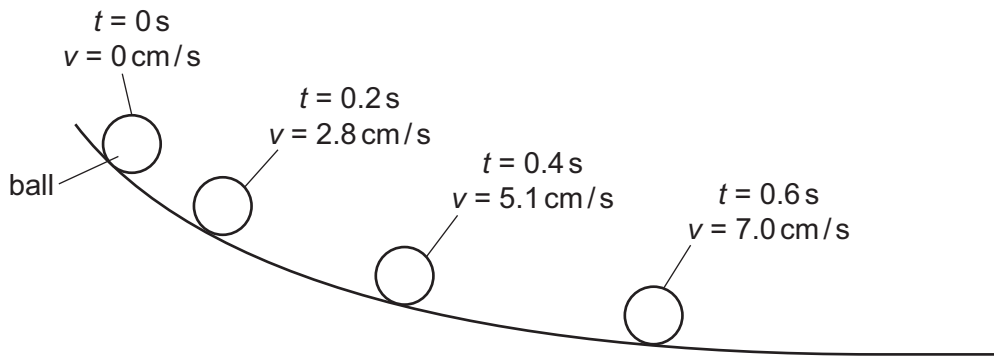
- 1 A pendulum makes 50 complete swings in 2 min 40 s.

What is the time period for 1 complete swing?

- A** 1.6 s **B** 3.2 s **C** 4.8 s **D** 6.4 s

- 2 A student investigates the motion of a ball rolling down a slope.

The diagram shows the speed v of the ball at different times t .



Which statement describes the motion of the ball?

- A** The acceleration is not constant.
B The acceleration is negative.
C The speed is decreasing.
D The velocity is constant.
- 3 Which statement about acceleration is correct?
- A** It is related to the changing speed of an object.
B It is the distance an object travels in one second.
C It is the force acting on an object divided by the distance it travels in one second.
D It is the force acting on an object when it is near to the Earth.
- 4 Which statement correctly describes the effects of placing a heavy load in a car?
- A** It is easier to accelerate the car and easier to bring the car to rest.
B It is easier to accelerate the car but more difficult to bring the car to rest.
C It is more difficult to accelerate the car and more difficult to bring the car to rest.
D It is more difficult to accelerate the car but easier to bring the car to rest.

- 5 A mass of 6.0 kg rests on the surface of a planet.

On this planet, $g = 20 \text{ N/kg}$.

What is the weight of the object?

- A** 0.30 N **B** 0.60 N **C** 60 N **D** 120 N

- 6 The mass of a measuring cylinder is 190 g.

400 cm³ of liquid is put into the measuring cylinder.

The total mass of the measuring cylinder and the liquid is 560 g.

Four solid objects are lowered in turn into the liquid. The densities of the objects are shown.

- 1 0.40 g/cm³
- 2 0.90 g/cm³
- 3 1.2 g/cm³
- 4 2.7 g/cm³

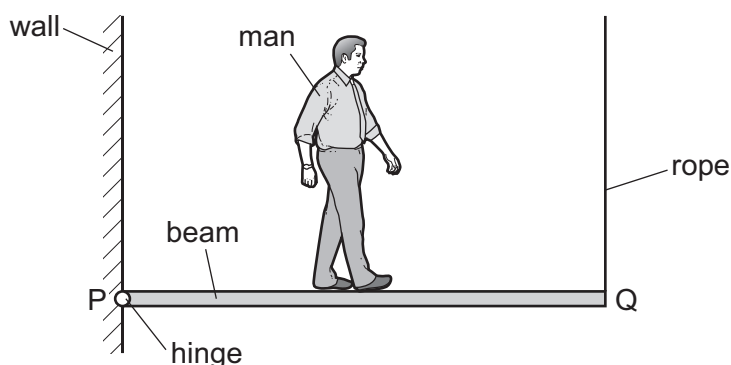
Which objects will float in the liquid?

- A** 1 only **B** 1 and 2 only **C** 1, 2 and 3 **D** 3 and 4 only

- 7 The diagram shows a wooden beam PQ, of negligible weight, which is attached to a wall by a hinge at P and kept in a horizontal position by a vertical rope attached at Q.

The beam is 3.0 m in length.

A man of weight 800 N walks along the beam from P to Q.



What is the distance of the man from P when the tension in the rope at Q becomes equal to 500 N?

- A** 0.53 m **B** 1.1 m **C** 1.9 m **D** 2.5 m

8 Which quantity is a vector?

- A acceleration
- B distance
- C speed
- D mass

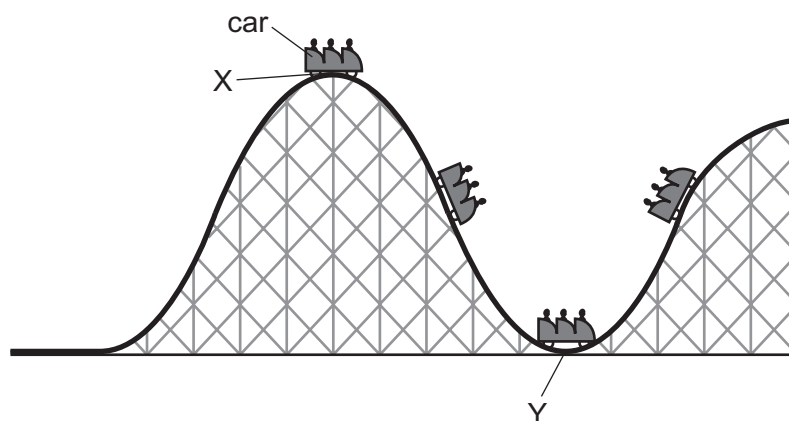
9 An object of mass 1.2 kg is moving with a velocity of 2.0 m/s when it is acted on by a force of 4.0 N . The velocity of the object increases to 5.0 m/s .

For what period of time does the force act on the object?

- A 0.90 s B 1.1 s C 1.5 s D 3.6 s

10 The diagram shows part of a rollercoaster ride with the car at different positions.

The car runs freely down from position X to position Y and up the hill on the other side.



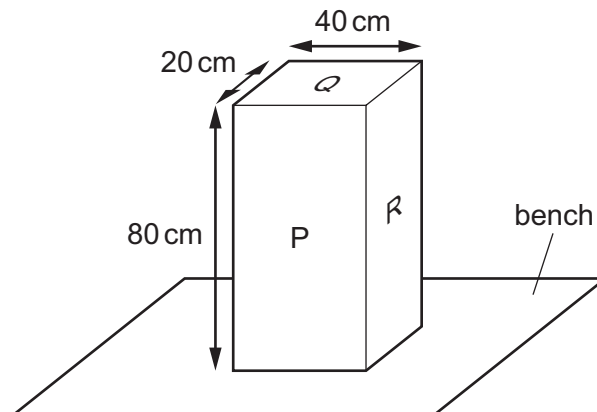
What happens to the kinetic energy and to the gravitational potential energy of the car as it moves from position X to position Y?

	kinetic energy	gravitational potential energy
A	decreases	decreases
B	decreases	increases
C	increases	decreases
D	increases	increases

11 Which energy resource does **not** have the Sun as the original source?

- A coal
- B geothermal
- C hydroelectric
- D wind

12 The diagram shows a solid block resting on a bench. The dimensions of the block are shown.



On which labelled surface should the block rest to produce the smallest pressure on the bench?

- A P
- B Q
- C R
- D P, Q and R produce the same pressure

13 The pressure due to the liquid on an object immersed in that liquid is 4500 Pa.

The density of the liquid is 900 kg/m^3 .

What is the depth of the object below the surface of the liquid?

- A 0.5 cm
- B 2.0 cm
- C 50 cm
- D 200 cm

14 A gas is heated in a sealed container.

The volume of the container does not change.

What happens to the molecules of the gas?

- A The average distance between molecules increases.
- B The average kinetic energy of the molecules increases.
- C The mass of each molecule increases.
- D The volume of each molecule increases.

- 15** Water in a beaker evaporates when left on a bench for a period of time.

Which three factors all affect the rate of evaporation of the water?

- A** wind speed, surface area, temperature
- B** wind speed, temperature, volume
- C** wind speed, surface area, volume
- D** surface area, temperature, volume

- 16** A solid is heated causing it to expand.

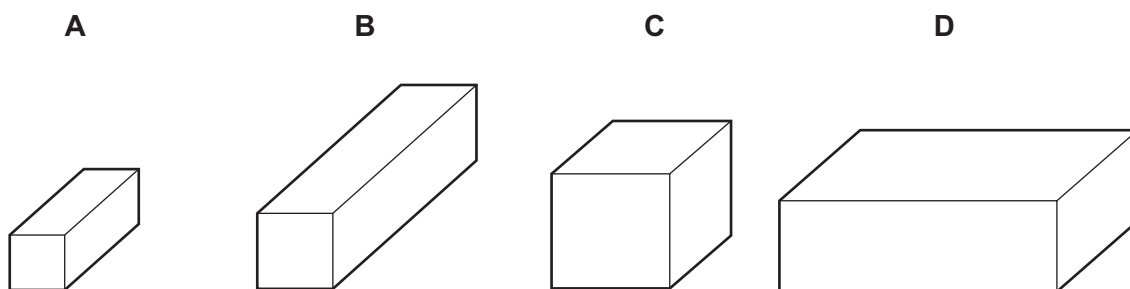
What effect does this have on its mass and on its density?

	mass	density
A	decreases	decreases
B	decreases	stays constant
C	stays constant	decreases
D	stays constant	stays constant

- 17** The diagrams show four blocks of steel. The blocks are all drawn to the same scale.

The same quantity of thermal energy is given to each block.

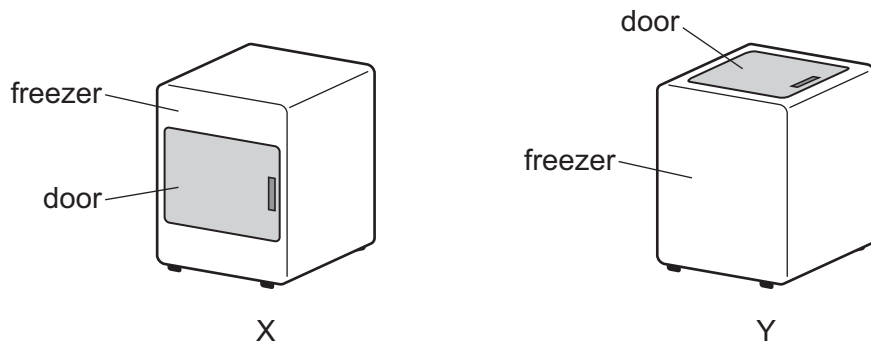
Which block shows the greatest rise in temperature?



- 18** Why are metals better thermal conductors than other solids?

- A** Metals contain free electrons which help transfer the energy.
- B** Molecules in metals are in fixed positions.
- C** Molecules in metals can move freely.
- D** Molecules in metals vibrate faster than those in other solids.

- 19** Two freezers X and Y are identical except that one has a door opening at the front and the other has a door opening at the top.



Both doors are the same size and are opened for the same amount of time.

Which freezer gains the least amount of thermal energy in this time and why?

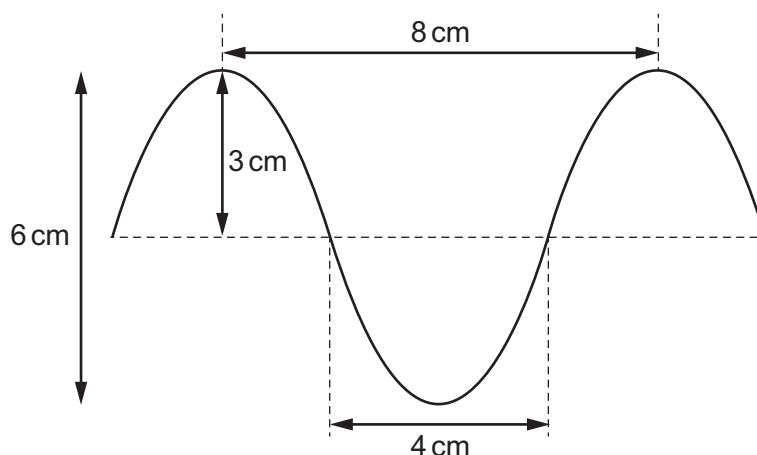
	freezer gaining the least thermal energy	reason
A	X	cold air falls
B	X	warm air falls
C	Y	cold air falls
D	Y	warm air falls

- 20** A wave of frequency 6600 Hz travels 1320 m in 4.0 s.

What is the wavelength?

- A** 0.050 m **B** 0.80 m **C** 1.3 m **D** 20 m

21 The diagram shows a wave.



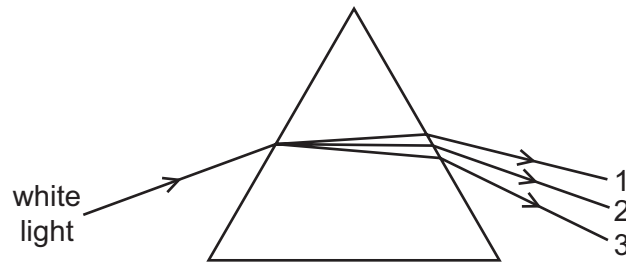
What are the amplitude and the wavelength of this wave?

	amplitude / cm	wavelength / cm
A	3	4
B	3	8
C	6	4
D	6	8

22 Which statement is correct?

- A** The speed of light in glass is equal to the speed of light in a vacuum multiplied by the refractive index of glass.
- B** The incident angle of a light ray at an air-glass surface is the angle between the ray and the glass surface.
- C** The sine of the critical angle at an air-glass surface is equal to $\frac{1}{\text{refractive index of glass}}$.
- D** The angle of refraction for light passing through an air-glass surface is proportional to the angle of incidence at that surface.

- 23** A narrow beam of white light passes through a prism and is dispersed into a spectrum.



Which row is correct?

	colour 1	colour 2	colour 3
A	blue	yellow	red
B	red	blue	yellow
C	red	yellow	blue
D	yellow	blue	red

- 24** An intruder alarm sensor detects that a person is warmer than his surroundings.

Which type of electromagnetic wave does the sensor detect?

- A** infrared
- B** radio
- C** ultraviolet
- D** visible light

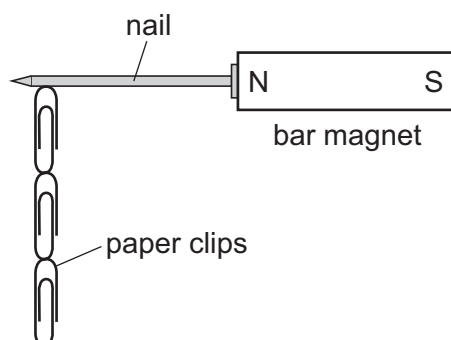
- 25** Sound travels through air as a series of compressions and rarefactions.

Which statement correctly compares a compression with a rarefaction?

- A** In a compression the wavelength is longer than in a rarefaction.
- B** In a compression the wavelength is shorter than in a rarefaction.
- C** In a compression the density of the air is greater than in a rarefaction.
- D** In a compression the density of the air is lower than in a rarefaction.

26 Four nails **A**, **B**, **C** and **D** are tested to find which makes the strongest permanent magnet.

One of the nails is placed against a bar magnet and the number of paper clips which the nail can support is recorded.

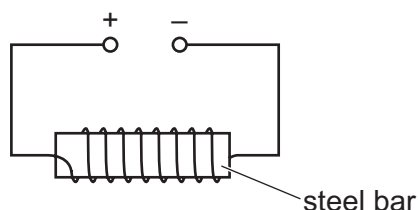


The bar magnet is then removed and the number of paper clips remaining attached to the nail is recorded. Each nail is tested individually.

Which nail becomes the strongest permanent magnet?

	number of paper clips attached to the nail	
	bar magnet present	bar magnet removed
A	2	0
B	2	1
C	4	3
D	5	2

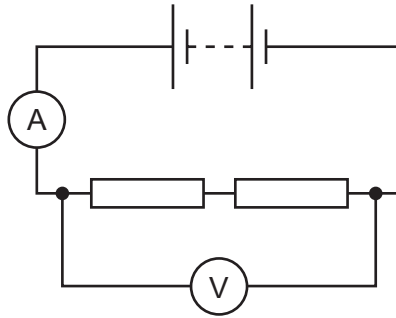
27 The circuit shows one method of magnetising a steel bar.



How can the circuit be altered so that it can be used to demagnetise the magnetised steel bar?

- A** remove the steel bar from the coil whilst the current is switched on
- B** replace the d.c. supply with an a.c. supply and gradually reduce the supply voltage to zero
- C** reverse the polarity of the d.c. supply
- D** reverse the polarity of the d.c. supply and gradually reduce the supply voltage to zero

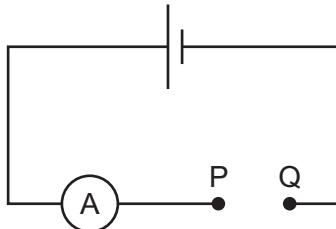
- 28 A student uses the circuit shown to determine the resistance of two identical resistors.



The voltmeter reading is 2.2 V and the ammeter reading is 0.25 A.

What is the resistance of each resistor?

- A** 0.275 Ω **B** 0.55 Ω **C** 4.4 Ω **D** 8.8 Ω
- 29 An electric fire is connected to a 240 V supply and transfers energy at a rate of 1.0 kW.
- How much charge passes through the fire in 1.0 h?
- A** 42 C **B** 250 C **C** 1.5×10^4 C **D** 2.4×10^5 C
- 30 The diagram shows an incomplete circuit. The temperature and light levels around the circuit remain unchanged.

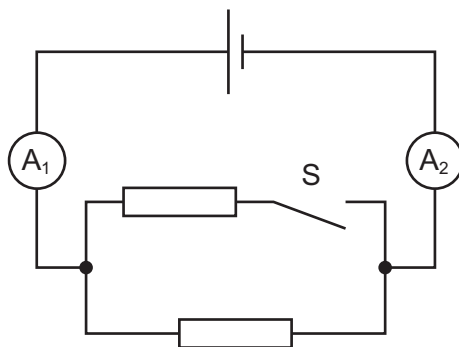


Four electrical components are connected in turn across PQ. The cell is reversed and the four electrical components are connected again in turn across PQ.

For which component is there a significant change in the magnitude of the current?

- A** diode
B light-dependent resistor
C resistor
D thermistor

31 In the circuit shown, A_1 and A_2 are ammeters.



Switch S is closed.

Which row is correct?

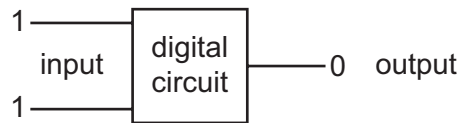
	the resistance of the whole circuit	reading of A_1	reading of A_2
A	decreases	stays the same	increases
B	decreases	increases	increases
C	increases	stays the same	stays the same
D	increases	decreases	decreases

32 Which symbol represents an OR gate?

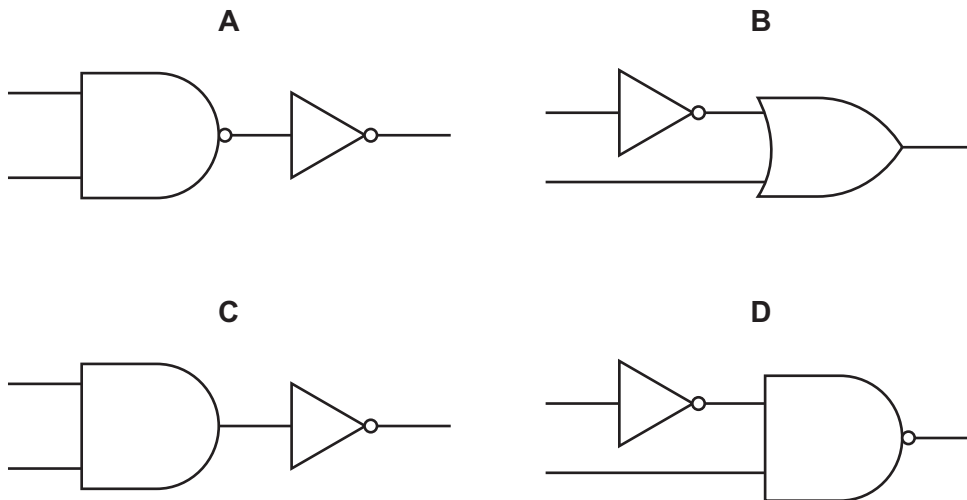


33 A digital circuit consists of two logic gates.

When the input to the circuit is 1 and 1, the output is 0.

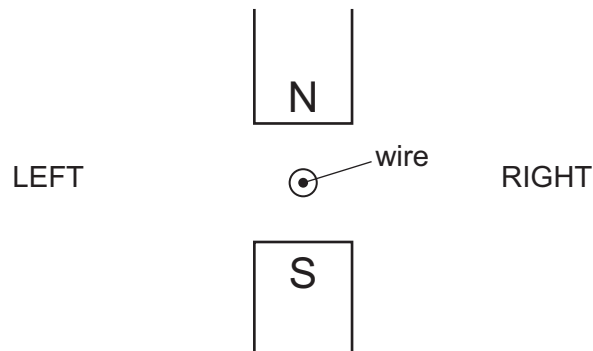


Which combination of logic gates gives this result?



34 The diagram shows a wire between the poles of a magnet.

The wire is perpendicular to the page.



The wire is moved and a current is induced upwards, out of the paper.

In which direction is the wire moved?

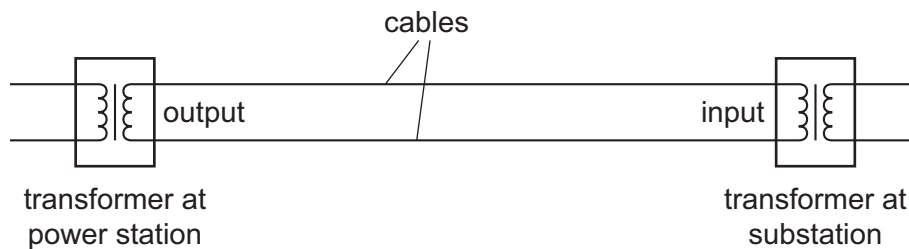
- A** left to right
- B** right to left
- C** up the page
- D** down the page

- 35** A 100% efficient transformer converts a 240 V input voltage to a 12 V output voltage. The output power of the transformer can be a maximum of 20 W.

The output is connected to two 0.30 A bulbs in parallel. One of the bulbs fails.

How does the current in the primary coil change?

- A** It decreases by 0.30 A.
B It decreases by 0.015 A.
C It increases by 0.15 A.
D It remains unchanged.
- 36** Cables transmit electrical power from the output of the transformer at a power station to the input of the transformer at a substation.

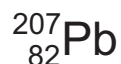


The power at the output of the transformer at the power station is 400 MW.

Which situation delivers the most power to the input of the transformer at the substation?

	potential difference at power station transformer output / kV	diameter of cables
A	200	large
B	200	small
C	400	large
D	400	small

- 37 Uranium-235 is a radioactive isotope. It undergoes a chain of decays and eventually forms the stable isotope lead-207. These two isotopes are represented as shown.



During this chain of decay, how many protons and how many neutrons are lost from a single nucleus of uranium-235 to form a single nucleus of lead-207?

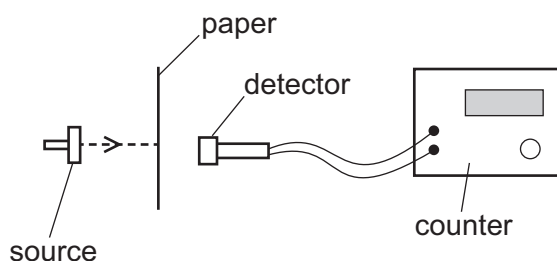
	protons	neutrons
A	10	18
B	10	28
C	18	10
D	28	10

- 38 A radioactive material has a half-life of 20 days.

A sample of the material contains 8.0×10^{10} atoms.

How many atomic nuclei have decayed after 60 days?

- A** 1.0×10^{10} **B** 4.0×10^{10} **C** 6.0×10^{10} **D** 7.0×10^{10}
- 39 A thin sheet of paper is placed between a radioactive source and a radiation detector. The count rate falls to a very low reading.



From this result, which type of radiation is the source emitting?

- A** α -particles
B β -particles
C γ -rays
D X-rays

40 α -particles, β -particles and γ -rays are emitted by radioactive nuclei when they decay.

Which emissions can be deflected by an electric field?

- A** α -particles and β -particles only
- B** β -particles and γ -rays only
- C** γ -rays and α -particles only
- D** α -particles, β -particles and γ -rays

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Cambridge IGCSE™ (9–1)

PHYSICS (9–1)

0972/21

Paper 2 Multiple Choice (Extended)

May/June 2020

MARK SCHEME

Maximum Mark: 40

Published

Students did not sit exam papers in the June 2020 series due to the Covid-19 global pandemic.

This mark scheme is published to support teachers and students and should be read together with the question paper. It shows the requirements of the exam.

Mark schemes should usually be read together with the Principal Examiner Report for Teachers. However, because students did not sit exam papers, there is no Principal Examiner Report for Teachers for the June 2020 series.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the June 2020 series for most Cambridge IGCSE™ and Cambridge International A & AS Level components, and some Cambridge O Level components.

This document consists of **3** printed pages.

Question	Answer	Marks
1	B	1
2	A	1
3	A	1
4	C	1
5	D	1
6	B	1
7	C	1
8	A	1
9	A	1
10	C	1
11	B	1
12	A	1
13	C	1
14	B	1
15	A	1
16	C	1
17	A	1
18	A	1
19	C	1
20	A	1
21	B	1
22	C	1
23	C	1
24	A	1
25	C	1
26	C	1
27	B	1
28	C	1

Question	Answer	Marks
29	C	1
30	A	1
31	B	1
32	A	1
33	C	1
34	B	1
35	B	1
36	C	1
37	A	1
38	D	1
39	A	1
40	A	1



Cambridge IGCSE™ (9–1)

PHYSICS

0972/22

Paper 2 Multiple Choice (Extended)

May/June 2020

45 minutes

You must answer on the multiple choice answer sheet.

You will need: Multiple choice answer sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)

INSTRUCTIONS

- There are **forty** questions on this paper. Answer **all** questions.
- For each question there are four possible answers **A**, **B**, **C** and **D**. Choose the **one** you consider correct and record your choice in soft pencil on the multiple choice answer sheet.
- Follow the instructions on the multiple choice answer sheet.
- Write in soft pencil.
- Write your name, centre number and candidate number on the multiple choice answer sheet in the spaces provided unless this has been done for you.
- Do **not** use correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.
- Take the weight of 1.0 kg to be 10 N (acceleration of free fall = 10 m/s^2).

INFORMATION

- The total mark for this paper is 40.
- Each correct answer will score one mark. A mark will not be deducted for a wrong answer.
- Any rough working should be done on this question paper.

This document has **16** pages. Blank pages are indicated.



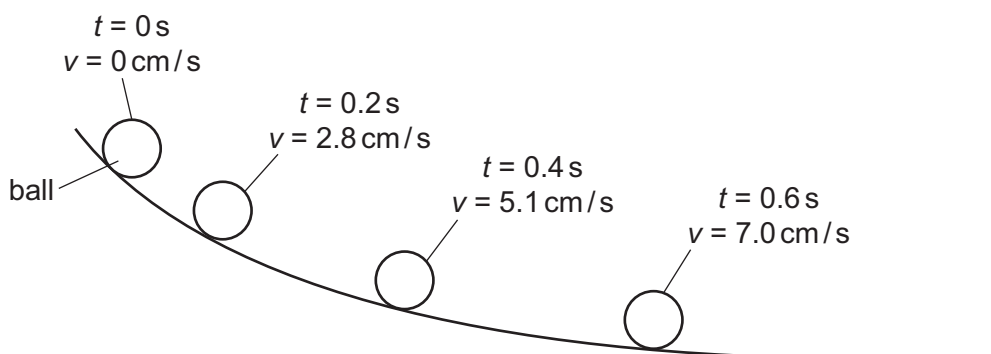
- 1 Five athletes P, Q, R, S and T compete in a race. The table shows the finishing times for the athletes.

athlete	P	Q	R	S	T
finishing time / s	22.50	24.40	25.20	26.50	23.20

Which statement is correct?

- A** Athlete P won the race and was 0.70 s ahead of the athlete in second place.
B Athlete P won the race and was 1.90 s ahead of the athlete in second place.
C Athlete S won the race and was 1.30 s ahead of the athlete in second place.
D Athlete S won the race and was 2.10 s ahead of the athlete in second place.
- 2 A student investigates the motion of a ball rolling down a slope.

The diagram shows the speed v of the ball at different times t .



Which statement describes the motion of the ball?

- A** The acceleration is not constant.
B The acceleration is negative.
C The speed is decreasing.
D The velocity is constant.
- 3 Which statement about acceleration is correct?
- A** It is related to the changing speed of an object.
B It is the distance an object travels in one second.
C It is the force acting on an object divided by the distance it travels in one second.
D It is the force acting on an object when it is near to the Earth.

4 Which statement correctly describes the effects of placing a heavy load in a car?

- A It is easier to accelerate the car and easier to bring the car to rest.
- B It is easier to accelerate the car but more difficult to bring the car to rest.
- C It is more difficult to accelerate the car and more difficult to bring the car to rest.
- D It is more difficult to accelerate the car but easier to bring the car to rest.

5 A space probe is taken from the Earth to Mars.

The force of gravity on the surface of Mars is less than the force of gravity on the surface of the Earth.

How do the weight and the mass of a space probe on the surface of Mars compare to their values when the probe is on the surface of the Earth?

	weight on Mars	mass on Mars
A	decreased	decreased
B	decreased	unchanged
C	unchanged	decreased
D	unchanged	unchanged

6 Water has a density of 1000 kg/m^3 .

A rectangular swimming pool has an average depth of 1.6 m.

The length of the pool is 25 m.

The width of the pool is 10 m.

What is the mass of the water in the swimming pool?

- A** 2.5 kg **B** 400 kg **C** 400 000 kg **D** 800 000 kg

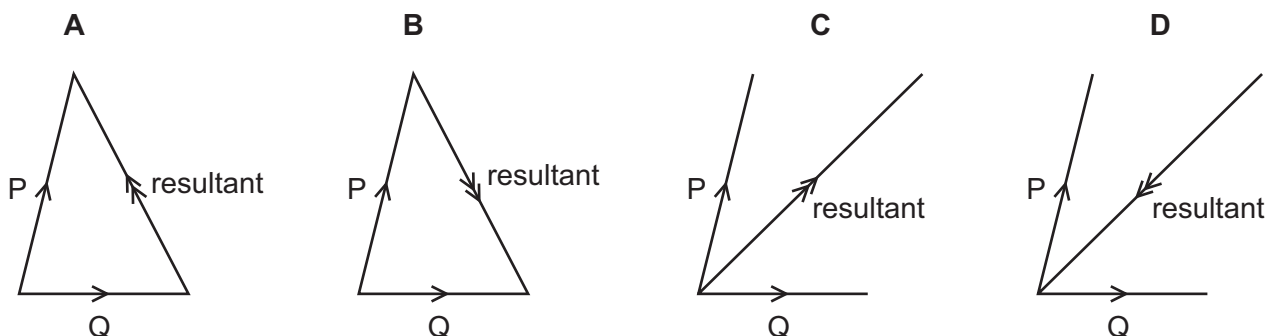
7 A satellite orbits the Earth at constant speed in a circular orbit.

Which statement is correct?

- A The resultant force on the satellite is zero.
- B The resultant force on the satellite is towards the Earth.
- C The resultant force on the satellite is away from the Earth.
- D The resultant force on the satellite is in the direction of motion.

- 8 Two forces P and Q act on an object.

Which diagram shows the resultant of these two forces?



- 9 An object is moving at $+3.0 \text{ m/s}$.

A force acts on the object.

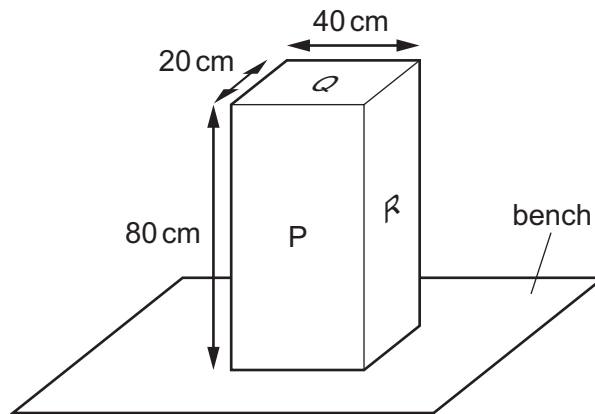
After a time, the object is moving at -4.0 m/s .

The mass of the object is 5.0 kg .

What is the change in momentum of the body?

- A** -35 kg m/s **B** -5.0 kg m/s **C** $+5.0 \text{ kg m/s}$ **D** $+35 \text{ kg m/s}$
- 10 Which energy resource is **not** renewable?
- A** geothermal
B nuclear fission
C solar
D wind
- 11 A car of mass 500 kg is moving at 10 m/s . The engine does work on the car and the speed increases to 16 m/s .
- How much work is done by the engine to increase the speed of the car?
- A** 3000 J **B** 9000 J **C** 39000 J **D** 78000 J

- 12 The diagram shows a solid block resting on a bench. The dimensions of the block are shown.



On which labelled surface should the block rest to produce the smallest pressure on the bench?

- A** P
B Q
C R
D P, Q and R produce the same pressure
- 13 An object is 60 cm below the surface of a liquid. The pressure due to the liquid at this depth is 9000 Pa.

What is the density of the liquid?

- A** 15 kg/m^3 **B** 540 kg/m^3 **C** 1500 kg/m^3 **D** $54\,000 \text{ kg/m}^3$
- 14 Which row describes the forces between the molecules and the motion of the molecules in a solid?

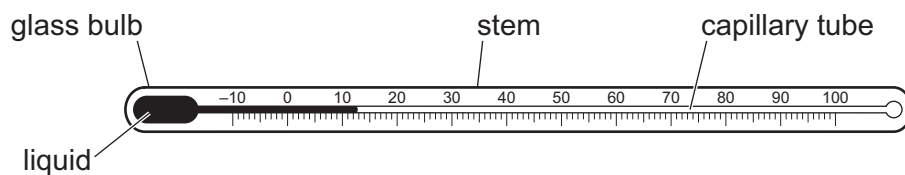
	forces between molecules	motion of molecules
A	strong	move freely
B	strong	vibrate only
C	weak	move freely
D	weak	vibrate only

15 Wet clothes are hanging outside to dry.

Which condition decreases the rate of evaporation of the water from the clothes?

- A** folded clothes
- B** higher temperature
- C** wetter clothes
- D** windy day

16 The diagram shows a liquid-in-glass thermometer.



The design of this thermometer includes the following features.

- 1 a liquid which expands linearly when it is heated
- 2 a glass bulb which has a thick glass wall
- 3 a capillary tube with a very small diameter

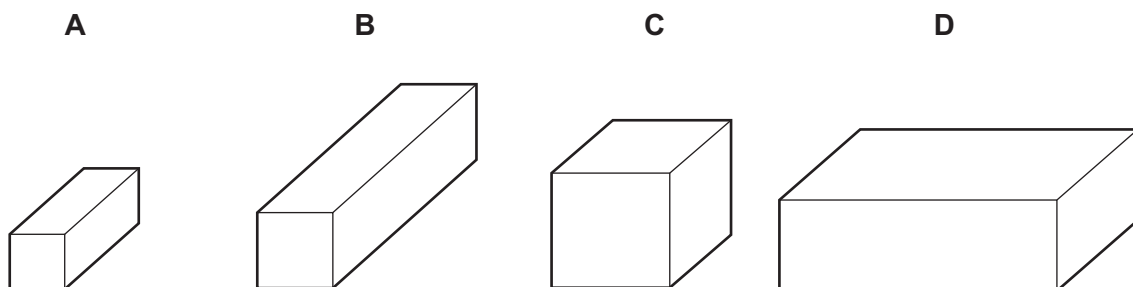
Which features increase the sensitivity of the thermometer?

- A** 1 only
- B** 1 and 2
- C** 2 and 3
- D** 3 only

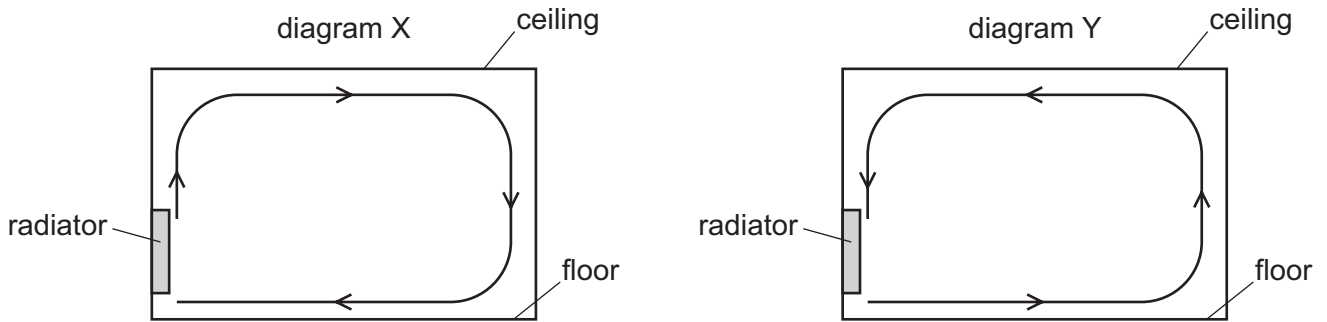
17 The diagrams show four blocks of steel. The blocks are all drawn to the same scale.

The same quantity of thermal energy is given to each block.

Which block shows the greatest rise in temperature?



- 18 A room is heated by a radiator. The diagrams X and Y show two possible circulations of hot air, which heat the room.

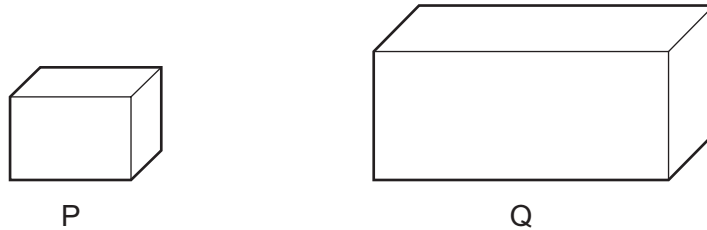


Which diagram and reason explain the heating of the room by convection?

	diagram	reason
A	X	air density decreases when air is heated
B	X	air density increases when air is heated
C	Y	air density decreases when air is heated
D	Y	air density increases when air is heated

- 19 Two copper containers P and Q are filled with hot water.

The diagrams are both drawn to the same scale.

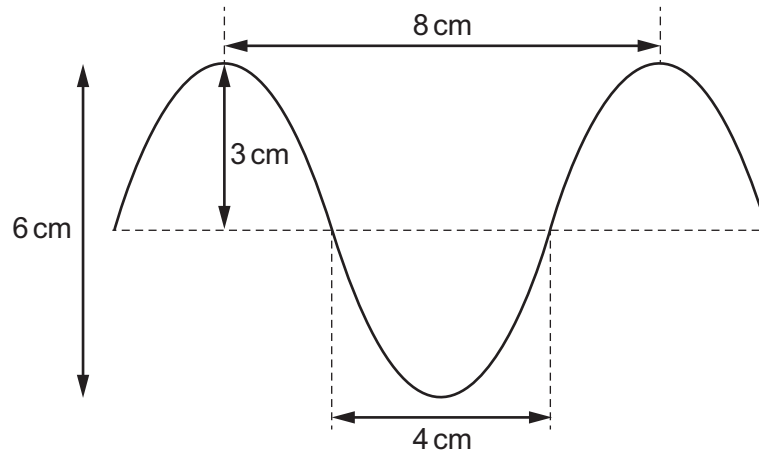


Container P emits more infrared radiation from its surfaces than container Q.

What is a possible reason for this?

- A** The surfaces of P are painted white and the surfaces of Q are painted black.
- B** The surfaces of P are shiny and the surfaces of Q are dull.
- C** The surfaces of P have a smaller area than the surfaces of Q.
- D** The water in P is hotter than the water in Q.

20 The diagram shows a wave.



What are the amplitude and the wavelength of this wave?

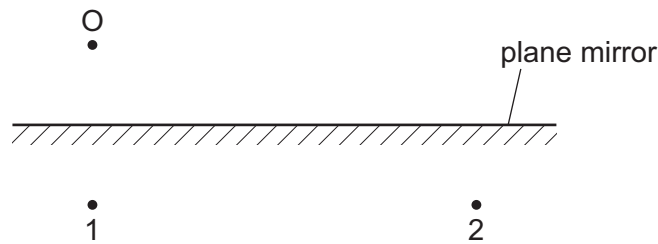
	amplitude / cm	wavelength / cm
A	3	4
B	3	8
C	6	4
D	6	8

21 The frequency of the microwaves used in a microwave oven is 2400 MHz.

What is the wavelength of these microwaves?

- A** 0.125 m **B** 8.00 m **C** 125 m **D** 7200 m

22 An object O is placed in front of a plane mirror as shown.



Which row is correct?

	position of the image	nature of the image
A	1	real
B	1	virtual
C	2	real
D	2	virtual

23 Which statement is correct?

- A** The speed of light in glass is equal to the speed of light in a vacuum multiplied by the refractive index of glass.
- B** The incident angle of a light ray at an air-glass surface is the angle between the ray and the glass surface.
- C** The sine of the critical angle at an air-glass surface is equal to $\frac{1}{\text{refractive index of glass}}$.
- D** The angle of refraction for light passing through an air-glass surface is proportional to the angle of incidence at that surface.

24 An intruder alarm sensor detects that a person is warmer than his surroundings.

Which type of electromagnetic wave does the sensor detect?

- A** infrared
- B** radio
- C** ultraviolet
- D** visible light

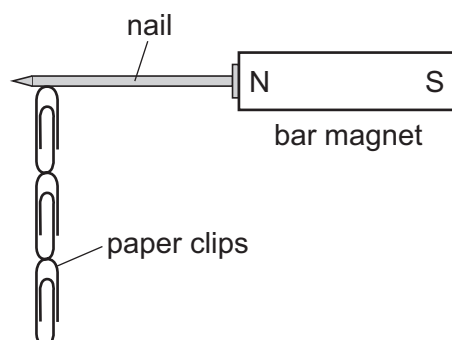
25 A dolphin sends out a sound wave. An echo returns 0.010 s later from a fish which is 7.5 m from the dolphin.

What is the speed of the sound wave in water?

- A** 0.075 m/s **B** 0.15 m/s **C** 750 m/s **D** 1500 m/s

26 Four nails **A**, **B**, **C** and **D** are tested to find which makes the strongest permanent magnet.

One of the nails is placed against a bar magnet and the number of paper clips which the nail can support is recorded.

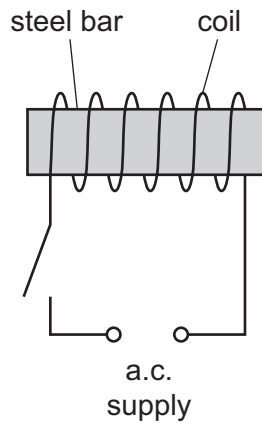


The bar magnet is then removed and the number of paper clips remaining attached to the nail is recorded. Each nail is tested individually.

Which nail becomes the strongest permanent magnet?

	number of paper clips attached to the nail	
	bar magnet present	bar magnet removed
A	2	0
B	2	1
C	4	3
D	5	2

- 27** A student wants to demagnetise a steel bar. He uses the apparatus shown. He switches on the circuit for a few seconds and then switches off. He finds that the steel bar is still magnetised.



What should he do to improve his method?

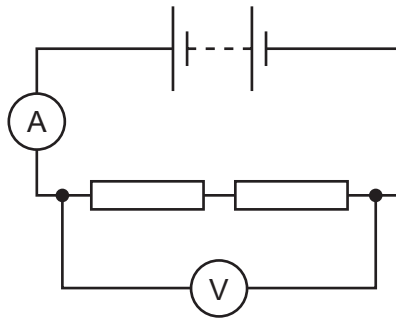
- A** change the supply from an alternating to a direct voltage
 - B** use a lower alternating voltage
 - C** remove the steel bar from the coil whilst the circuit is switched on
 - D** use a coil that has fewer turns on it
- 28** A cloth is used to rub an uncharged plastic rod.

Both the rod and the cloth become charged.

Why does the plastic rod become negatively charged and the cloth become positively charged?

- A** The rod gains electrons and the cloth gains positive charges.
- B** The rod gains electrons and the cloth loses electrons.
- C** The rod loses electrons and the cloth gains electrons.
- D** The rod loses electrons and the cloth loses positive charges.

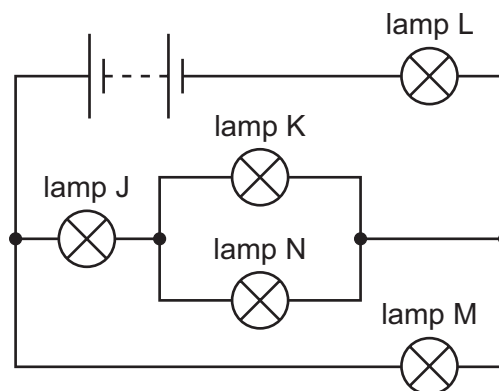
- 29 A student uses the circuit shown to determine the resistance of two identical resistors.



The voltmeter reading is 2.2 V and the ammeter reading is 0.25 A.

What is the resistance of each resistor?

- A** 0.275 Ω **B** 0.55 Ω **C** 4.4 Ω **D** 8.8 Ω
- 30 A cell passes a current of 2.0 A in a circuit for 30 s. In this time the cell transfers 120 J of energy.
- What is the electromotive force (e.m.f.) of the cell?
- A** 0.50 V **B** 1.5 V **C** 2.0 V **D** 8.0 V
- 31 The circuit shown contains five lamps J, K, L, M and N. All the lamps are glowing.

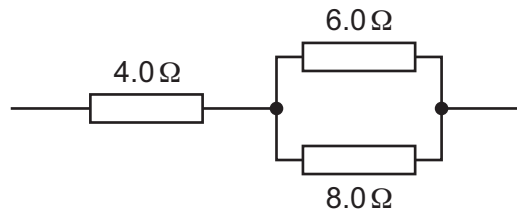


One lamp is removed and two other lamps go out.

Which lamp is removed?

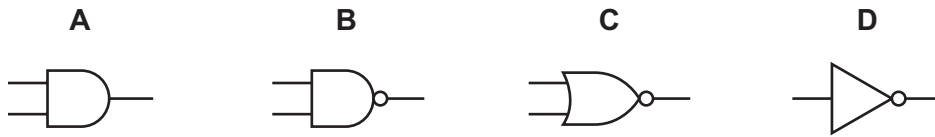
- A** lamp J
B lamp K
C lamp L
D lamp M

32 What is the effective resistance of the following combination of resistors?

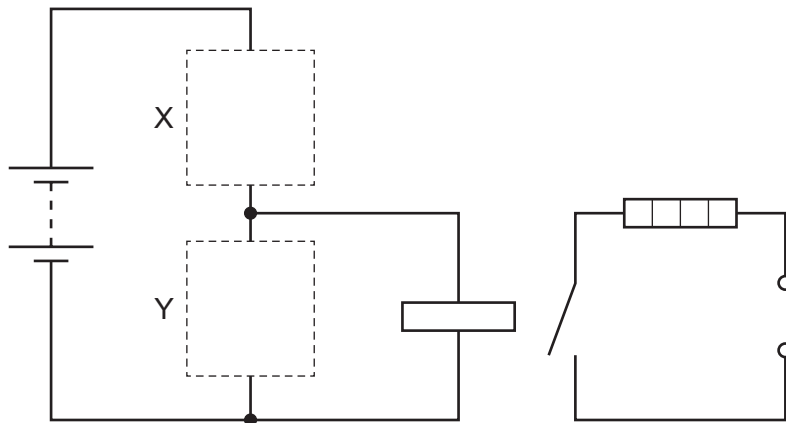


- A 1.8 Ω B 7.4 Ω C 11 Ω D 18 Ω

33 Which symbol represents a NAND gate?



34 The diagram shows a circuit used to switch on a heater when the temperature drops below a certain value.

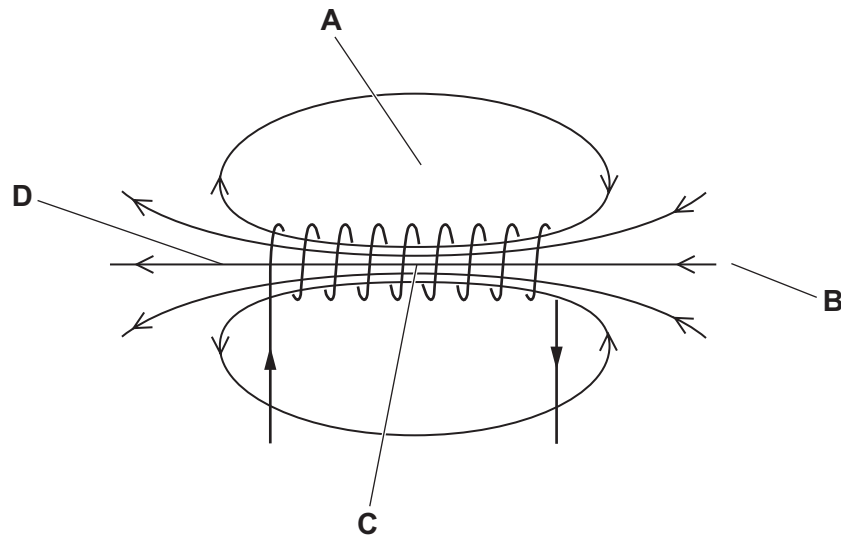


Which row shows the components that should be connected at X and at Y?

	X	Y
A		
B		
C		
D		

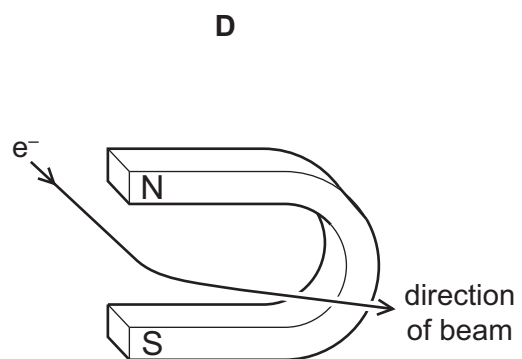
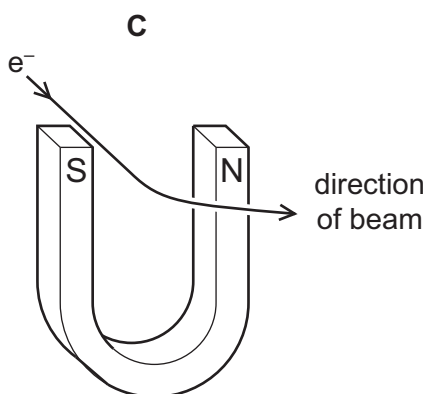
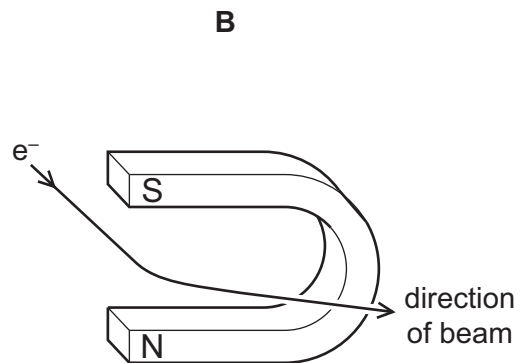
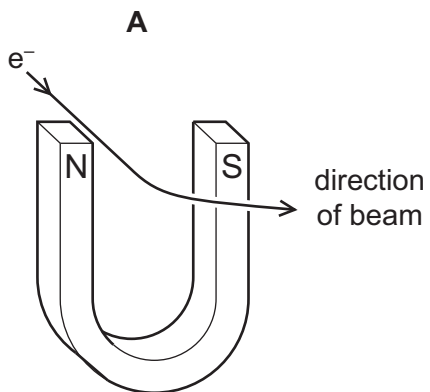
35 The diagram shows the magnetic field due to a current in a solenoid.

Where is the magnetic field the strongest?



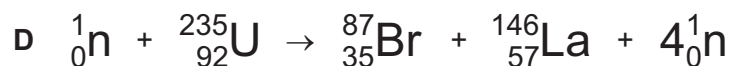
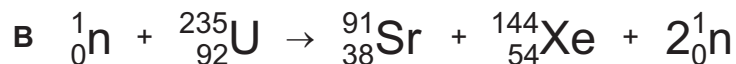
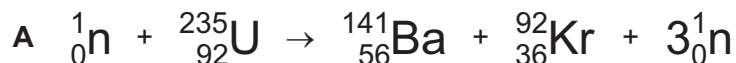
36 A beam of electrons is passed through the magnetic field of a magnet.

How must the magnet be positioned to deflect the beam in the direction shown?



37 Uranium-235 can undergo nuclear fission in many ways.

Which equation correctly shows a possible fission reaction for uranium-235?



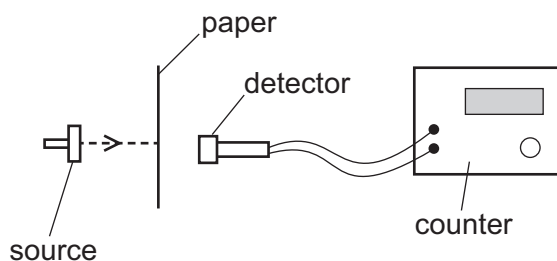
38 A radioactive material has a half-life of 20 days.

A sample of the material contains 8.0×10^{10} atoms.

How many atomic nuclei have decayed after 60 days?

- A 1.0×10^{10} B 4.0×10^{10} C 6.0×10^{10} D 7.0×10^{10}

39 A thin sheet of paper is placed between a radioactive source and a radiation detector. The count rate falls to a very low reading.



From this result, which type of radiation is the source emitting?

- A α -particles
 B β -particles
 C γ -rays
 D X-rays

40 α -particles, β -particles and γ -rays are emitted by radioactive nuclei when they decay.

Which emissions can be deflected by an electric field?

- A** α -particles and β -particles only
- B** β -particles and γ -rays only
- C** γ -rays and α -particles only
- D** α -particles, β -particles and γ -rays

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Cambridge IGCSE™ (9–1)

PHYSICS (9–1)

0972/22

Paper 2 Multiple Choice (Extended)

May/June 2020

MARK SCHEME

Maximum Mark: 40

Published

Students did not sit exam papers in the June 2020 series due to the Covid-19 global pandemic.

This mark scheme is published to support teachers and students and should be read together with the question paper. It shows the requirements of the exam.

Mark schemes should usually be read together with the Principal Examiner Report for Teachers. However, because students did not sit exam papers, there is no Principal Examiner Report for Teachers for the June 2020 series.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the June 2020 series for most Cambridge IGCSE™ and Cambridge International A & AS Level components, and some Cambridge O Level components.

This document consists of **3** printed pages.

Question	Answer	Marks
1	A	1
2	A	1
3	A	1
4	C	1
5	B	1
6	C	1
7	B	1
8	C	1
9	A	1
10	B	1
11	C	1
12	A	1
13	C	1
14	B	1
15	A	1
16	D	1
17	A	1
18	A	1
19	D	1
20	B	1
21	A	1
22	B	1
23	C	1
24	A	1
25	D	1
26	C	1
27	C	1
28	B	1

Question	Answer	Marks
29	C	1
30	C	1
31	A	1
32	B	1
33	B	1
34	B	1
35	C	1
36	B	1
37	A	1
38	D	1
39	A	1
40	A	1



Cambridge IGCSE™ (9–1)

PHYSICS

0972/21

Paper 2 Multiple Choice (Extended)

October/November 2020

45 minutes

You must answer on the multiple choice answer sheet.

You will need: Multiple choice answer sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)

INSTRUCTIONS

- There are **forty** questions on this paper. Answer **all** questions.
- For each question there are four possible answers **A**, **B**, **C** and **D**. Choose the **one** you consider correct and record your choice in soft pencil on the multiple choice answer sheet.
- Follow the instructions on the multiple choice answer sheet.
- Write in soft pencil.
- Write your name, centre number and candidate number on the multiple choice answer sheet in the spaces provided unless this has been done for you.
- Do **not** use correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.
- Take the weight of 1.0 kg to be 10 N (acceleration of free fall = 10 m/s^2).

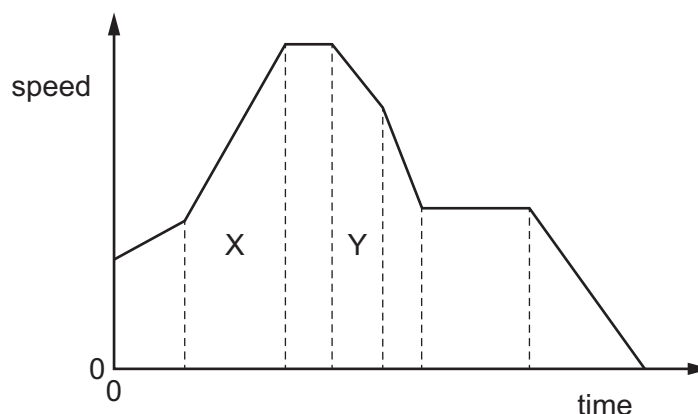
INFORMATION

- The total mark for this paper is 40.
- Each correct answer will score one mark. A mark will not be deducted for a wrong answer.
- Any rough working should be done on this question paper.

This document has **16** pages. Blank pages are indicated.



- 1 For which one of the following measurements would a micrometer screw gauge be most suitable?
- A length of this page
 - B length of a pencil
 - C diameter of a wire
 - D diameter of an atom
- 2 The speed–time graph represents a journey.



How does the graph show that the distance travelled in section X of the journey is greater than the distance travelled in section Y?

- A The area below section X of the graph is greater than the area below section Y.
 - B The gradient of section X of the graph is greater than the gradient of section Y.
 - C The speed at the end of section X of the journey is greater than the speed at the end of section Y.
 - D The time for section X of the journey is greater than the time for section Y.
- 3 A car is travelling at a velocity of 2.0 m/s . It accelerates at a constant 0.20 m/s^2 for 2.5 minutes.
- What is the final velocity of the car?
- A 2.5 m/s
 - B 5.2 m/s
 - C 30 m/s
 - D 32 m/s
- 4 Which quantity is weight an example of?
- A acceleration
 - B force
 - C mass
 - D pressure

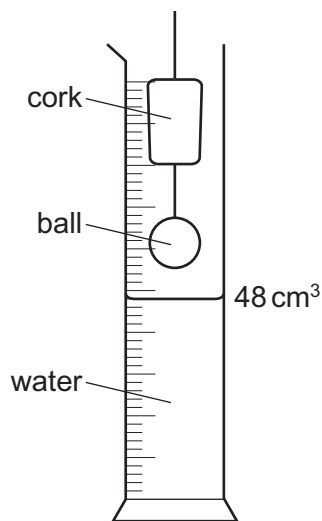
- 5 A sphere P, made of steel, has a weight of 10 N on Earth.

Another sphere Q, also made of steel, has a weight of 10 N on Mars.

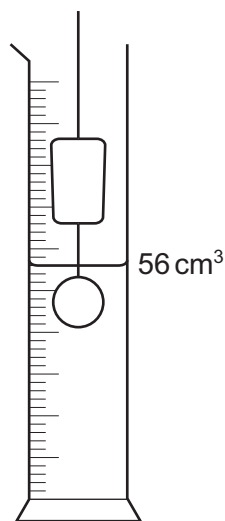
The gravitational field strength on Earth is greater than the gravitational field strength on Mars.

Which statement is correct?

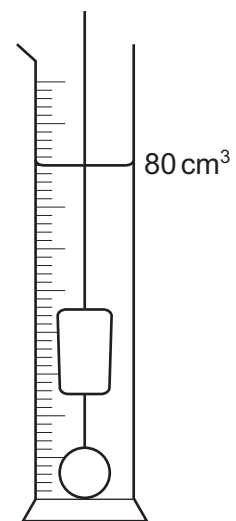
- A** The mass of sphere P is the same as the mass of sphere Q.
B The mass of sphere P is less than the mass of sphere Q.
C On Mars, the weight of sphere P is more than 10 N.
D On Earth, the weight of sphere Q is less than 10 N.
- 6 A metal ball is attached to a cork and is lowered into a measuring cylinder, pulling the cork into the water, as shown.



ball and cork
above the water



ball fully submerged
but cork above the water



both ball and cork
fully submerged

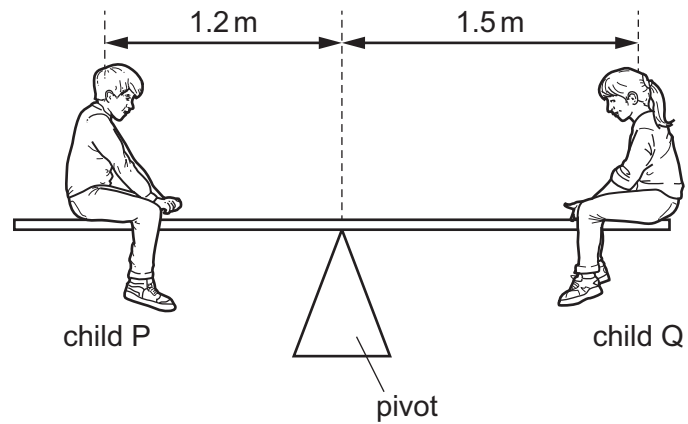
The mass of the cork is 4.8 g.

What is the density of the cork?

- A** 0.15 g/cm^3 **B** 0.20 g/cm^3 **C** 0.60 g/cm^3 **D** 5.0 g/cm^3

- 7 A uniform plank rests on a pivot at its centre.

Two children P and Q sit on the plank in the positions shown.



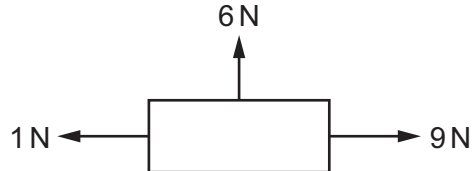
The mass of child P is 25 kg.

The plank is balanced.

What is the mass of child Q?

- A** 20 kg **B** 25 kg **C** 31 kg **D** 45 kg

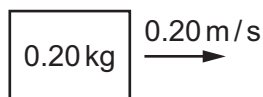
- 8 The diagram shows three forces acting on an object.



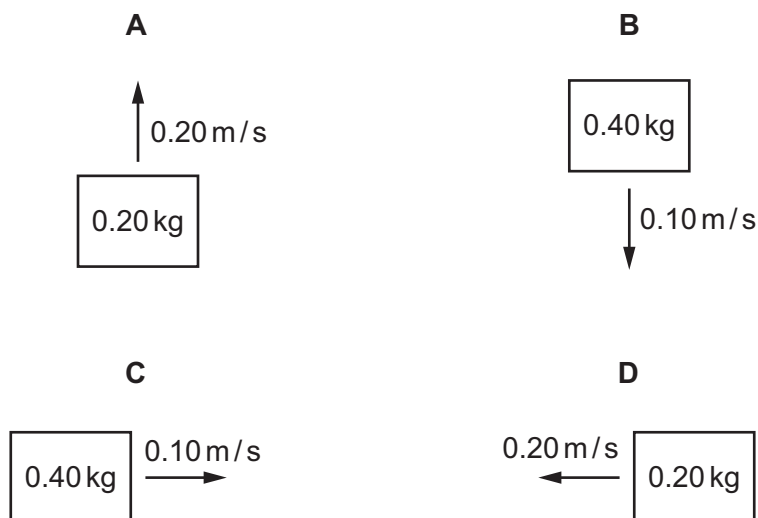
What is the value of the resultant force acting on the object?

- A** 2 N **B** 10 N **C** 14 N **D** 16 N

- 9 An object with a mass of 0.20 kg moves at 0.20 m/s, as shown.



Which other object has a momentum that is identical to the momentum of this object?



- 10 A stone is dropped from rest at a height of 2.0 m above the surface of a planet.

The planet has no atmosphere.

The speed of the stone just before reaching the surface of the planet is 3.8 m/s.

What is the acceleration of free fall on the planet?

- A** zero **B** 1.9 m/s^2 **C** 3.6 m/s^2 **D** 7.2 m/s^2

- 11 An electric motor uses 1000 J of electrical energy. It provides 450 J of useful output energy.

What is the efficiency of the motor?

- A** 4.5% **B** 5.5% **C** 45% **D** 55%

- 12 To calculate the power produced by a force, the size of the force must be known.

What else needs to be known to calculate the power?

	the distance that the force moves the object	the time for which the force acts on the object
A	✓	✓
B	✓	✗
C	✗	✓
D	✗	✗

key

✓ = needed

✗ = not needed

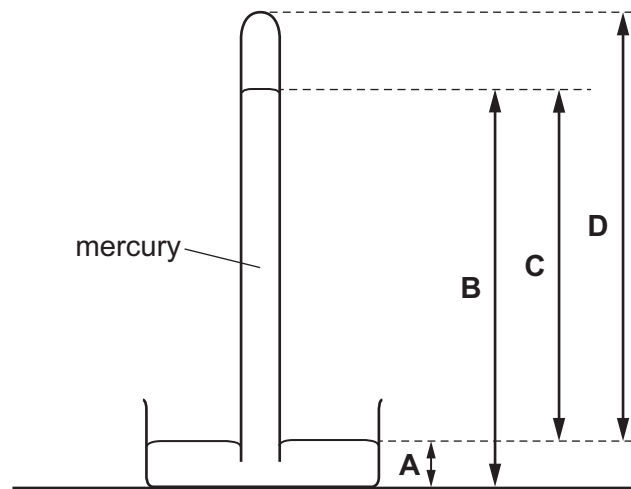
- 13** A barometer reads 780 mm Hg. Mercury has a density of $1.36 \times 10^4 \text{ kg/m}^3$.

What is the pressure of the atmosphere in N/m^2 ?

- A** $1.1 \times 10^4 \text{ N/m}^2$
- B** $1.1 \times 10^5 \text{ N/m}^2$
- C** $1.1 \times 10^7 \text{ N/m}^2$
- D** $1.1 \times 10^8 \text{ N/m}^2$

- 14** The diagram shows a mercury barometer.

Which height is used as a measurement of atmospheric pressure?



- 15** A student splashes water on to her face. Here are three statements about the effects.

- P** The water uses energy to evaporate.
- Q** The water gains energy from the student.
- R** The face of the student cools.

Which statements are correct?

- A** P and Q only
- B** P and R only
- C** Q and R only
- D** P, Q and R

- 16** When a bridge is built, a gap is left between each concrete slab.

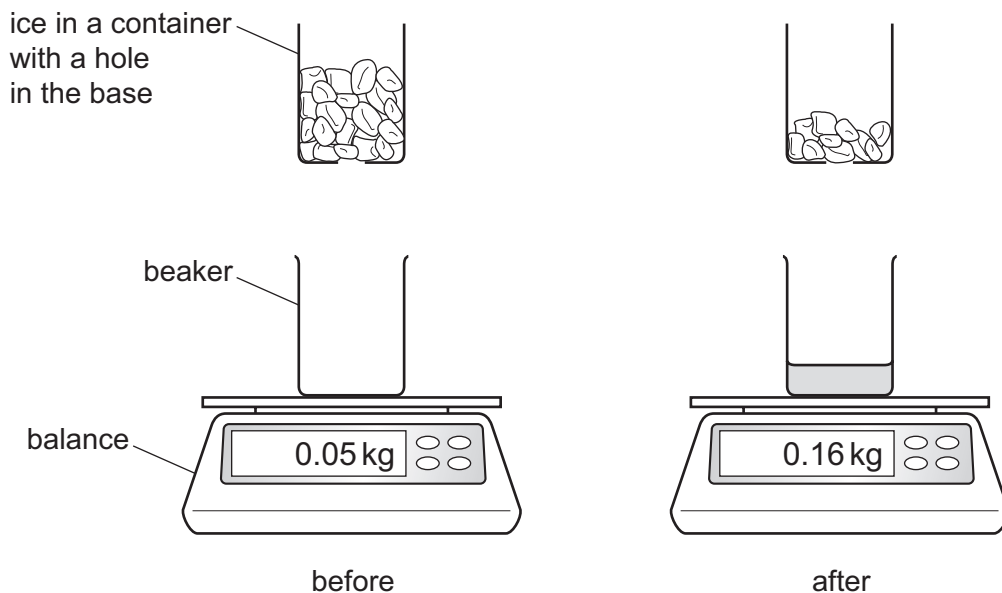
Why are these gaps left?

- A** Concrete expands on warm days.
- B** Concrete contracts on warm days.
- C** The gaps expand on warm days.
- D** The gaps contract on cold days.

- 17 The specific heat capacity of solid P is greater than that of solid Q.

What does this statement mean?

- A Less energy is needed to raise the temperature by 1°C of unit mass of solid P than unit mass of solid Q.
 - B Less energy is needed to melt unit mass of solid P than unit mass of solid Q.
 - C More energy is needed to raise the temperature by 1°C of unit mass of solid P than unit mass of solid Q.
 - D More energy is needed to melt unit mass of solid P than unit mass of solid Q.
- 18 A student placed a number of ice cubes in a container with a hole in the base. He left them to melt so that the water dripped into a beaker placed on a balance. The student recorded the initial mass of the beaker and the final mass of the beaker and water after five minutes.

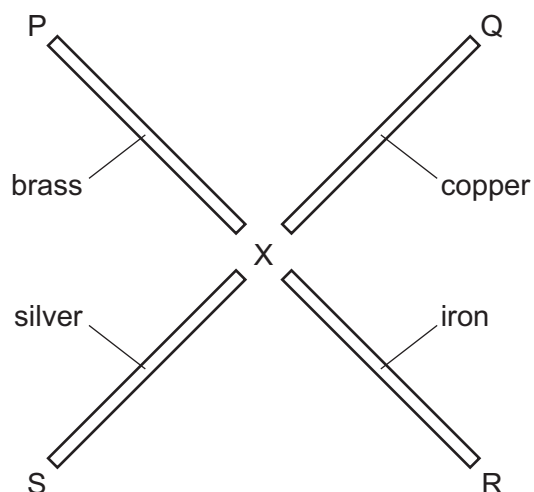


The specific latent heat of fusion for water is 334 J/g .

How much energy was absorbed from the surroundings in order to melt the ice?

- A 37 J
- B 54 J
- C 37 000 J
- D 54 000 J

- 19 The diagram shows four rods. Each rod is made of a different metal.



Wax is used to attach small metal balls at the rod ends P, Q, R and S.

Each rod is the same size. They are heated uniformly by a Bunsen burner at point X.

As the rods warm up, the wax melts and the balls fall off.

Why does the ball on the silver rod fall first?

- A** Silver is the best conductor of heat.
- B** Silver is the worst conductor of heat.
- C** Silver is the best radiator of heat.
- D** Silver is the worst radiator of heat.

- 20 Four cups **A**, **B**, **C** and **D** contain hot coffee.

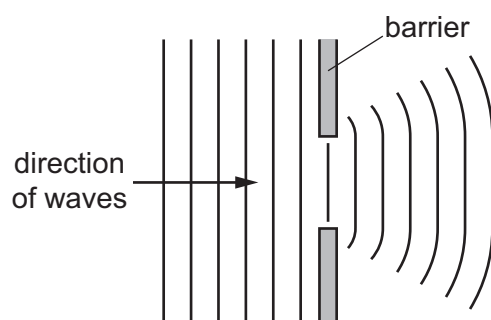
Which cup keeps the coffee warm the longest?

	the outside surface of the cup	the top of the cup
A	black	covered with a lid
B	black	no lid
C	white	covered with a lid
D	white	no lid

21 Which row correctly describes light waves?

	wave type	direction of vibrations
A	longitudinal	parallel to direction of wave travel
B	longitudinal	perpendicular to direction of wave travel
C	transverse	parallel to direction of wave travel
D	transverse	perpendicular to direction of wave travel

22 The diagram shows part of a diffracted wave pattern.



Changes are made to the wavelength and to the gap size to produce a semicircular diffracted wave pattern.

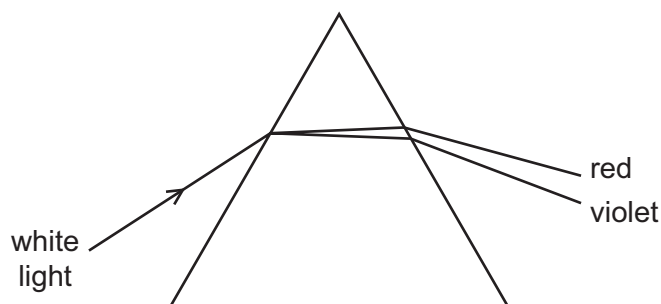
Which row produces the required semicircular diffracted wave pattern?

	gap in barrier	wavelength
A	larger	same
B	larger	smaller
C	same	larger
D	same	smaller

23 Which statement about a thin converging lens is correct?

- A** All rays of light refracted by the lens pass through the principal focus.
- B** All rays initially parallel to the principal axis of the lens are refracted through the principal focus.
- C** The focal length of the lens is the distance between the image and the principal focus.
- D** The focal length of the lens is the distance between the object and the image.

- 24 The diagram shows white light passing through a prism.



Which description of what happens as the light passes into the prism is correct?

- A** The speed of the red light is less than the speed of the violet light and the red light is the least refracted.
- B** The speed of the red light is greater than the speed of the violet light and the red light is the least refracted.
- C** The speed of the violet light is less than the speed of the red light and the violet light is the least refracted.
- D** The speed of the violet light is greater than the speed of the red light and the violet light is the least refracted.

- 25 Which row gives possible values for the speed of sound?

	<u>speed in gas</u> m/s	<u>speed in liquid</u> m/s	<u>speed in solid</u> m/s
A	972	1450	3560
B	972	3560	1450
C	1450	3560	972
D	3560	972	1450

- 26 A police car with its siren sounding is stationary in heavy traffic. A pedestrian notices that, although the loudness of the sound produced does not change, the pitch varies.

Which row describes the amplitude and the frequency of the sound?

	amplitude	frequency
A	constant	constant
B	constant	varying
C	varying	constant
D	varying	varying

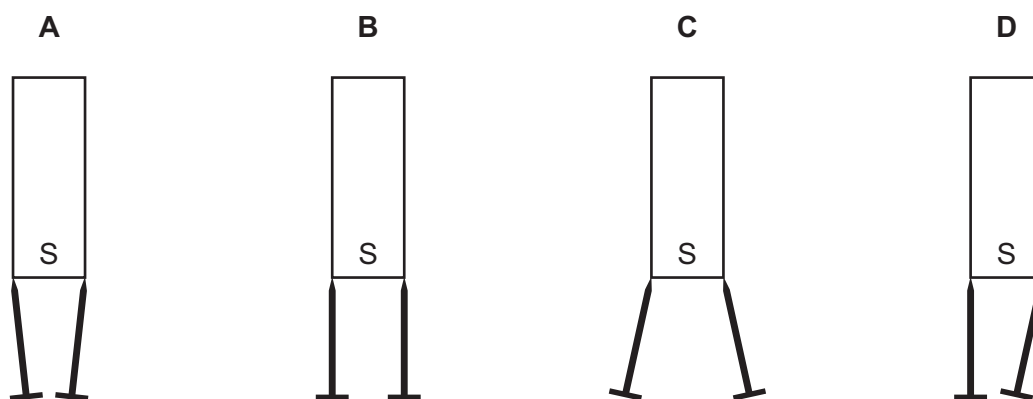
- 27 A piece of steel is slightly magnetised. It is hit several times with a hammer.

What effect will this have on the steel?

	the steel is parallel to a strong magnetic field	the steel is at right-angles to a weak magnetic field
A	it becomes magnetised more strongly	it becomes magnetised more strongly
B	it becomes magnetised more strongly	it loses its magnetism
C	it loses its magnetism	it becomes magnetised more strongly
D	it loses its magnetism	it loses its magnetism

- 28 Two soft-iron pins are suspended from the S pole of a bar magnet.

Which diagram shows how the pins are deflected?



- 29 A negatively charged plastic rod is brought near to an uncharged metal sphere and held there.

What happens when the metal sphere is earthed?

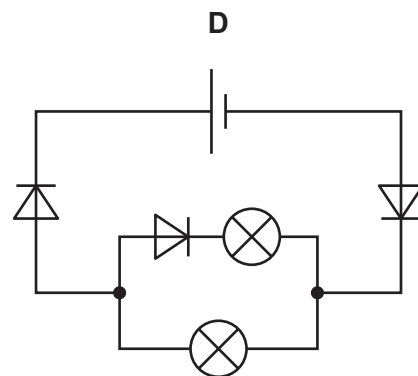
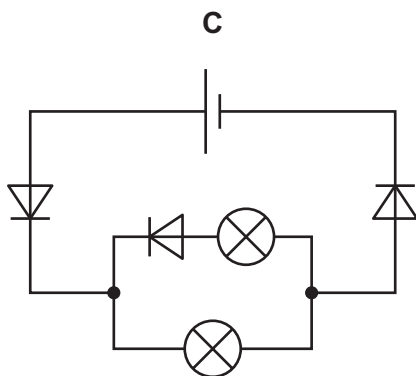
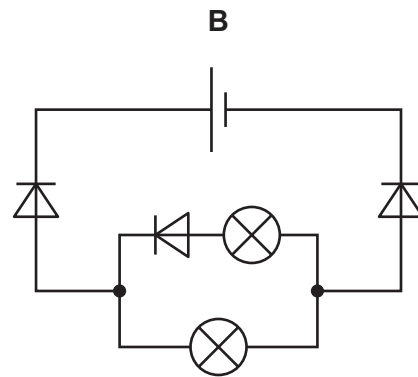
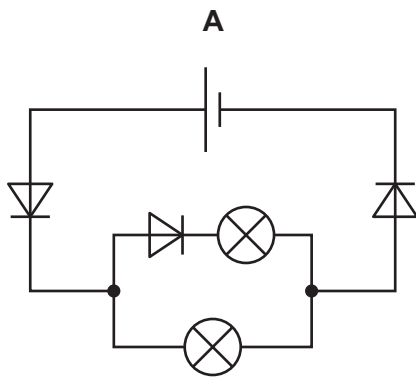
- A** Electrons flow from the metal sphere to earth.
B Electrons flow from earth to the metal sphere.
C Positive charge flows from the metal sphere to earth.
D Positive charge flows from earth to the metal sphere.
- 30 Which statement defines the electromotive force (e.m.f.) of a cell?
- A** the current in the cell when 1.0 C of charge flows in 1.0 s
B the current supplied by the cell to drive 1.0 C of charge around a complete circuit
C the energy supplied by the cell to drive 1.0 C of charge around a complete circuit
D the energy supplied by the cell to drive 1.0 A of current around a complete circuit

31 Four wires are made of the same metal.

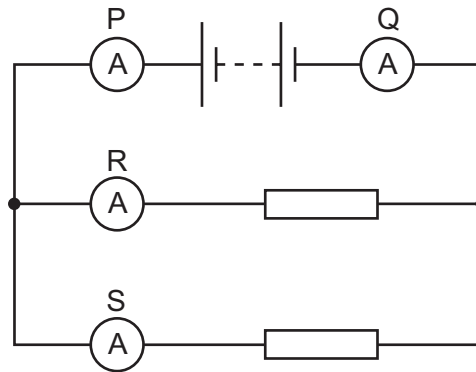
Which wire has the greatest resistance?

- A** a 100 cm long wire with a diameter of 3.0 mm
- B** a 100 cm long wire with a diameter of 6.0 mm
- C** a 10 cm long wire with a diameter of 3.0 mm
- D** a 10 cm long wire with a diameter of 6.0 mm

32 In which circuit is there just a single lamp lit?



- 33 A student uses four ammeters P, Q, R and S to measure the current in different parts of the circuit shown.

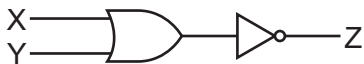


Which two ammeters read the largest current?

- A** P and Q **B** P and R **C** R and Q **D** R and S
- 34 Which combination of logic gates gives the truth table shown?

inputs		output
X	Y	Z
0	0	1
0	1	1
1	0	1
1	1	0

A



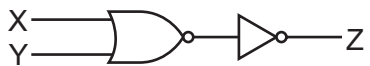
B



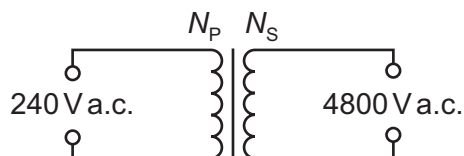
C



D



- 35 A transformer is needed to convert a supply of 240 V a.c. into 4800 V a.c..

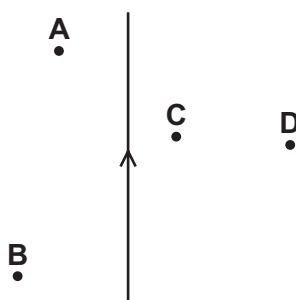


Which pair of coils would be suitable for this transformer?

	number of turns on primary coil N_P	number of turns on secondary coil N_S
A	50	1 000
B	240	48 000
C	480	24
D	2000	100

- 36 The diagram shows part of a long current-carrying conductor.

At which point is the magnetic field strongest?

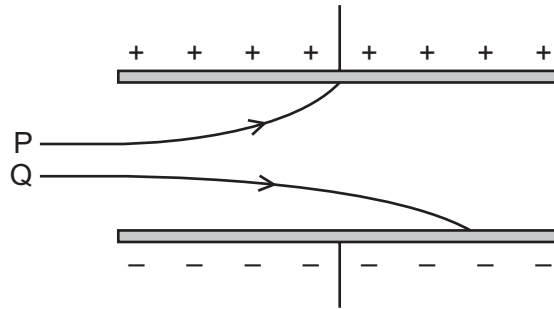


- 37 A beam of particles moves through a magnetic field.

In which situation do the particles experience a magnetic force?

- A** a beam of α -particles moving parallel to the magnetic field lines
 - B** a beam of electrons moving parallel to the magnetic field lines
 - C** a beam of β -particles moving perpendicularly across the magnetic field lines
 - D** a beam of neutrons moving perpendicularly across the magnetic field lines
- 38 Which statement is correct for the nucleus of **any** atom?
- A** The nucleus contains electrons, neutrons and protons.
 - B** The nucleus contains the same number of protons as neutrons.
 - C** The nucleus has a total charge of zero.
 - D** The nucleus is very small compared with the size of the atom.

39 Two beams of radiation, P and Q, enter an electric field as shown.



Which type of radiations are P and Q?

	P	Q
A	beta (β)	alpha (α)
B	beta (β)	gamma (γ)
C	gamma (γ)	alpha (α)
D	gamma (γ)	gamma (γ)

40 Which equation represents the β -decay of lead-209?

- A** ${}^{209}_{82}\text{Pb} + {}^0_{-1}\text{e} \rightarrow {}^{209}_{83}\text{Bi}$
- B** ${}^{209}_{82}\text{Pb} + {}^0_{-1}\text{e} \rightarrow {}^{209}_{81}\text{Tl}$
- C** ${}^{209}_{82}\text{Pb} \rightarrow {}^{209}_{83}\text{Bi} + {}^0_{-1}\text{e}$
- D** ${}^{209}_{82}\text{Pb} \rightarrow {}^{209}_{81}\text{Tl} + {}^0_{-1}\text{e}$

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Cambridge IGCSE™ (9–1)

PHYSICS

0972/21

Paper 2 Multiple Choice (Extended)

October/November 2020

MARK SCHEME

Maximum Mark: 40

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

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This document consists of **3** printed pages.

Question	Answer	Marks
1	C	1
2	A	1
3	D	1
4	B	1
5	B	1
6	B	1
7	A	1
8	B	1
9	C	1
10	C	1
11	C	1
12	A	1
13	B	1
14	C	1
15	D	1
16	A	1
17	C	1
18	C	1
19	A	1
20	C	1
21	D	1
22	C	1
23	B	1
24	B	1
25	A	1
26	B	1
27	B	1
28	C	1

Question	Answer	Marks
29	A	1
30	C	1
31	A	1
32	C	1
33	A	1
34	B	1
35	A	1
36	C	1
37	C	1
38	D	1
39	A	1
40	C	1



Cambridge IGCSE™ (9–1)

PHYSICS

0972/21

Paper 2 Multiple Choice (Extended)

May/June 2021

45 minutes

You must answer on the multiple choice answer sheet.

You will need: Multiple choice answer sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)

INSTRUCTIONS

- There are **forty** questions on this paper. Answer **all** questions.
- For each question there are four possible answers **A**, **B**, **C** and **D**. Choose the **one** you consider correct and record your choice in soft pencil on the multiple choice answer sheet.
- Follow the instructions on the multiple choice answer sheet.
- Write in soft pencil.
- Write your name, centre number and candidate number on the multiple choice answer sheet in the spaces provided unless this has been done for you.
- Do **not** use correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.
- Take the weight of 1.0 kg to be 10 N (acceleration of free fall = 10 m/s^2).

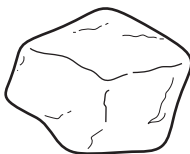
INFORMATION

- The total mark for this paper is 40.
- Each correct answer will score one mark.
- Any rough working should be done on this question paper.

This document has **16** pages.



- 1 The diagram shows a stone of irregular shape.



Which property of the stone can be found by lowering it into a measuring cylinder half-filled with water?

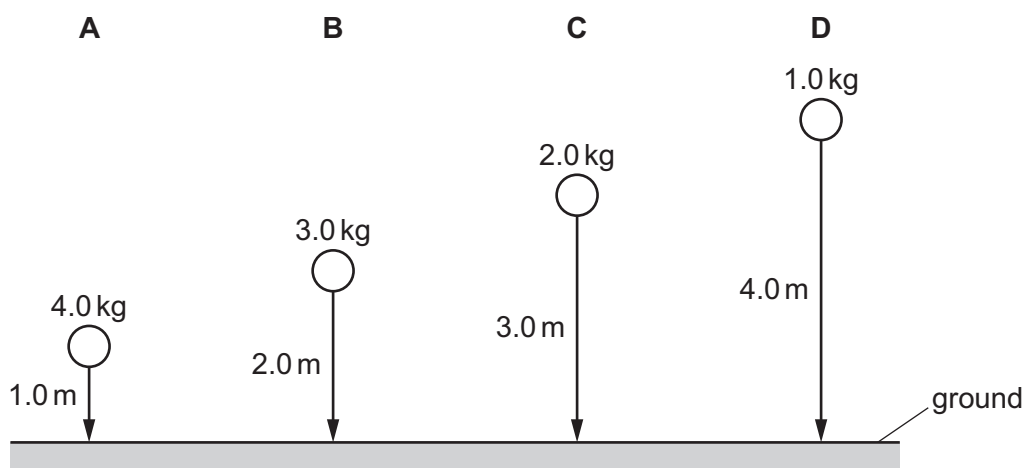
- A** length
B mass
C volume
D weight
- 2 Which row describes speed and velocity?

	speed	velocity
A	scalar	scalar
B	scalar	vector
C	vector	scalar
D	vector	vector

- 3 Four balls with different masses are dropped from the heights shown.

Air resistance may be ignored.

Which ball has the smallest average speed?



- 4 An object of mass 2.0 kg is taken from the Earth, where the gravitational field strength is 10 N/kg, to the Moon, where the gravitational field strength is 1.6 N/kg.

Which row is correct?

	weight on the Earth / N	weight on the Moon / N
A	0.20	0.80
B	0.20	3.2
C	20	0.80
D	20	3.2

- 5 The mass of an empty flask is 34 g.

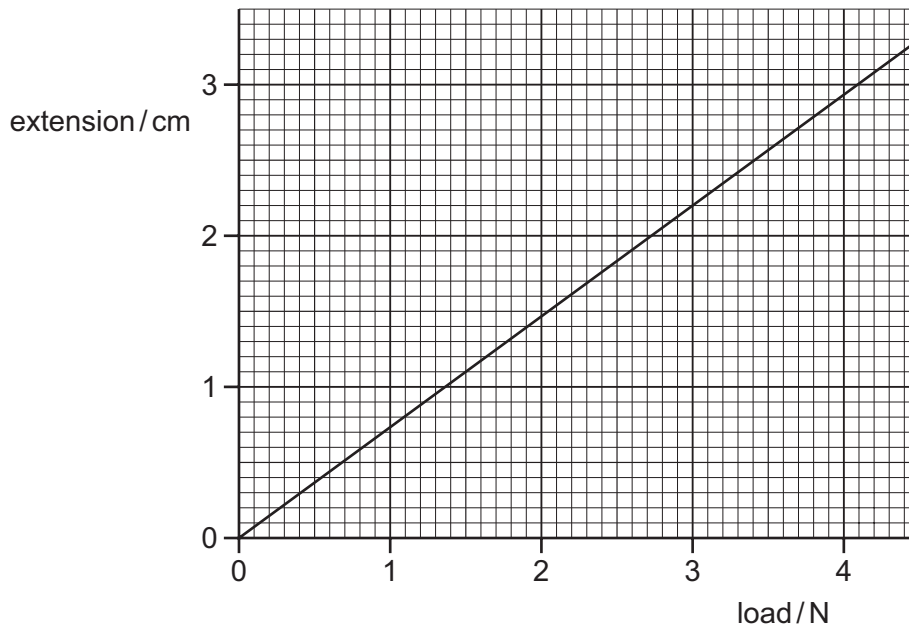
The volume of liquid added to the flask is 20 cm³.

The total mass of the flask and the liquid is 50 g.

What is the density of the liquid?

- A** 0.80 g/cm³ **B** 1.25 g/cm³ **C** 2.50 g/cm³ **D** 4.20 g/cm³

- 6 The extension–load graph for a spring is shown. The unstretched length of the spring is 17.0 cm.



When an object is suspended from the spring, the length of the spring is 19.2 cm.

What is the weight of the object?

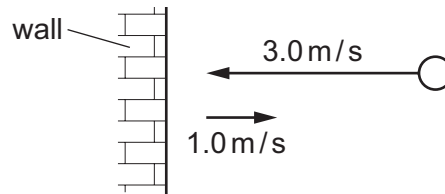
- A** 1.4 N **B** 1.6 N **C** 2.6 N **D** 3.0 N

- 7 A cart has a mass of 10 kg. A boy pushes on the cart horizontally with a force of 50 N. The cart accelerates at 0.50 m/s^2 .

What is the frictional force acting on the cart?

- A** 5.0 N **B** 20 N **C** 30 N **D** 45 N

- 8 A ball has a mass of 2.0 kg. The ball approaches a wall at a speed of 3.0 m/s and rebounds at a speed of 1.0 m/s .



What is the impulse on the wall?

- A** 4.0 N **B** 4.0 Ns **C** 8.0 N **D** 8.0 Ns

- 9 Which situation involves no work being done and no energy being transferred?

- A** a car skidding to a stop on a road
B a crane lifting a load
C a heavy load hanging from a strong bar
D a student dragging a big box over a rough floor

- 10 A student suggests that there are several ways of transferring energy to a small, stationary block of iron on a smooth table. He makes the following suggestions.

- 1 Heat it.
- 2 Shine light on it.
- 3 Pass a current through it.

Which suggestions are correct?

- A** 1 and 2 only **B** 1 and 3 only **C** 2 and 3 only **D** 1, 2 and 3

- 11 An engine produces 240 kJ of energy in 2.0 minutes.

What is the power output of the engine?

- A** 2.0 kW **B** 120 kW **C** 480 kW **D** 28 800 kW

- 12** A book has a mass of 400 g.

The surface of the book in contact with a table has dimensions $0.10 \text{ m} \times 0.20 \text{ m}$.

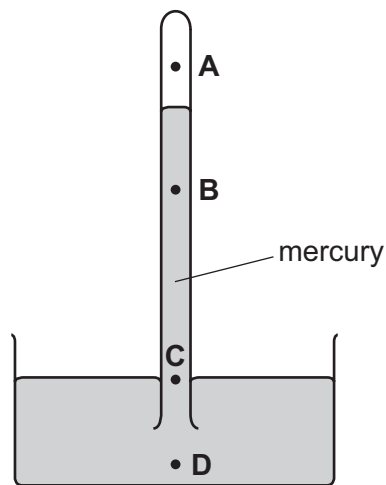
The gravitational field strength g is 10 N/kg .

What is the pressure exerted on the table due to the book?

- A** 0.08 N/m^2 **B** 8.0 N/m^2 **C** 20 N/m^2 **D** 200 N/m^2

- 13** The diagram shows a mercury barometer.

At which point is the pressure greater than atmospheric pressure?



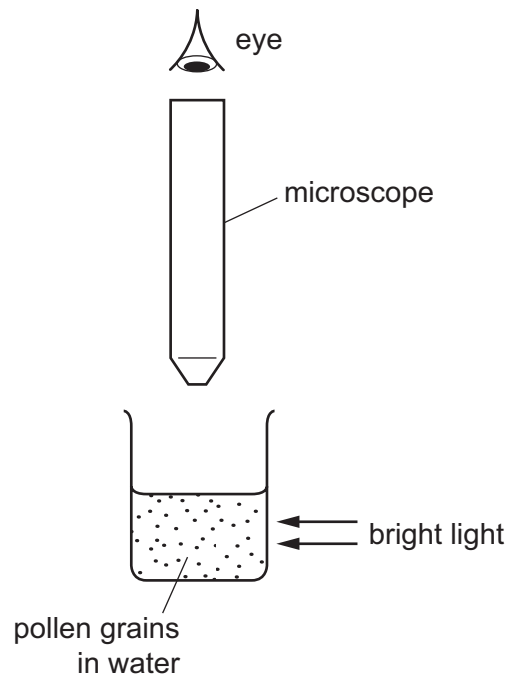
- 14** The table gives information about molecules.

Which row describes a gas?

	force between molecules	distance between molecules
A	strong	close together
B	strong	far apart
C	negligible	far apart
D	negligible	close together

- 15** Very small pollen grains are suspended in water. A bright light shines from the side.

When looked at through a microscope, small specks of light are seen to be moving in a random, jerky manner.



What are the moving specks of light?

- A** pollen grains being hit by other pollen grains
 - B** pollen grains being hit by water molecules
 - C** water molecules being hit by other water molecules
 - D** water molecules being hit by pollen grains
- 16** Why are small gaps left between the metal rails of a railway track?
- A** to allow for expansion of the rails on a hot day
 - B** to allow for contraction of the rails on a hot day
 - C** to allow for expansion of the rails on a cold day
 - D** to allow for contraction of the rails on a cold day

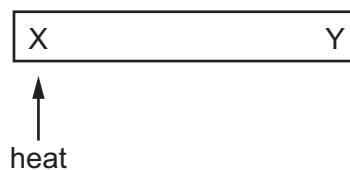
- 17 A block of metal absorbs 2000 J of thermal energy.

The temperature of the block rises from 10 °C to 20 °C.

The mass of the block is 2.0 kg.

What is the specific heat capacity of the metal?

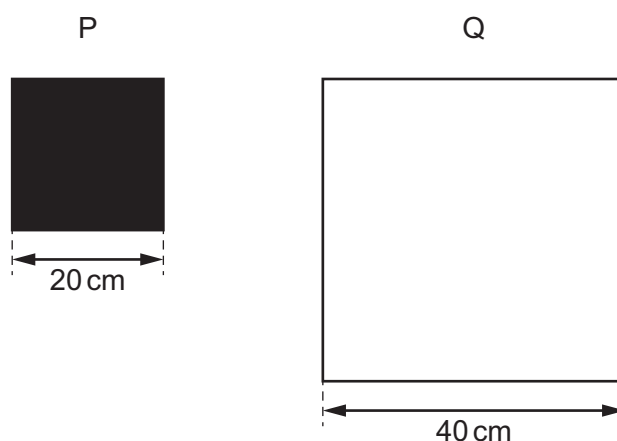
- A** 50 J/(kg °C) **B** 100 J/(kg °C) **C** 200 J/(kg °C) **D** 400 J/(kg °C)
- 18 Which statement about boiling and evaporation is correct?
- A** Boiling requires a supply of thermal energy but evaporation does not.
- B** Evaporation takes place at the surface of a liquid but boiling takes place throughout the liquid.
- C** When water boils in a kettle, its temperature decreases.
- D** When water evaporates, its temperature increases.
- 19 A metal rod is heated at end X.



Why does end Y of the metal rod become hot?

- A** Energy is transferred from end X of the rod to end Y by vibration of positive ions and by movement of electrons.
- B** Energy is transferred from end X of the rod to end Y by movement of positive ions only.
- C** Energy is transferred from end X of the rod to end Y by vibration of positive ions only.
- D** Energy is transferred from end X of the rod to end Y by movement of electrons only.

- 20** Two square sheets of metal, P and Q, are heated to the same temperature. The metal sheets are shown.



Sheet Q is emitting more radiation than sheet P.

Which statement explains this?

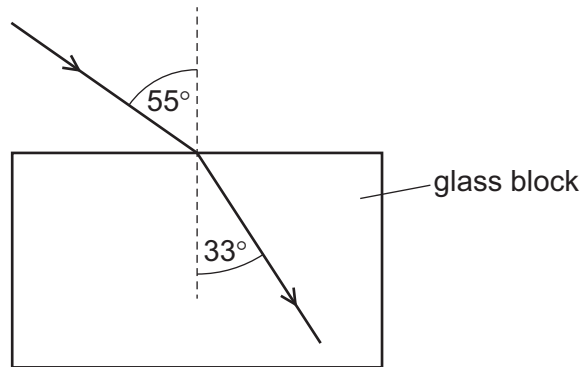
- A** Dull black surfaces are better conductors of radiation.
 - B** Dull black surfaces are better emitters of radiation.
 - C** The surface area of Q is larger than that of P.
 - D** White surfaces are better absorbers of radiation.
- 21** A water wave passes into a region where the wave travels more slowly.

As it passes into the slow region, what happens to the frequency and what happens to the wavelength of the wave?

	frequency	wavelength
A	decreases	remains the same
B	increases	remains the same
C	remains the same	decreases
D	remains the same	increases

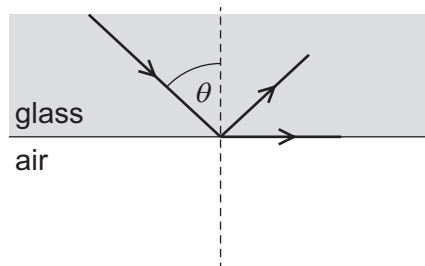
- 22** Light travelling at a speed of 3.0×10^8 m/s strikes the surface of a glass block and undergoes refraction as it enters the block.

The diagram shows a ray of this light before and after it enters the block.



What is the speed of light in the glass?

- A** 1.8×10^8 m/s
B 2.0×10^8 m/s
C 4.5×10^8 m/s
D 5.0×10^8 m/s
- 23** The diagram shows a narrow beam of light incident on a glass-air boundary. Some of the light emerges along the surface of the glass and some is reflected back into the glass.

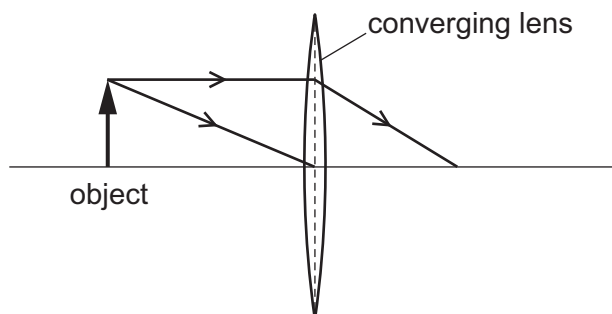


Which row is correct?

	this is an example of total internal reflection	angle θ is the critical angle
A	no	yes
B	no	no
C	yes	no
D	yes	yes

- 24** An object is placed in front of a thin converging lens.

The diagram shows the paths of two rays from the top of the object.



An image of the object is formed on a screen to the right of the lens.

How does this image compare with the object?

- A** It is larger and inverted.
B It is larger and the same way up.
C It is smaller and inverted.
D It is smaller and the same way up.
- 25** A remote-controlled vehicle is travelling on the surface of a planet. The vehicle senses an obstacle ahead. It sends a radio message to the control room from where it is being controlled. The control room is 2.4×10^6 km away from the vehicle. The control room sends a message back to the vehicle telling it to stop.

What is the minimum time that elapses between the vehicle sensing the obstacle and receiving the message back from the control room?

- A** 8.0 ms **B** 16 ms **C** 8.0 s **D** 16 s
- 26** A sound wave is travelling outwards from a loudspeaker into the surrounding air.

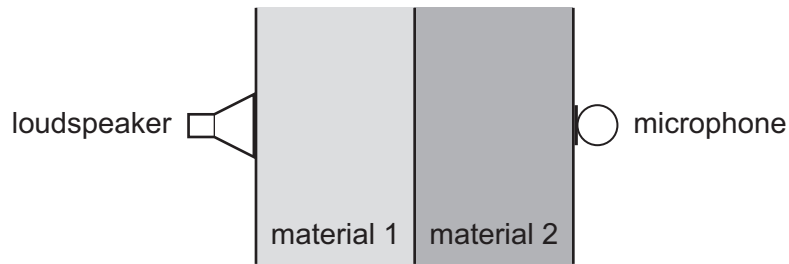
Here are three statements.

- 1 The air pressure is lower at a rarefaction compared with undisturbed air.
- 2 The density of the air is less at a compression compared with undisturbed air.
- 3 The distance from a compression to a rarefaction equals half a wavelength.

Which statements about the sound wave are correct?

- A** 1 and 2 only **B** 1 and 3 only **C** 2 and 3 only **D** 1, 2 and 3

- 27 The sound from a loudspeaker must pass through two materials to reach a microphone.



Which combination of materials gives the shortest time for the sound to reach the microphone?

	material 1	material 2
A	air	hydrogen
B	air	water
C	copper	aluminium
D	water	oil

- 28 Diagram 1 shows a small compass needle with its poles marked. It is not near any magnetic materials.

Diagram 2 shows a bar magnet with its poles marked. The compass needle is placed at point P.

diagram 1

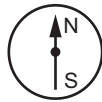


diagram 2



• P

In which direction will the N pole of the compass needle point?

- A** to the left
- B** to the right
- C** down the page
- D** up the page

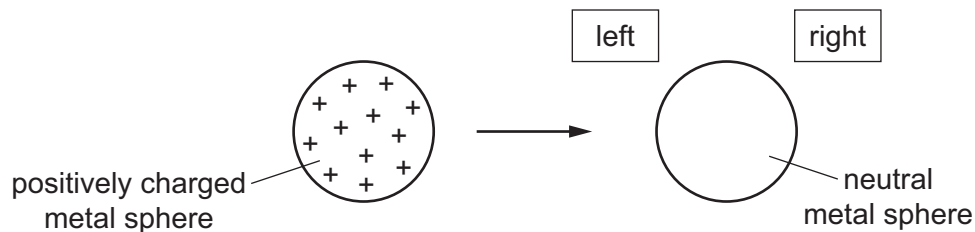
- 29** A student rubs a plastic rod with a cloth.

The rod becomes positively charged.

What has happened to the rod?

- A** It has gained electrons.
 - B** It has gained protons.
 - C** It has lost electrons.
 - D** It has lost protons.
- 30** An isolated metal sphere is positively charged.

It is then brought near to another isolated metal sphere that is neutral.

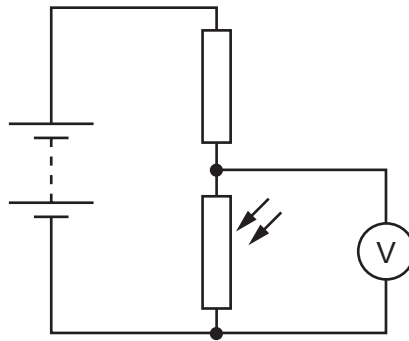


What happens to the charges on the neutral sphere as the positively charged sphere is brought close to it?

- A** Some positive charges move to the left and some negative charges move to the right.
 - B** Some positive charges move to the right and some negative charges move to the left.
 - C** Some positive charges move to the right, but the negative charges do not move.
 - D** The positive charges do not move, but some negative charges move to the left.
- 31** Which statement about the resistance of a metal wire is correct?
- A** The resistance is directly proportional to its length and directly proportional to its cross-sectional area.
 - B** The resistance is directly proportional to its length and inversely proportional to its cross-sectional area.
 - C** The resistance is inversely proportional to its length and directly proportional to its cross-sectional area.
 - D** The resistance is inversely proportional to its length and inversely proportional to its cross-sectional area.

32 The circuit diagram shows a light-dependent resistor (LDR) in a potential divider.

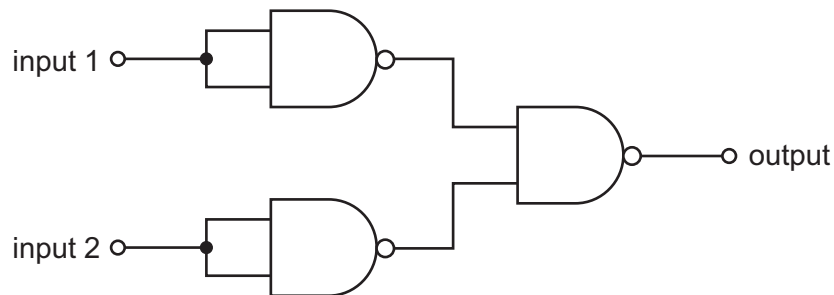
A voltmeter is connected across the LDR.



Which row shows the resistance of the LDR and the potential difference (p.d.) shown on the voltmeter at a specific light level?

	light level	resistance of LDR	p.d. shown on the voltmeter
A	bright	low	high
B	bright	high	low
C	dim	high	high
D	dim	low	low

33 Three NAND gates are connected in a single chip as shown.

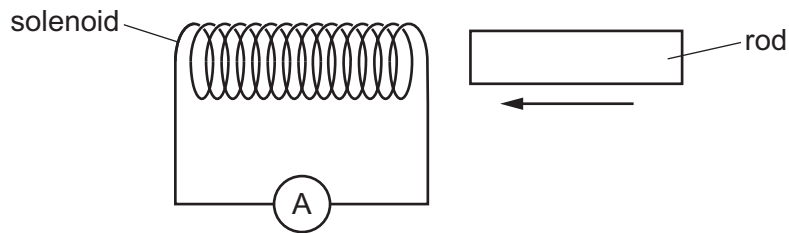


The whole chip behaves as a single logic gate.

Which type of logic gate does the chip act as?

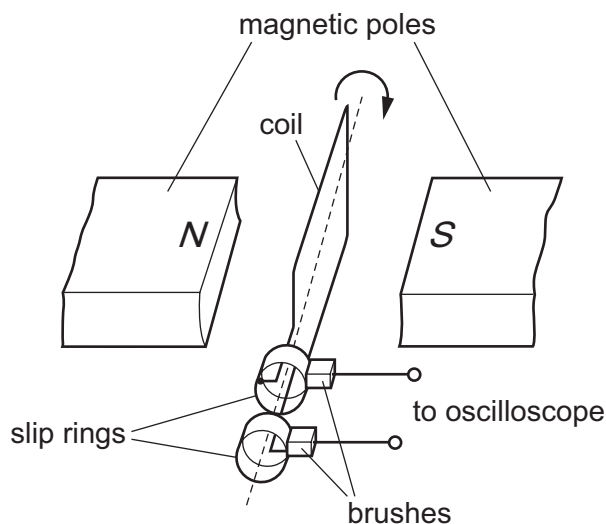
- A** AND gate
- B** NAND gate
- C** NOR gate
- D** OR gate

- 34 A solenoid is connected to a very sensitive ammeter. A rod is inserted into one end of the solenoid. The ammeter shows that there is a small electric current in the solenoid while the rod is moving.



Which rod is being inserted?

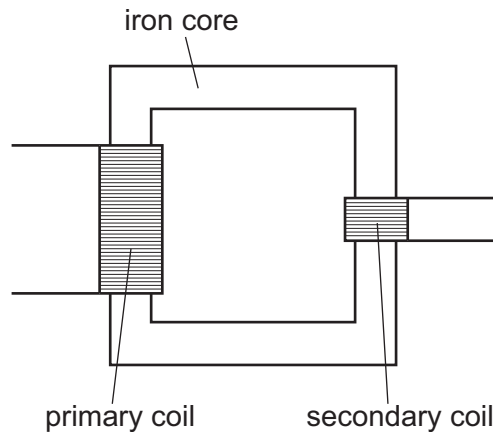
- A a heated copper rod
 - B a magnetised steel rod
 - C an uncharged nylon rod
 - D a radioactive uranium rod
- 35 The diagram shows an electric generator with the coil in a vertical position.



Which row describes the generator?

	the generator produces	the voltage output will be zero when
A	a.c.	the coil is horizontal
B	a.c.	the coil is vertical
C	d.c.	the coil is horizontal
D	d.c.	the coil is vertical

- 36 The diagram shows a transformer. There are 460 turns on the primary coil and 24 turns on the secondary coil. The primary voltage is 230 V.



What is the secondary voltage?

- A** 2.0 V **B** 12 V **C** 48 V **D** 4400 V
- 37 The charge on a proton is e .

What is the charge on an electron and what is the charge on a neutron?

	electron	neutron
A	e	e
B	e	0
C	$-e$	$-e$
D	$-e$	0

- 38 Four students are asked to comment on the processes of nuclear fission and nuclear fusion.

Their comments are recorded in the table.

Which row is correct?

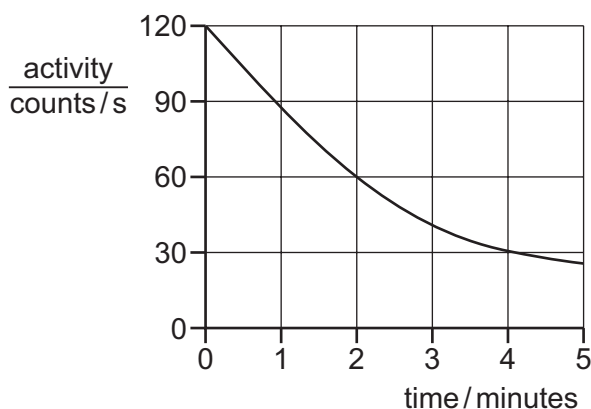
	fission	fusion
A	energy is absorbed	a large unstable nucleus splits
B	a large unstable nucleus splits	energy is absorbed
C	two light nuclei join	energy is absorbed
D	energy is released	two light nuclei join

- 39 Radon $^{219}_{86}\text{Rn}$ decays by emitting an α -particle.

Which nuclide is formed in this decay?

- A $^{215}_{84}\text{Po}$ B $^{223}_{88}\text{Ra}$ C $^{219}_{87}\text{Fr}$ D $^{219}_{85}\text{At}$

- 40 The graph shows the activity of a radioactive source over a period of time.



What is the half-life of the source?

- A 1.0 minute B 2.0 minutes C 2.5 minutes D 4.0 minutes

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PHYSICS (9–1)

0972/21

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3	A	1
4	D	1
5	A	1
6	D	1
7	D	1
8	D	1
9	C	1
10	D	1
11	A	1
12	D	1
13	D	1
14	C	1
15	B	1
16	A	1
17	B	1
18	B	1
19	A	1
20	C	1
21	C	1
22	B	1
23	A	1
24	A	1
25	D	1
26	B	1
27	C	1
28	B	1

Question	Answer	Marks
29	C	1
30	D	1
31	B	1
32	C	1
33	D	1
34	B	1
35	B	1
36	B	1
37	D	1
38	D	1
39	A	1
40	B	1



Cambridge IGCSE™ (9–1)

PHYSICS

0972/21

Paper 2 Multiple Choice (Extended)

October/November 2021

45 minutes

You must answer on the multiple choice answer sheet.

You will need: Multiple choice answer sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)

INSTRUCTIONS

- There are **forty** questions on this paper. Answer **all** questions.
- For each question there are four possible answers **A**, **B**, **C** and **D**. Choose the **one** you consider correct and record your choice in soft pencil on the multiple choice answer sheet.
- Follow the instructions on the multiple choice answer sheet.
- Write in soft pencil.
- Write your name, centre number and candidate number on the multiple choice answer sheet in the spaces provided unless this has been done for you.
- Do **not** use correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.
- Take the weight of 1.0 kg to be 10 N (acceleration of free fall = 10 m/s^2).

INFORMATION

- The total mark for this paper is 40.
- Each correct answer will score one mark.
- Any rough working should be done on this question paper.

This document has **16** pages.

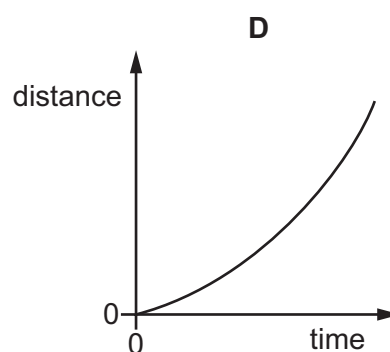
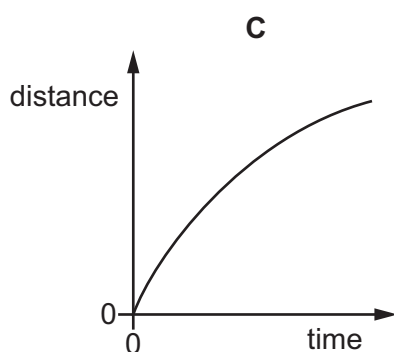
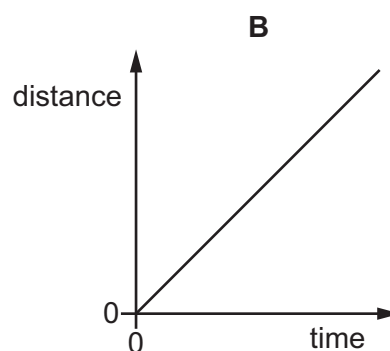
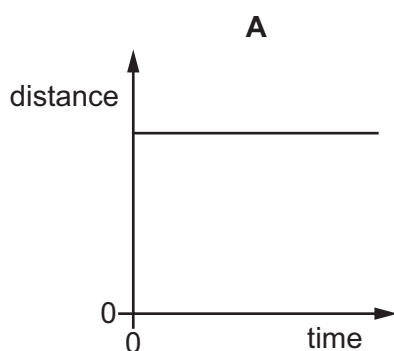


- 1 Which instrument is most suitable for measuring the thickness of a single sheet of paper?

A 15 cm rule
B balance
C metre rule
D micrometer screw gauge

- 2 The diagrams show distance–time graphs for four objects.

Which graph represents an object moving with an increasing speed?



- 3 An object has a weight of 6.4 N on the Earth where the gravitational field strength is 10 N/kg.

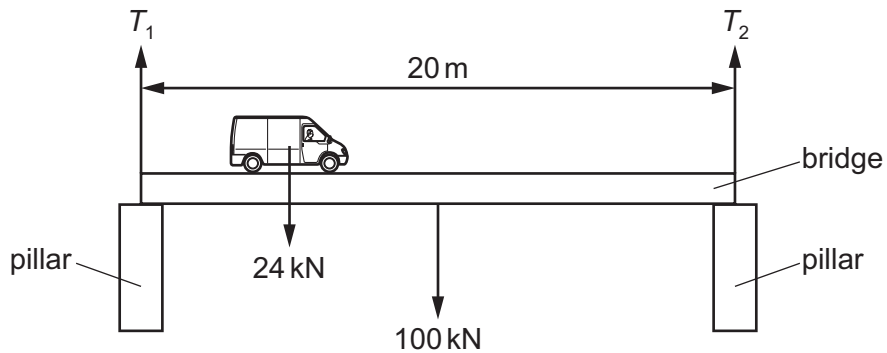
Which row states the mass and the weight of the object on the Moon where the gravitational field strength is 1.6 N/kg?

	mass / kg	weight on the Moon / N
A	0.64	1.0
B	0.64	6.4
C	4.0	1.0
D	4.0	6.4

- 4 Which substance in the table has the lowest density?

	substance	mass / g	volume / cm ³
A	nylon	1.2	1.0
B	cotton	1.5	1.0
C	olive oil	1.8	2.0
D	water	2.0	2.0

- 5 A 20 m long, uniform bridge of weight 100 kN is supported at each end by pillars, as shown.



The pillars exert forces T_1 and T_2 on the ends of the bridge.

What are the values of T_1 and T_2 when a van of weight 24 kN is on the bridge, 5 m from the left-hand pillar?

	T_1 / kN	T_2 / kN
A	56	68
B	62	62
C	68	56
D	74	50

- 6 A spring, which obeys Hooke's law, has an unstretched length of 10 cm.

A load of 20 N is suspended from the spring.

The new length of the spring is 36 cm.

What is the spring constant k of the spring?

- A** 0.56 N/cm **B** 0.77 N/cm **C** 1.3 N/cm **D** 1.8 N/cm

- 7 What is the relationship between the impulse acting on an object and the change in momentum of the object?

- A impulse = change in momentum
 B impulse = change in momentum $\times$ time
 C impulse = $\frac{\text{change in momentum}}{\text{time}}$
 D impulse = $\frac{\text{change in momentum}}{\text{mass}}$

- 8 Electrical energy may be obtained from nuclear fission.

In which order is the energy transferred in this process?

- A nuclear fuel $\rightarrow$ generator $\rightarrow$ reactor and boiler $\rightarrow$ turbines
 B nuclear fuel $\rightarrow$ generator $\rightarrow$ turbines $\rightarrow$ reactor and boiler
 C nuclear fuel $\rightarrow$ reactor and boiler $\rightarrow$ generator $\rightarrow$ turbines
 D nuclear fuel $\rightarrow$ reactor and boiler $\rightarrow$ turbines $\rightarrow$ generator

- 9 A motor of power P exerts a force F on an object. The object moves a distance d during the time t that the force acts.

Which equation is used to calculate the time t ?

- A $t = \frac{F}{Pd}$ B $t = \frac{Fd}{P}$ C $t = \frac{Pd}{F}$ D $t = \frac{P}{Fd}$

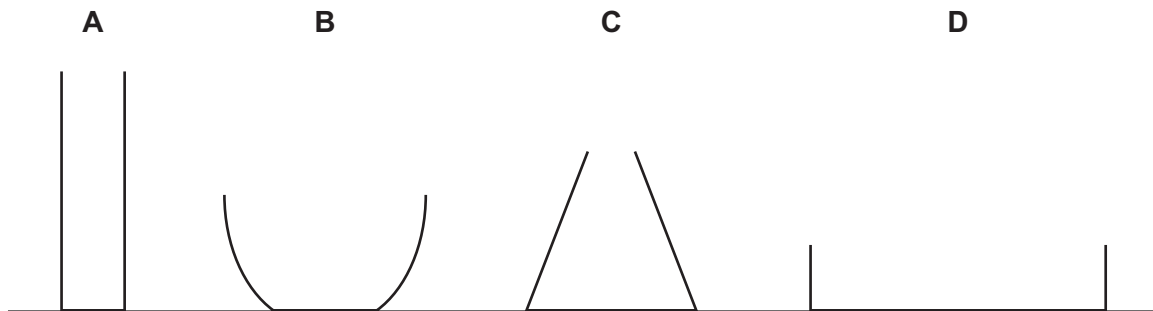
- 10 A scientist uses an electric motor to lift a load through a vertical distance of 2.0 m.

He then increases the input power to the motor and repeats the experiment. The efficiency of the motor does not change.

Which row correctly describes the effect that this has on the useful work done lifting the load and the time taken to lift it?

	work done	time taken
A	decreases	decreases
B	stays the same	decreases
C	decreases	stays the same
D	stays the same	stays the same

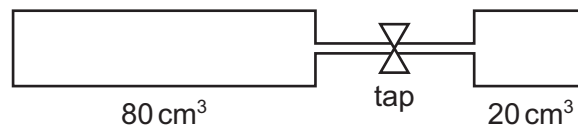
- 11** Four containers are filled to the top with the same liquid. The base of each container is circular.
- Which container has the greatest pressure exerted by the liquid at its base?



- 12** A liquid is evaporating. The liquid is not boiling.

Which statement about the liquid is correct at an instant in time?

- A** Any molecule can escape, and from any part of the liquid.
 - B** Any molecule can escape, but only from the liquid's surface.
 - C** Only molecules with enough energy can escape, and only from the liquid's surface.
 - D** Only molecules with enough energy can escape, but from any part of the liquid.
- 13** The diagram shows two cylinders connected by a narrow tube fitted with a tap.



One cylinder contains 80 cm^3 of gas at a pressure of $2.0 \times 10^5 \text{ Pa}$. The other cylinder contains a vacuum.

The volume of the evacuated cylinder is 20 cm^3 . The tap is opened so that the gas can flow to fill both cylinders.

The temperature of the gas remains constant.

What is the new pressure of the gas?

- A** $0.50 \times 10^5 \text{ Pa}$
- B** $1.6 \times 10^5 \text{ Pa}$
- C** $2.5 \times 10^5 \text{ Pa}$
- D** $8.0 \times 10^5 \text{ Pa}$

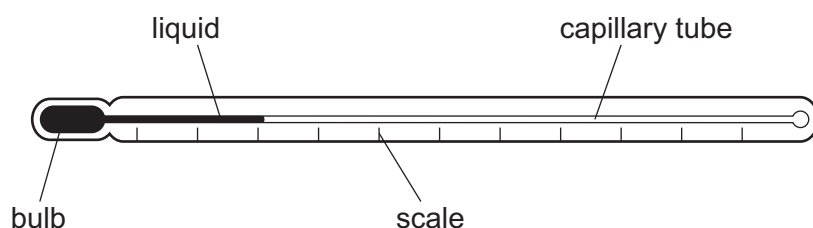
- 14 An aluminium block has a mass of 200 g.

The specific heat capacity of aluminium is $900 \text{ J/(kg } ^\circ\text{C)}$.

How much energy is needed to increase the temperature of the block from 20°C to 110°C ?

- A** 2.0 J **B** 2000 J **C** 16 200 J **D** 16 200 000 J

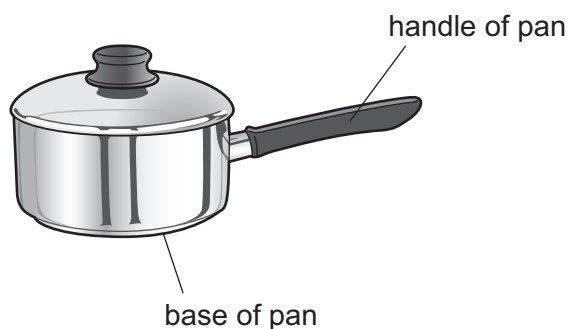
- 15 The diagram shows a liquid-in-glass thermometer.



Which change to the design would result in a more sensitive thermometer?

- A** Increase the density of the liquid.
B Increase the diameter of the capillary tube.
C Increase the number of scale markings.
D Increase the volume of the bulb.

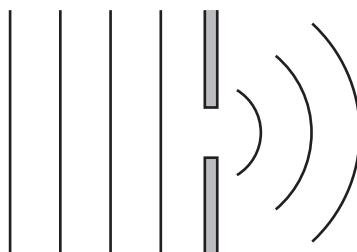
- 16 The diagram shows a pan used for cooking food.



Which row is correct for the materials used to make the base and the handle of the pan?

	base of pan	handle of pan
A	good thermal conductor	good thermal conductor
B	good thermal conductor	poor thermal conductor
C	poor thermal conductor	good thermal conductor
D	poor thermal conductor	poor thermal conductor

- 17 The diagram shows waves in a ripple tank containing water.



The waves approach a barrier and pass through the gap in the barrier.

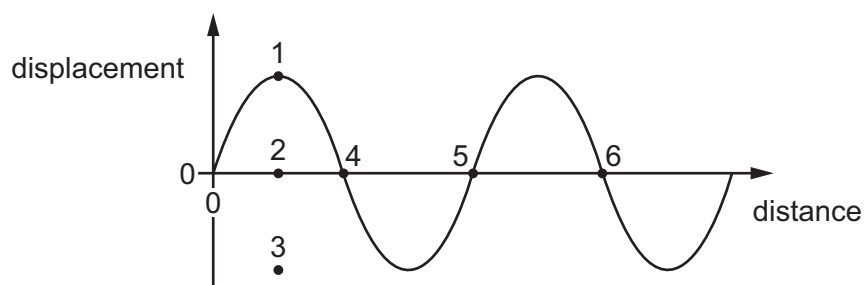
The size of the gap is about the same size as the wavelength of the ripples.

The gap size is increased.

What happens to the ripple pattern to the right of the barrier?

- A** The ripples are closer together.
- B** The ripples are further apart.
- C** The ripples are more curved.
- D** The ripples are less curved.

- 18 The diagram shows a wave.



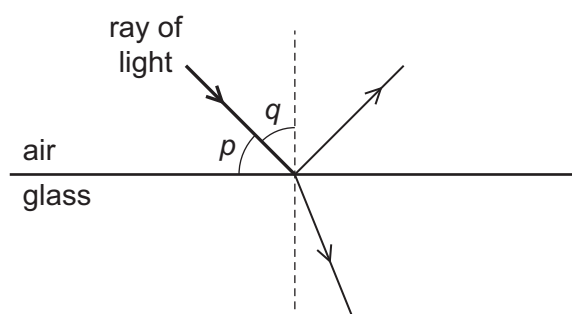
Which row correctly indicates the amplitude and the wavelength of the wave?

	amplitude	wavelength
A	the distance between 1 and 2	the distance between 4 and 5
B	the distance between 1 and 2	the distance between 4 and 6
C	the distance between 1 and 3	the distance between 4 and 5
D	the distance between 1 and 3	the distance between 4 and 6

- 19 Two beams of light are both the same colour of red. One beam is travelling through air. The other beam is travelling through water. Each beam has a different brightness.

Which quantity is the same for both sets of waves?

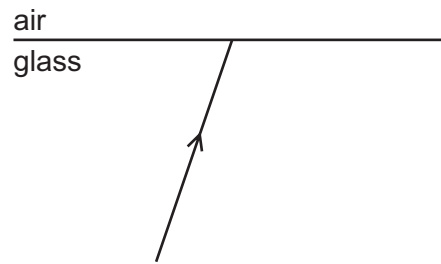
- A amplitude
 - B frequency
 - C speed
 - D wavelength
- 20 The diagram shows a ray of light in air incident on a glass block. Some of the light is refracted and some of the light is reflected. Two angles, p and q , are marked on the diagram.



Which row gives the angle of incidence and states whether total internal reflection occurs?

	angle of incidence	total internal reflection
A	p	no
B	p	yes
C	q	no
D	q	yes

- 21 The diagram shows a ray of light in glass incident on the surface between the glass and air.



What happens if the angle of incidence is made larger than the critical angle for the glass?

- A The angle of refraction becomes equal to 90° .
- B There is a refracted ray and a ray reflected inside the glass.
- C There is a refracted ray only.
- D There is only a ray reflected inside the glass.

- 22 The Sun emits infrared radiation and light.

Light from the Sun reaches the Earth in 8 minutes.

Which row gives correct information about the infrared radiation?

	wavelength of infrared radiation	time taken for infrared radiation to reach the Earth
A	longer than wavelength of light	8 minutes
B	longer than wavelength of light	much less than 8 minutes
C	shorter than wavelength of light	8 minutes
D	shorter than wavelength of light	much more than 8 minutes

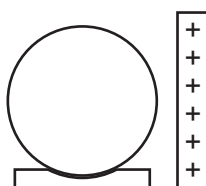
- 23 Which list shows regions of the electromagnetic spectrum in order of increasing frequency?

- A X-ray $\rightarrow$ ultraviolet $\rightarrow$ visible light $\rightarrow$ infrared
- B X-ray $\rightarrow$ infrared $\rightarrow$ visible light $\rightarrow$ ultraviolet
- C infrared $\rightarrow$ visible light $\rightarrow$ ultraviolet $\rightarrow$ X-ray
- D ultraviolet $\rightarrow$ visible light $\rightarrow$ infrared $\rightarrow$ X-ray

- 24 What is ultrasound?

- A sound waves that are so loud that they damage human hearing
- B sound waves that are too high-pitched for humans to hear
- C sound waves that are too low-pitched for humans to hear
- D sound waves that are too quiet for humans to hear

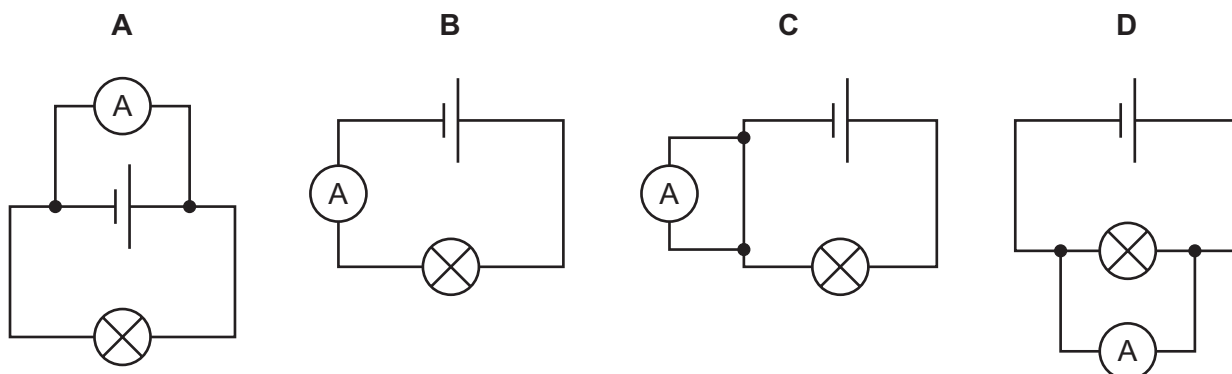
- 25** An uncharged, metal sphere is placed on an insulating support. A positively charged rod is brought close to the sphere, but does not touch it.



How do the charges in the sphere move and what is now the charge on the sphere?

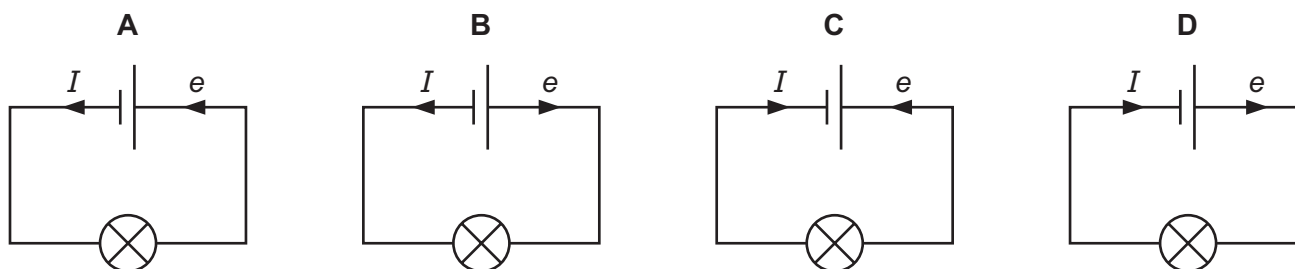
	movement of charges	charge on sphere
A	negative charges move to the right of the sphere	positive
B	negative charges move to the right of the sphere	neutral
C	positive charges move to the left of the sphere	positive
D	positive charges move to the left of the sphere	neutral

- 26** In which circuit is the ammeter measuring the flow of charge through the lamp?



- 27** A lamp is connected to a cell.

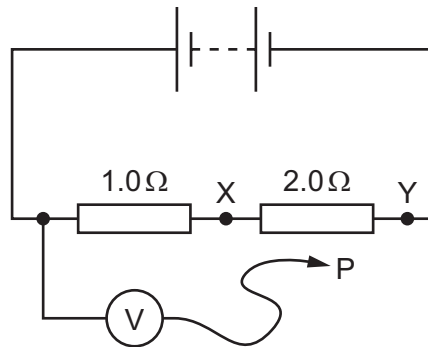
Which circuit diagram shows the direction of conventional current I and also the direction of flow of electrons e ?



- 28** The diagram shows a circuit containing two resistors of resistance $1.0\ \Omega$ and $2.0\ \Omega$.

A voltmeter is connected across the $1.0\ \Omega$ resistor by connecting P to X.

The reading on the voltmeter is $6.0\ \text{V}$.

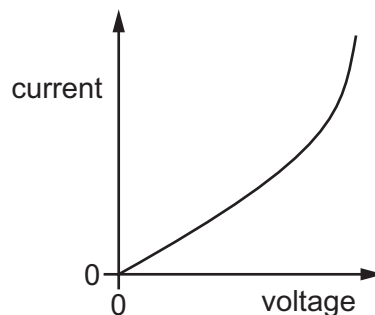


P is moved to point Y in the circuit.

What is the new reading on the voltmeter?

- A** $3.0\ \text{V}$ **B** $6.0\ \text{V}$ **C** $12\ \text{V}$ **D** $18\ \text{V}$

- 29** The graph shows the current–voltage relationship for a circuit component X.



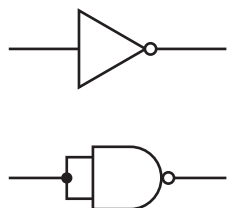
What happens to the resistance of X and what happens to the temperature of X as the voltage increases?

	resistance of X	temperature of X
A	decreases	decreases
B	decreases	increases
C	increases	decreases
D	increases	increases

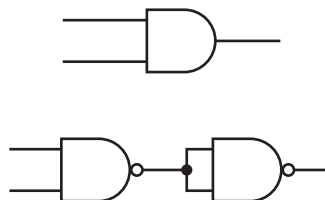
30 The diagrams show pairs of circuits containing logic gates.

In which diagram does the lower circuit of the pair behave differently from the upper circuit?

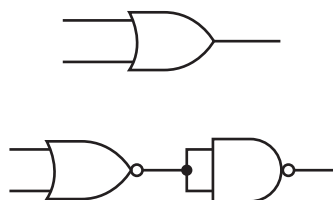
A



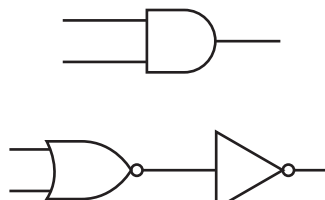
B



C

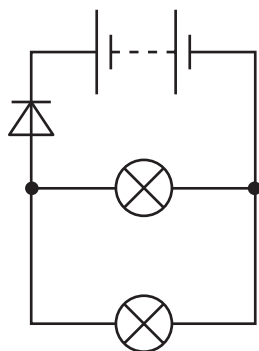


D

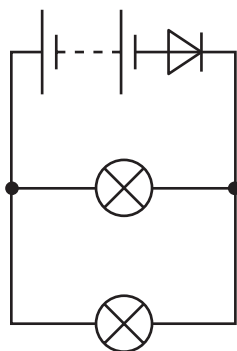


31 In which circuit do both lamps light?

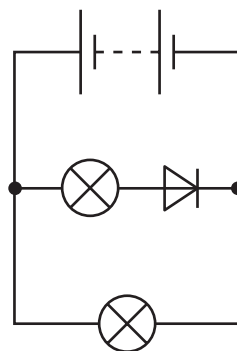
A



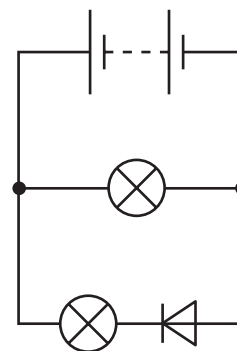
B



C

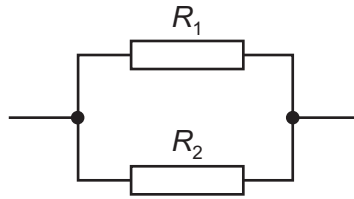


D



- 32 Two resistors, with resistances R_1 and R_2 , are connected in parallel.

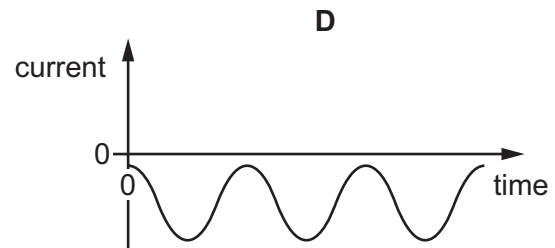
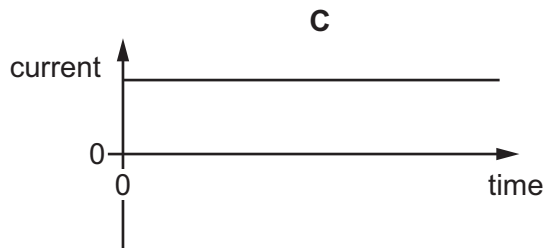
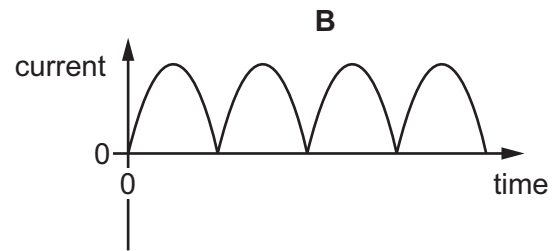
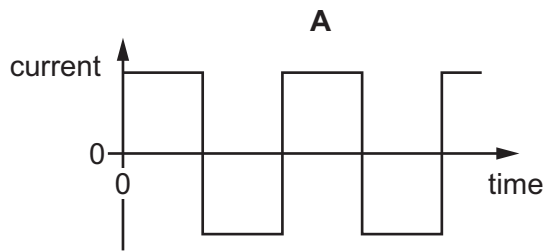
The resistance R_1 is greater than the resistance R_2 .



What is the resistance of the parallel combination?

- A less than either R_1 or R_2
 - B equal to R_1
 - C equal to R_2
 - D the average of R_1 and R_2
- 33 The metal cases of electrical appliances are connected to an earth wire.
- Which statement is **not** correct?
- A The live wire may become loose and touch the metal case.
 - B If the metal case becomes live, the earth wire conducts current to the ground.
 - C The earth wire needs to have a high resistance.
 - D Earthing metal cases helps prevent a person from receiving an electric shock.
- 34 What is the function of the split-ring commutator in an electric motor with a single rotating coil?
- A to enable the motor to function with an a.c. source
 - B to reverse the current in the coil once every revolution
 - C to reverse the current in the coil whenever its plane becomes perpendicular to the magnetic field
 - D to reverse the current in the coil whenever its plane is parallel with the magnetic field

35 Which graph represents an alternating current (a.c.)?

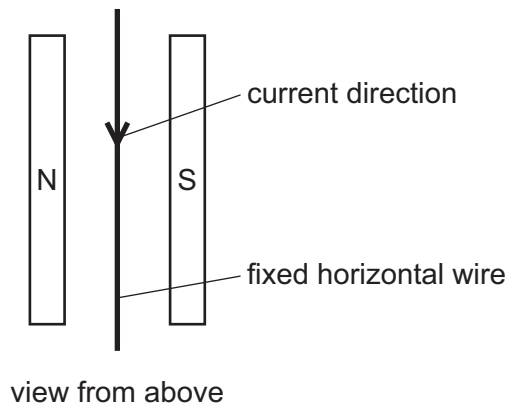
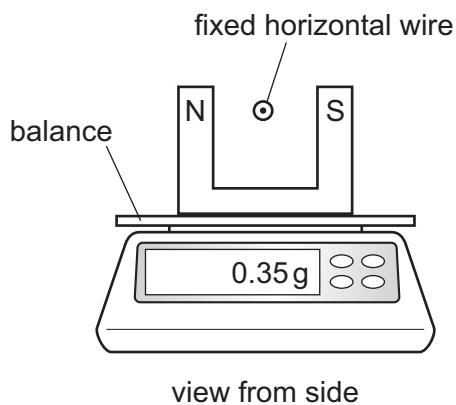


36 The diagrams show a horizontal wire in a magnetic field.

The horizontal wire is firmly held at each end (not shown) and cannot move.

The magnets and holder are on a balance.

When there is no current in the wire, the reading on the balance is 0.35 g.



There is a d.c. current in the wire, as shown.

What happens to the reading on the balance?

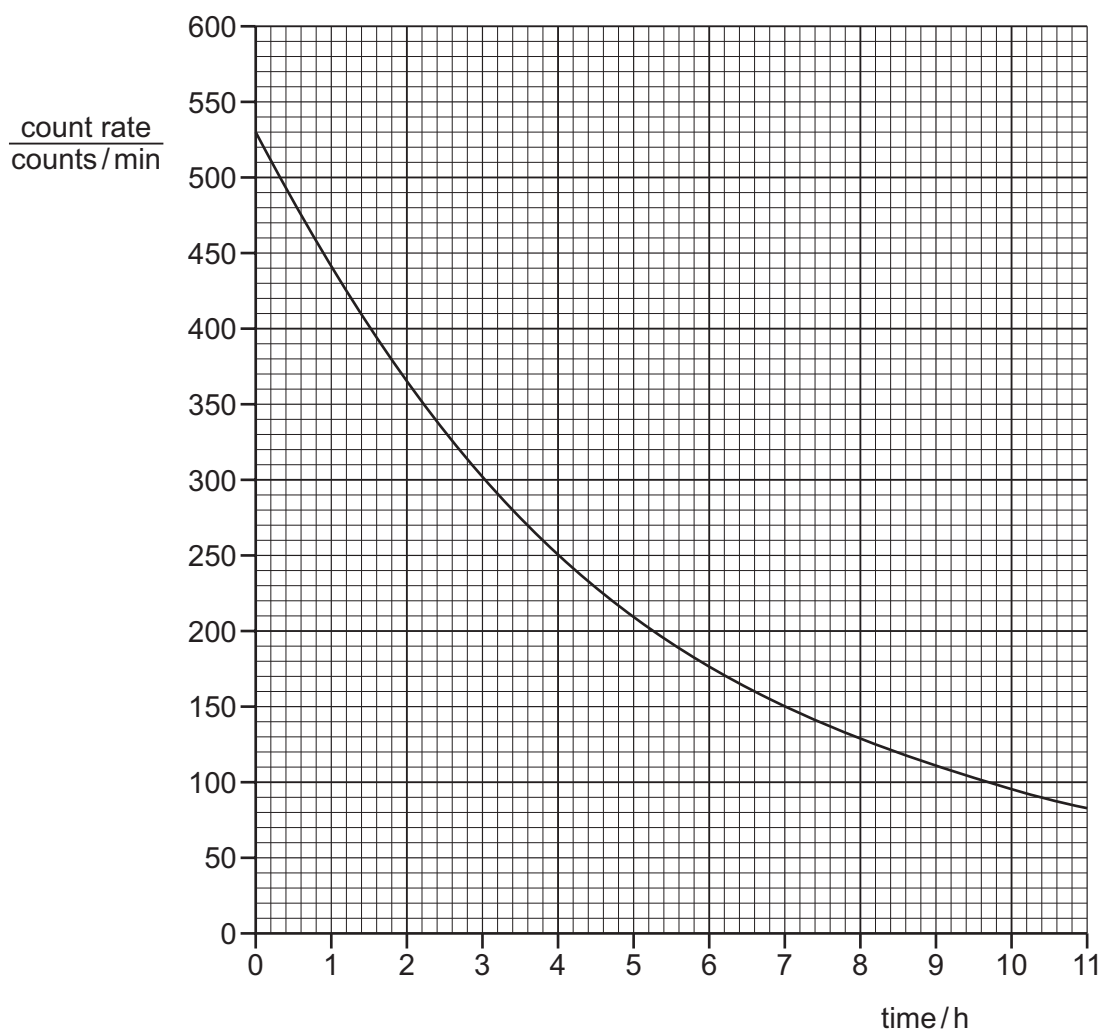
- A** smaller than 0.35 g
- B** no change
- C** changing from smaller to larger than 0.35 g repeatedly
- D** larger than 0.35 g

- 37 The nucleus of an americium atom contains 146 neutrons and 95 protons. It decays by emitting an α -particle.

How many neutrons and how many protons remain in the nucleus when this form of americium decays?

	number of neutrons remaining	number of protons remaining
A	142	93
B	142	95
C	144	93
D	144	95

- 38 The graph shows how the count rate measured by a radioactivity detector placed near a radioactive sample changed with time.



Given that the background count rate is 30 counts/min, what is the half-life of this sample?

- A** 3.4 h **B** 3.6 h **C** 4.0 h **D** 5.5 h

- 39 A teacher holds a radioactive source near a detector.

The reading on the detector is 320 counts/min.

The detector is switched on again after the source has been removed and it shows a reading of 20 counts/min.

What is the counts/min solely due to the source and why is there a reading on the detector when there is no radioactive source present?

	counts/min due to the source	reason for reading with no source
A	300	zero error on detector
B	300	background radiation
C	340	zero error on detector
D	340	background radiation

- 40 Which statement is **not** correct?

- A** α -particles are used to detect cracks in metallic structures.
- B** β -particles are used in the measurement of the thickness of paper.
- C** γ -rays may be used to treat cancer patients.
- D** Smoke alarms contain a weak source of α -particles.

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Cambridge IGCSE™ (9–1)

PHYSICS

0972/21

Paper 2 Multiple Choice (Extended)

October/November 2021

MARK SCHEME

Maximum Mark: 40

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the October/November 2021 series for most Cambridge IGCSE™, Cambridge International A and AS Level components and some Cambridge O Level components.

This document consists of **3** printed pages.

Question	Answer	Marks
1	D	1
2	D	1
3	A	1
4	C	1
5	C	1
6	B	1
7	A	1
8	D	1
9	B	1
10	B	1
11	A	1
12	C	1
13	B	1
14	C	1
15	D	1
16	B	1
17	D	1
18	B	1
19	B	1
20	C	1
21	D	1
22	A	1
23	C	1
24	B	1
25	B	1
26	B	1
27	C	1
28	D	1

Question	Answer	Marks
29	B	1
30	D	1
31	C	1
32	A	1
33	C	1
34	C	1
35	A	1
36	D	1
37	C	1
38	A	1
39	B	1
40	A	1



Cambridge IGCSE™ (9–1)

PHYSICS

0972/21

Paper 2 Multiple Choice (Extended)

May/June 2022

45 minutes

You must answer on the multiple choice answer sheet.

You will need: Multiple choice answer sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)

INSTRUCTIONS

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- Write your name, centre number and candidate number on the multiple choice answer sheet in the spaces provided unless this has been done for you.
- Do **not** use correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.
- Take the weight of 1.0 kg to be 10 N (acceleration of free fall = 10 m/s^2).

INFORMATION

- The total mark for this paper is 40.
- Each correct answer will score one mark.
- Any rough working should be done on this question paper.

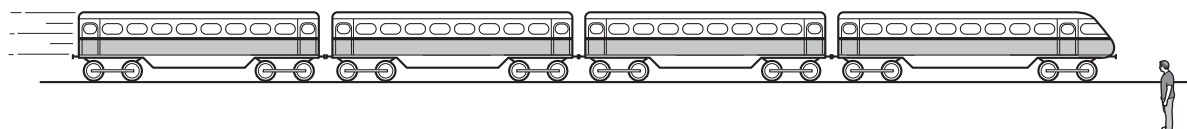
This document has **16** pages.



- 1 Which measuring devices are most suitable for determining the length of a swimming pool and the thickness of aluminium foil?

	length of a swimming pool	thickness of aluminium foil
A	ruler	measuring cylinder
B	tape measure	micrometer screw gauge
C	tape measure	ruler
D	ruler	micrometer screw gauge

- 2 A man stands next to a railway track.



A train travelling at 40 m/s takes 2.0 s to pass the man.

What is the length of the train?

- A** 20 m **B** 38 m **C** 40 m **D** 80 m

- 3 A speed–time graph is used to describe the motion of an object.

Which quantities are calculated from the gradient of the graph and from the area under the graph?

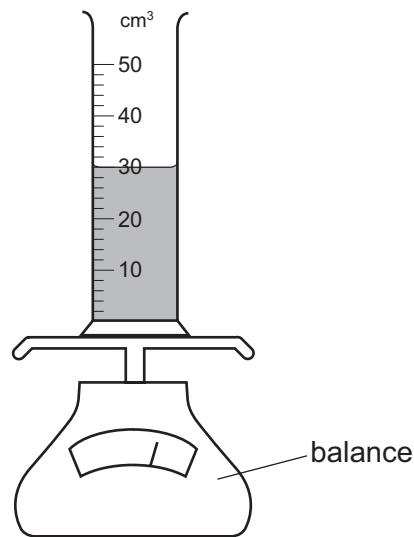
	gradient of the graph	area under the graph
A	acceleration	distance travelled
B	acceleration	total journey time
C	distance travelled	acceleration
D	total journey time	distance travelled

- 4 On the Moon, all objects fall with the same acceleration.

Which statement explains this?

- A** On the Moon, all objects have the same weight.
B The Moon has a smaller gravitational field strength than the Earth.
C The weight of an object is directly proportional to its mass.
D The weight of an object is inversely proportional to its mass.

- 5 A measuring cylinder contains 30 cm^3 of a liquid.



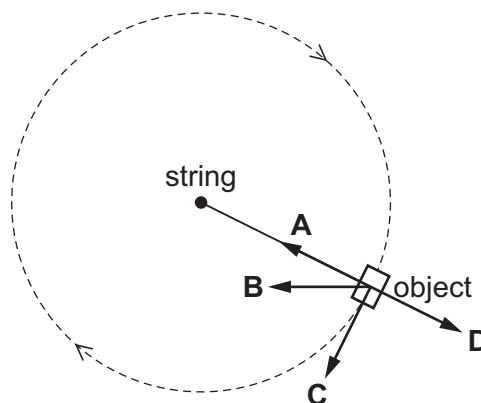
Some more of the liquid is added until the liquid level reaches the 50 cm^3 mark.

The reading on the balance increases by 30 g.

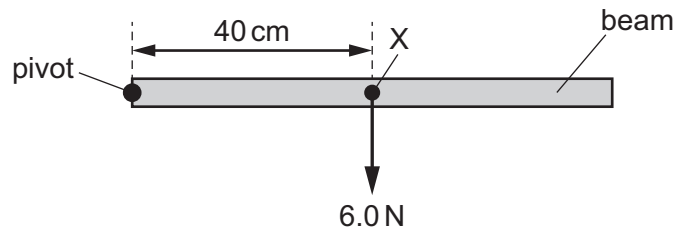
What is the density of the liquid?

- A 0.60 g/cm^3 B 0.67 g/cm^3 C 1.5 g/cm^3 D 1.7 g/cm^3
- 6 An object on the end of a string moves in a clockwise circular path at constant speed. The diagram shows the object as viewed from above.

What is the direction of the resultant force on the object when it is in the position shown?



- 7 A beam is pivoted at one end, as shown.

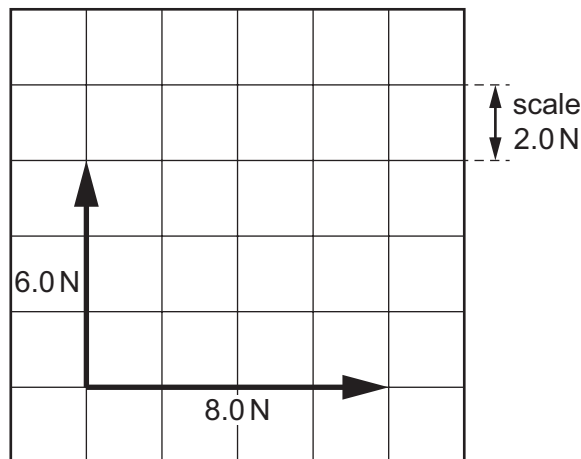


The beam weighs 6.0 N and its weight acts at a point X 40 cm from the pivot.

A force of 4.0 N is applied to the beam causing it to balance horizontally.

In which direction and where is the 4.0 N force applied?

- A vertically downwards at 20 cm to the left of X
 - B vertically downwards at 20 cm to the right of X
 - C vertically upwards at 20 cm to the left of X
 - D vertically upwards at 20 cm to the right of X
- 8 On the diagram shown, what is the magnitude of the resultant force of the two vectors?



- A 2.0 N
 - B 7.0 N
 - C 10 N
 - D 14 N
- 9 Three situations are listed.
- 1 An object has a resultant force acting on it.
 - 2 A moving object experiences an impulse.
 - 3 An object is decelerating.

In which situations is the momentum of the object changing?

- A 1 and 2 only
- B 1 and 3 only
- C 2 and 3 only
- D 1, 2 and 3

- 10** A ball of mass 0.16 kg is moving forwards at a speed of 0.50 m/s. A second ball of mass 0.10 kg is stationary. The first ball strikes the second ball. The second ball moves forwards at a speed of 0.50 m/s.

What is the speed of the first ball after the collision?

- A** 0.0 m/s **B** 0.19 m/s **C** 0.31 m/s **D** 0.50 m/s

- 11** A mass hangs vertically from a spring.

The mass is raised to a point P and is then released.

The mass oscillates repeatedly between point P and a lower point Q.

Which energies alternately increase and decrease throughout the oscillations?

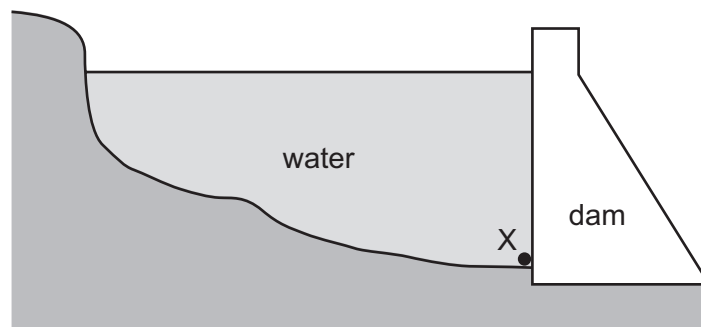
- A** gravitational potential energy, kinetic energy and elastic energy
B gravitational potential energy and kinetic energy only
C gravitational potential energy, kinetic energy and internal energy
D internal energy and elastic energy

- 12** A car has 620 kJ of kinetic energy. The car brakes and stops in a distance of 91 m.

What is the average braking force acting on the car?

- A** 0.15 N **B** 6.8 N **C** 6800 N **D** 56 000 N

- 13** The diagram shows a deep reservoir formed by a dam.



On what does the pressure at X depend?

- A** the depth of the water at X
B the length of the reservoir
C the surface area of the water
D the thickness of the dam wall

- 14** A sealed rigid container has a fixed volume. The container is filled with air.

The container is placed in a freezer cabinet and the temperature of the air in the container decreases.

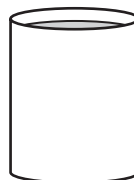
Which row correctly describes what happens to the air in the container?

	average distance between air particles	average speed of air particles
A	decreases	increases
B	decreases	decreases
C	no change	increases
D	no change	decreases

- 15** Two open containers are filled with water at room temperature. The containers have different shapes.



container 1

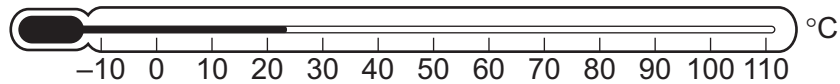


container 2

From which container does the water evaporate at the greater rate and how can the rate of evaporation be increased?

	container with the greater rate of evaporation	how the rate of evaporation can be increased
A	1	decrease the water temperature
B	1	increase the water temperature
C	2	decrease the water temperature
D	2	increase the water temperature

- 16 The diagram shows a liquid-in-glass thermometer.



A student wishes to check the marking of the upper fixed point on this thermometer.

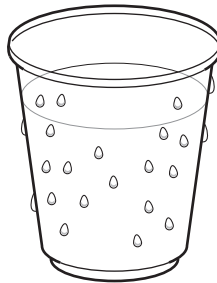
What should she do?

- A Put the bulb in a beaker of boiling sea water.
 - B Put the bulb in a beaker of boiling pure water.
 - C Put the bulb in a beaker of ice and salt.
 - D Put the bulb in a beaker of pure melting ice.
- 17 Water in a beaker gains thermal energy at a rate of 3000 W. The water is at its boiling point.

The specific latent heat of vaporisation of water is 2260 J/g.

How long does it take for 250 g of the water to vaporise?

- A 12 s
 - B 188 s
 - C 332 s
 - D 750 s
- 18 A glass contains an iced drink on a warm and humid day. Water starts to form on the outside of the glass.



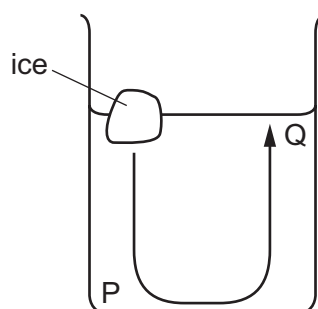
What is the name of the effect by which the water forms?

- A condensation
- B conduction
- C convection
- D evaporation

- 19** One end of a copper bar is heated to a high temperature.

Which mechanism is responsible for the transfer of thermal energy to the other end of the copper bar?

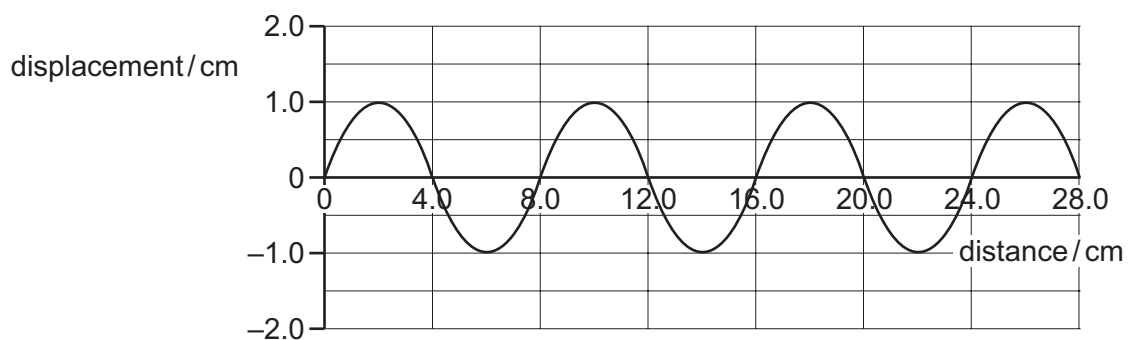
- A** the lattice vibrations of copper ions only
 - B** the lattice vibrations of copper ions and the movement of high energy electrons along the bar
 - C** the movement of high energy copper ions along the bar
 - D** the movement of high energy electrons along the bar only
- 20** The diagram shows a convection current caused by a piece of ice placed in a beaker of water at room temperature.



Which row correctly compares the temperatures and densities at water points P and Q?

	temperature at P	density at P
A	higher than at Q	higher than at Q
B	higher than at Q	lower than at Q
C	lower than at Q	higher than at Q
D	lower than at Q	lower than at Q

21 The diagram shows a wave.



Which row is correct?

	amplitude of the wave / cm	wavelength of the wave / cm
A	1.0	4.0
B	1.0	8.0
C	2.0	4.0
D	2.0	8.0

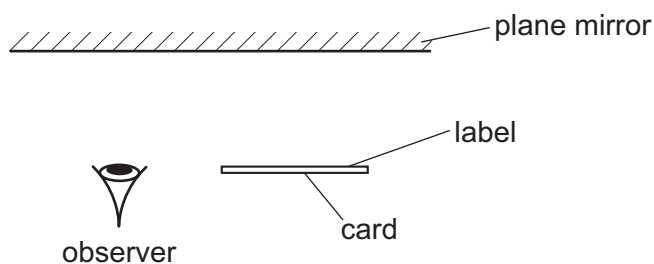
22 A sound wave is created by a loudspeaker that vibrates backwards and forwards 96 000 times per minute.

The speed of sound is 320 m/s.

What is the wavelength of the sound wave?

- A** 0.20 m **B** 5.0 m **C** 300 m **D** 18 000 m

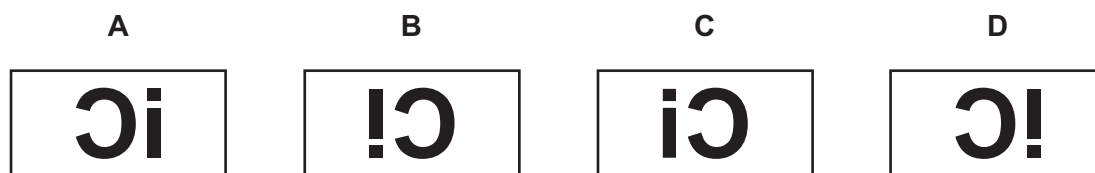
- 23** A card is placed in front of a plane mirror so that its label is facing the mirror, as shown.



The label is shown.



How does the image of the label formed by the mirror appear to the observer?



- 24** A thin converging lens can produce both real and virtual images.

Which row describes a real and a virtual image?

	real image	virtual image
A	rays converge to form the image	image can be projected onto a screen
B	rays converge to form the image	image cannot be projected onto a screen
C	rays diverge to form the image	image can be projected onto a screen
D	rays diverge to form the image	image cannot be projected onto a screen

- 25** The speed of light in air is $3.0 \times 10^8 \text{ m/s}$.

The critical angle for light in a transparent plastic material placed in air is 37° .

What is the speed of light in the plastic material?

- A** $1.8 \times 10^8 \text{ m/s}$
- B** $2.4 \times 10^8 \text{ m/s}$
- C** $3.8 \times 10^8 \text{ m/s}$
- D** $5.0 \times 10^8 \text{ m/s}$

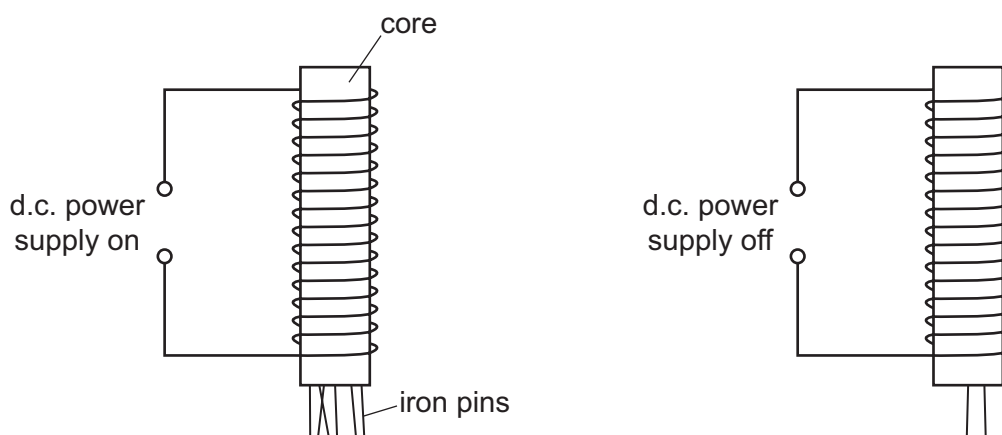
26 Which part of the electromagnetic spectrum is used by a remote controller for a television?

- A infrared waves
- B microwaves
- C radio waves
- D visible light

27 Which statement correctly compares radio waves and X-rays?

- A Radio waves have a longer wavelength and a greater speed in a vacuum.
- B Radio waves have a longer wavelength and the same speed in a vacuum.
- C Radio waves have a shorter wavelength and a greater speed in a vacuum.
- D Radio waves have a shorter wavelength and the same speed in a vacuum.

28 A student counts how many iron pins an electromagnet picks up when its power supply is switched on. Then, she counts how many pins are picked up when the power supply is switched off.



She repeats the experiment using cores made of different materials. The results are shown.

Which core is made out of soft iron?

	pins picked up with the power supply on	pins picked up with the power supply off
A	0	0
B	2	7
C	8	5
D	12	0

- 29** A plastic rod is brought near to a small plastic sphere suspended from a stand. The sphere is repelled by the rod.

Why is this?

- A** The rod and the sphere have like charges.
- B** The rod and the sphere have unlike charges.
- C** The rod is charged and the sphere is uncharged.
- D** The rod is uncharged and the sphere is charged.

- 30** Which unit is equivalent to a volt (V)?

- A** A/Ω **B** J/C **C** J/s **D** W/C

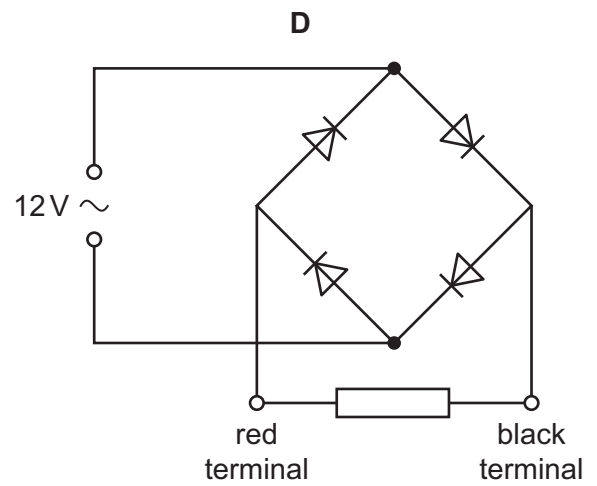
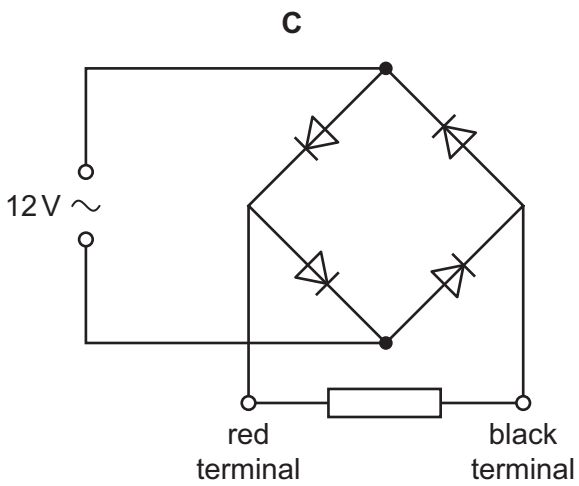
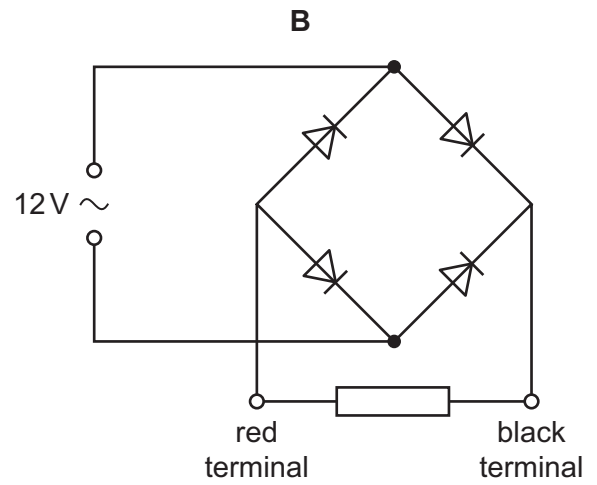
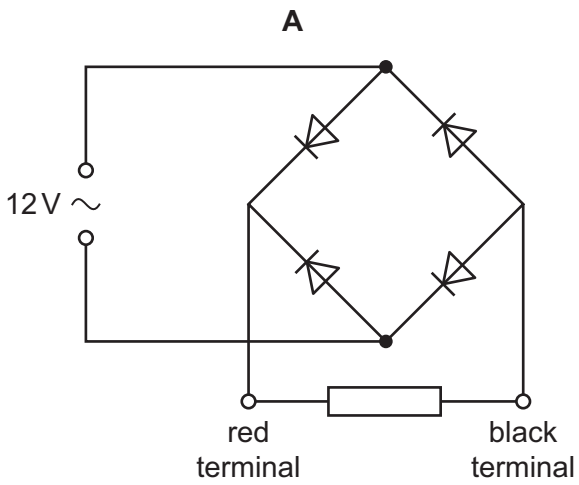
- 31** A resistor converts 360 J of energy when there is a current of 3.0 A in it. The potential difference across the resistor is 6.0 V.

For how long is there this current in the resistor?

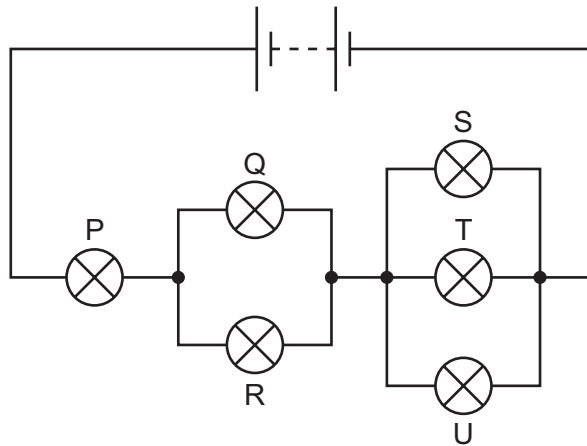
- A** 0.05 s **B** 20 s **C** 180 s **D** 720 s

32 The four circuits shown each contain four diodes.

In which circuit is the direction of the current in the resistor always from the red terminal to the black terminal?



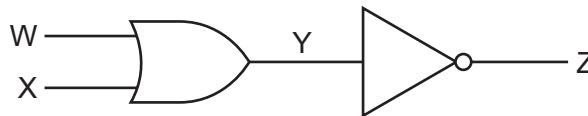
33 The diagram shows a circuit of six identical lamps connected to a battery.



Which lamps are brightest?

- A P only
- B Q and R only
- C S, T and U only
- D P, Q, R, S, T and U are equally bright

34 A digital circuit is made of two logic gates.



Which row is correct for this digital circuit?

	W	X	Y	Z
A	0	0	1	0
B	0	1	0	1
C	1	0	1	0
D	1	1	0	1

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- 35** A magnet is dropped vertically through a solenoid. This induces magnetic poles at both ends of the solenoid.

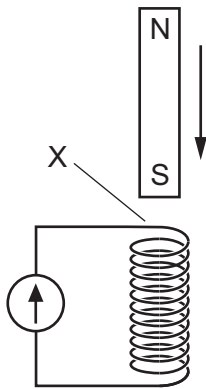


diagram 1

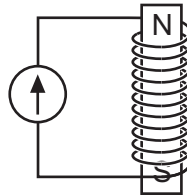


diagram 2

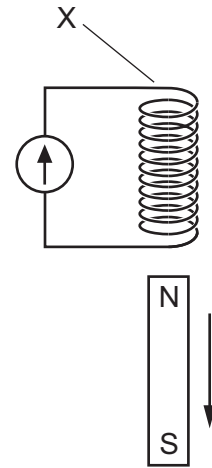
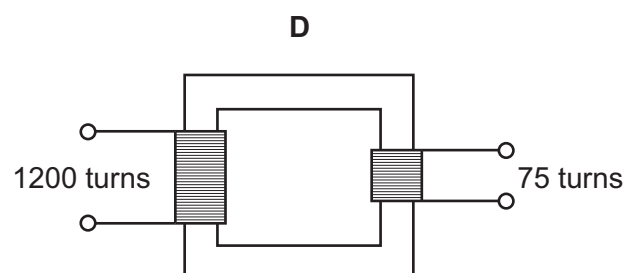
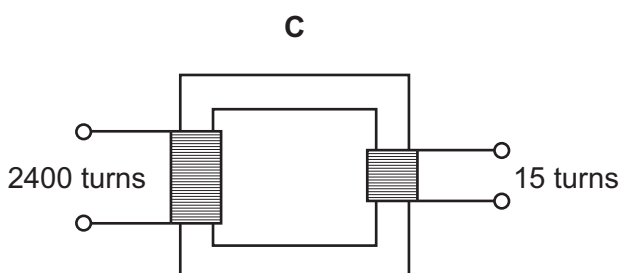
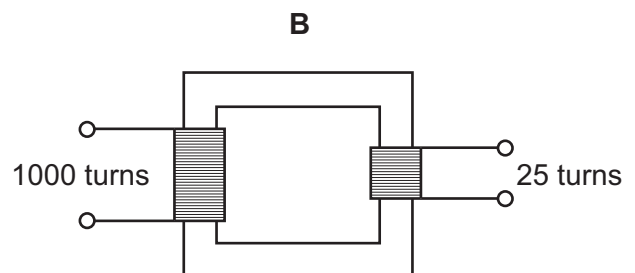
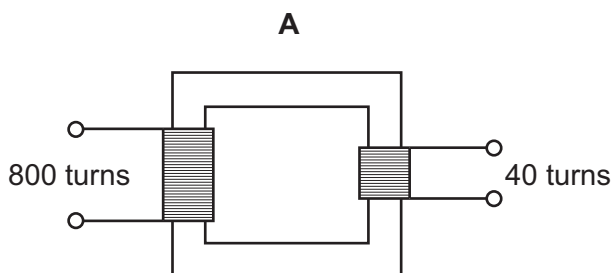


diagram 3

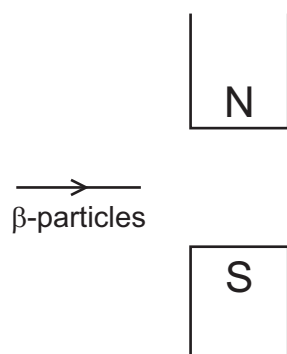
Which magnetic poles are induced at position X in diagram 1 and diagram 3?

	diagram 1	diagram 3
A	N pole	N pole
B	N pole	S pole
C	S pole	N pole
D	S pole	S pole

- 36** Which transformer can change a 240 V a.c. input into a 15 V a.c. output?



- 37 What is the purpose of the split-ring commutator in an electric motor?
- A to ensure that the magnetic field in the motor changes direction every half rotation
 - B to ensure that the magnetic field in the motor stays in the same direction at all times
 - C to ensure that the turning effect on the motor changes direction every half rotation
 - D to ensure that the turning effect on the motor stays in the same direction at all times
- 38 How do the sizes of the two nuclei produced in a nuclear fission reaction compare to the size of the original nucleus?
- A both larger than the original nucleus
 - B one larger and one smaller than the original nucleus
 - C both smaller than the original nucleus
 - D one smaller and one the same size as the original nucleus
- 39 Which statement about the radioactive decay of a substance is correct?
- A It cannot be predicted when a particular nucleus will decay.
 - B Placing a radioactive substance inside a lead-lined box prevents it from decaying.
 - C The decay always produces poisonous gases.
 - D The rate of decay increases if the substance is dissolved in water.
- 40 The diagram shows a stream of β -particles travelling in a line that passes between the poles of a magnet.



In which direction will the β -particles be deflected by the magnet?

- A towards the N pole
- B towards the S pole
- C into the page
- D out of the page



Cambridge IGCSE™ (9–1)

PHYSICS (9–1)

0972/21

Paper 2 Multiple Choice (Extended)

May/June 2022

MARK SCHEME

Maximum Mark: 40

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the May/June 2022 series for most Cambridge IGCSE, Cambridge International A and AS Level and Cambridge Pre-U components, and some Cambridge O Level components.

This document consists of **3** printed pages.

Question	Answer	Marks
1	B	1
2	D	1
3	A	1
4	C	1
5	C	1
6	A	1
7	D	1
8	C	1
9	D	1
10	B	1
11	A	1
12	C	1
13	A	1
14	D	1
15	B	1
16	B	1
17	B	1
18	A	1
19	B	1
20	C	1
21	B	1
22	A	1
23	C	1
24	B	1
25	A	1
26	A	1
27	B	1
28	D	1

Question	Answer	Marks
29	A	1
30	B	1
31	B	1
32	A	1
33	A	1
34	C	1
35	C	1
36	D	1
37	D	1
38	C	1
39	A	1
40	D	1



Cambridge IGCSE™ (9–1)

PHYSICS

0972/22

Paper 2 Multiple Choice (Extended)

May/June 2022

45 minutes

You must answer on the multiple choice answer sheet.

You will need: Multiple choice answer sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)

INSTRUCTIONS

- There are **forty** questions on this paper. Answer **all** questions.
- For each question there are four possible answers **A**, **B**, **C** and **D**. Choose the **one** you consider correct and record your choice in soft pencil on the multiple choice answer sheet.
- Follow the instructions on the multiple choice answer sheet.
- Write in soft pencil.
- Write your name, centre number and candidate number on the multiple choice answer sheet in the spaces provided unless this has been done for you.
- Do **not** use correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.
- Take the weight of 1.0 kg to be 10 N (acceleration of free fall = 10 m/s^2).

INFORMATION

- The total mark for this paper is 40.
- Each correct answer will score one mark.
- Any rough working should be done on this question paper.

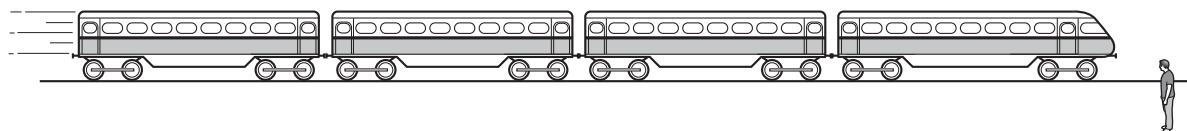
This document has **20** pages. Any blank pages are indicated.



1 Very small values of which quantity are measured using a micrometer screw gauge?

- A time
- B pressure
- C moment
- D distance

2 A man stands next to a railway track.



A train travelling at 40 m/s takes 2.0 s to pass the man.

What is the length of the train?

- A 20 m B 38 m C 40 m D 80 m

3 Which quantity is equal to acceleration?

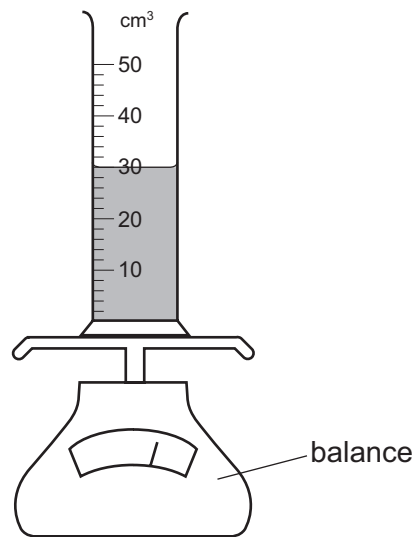
- A area under a distance–time graph
- B area under a speed–time graph
- C gradient of a distance–time graph
- D gradient of a speed–time graph

4 On the Moon, all objects fall with the same acceleration.

Which statement explains this?

- A On the Moon, all objects have the same weight.
- B The Moon has a smaller gravitational field strength than the Earth.
- C The weight of an object is directly proportional to its mass.
- D The weight of an object is inversely proportional to its mass.

- 5 A measuring cylinder contains 30 cm^3 of a liquid.



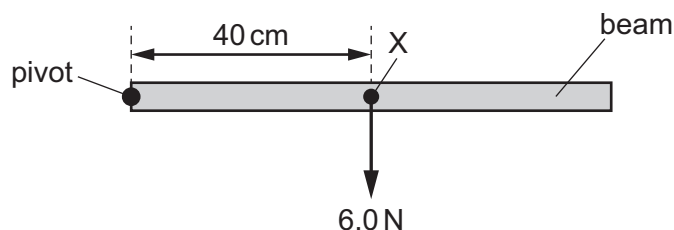
Some more of the liquid is added until the liquid level reaches the 50 cm^3 mark.

The reading on the balance increases by 30 g.

What is the density of the liquid?

- A** 0.60 g/cm^3 **B** 0.67 g/cm^3 **C** 1.5 g/cm^3 **D** 1.7 g/cm^3
- 6 An object moves at constant speed around a circular path.
- Which statement is correct?
- A** A resultant force acts on the object outwards from the centre of the circle.
B A resultant force acts on the object in the direction it is travelling.
C A resultant force acts on the object towards the centre of the circle.
D There is no resultant force acting on the object because it is moving at constant speed.

- 7 A beam is pivoted at one end, as shown.

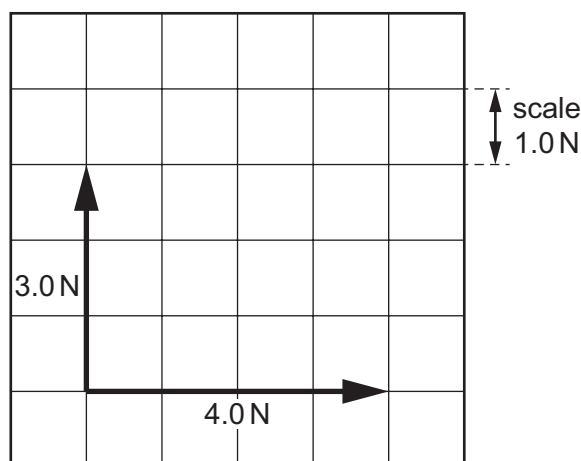


The beam weighs 6.0 N and its weight acts at a point X 40 cm from the pivot.

A force of 4.0 N is applied to the beam causing it to balance horizontally.

In which direction and where is the 4.0 N force applied?

- A vertically downwards at 20 cm to the left of X
 - B vertically downwards at 20 cm to the right of X
 - C vertically upwards at 20 cm to the left of X
 - D vertically upwards at 20 cm to the right of X
- 8 The diagram shows two forces acting at right angles to each other.



What is the resultant of the two forces?

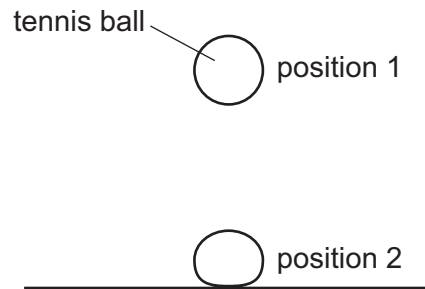
- A 1.0 N
 - B 5.0 N
 - C 7.0 N
 - D 12.0 N
- 9 Which statement describes the impulse acting on an object?
- A Impulse is the change in kinetic energy of the object.
 - B Impulse is the change in momentum of the object.
 - C Impulse is the rate of change of force acting on the object.
 - D Impulse is the rate of change of momentum of the object.

- 10** A ball of mass 0.16 kg is moving forwards at a speed of 0.50 m/s. A second ball of mass 0.10 kg is stationary. The first ball strikes the second ball. The second ball moves forwards at a speed of 0.50 m/s.

What is the speed of the first ball after the collision?

- A** 0.0 m/s **B** 0.19 m/s **C** 0.31 m/s **D** 0.50 m/s

- 11** A tennis ball is dropped from position 1. It falls vertically onto a hard surface at position 2.



Which energy changes have taken place between position 1 and position 2?

- A** gravitational potential → kinetic → chemical
B gravitational potential → kinetic → elastic (strain)
C kinetic → gravitational potential → chemical
D kinetic → gravitational potential → elastic (strain)
- 12** A boy holds onto a bar and pulls himself up until his chin is level with the bar.

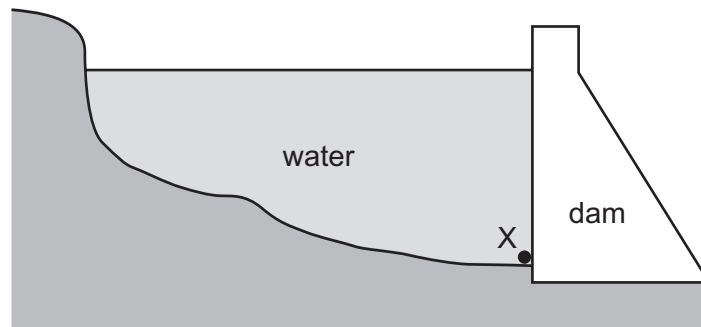
He raises himself through 40 cm in 0.5 s.

The weight of the boy is 500 N.

What is the average power he produces as he raises himself?

- A** 40 W **B** 400 W **C** 4000 W **D** 40 000 W

- 13 The diagram shows a deep reservoir formed by a dam.



On what does the pressure at X depend?

- A the depth of the water at X
 - B the length of the reservoir
 - C the surface area of the water
 - D the thickness of the dam wall
- 14 The conditions of a sample of gas change in two stages.
- stage 1 increase of temperature at constant volume
- stage 2 increase of volume with no further change of temperature

Which row about the pressure after each stage is correct?

	stage 1 pressure compared to original pressure	stage 2 pressure compared to pressure after stage 1
A	decreases	decreases
B	decreases	increases
C	increases	decreases
D	increases	increases

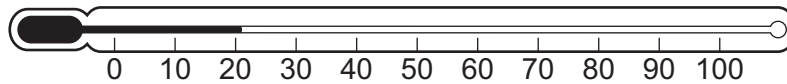
- 15 Small pollen particles are suspended in water.

When viewed with a microscope, the pollen particles can be seen to be moving about irregularly.

What causes this movement?

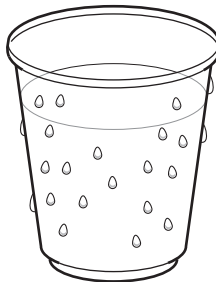
- A The pollen particles are being bombarded by the heavier particles of the water.
- B The pollen particles are being bombarded by the lighter particles of the water.
- C The temperature of the water is higher than that of the pollen particles.
- D The temperature of the water is lower than that of the pollen particles.

- 16 The diagram shows a liquid-in-glass thermometer.



Which physical property of the thermometer is used to measure temperature?

- A expansion of glass
 - B expansion of liquid
 - C mass of glass
 - D mass of liquid
- 17 A block of lead of mass 500 g is at its melting point.
- The specific latent heat of fusion of lead = 23 kJ/kg.
- How much energy is required to completely melt the block?
- A 46 J B 12 000 J C 46 000 J D 12 000 000 J
- 18 A glass contains an iced drink on a warm and humid day. Water starts to form on the outside of the glass.



What is the name of the effect by which the water forms?

- A condensation
- B conduction
- C convection
- D evaporation

- 19** One end of a copper bar is heated to a high temperature.

Which mechanism is responsible for the transfer of thermal energy to the other end of the copper bar?

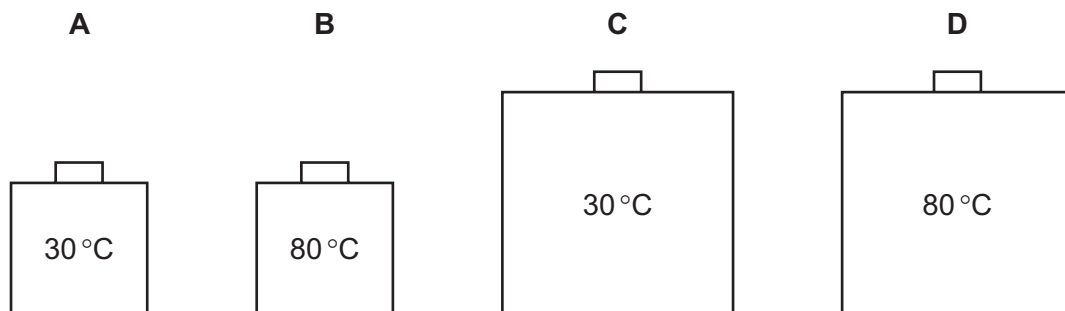
- A** the lattice vibrations of copper ions only
- B** the lattice vibrations of copper ions and the movement of high energy electrons along the bar
- C** the movement of high energy copper ions along the bar
- D** the movement of high energy electrons along the bar only

- 20** Four cubic copper containers are filled with water. The surfaces of all the containers are painted black.

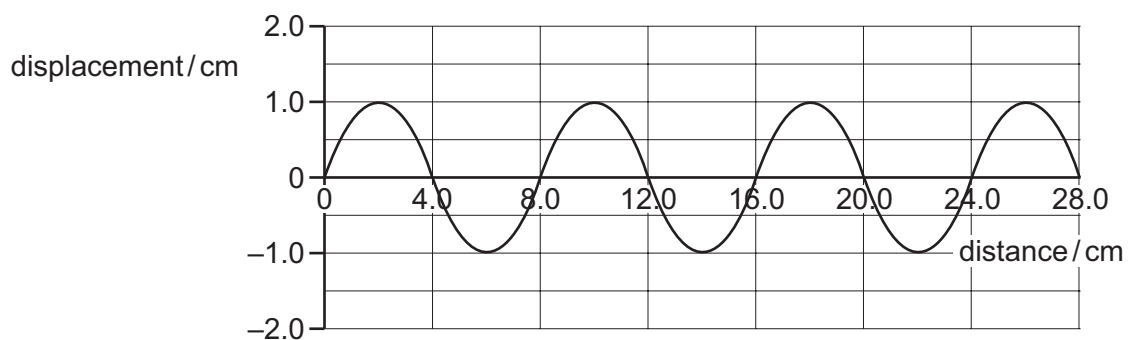
Two of the containers have sides of length 20 cm and the other two containers have sides of length 40 cm.

Two of the containers contain water at 80 °C and the other two contain water at 30 °C.

Which container radiates energy at the lowest rate?



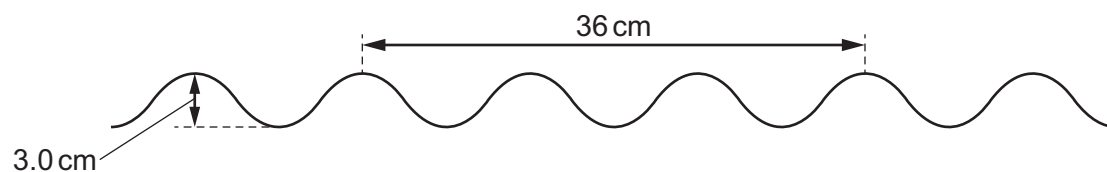
21 The diagram shows a wave.



Which row is correct?

	amplitude of the wave / cm	wavelength of the wave / cm
A	1.0	4.0
B	1.0	8.0
C	2.0	4.0
D	2.0	8.0

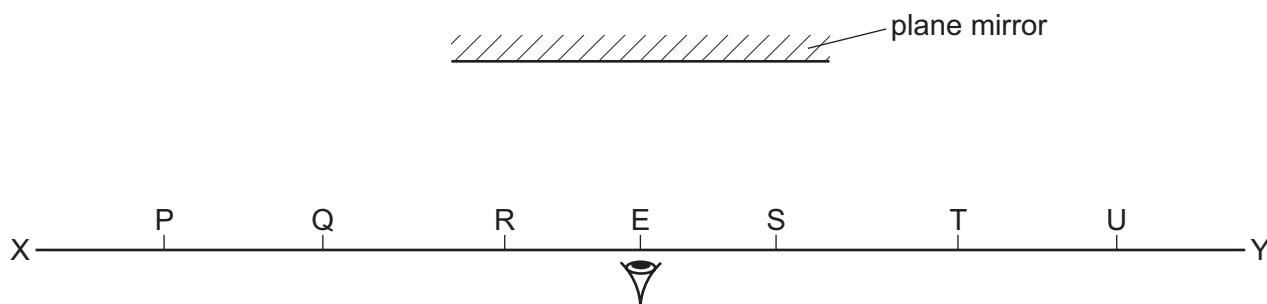
22 The water wave shown has a frequency of 4.0 Hz.



What is the speed of the wave?

- A** 3.0 cm/s **B** 12 cm/s **C** 48 cm/s **D** 72 cm/s

- 23 A student uses one eye to look at images in a plane mirror.



Objects are placed on the line XY.

Which objects give rise to images that can be seen by the eye at E?

- A** P, Q, R, S, T and U
B Q, R, S and T only
C P and U only
D R and S only
- 24 An object is placed in front of a converging lens of focal length 15 cm.

Which row describes the image of the object?

	distance of object from lens / cm	nature of the image
A	40	real, upright, diminished
B	30	virtual, inverted, enlarged
C	20	real, inverted, diminished
D	10	virtual, upright, enlarged

- 25 The speed of light in air is 3.0×10^8 m/s.

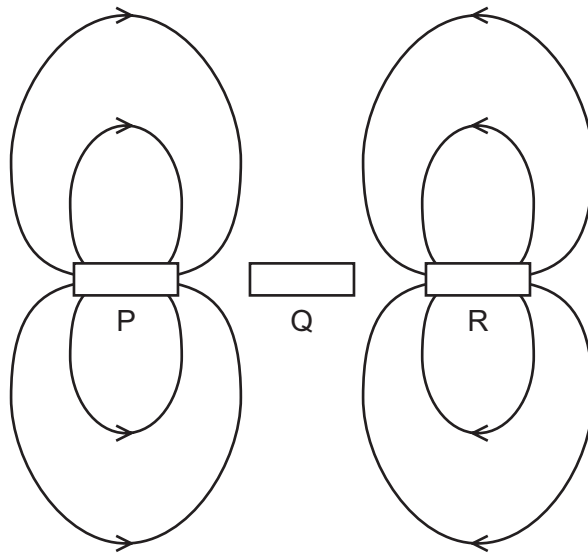
The critical angle for light in a transparent plastic material placed in air is 37° .

What is the speed of light in the plastic material?

- A** 1.8×10^8 m/s
B 2.4×10^8 m/s
C 3.8×10^8 m/s
D 5.0×10^8 m/s

- 26** Which type of waves are produced by a television remote controller?
- A** infrared waves
 - B** radio waves
 - C** ultraviolet waves
 - D** visible light
- 27** Which statement correctly compares radio waves and X-rays?
- A** Radio waves have a longer wavelength and a greater speed in a vacuum.
 - B** Radio waves have a longer wavelength and the same speed in a vacuum.
 - C** Radio waves have a shorter wavelength and a greater speed in a vacuum.
 - D** Radio waves have a shorter wavelength and the same speed in a vacuum.

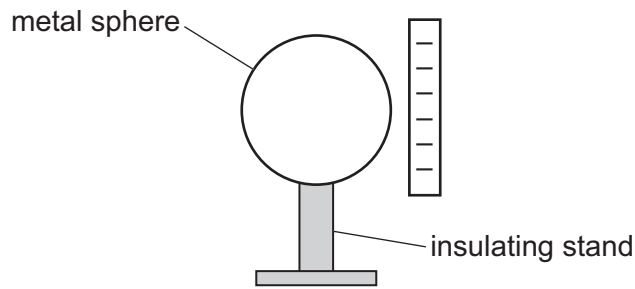
- 28 The diagram shows the magnetic fields around three objects, P, Q and R, placed close to each other.



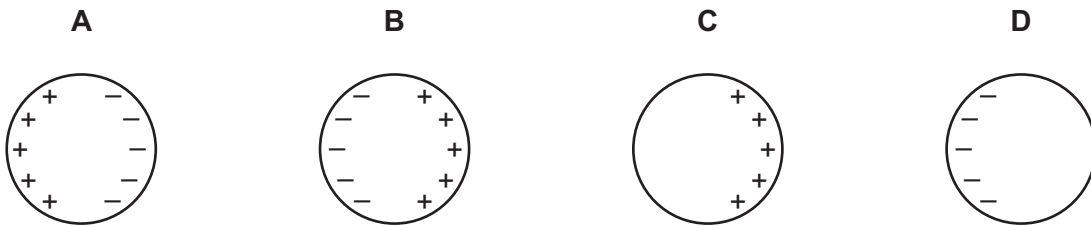
Which row shows the nature of each of the objects?

	P	Q	R
A	permanent magnet N S	copper rod	permanent magnet N S
B	permanent magnet N S	copper rod	permanent magnet S N
C	permanent magnet N S	iron rod	permanent magnet N S
D	permanent magnet N S	iron rod	permanent magnet S N

- 29 A negatively charged rod is brought near to an uncharged metal sphere that is placed on an insulating stand.



Which diagram shows the distribution of charges on the sphere?



- 30 A resistor has a potential difference (p.d.) of 12 V across it and a current of 0.60 A in it.

What is the resistance of the resistor?

- A** $0.050\ \Omega$ **B** $2.0\ \Omega$ **C** $7.2\ \Omega$ **D** $20\ \Omega$

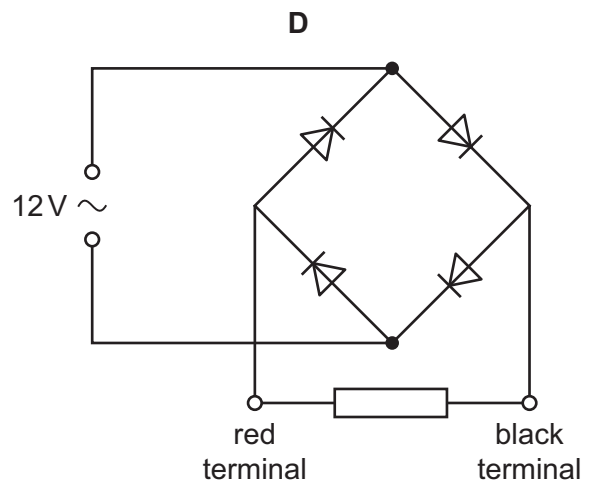
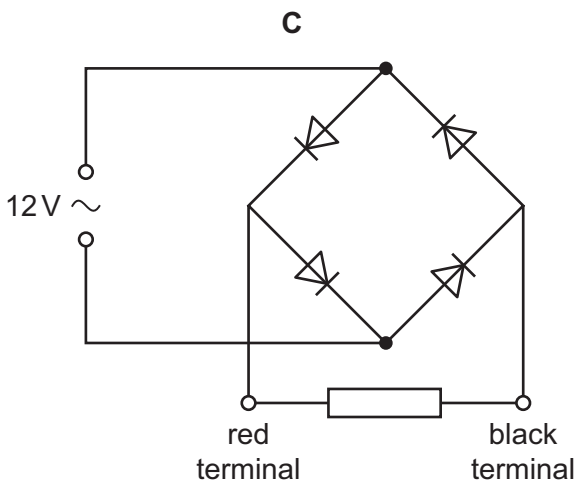
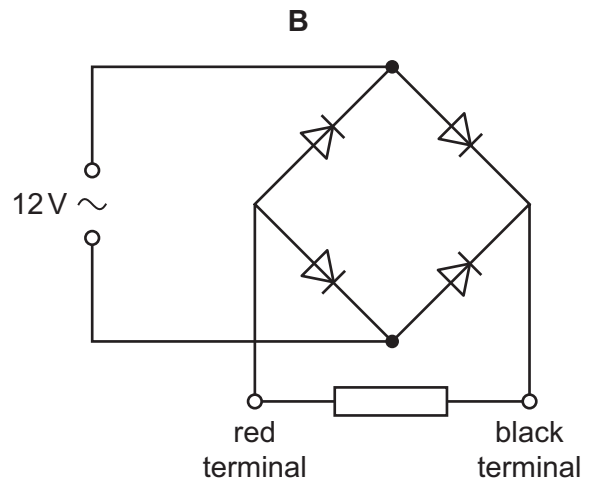
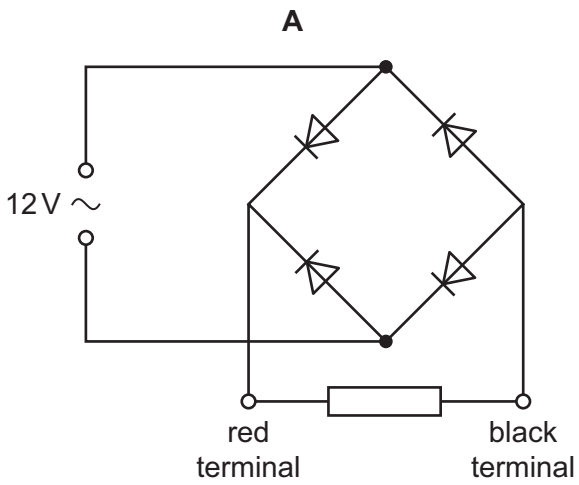
- 31 There is a current I in a resistor of resistance R for a time t . The potential difference (p.d.) across the resistor is V .

Which equation gives the power P dissipated in the resistor?

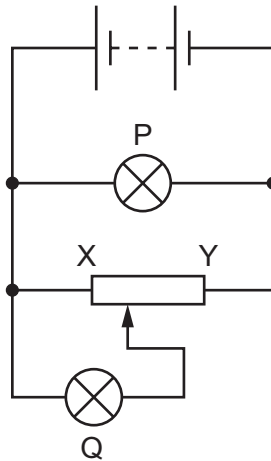
- A** $P = IR$ **B** $P = It$ **C** $P = IV$ **D** $P = \frac{IV}{t}$

32 The four circuits shown each contain four diodes.

In which circuit is the direction of the current in the resistor always from the red terminal to the black terminal?



- 33 The diagram shows a battery connected to a potential divider and to two lamps, P and Q.

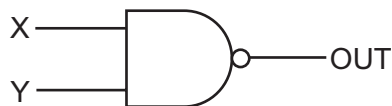


The slider on the potential divider is moved from end X to end Y of the resistor.

Which row shows the effect on the brightness of each lamp?

	brightness of P	brightness of Q
A	brighter	brighter
B	brighter	dimmer
C	unchanged	brighter
D	unchanged	dimmer

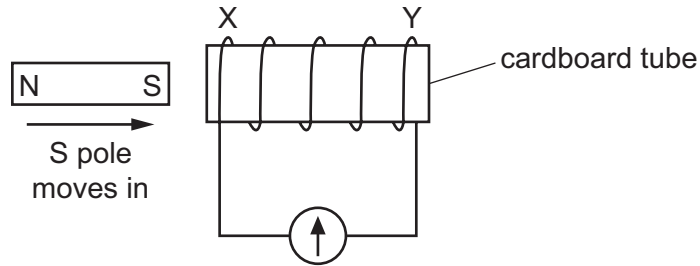
- 34 What is the truth table for the logic gate shown by the symbol?



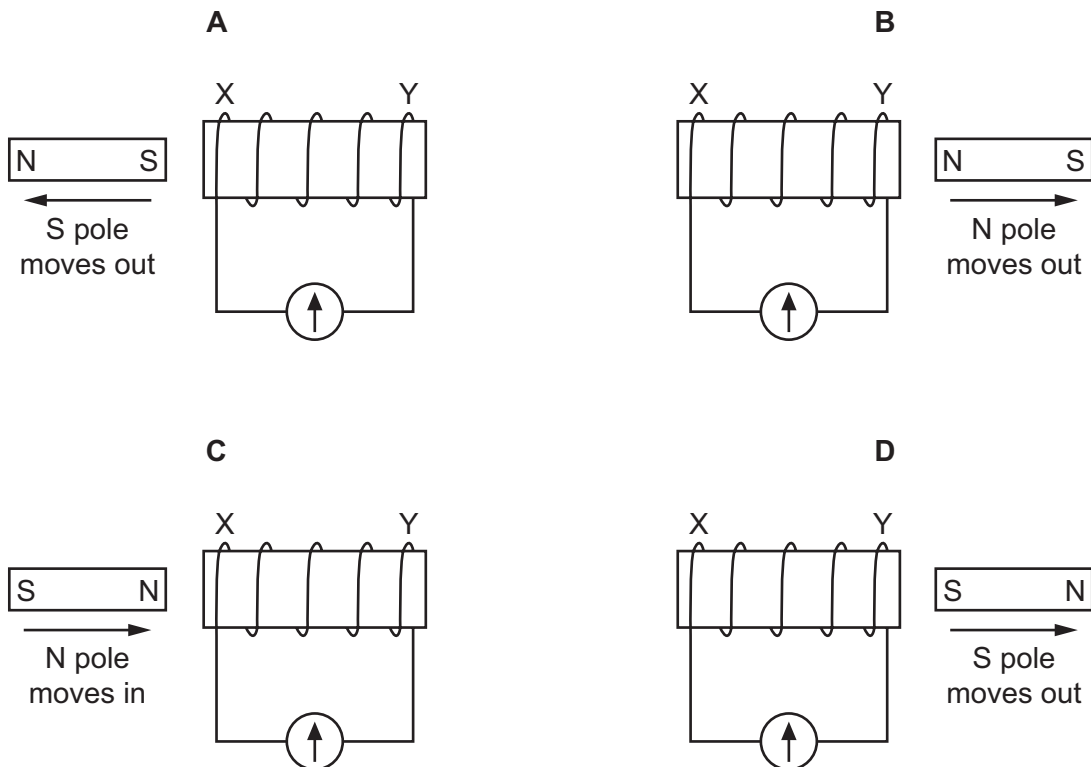
A			B			C			D		
X	Y	OUT	X	Y	OUT	X	Y	OUT	X	Y	OUT
0	0	0	0	0	0	0	0	1	0	0	1
0	1	0	0	1	1	0	1	0	0	1	1
1	0	0	1	0	1	1	0	0	1	0	1
1	1	1	1	1	1	1	1	0	1	1	0

35 A coil XY is wound around a cardboard tube.

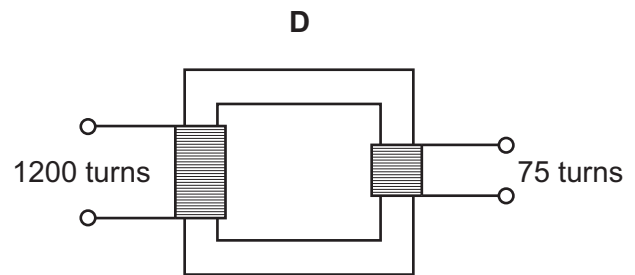
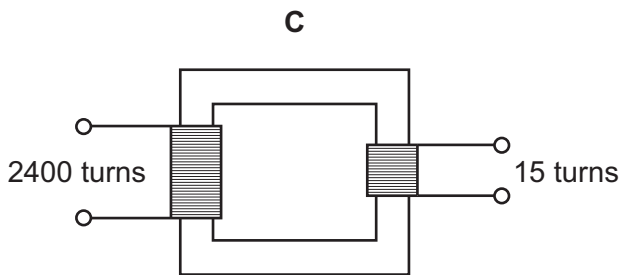
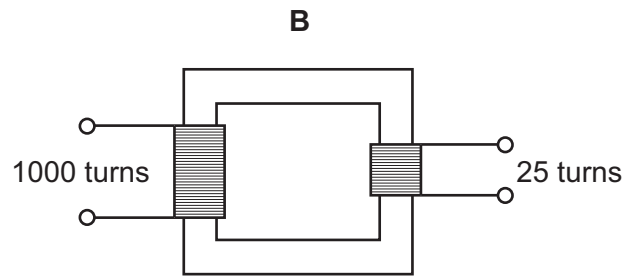
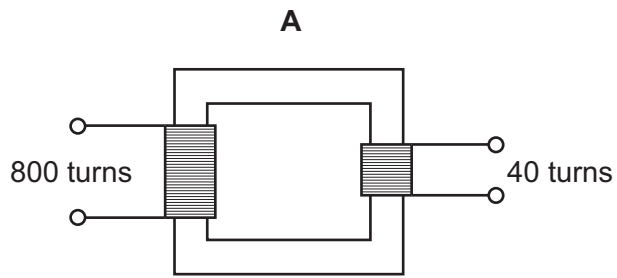
When the S pole of a magnet is pushed into the coil XY, the galvanometer deflects to the left.



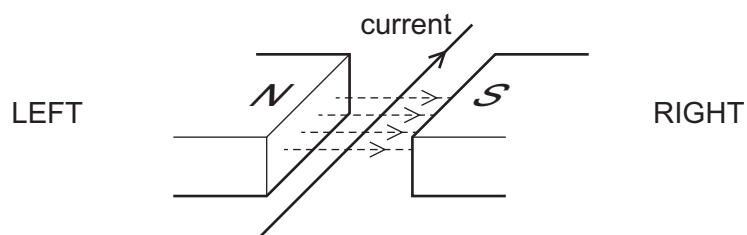
What other movement of the magnet will produce a deflection to the left?



36 Which transformer can change a 240 V a.c. input into a 15 V a.c. output?



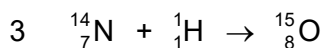
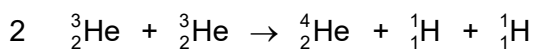
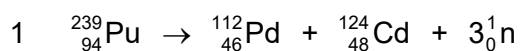
37 The diagram shows a current-carrying wire in a magnetic field.



In which direction is the force acting on the wire?

- A** towards the bottom of the page
- B** to the left
- C** to the right
- D** towards the top of the page

- 38 A scientist was asked to separate the following equations into two categories: nuclear fission and nuclear fusion.



Which equations show nuclear fission?

- A** 1 and 2 **B** 1 and 3 **C** 1 and 4 **D** 2 and 4

- 39 Which radioactive source is used in a smoke alarm system and what is the reason for this?

	source	reason
A	α	causes least ionisation of air
B	α	causes most ionisation of air
C	γ	causes least ionisation of air
D	γ	causes most ionisation of air

- 40 A beam of α -particles and β -particles is incident at right angles to an electric field.

Which statement about the deflection of the particles in the field is correct?

- A** α -particles deflect, but β -particles do not deflect.
B α -particles deflect in the opposite direction to β -particles.
C β -particles deflect, but α -particles do not deflect.
D Both α -particles and β -particles deflect in the same direction.

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Cambridge IGCSE™ (9–1)

PHYSICS (9–1)

0972/22

Paper 2 Multiple Choice (Extended)

May/June 2022

MARK SCHEME

Maximum Mark: 40

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the May/June 2022 series for most Cambridge IGCSE, Cambridge International A and AS Level and Cambridge Pre-U components, and some Cambridge O Level components.

This document consists of **3** printed pages.

Question	Answer	Marks
1	D	1
2	D	1
3	D	1
4	C	1
5	C	1
6	C	1
7	D	1
8	B	1
9	B	1
10	B	1
11	B	1
12	B	1
13	A	1
14	C	1
15	B	1
16	B	1
17	B	1
18	A	1
19	B	1
20	A	1
21	B	1
22	C	1
23	B	1
24	D	1
25	A	1
26	A	1
27	B	1
28	B	1

Question	Answer	Marks
29	B	1
30	D	1
31	C	1
32	A	1
33	C	1
34	D	1
35	D	1
36	D	1
37	A	1
38	C	1
39	B	1
40	B	1



Cambridge IGCSE™ (9–1)

PHYSICS

0972/21

Paper 2 Multiple Choice (Extended)

October/November 2022

45 minutes

You must answer on the multiple choice answer sheet.

You will need: Multiple choice answer sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)

INSTRUCTIONS

- There are **forty** questions on this paper. Answer **all** questions.
- For each question there are four possible answers **A**, **B**, **C** and **D**. Choose the **one** you consider correct and record your choice in soft pencil on the multiple choice answer sheet.
- Follow the instructions on the multiple choice answer sheet.
- Write in soft pencil.
- Write your name, centre number and candidate number on the multiple choice answer sheet in the spaces provided unless this has been done for you.
- Do **not** use correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.
- Take the weight of 1.0 kg to be 10 N (acceleration of free fall = 10 m/s^2).

INFORMATION

- The total mark for this paper is 40.
- Each correct answer will score one mark.
- Any rough working should be done on this question paper.

This document has **16** pages.



1 Which measuring instrument is used to measure the diameter of a thin metal wire?

- A 30 cm rule
- B measuring tape
- C metre rule
- D micrometre screw gauge

2 A car starts from rest.

The table shows the readings from its speedometer every 10 s.

time / s	0	10	20	30	40	50	60
$\frac{\text{speed}}{\text{m/s}}$	0	4	8	12	12	12	12

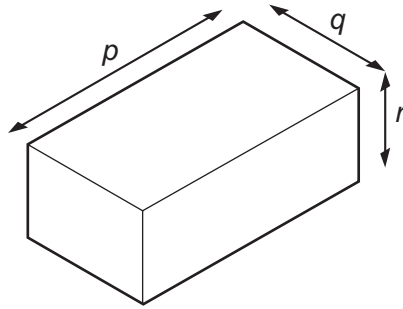
Which row describes the car's motion in the first 30 seconds and in the last 30 seconds?

	motion during first 30 s	motion during last 30 s
A	non-zero acceleration	at rest
B	zero acceleration	constant speed
C	zero acceleration	at rest
D	non-zero acceleration	constant speed

3 Which property of an object is a consequence of the effect of a gravitational field acting on it?

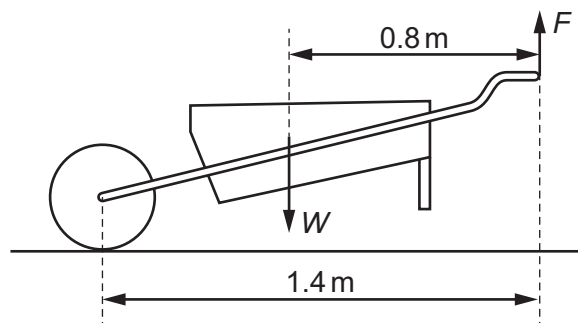
- A density
- B mass
- C volume
- D weight

- 4 The diagram shows the dimensions of a solid rectangular block of metal of mass m .



Which expression is used to calculate the density of the metal?

- A $\frac{m}{(p \times q)}$
 B $\frac{m}{(p \times q \times r)}$
 C $m \times p \times q$
 D $m \times p \times q \times r$
- 5 A wheelbarrow has a weight W of 140 N.

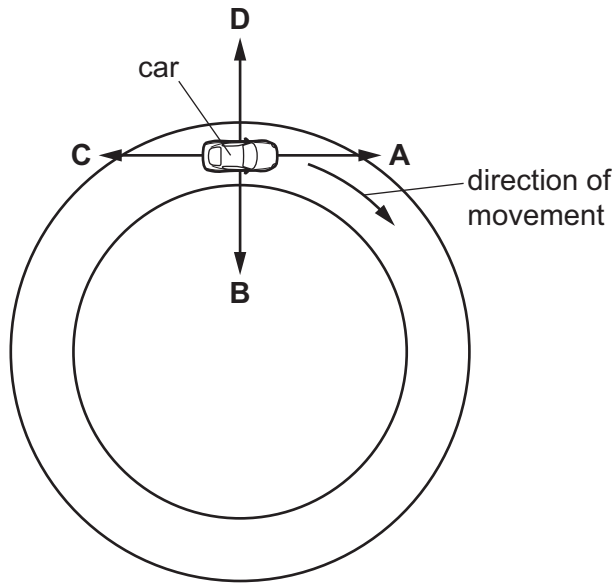


Which vertical force F is needed to support the wheelbarrow in the position shown?

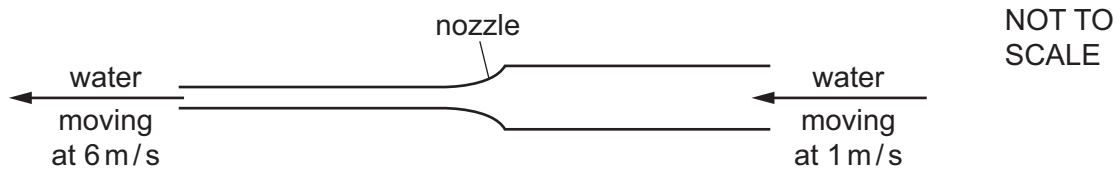
- A 60 N B 80 N C 140 N D 245 N

- 6 A car is travelling around a circular track at a constant speed, as shown.

In which direction is the resultant force on the car?



- 7 The diagram shows part of a hose used by a firefighter.



15 kg of water flows through the hose each second.

Which force is applied to the hose by the water?

- A** 15 N **B** 75 N **C** 90 N **D** 105 N
- 8 Brakes are used to slow down a moving car.

Into which form of energy is most of the kinetic energy converted as the car slows down?

- A** chemical
B elastic
C thermal
D sound

- 9 An object has kinetic energy of 200 J.

A constant resultant force of 190 N is applied in the direction of its motion through a distance of 10 m.

What is the final kinetic energy of the object?

- A** 390 J **B** 1700 J **C** 2000 J **D** 2100 J

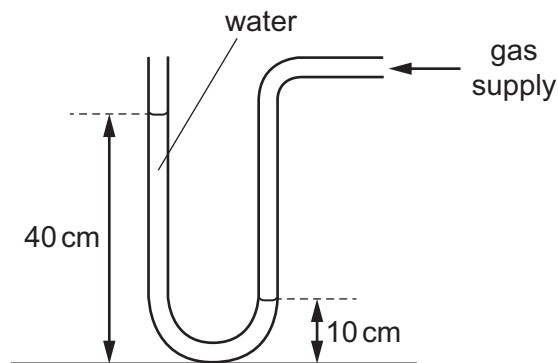
- 10 The statements describe what happens when the power of a machine is increased.

- 1 The work done in a given time decreases.
- 2 The work done in a given time increases.
- 3 The time taken to do a given quantity of work decreases.
- 4 The time taken to do a given quantity of work increases.

Which statements are correct?

- A** 1 and 3 **B** 1 and 4 **C** 2 and 3 **D** 2 and 4

- 11 A manometer containing water is used to measure the pressure of a gas supply, as shown.

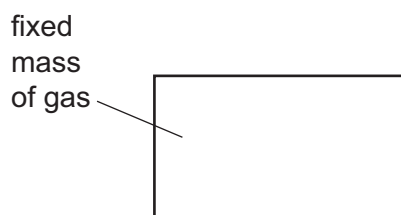


The density of water is 1000 kg/m^3 .

What is the pressure of the gas supply?

- A** 300 Pa above atmospheric pressure
B 400 Pa above atmospheric pressure
C 3000 Pa above atmospheric pressure
D 4000 Pa above atmospheric pressure

- 12** A fixed mass of gas is trapped in a container. The temperature of the gas is increased but the volume of the gas is kept constant.



How does this change affect the average kinetic energy of the molecules and the pressure on the walls of the container?

	average kinetic energy	pressure
A	increases	increases
B	stays the same	increases
C	increases	decreases
D	decreases	increases

- 13** Which row correctly describes boiling and evaporation of water?

	boiling	evaporation
A	bubbles seen	occurs at surface only
B	bubbles seen	occurs throughout the water
C	no bubbles	occurs at surface only
D	no bubbles	occurs throughout the water

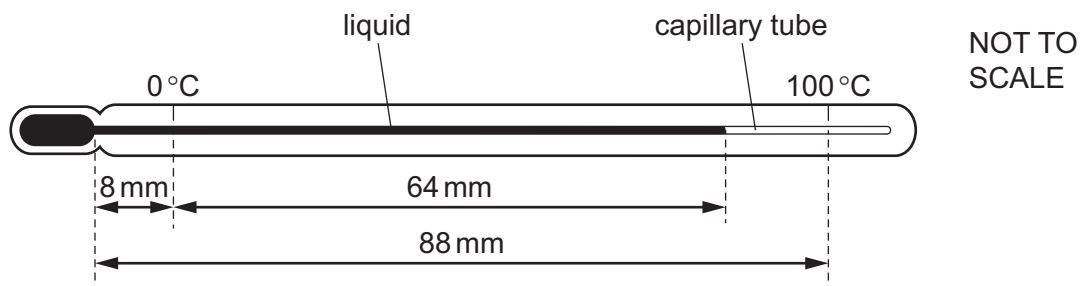
- 14** The diagram shows a liquid-in-glass thermometer.



Which change increases the sensitivity of the thermometer?

- A** a narrower capillary tube
- B** a wider capillary tube
- C** thicker glass around the bulb
- D** thinner glass around the bulb

- 15 The diagram shows a liquid-in-glass thermometer with a uniform capillary tube.



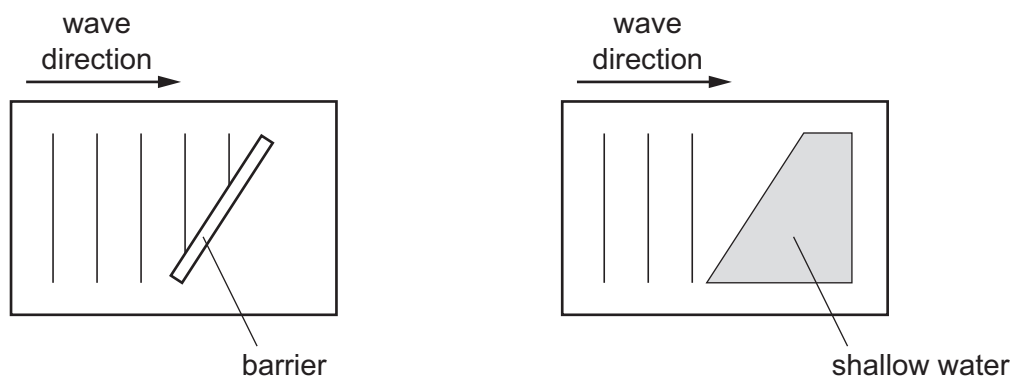
Which temperature is indicated by the thermometer?

- A** 73°C **B** 80°C **C** 82°C **D** 90°C
- 16 Four thermometers, with their bulbs painted different colours, are placed at equal distances from a radiant heater.
- Which thermometer shows the slowest temperature rise when the heater is first switched on?
- A** dull black
B dull white
C shiny black
D shiny white
- 17 A wave source produces 3000 crests every minute. The wave has a speed of 300 m/s.

What is the wavelength of the wave?

- A** 0.10 m **B** 0.17 m **C** 6.0 m **D** 10.0 m

18 The diagrams show two sets of wavefronts in a ripple tank.



A student makes two statements about the waves.

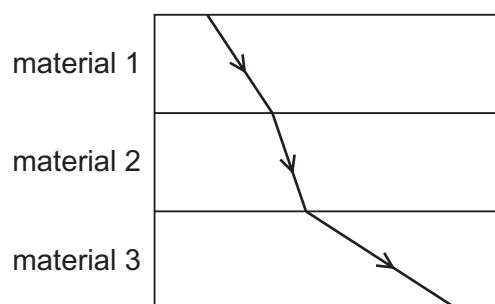
- 1 When the waves reflect from the barrier the direction changes but the wavelength remains the same.
- 2 When the waves refract as they enter the shallow water the direction remains the same, but the wavelength changes.

Which statements are correct?

- A** statement 1 and statement 2
B statement 1 only
C statement 2 only
D neither statement 1 nor statement 2

19 A composite block is made by joining together three transparent materials.

The diagram shows a ray of light passing through the composite block.



Which list gives the three materials in order of the speeds of light in the materials, from slowest to fastest?

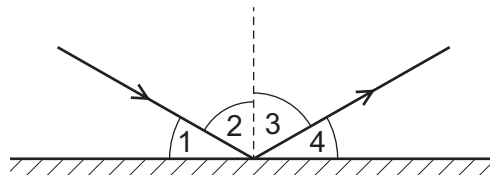
- A** 1 → 2 → 3 **B** 1 → 3 → 2 **C** 2 → 1 → 3 **D** 2 → 3 → 1

- 20** An object is placed 30 cm in front of a plane mirror.

Which statement describes the image of the object?

- A** The image is the same size and 30 cm from the object.
- B** The image is the same size and 60 cm from the object.
- C** The image is smaller and 30 cm from the object.
- D** The image is smaller and 60 cm from the object.

- 21** A ray of light is shone onto the surface of a mirror.



Which two angles represent the angle of incidence and the angle of reflection?

- A** 1 and 2 **B** 1 and 4 **C** 2 and 3 **D** 3 and 4

- 22** The speed of sound in air is 330 m/s.

How do the speeds of sound in concrete and water compare with this speed?

	speed in concrete	speed in water
A	greater	greater
B	greater	less
C	less	greater
D	less	less

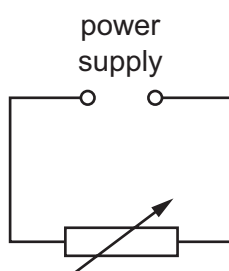
- 23** What is the speed of microwaves in air?

- A** $3 \times 10^8 \mu\text{m/s}$
- B** $3 \times 10^8 \text{cm/s}$
- C** $3 \times 10^8 \text{m/s}$
- D** $3 \times 10^8 \text{km/s}$

- 24 An observer stands at the finish line of a 100 m race. He wants to time the winner's run. He starts his stop-watch as soon as he sees the smoke from the starting gun instead of when he hears the bang.

What is the reason for doing this?

- A Light travels much faster than sound.
 B There is a risk he might respond to an echo from a wall.
 C Humans react slower to sound than to light.
 D Humans react more quickly to sound than to light.
- 25 Which statement is **not** a method for magnetising an iron bar?
- A Place the bar next to a magnet and hammer the bar.
 B Place the bar inside a solenoid. Switch on an alternating current (a.c.) in the coil and gradually reduce the current.
 C Place the bar inside a solenoid. Switch on a direct current (d.c.) in the coil and gradually reduce the current.
 D Stroke the bar repeatedly with a magnet.
- 26 The diagram shows a circuit containing a variable resistor connected to a variable power supply.



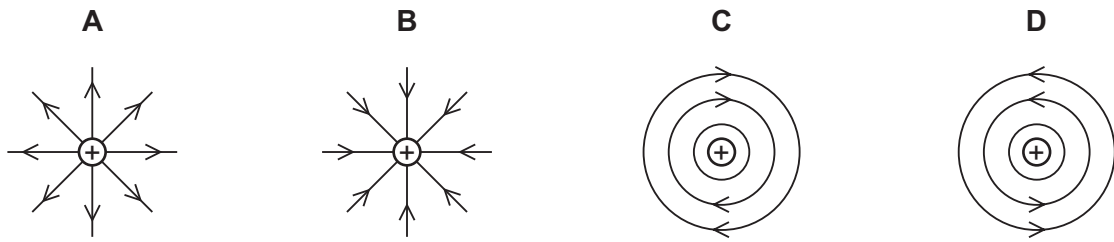
The table shows the currents for different values of the potential difference (p.d.) and the resistance.

p.d. / V	resistance / Ω	current
3.6	12	I_1
1.2	12	I_2
3.6	6	I_3

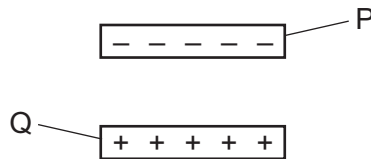
What is the order of the currents from smallest to largest?

- A $I_1 \rightarrow I_2 \rightarrow I_3$ B $I_1 \rightarrow I_3 \rightarrow I_2$ C $I_2 \rightarrow I_1 \rightarrow I_3$ D $I_3 \rightarrow I_1 \rightarrow I_2$

27 Which diagram shows the electric field pattern around a positive charge?



28 A negatively charged plastic rod P is placed above a positively charged plastic rod Q.



What are the directions of the electrostatic forces on rod P and on rod Q?

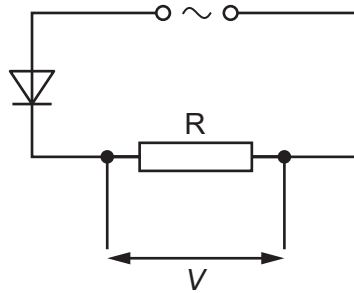
	electrostatic force on rod P	electrostatic force on rod Q
A	downwards	downwards
B	downwards	upwards
C	upwards	downwards
D	upwards	upwards

29 A lamp rated 12 V, 2.0 A is switched on for one minute.

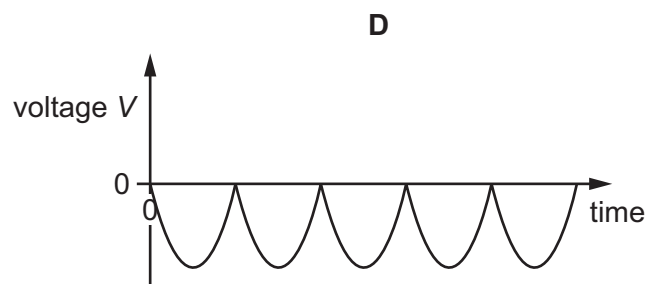
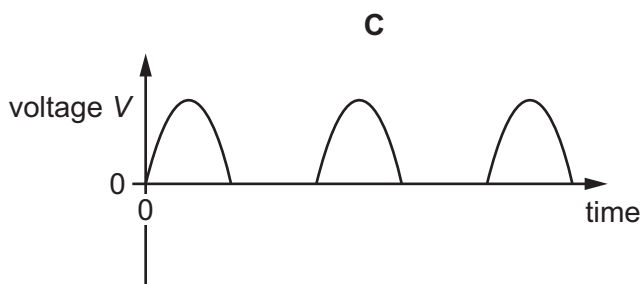
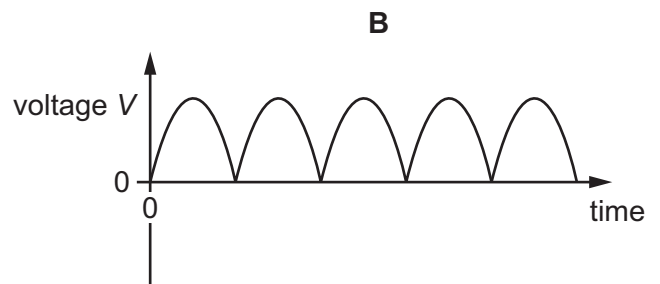
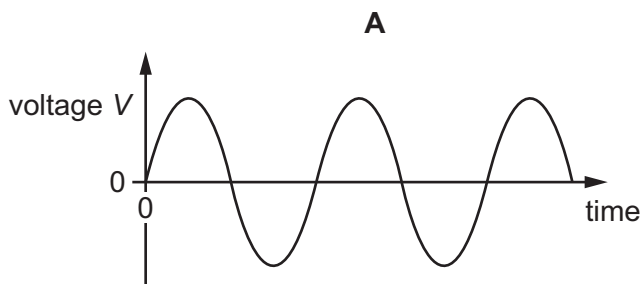
How much energy is transferred by the lamp?

- A** 6.0 J **B** 24 J **C** 360 J **D** 1440 J

- 30** An alternating current (a.c.) power supply is connected in series with a resistor R and a diode.



Which graph shows how the voltage V across the resistor R varies with time?

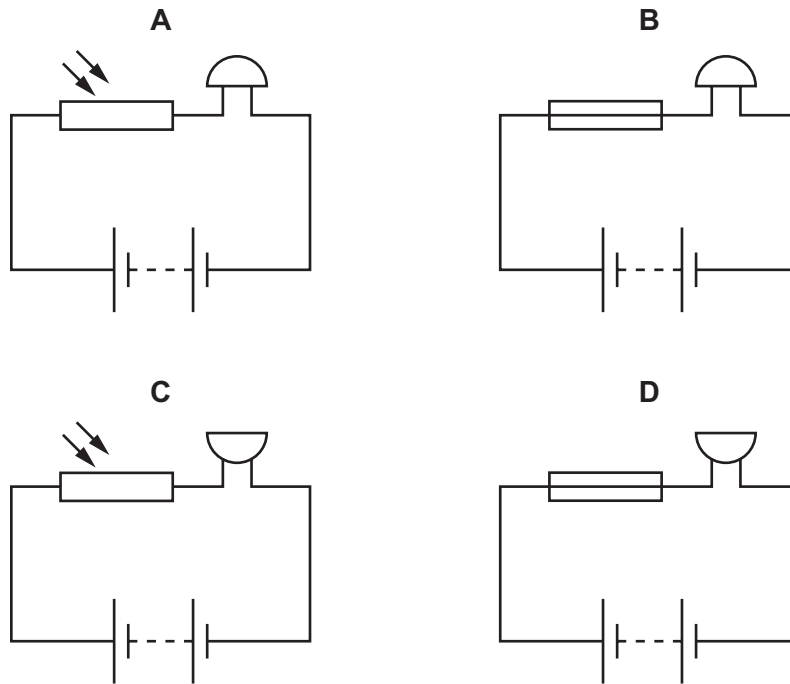


- 31** A student makes four resistors using different pieces of wire. The wires have different diameters and lengths. All the pieces of wire are made of the same material.

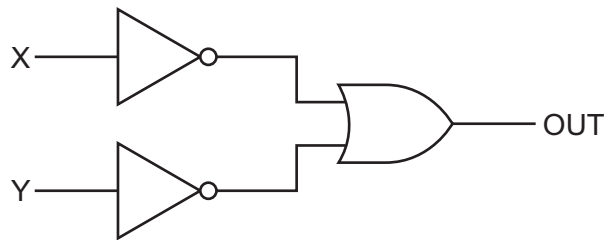
Which piece of wire will make the resistor with the largest resistance?

	diameter / mm	length / cm
A	0.8	10
B	0.8	17
C	2.0	10
D	2.0	17

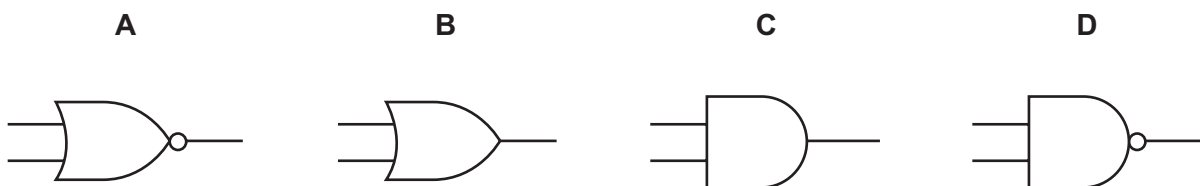
32 Which diagram shows a circuit containing a battery, a fuse and a buzzer?



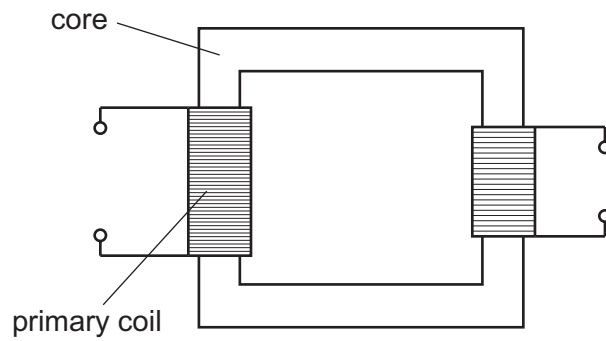
33 The diagram shows a network of logic gates.



Which single logic gate, placed between X, Y and OUT, would have the same effect as the network?



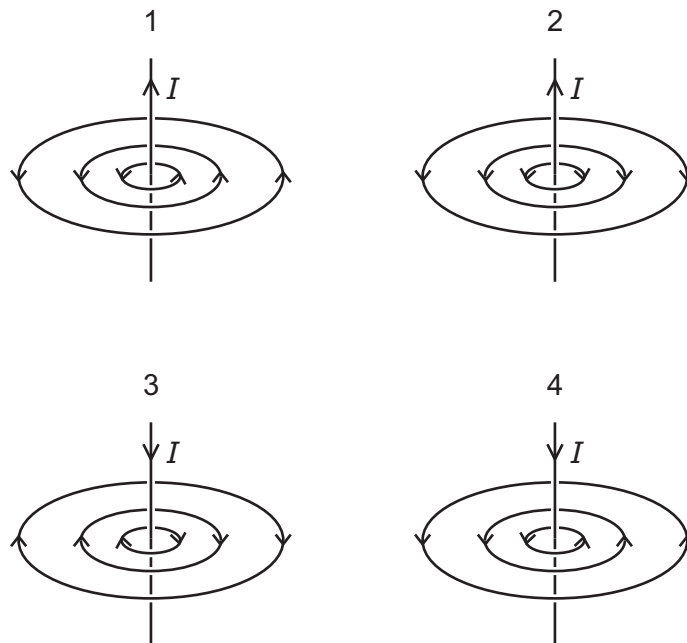
34 The diagram represents a transformer.



Which row shows materials suitable for making the core and the primary coil?

	core	primary coil
A	iron	copper
B	iron	plastic
C	steel	copper
D	steel	plastic

35 The diagrams show the magnetic field lines around a wire carrying a current, I .



Which diagrams are correct?

- A** 1 only **B** 2 and 3 **C** 4 only **D** 1 and 3

36 Which statement about electromagnetic induction is correct?

- A A conductor moving parallel to magnetic field lines induces an electromotive force (e.m.f.) in the conductor.
- B The direction of the induced electromotive force (e.m.f.) opposes the change which produces it.
- C The induced current is in a direction parallel to the direction of motion producing it.
- D The induced electromotive force (e.m.f.) decreases when the speed of the conductor moving across the field lines increases.

37 A very important experiment increased scientists' understanding of the structure of matter.

In the experiment, particles scattered as they passed through a thin metal foil.

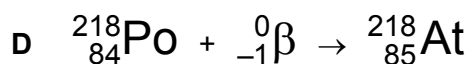
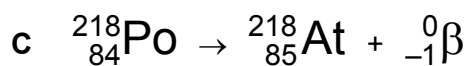
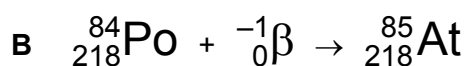
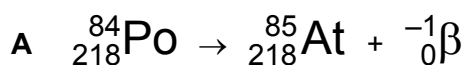
Which particles were used, and to which conclusion did the experiment lead?

	particles	conclusion
A	alpha-particles	matter is made up of atoms
B	alpha-particles	atoms have a very small nucleus
C	beta-particles	matter is made up of atoms
D	beta-particles	atoms have a very small nucleus

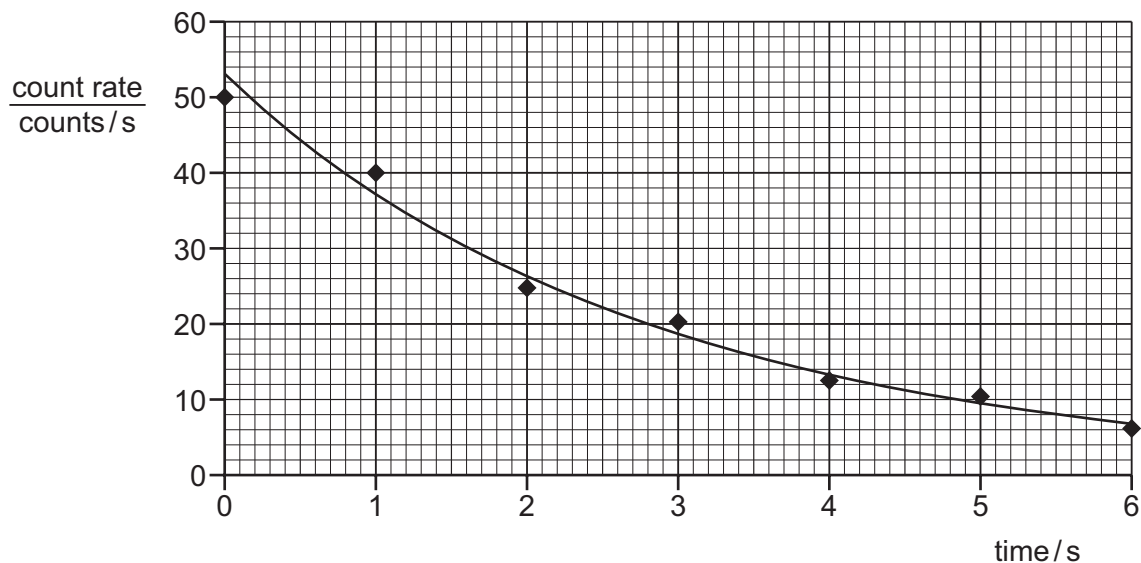
38 Polonium, Po, has a proton number equal to 84 and a nucleon number equal to 218.

Polonium changes into astatine, At, by emitting a β -particle.

Which equation represents this decay?



39 The graph shows how the count rate from a radioactive isotope changes with time.



What is the half-life of this isotope?

- A** 2.0 s **B** 6.0 s **C** 12 s **D** 53 s

40 What is the nature of α -emission?

- A** electromagnetic waves
B negatively charged particles
C positively charged particles
D uncharged particles

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Cambridge IGCSE™ (9–1)

PHYSICS

0972/21

Paper 2 Multiple Choice (Extended)

October/November 2022

MARK SCHEME

Maximum Mark: 40

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

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Cambridge International is publishing the mark schemes for the October/November 2022 series for most Cambridge IGCSE™, Cambridge International A and AS Level components and some Cambridge O Level components.

This document consists of **3** printed pages.

Question	Answer	Marks
1	D	1
2	D	1
3	D	1
4	B	1
5	A	1
6	B	1
7	B	1
8	C	1
9	D	1
10	C	1
11	C	1
12	A	1
13	A	1
14	A	1
15	B	1
16	D	1
17	C	1
18	B	1
19	C	1
20	B	1
21	C	1
22	A	1
23	C	1
24	A	1
25	B	1
26	C	1
27	A	1
28	B	1

Question	Answer	Marks
29	D	1
30	C	1
31	B	1
32	D	1
33	D	1
34	A	1
35	D	1
36	B	1
37	B	1
38	C	1
39	A	1
40	C	1



Cambridge IGCSE™ (9–1)

PHYSICS

0972/21

Paper 2 Multiple Choice (Extended)

May/June 2023

45 minutes

You must answer on the multiple choice answer sheet.

You will need: Multiple choice answer sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)

INSTRUCTIONS

- There are **forty** questions on this paper. Answer **all** questions.
- For each question there are four possible answers **A**, **B**, **C** and **D**. Choose the **one** you consider correct and record your choice in soft pencil on the multiple choice answer sheet.
- Follow the instructions on the multiple choice answer sheet.
- Write in soft pencil.
- Write your name, centre number and candidate number on the multiple choice answer sheet in the spaces provided unless this has been done for you.
- Do **not** use correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.
- Take the weight of 1.0 kg to be 9.8 N (acceleration of free fall = 9.8 m/s^2).

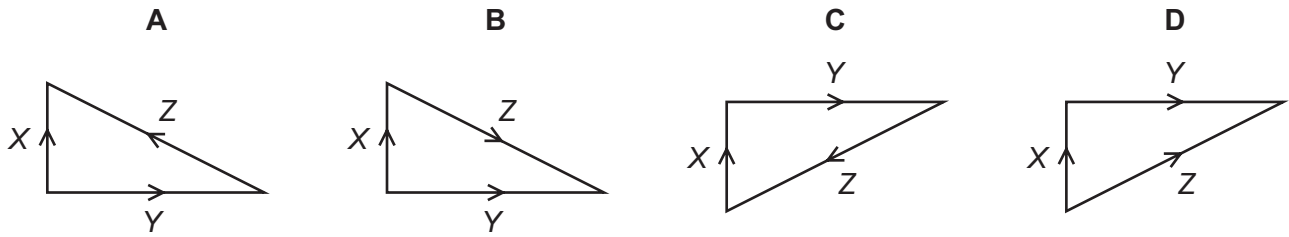
INFORMATION

- The total mark for this paper is 40.
- Each correct answer will score one mark.
- Any rough working should be done on this question paper.

This document has **16** pages. Any blank pages are indicated.



- 1 Which vector diagram correctly shows the force Z as the resultant of forces X and Y ?

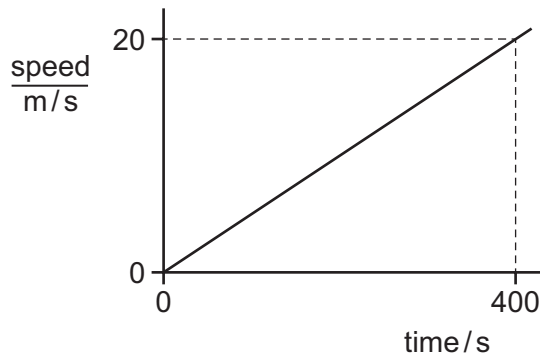


- 2 An object falls towards the surface of the Earth.

The object is falling at its terminal velocity.

Which statement is correct?

- A** There is air resistance and the acceleration of the object is negative.
B There is air resistance and the acceleration of the object is zero.
C There is no air resistance and the acceleration of the object is negative.
D There is no air resistance and the acceleration of the object is zero.
- 3 The graph represents the motion of a vehicle.



What is the distance travelled by the vehicle in 400 s?

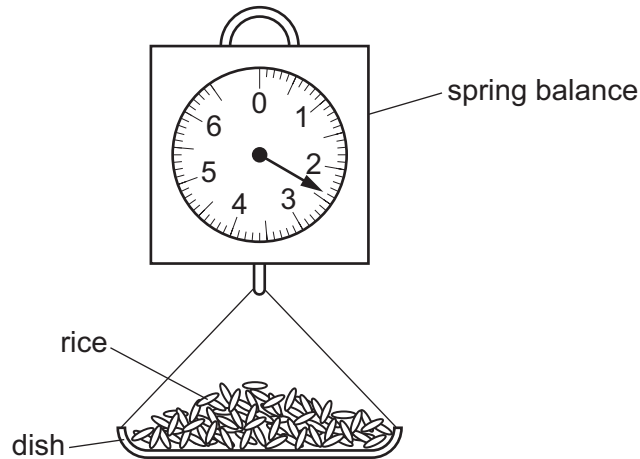
- A** 20 m **B** 400 m **C** 4000 m **D** 8000 m
- 4 On the Earth, a spring stretches by 5.0 cm when a mass of 3.0 kg is suspended from one end.

The gravitational field strength on the Moon is $\frac{1}{6}$ of that on the Earth.

Which mass, on the Moon, would stretch the spring by the same extension?

- A** 0.50 kg **B** 3.0 kg **C** 5.0 kg **D** 18 kg

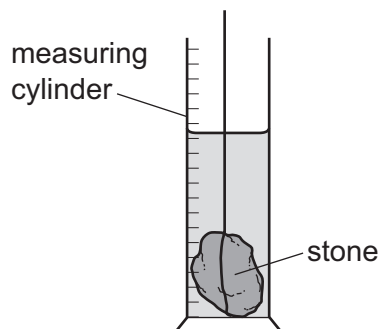
- 5 A shopkeeper pours rice into a dish that hangs from a spring balance. He records the reading.



A customer buys some pasta. The shopkeeper notices that the reading on the spring balance, with just pasta in the dish, is the same as it was with just rice in the dish.

Which quantity **must** be the same for the rice and for the pasta?

- A density
 - B temperature
 - C volume
 - D weight
- 6 A student determines the density of an irregularly shaped stone. The stone is slowly lowered into a measuring cylinder partly filled with water.

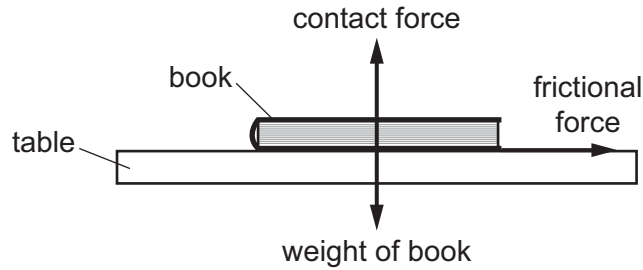


Which other apparatus does the student need to calculate the density of the irregularly shaped stone?

- A a balance
- B a thermometer
- C a metre rule
- D a stop-watch

- 7 A train is travelling horizontally in a straight line. A book is on a table in the train.

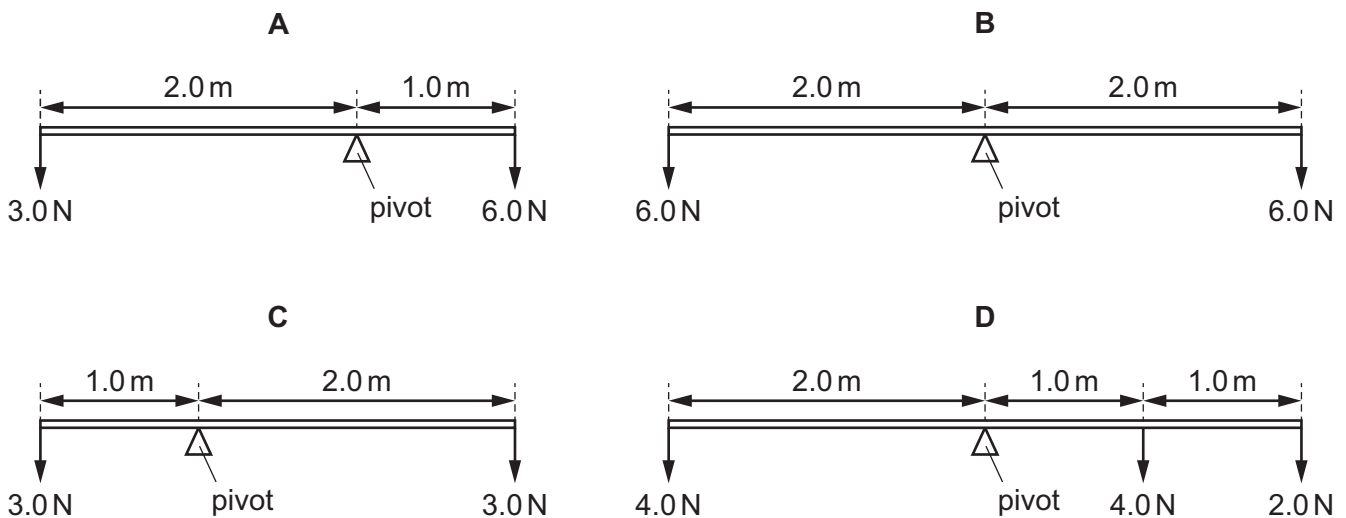
The diagram shows all the forces acting on the book.



How is the train moving?

- A** accelerating to the left of the diagram
B accelerating to the right of the diagram
C moving at uniform speed to the left of the diagram
D moving at uniform speed to the right of the diagram
- 8 The diagrams show four beams, each of negligible weight and freely pivoted.

Which beam is **not** in equilibrium?



- 9 An object of mass 1.2 kg is moving with a velocity of 2.0 m/s when it is acted on by a force of 4.0 N . The velocity of the object increases to 5.0 m/s in the same direction.

For which period of time does the force act on the object?

- A** 0.90 s **B** 1.1 s **C** 1.5 s **D** 3.6 s

10 Which row about the change of energy in the energy store must be correct?

	process	energy store	change of energy in store
A	water pumped up to a high-altitude dam	gravitational potential energy of water	increases
B	water pumped up to a high-altitude dam	kinetic energy of water	decreases
C	air passes through a wind turbine	gravitational potential energy of air	increases
D	air passes through a wind turbine	kinetic energy of air	increases

11 A woman of mass 50 kg has 81 J of kinetic energy.

What is her speed?

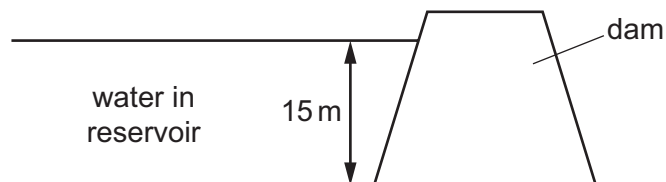
- A** 1.3 m/s **B** 1.6 m/s **C** 1.8 m/s **D** 3.2 m/s

12 A child runs up a set of stairs four times. The time taken for each run is recorded.

Which time is measured when the child's useful power is greatest?

- A** 10 s **B** 20 s **C** 30 s **D** 40 s

13 A dam holds water in a reservoir. The height of the water in the reservoir is 15 m.



The density of water is 1000 kg/m^3 .

What is the pressure due to the water at the bottom of the dam?

- A** 6.8 Pa **B** 1500 Pa **C** 15 000 Pa **D** 150 000 Pa

14 A student uses a microscope to observe pollen moving on the surface of water.

Which statement describes the reason for this movement?

- A** Water molecules are moved by microscopic pollen particles.
B Water molecules are moved by pollen molecules.
C Microscopic pollen particles are moved by water molecules.
D Pollen molecules are moved by water molecules.

15 Which statements about evaporation of water are correct?

- 1 Evaporation causes the remaining liquid to cool.
- 2 During evaporation, the more energetic particles escape from the surface of the liquid.
- 3 Evaporation only happens at 100°C .

A 1, 2 and 3 **B** 1 and 2 only **C** 1 and 3 only **D** 2 and 3 only

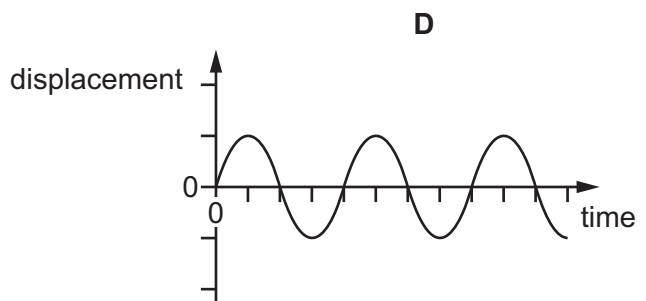
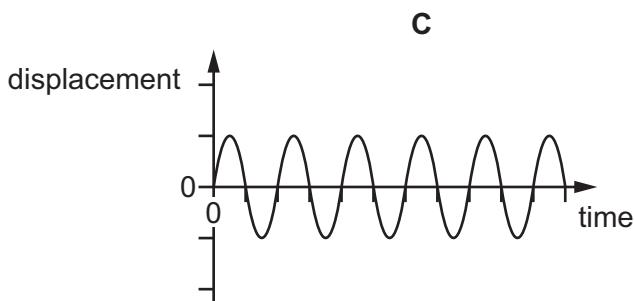
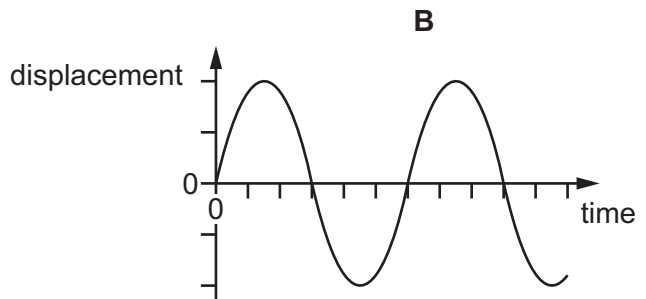
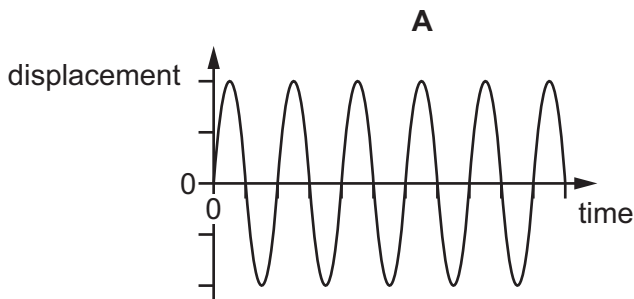
16 Some hot water is sealed inside a metal can. The can is in a vacuum in outer space. The hot water slowly cools down.

How does the thermal energy escape into space?

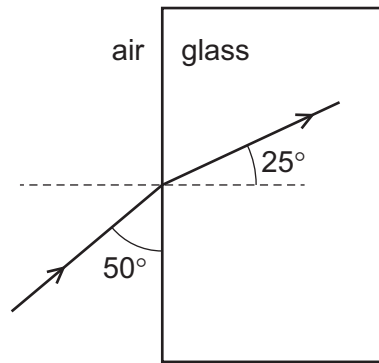
- A** by conduction then convection
B by conduction then radiation
C by evaporation then convection
D by evaporation then radiation

17 The diagrams show graphs of displacement against time for four waves. All the graphs are drawn to the same scale.

Which wave has the largest amplitude and the highest frequency?

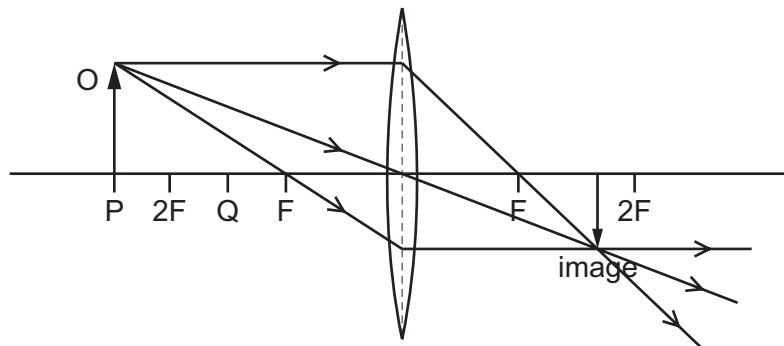


- 18 The diagram shows a ray of light entering a glass block.



Which calculation gives the refractive index of the glass?

- A $\frac{\sin 40^\circ}{\sin 25^\circ}$ B $\frac{\sin 40^\circ}{\sin 65^\circ}$ C $\frac{\sin 50^\circ}{\sin 25^\circ}$ D $\frac{\sin 50^\circ}{\sin 65^\circ}$
- 19 An object O is placed at point P near to a thin converging lens. The diagram shows three rays from the top of O passing through the lens. Each point F is one focal length from the centre of the lens. Each point 2F is two focal lengths from the centre of the lens.

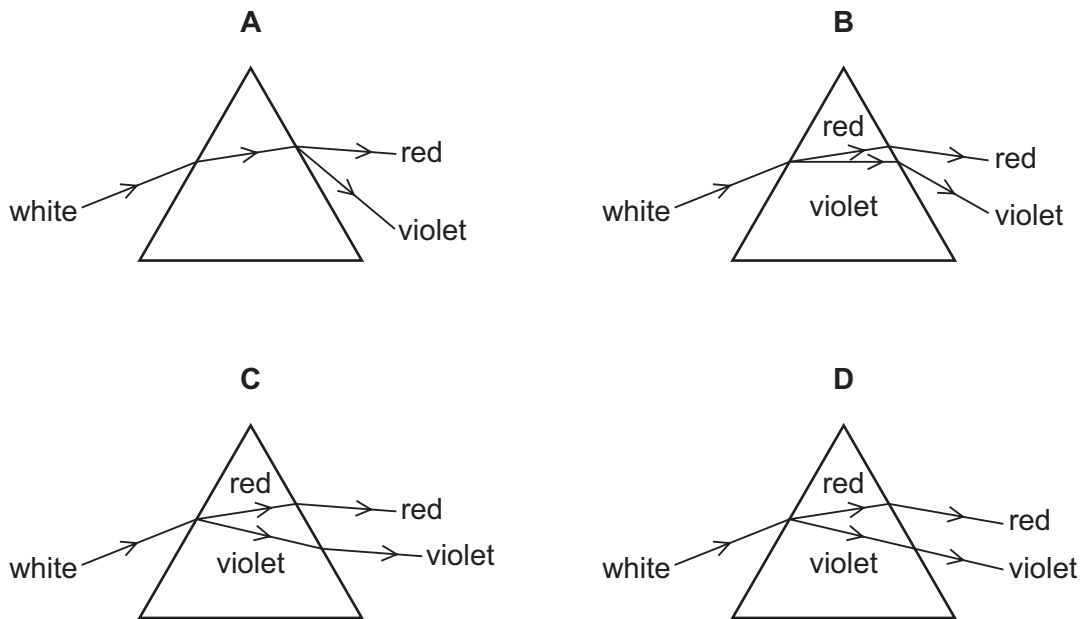


The object O is moved to point Q on the diagram.

Which type of image is produced when the object O is at point Q?

- A inverted and the same size as the object
 B inverted and enlarged
 C upright and the same size as the object
 D upright and enlarged

20 Which diagram shows the dispersion of white light by a glass prism?



21 Visible light has wavelengths in the range $4.0 \times 10^{-7} \text{ m}$ to $7.0 \times 10^{-7} \text{ m}$.

What is the range of the frequencies of visible light?

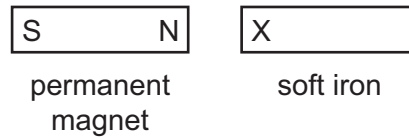
- A 0.12 Hz to 0.21 Hz
- B 120 Hz to 210 Hz
- C $4.3 \times 10^{11} \text{ Hz}$ to $7.5 \times 10^{11} \text{ Hz}$
- D $4.3 \times 10^{14} \text{ Hz}$ to $7.5 \times 10^{14} \text{ Hz}$

22 Student X fires a starting pistol which produces smoke and sound. Student Y is standing 100 m away and sees the smoke the instant it is produced. The speed of sound in air is 340 m/s.

What is the time delay between student Y seeing the smoke and hearing the sound?

- A 0.29 s
- B 0.59 s
- C 1.7 s
- D 3.4 s

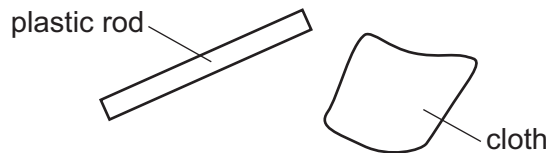
- 23** An unmagnetised piece of soft iron is placed close to a strong permanent magnet, as shown.



What is the induced polarity of end X of the soft iron and in which direction does the magnetic force act on the soft iron?

	polarity of end X	direction of force on the soft iron
A	N	to the left
B	N	to the right
C	S	to the left
D	S	to the right

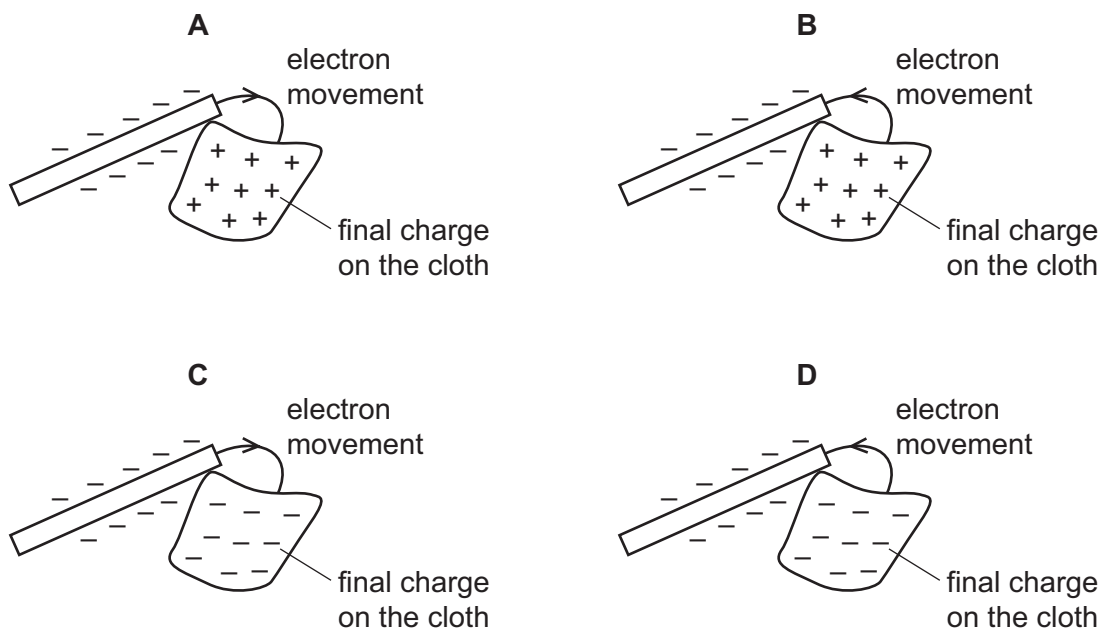
- 24** A plastic rod is rubbed with a cloth.



The rod and the cloth both become charged as electrons move between them.

The rod becomes negatively charged.

Which diagram shows how the rod becomes negatively charged and shows the final charge on the cloth?

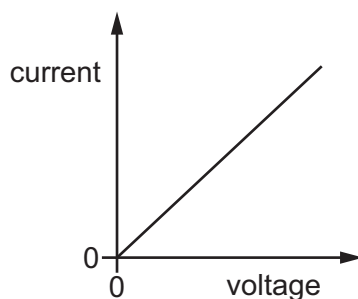


25 A wire has a uniform circular cross-sectional area.

Which statement is correct?

- A** The resistance of the wire is directly proportional to its cross-sectional area and inversely proportional to its diameter.
- B** The resistance of the wire is directly proportional to its cross-sectional area and inversely proportional to its length.
- C** The resistance of the wire is directly proportional to its length and inversely proportional to its cross-sectional area.
- D** The resistance of the wire is directly proportional to its length and inversely proportional to its diameter.

26 The diagram shows the current–voltage graph for a metal wire.



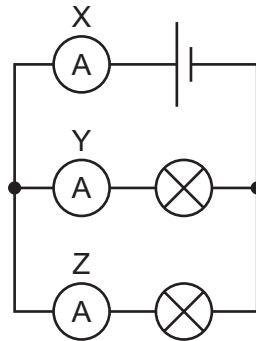
What can be deduced from the graph?

- A** As voltage increases, the temperature of the wire increases.
 - B** As voltage increases, the temperature of the wire decreases.
 - C** As voltage increases, the resistance of the wire increases.
 - D** As voltage increases, the resistance of the wire remains constant.
- 27** A battery is connected to a circuit. It is switched on for 1.0 minute. During that time, there is a current of 0.40 A in the circuit and the battery supplies a total of 48 J of energy.

Which row gives the charge that passes and the electromotive force (e.m.f.) of the battery?

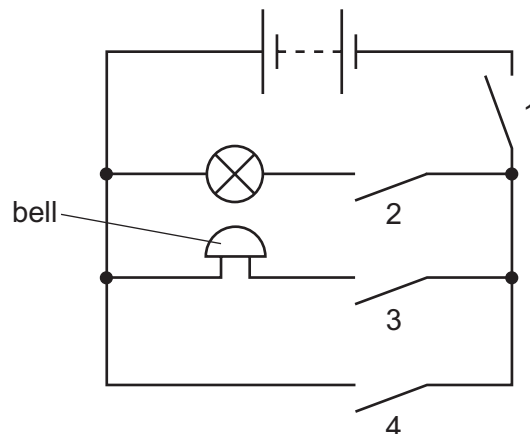
	charge that passes in 1.0 minute / C	e.m.f. of the battery / V
A	0.40	2.0
B	0.40	120
C	24	2.0
D	24	120

- 28 The circuit diagram shows two identical lamps connected in parallel to a cell. Three ammeters, X, Y and Z, are also connected in the circuit, as shown.



Which statement about the current in X is correct?

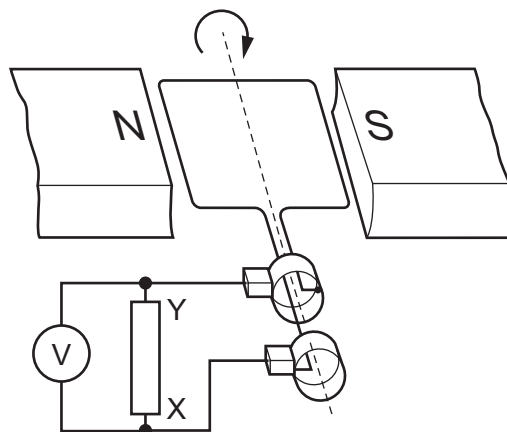
- A It is equal to the current in Y and to the current in Z.
 - B It is less than either the current in Y or the current in Z.
 - C It is equal to the sum of the current in Y and the current in Z.
 - D It is equal to the difference between the current in Y and the current in Z.
- 29 A student connects the circuit shown.



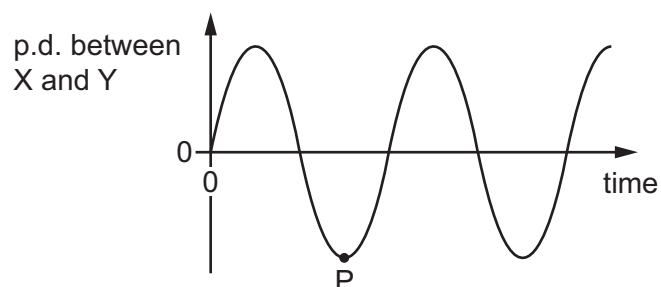
Which switches must be closed for the bell to ring without lighting the lamp?

- A 1 and 2 only
- B 1 and 3 only
- C 1, 3 and 4
- D 2, 3 and 4

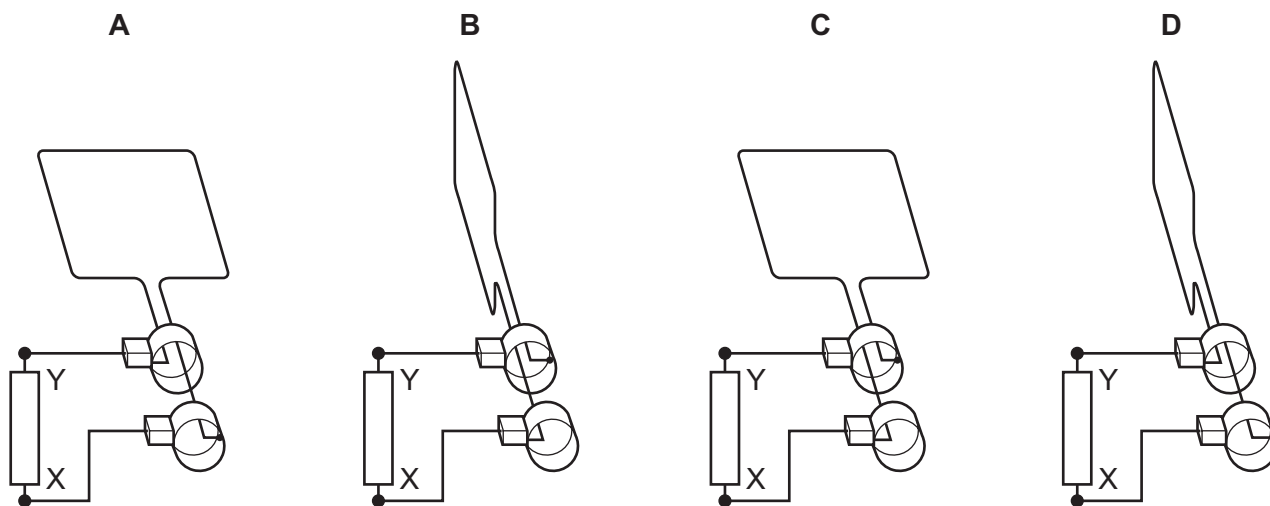
30 The diagram shows an a.c. generator.



The graph shows the potential difference (p.d.) between points X and Y plotted against time. A positive value of p.d. indicates that X is more positive than Y.



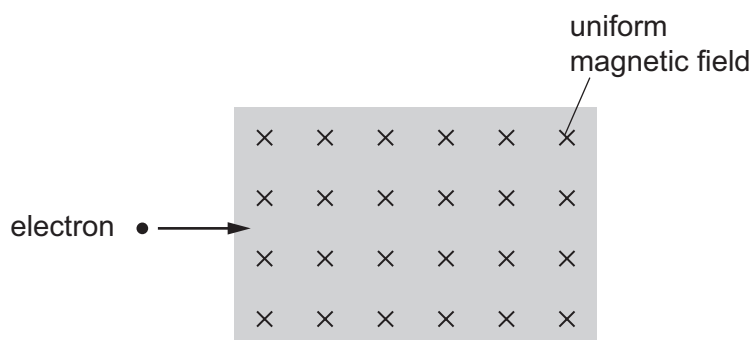
Which diagram shows the position of the coil at point P on the graph?



31 An electron moves into a uniform magnetic field.

The arrow shows the initial direction of motion of the electron.

The direction of the magnetic field is into the plane of the page.



In which direction does a force act on the electron when it enters the magnetic field?

- A** into the page
- B** out of the page
- C** towards the bottom of the page
- D** towards the top of the page

32 Which statement about a transformer is correct?

- A** There is an alternating current in the iron core from the primary coil to the secondary coil.
- B** An alternating current in the iron core induces an alternating current in the secondary coil.
- C** An alternating current in the primary coil induces a direct current in the secondary coil.
- D** An alternating current in the primary coil induces an alternating voltage across the secondary coil.

33 How are positive and negative ions formed from atoms?

	positive ion	negative ion
A	add positive charge to the nucleus	add an electron to the atom
B	add positive charge to the nucleus	remove positive charge from the nucleus
C	remove an electron from the atom	add an electron to the atom
D	remove an electron from the atom	remove positive charge from the nucleus

- 34 A nucleus of element X is represented as $^{56}_{26}\text{X}$.

Which is an isotope of element X?

- A $^{26}_{56}\text{X}$ B $^{54}_{26}\text{X}$ C $^{56}_{24}\text{X}$ D $^{54}_{28}\text{X}$

- 35 A radioactive source is placed near a detector connected to a counter.

210 counts are recorded by the counter in 3 minutes.

The background count rate is 20 counts per minute (cpm).

What is the corrected count rate for the radioactive source?

- A 50 cpm B 70 cpm C 190 cpm D 270 cpm

- 36 The background count rate measured by a radiation counter is 40 counts per minute (cpm).

With the counter close to a radioactive source, the counter reading is 960 cpm.

The half-life of the source is 20 minutes.

What is the counter reading one hour later?

- A 115 cpm B 120 cpm C 155 cpm D 160 cpm

- 37 Which planet in our Solar System is nearest to the Sun and what is the nature of the planet?

	planet	nature
A	Mercury	rocky
B	Mercury	gaseous
C	Venus	rocky
D	Venus	gaseous

- 38 A space station orbits the Earth at a distance of 7000 km from the Earth's centre. It makes 15 orbits in every 24-hour period.

What is the speed of the space station in its orbit?

- A 2900 km/h B 4400 km/h C 8800 km/h D 27 000 km/h

- 39** Which nuclear reaction powers a stable star?
- A** nuclear fission of nuclei producing hydrogen
 - B** nuclear fission of a uranium nucleus into a krypton nucleus and a barium nucleus
 - C** nuclear fusion of a krypton nucleus and a barium nucleus into a uranium nucleus
 - D** nuclear fusion of hydrogen nuclei producing helium
- 40** Which stages in the life cycle of a star are listed in the order that they occur?
- A** interstellar dust cloud → stable star → protostar
 - B** protostar → red giant → stable star
 - C** red giant → white dwarf → protostar
 - D** stable star → red giant → white dwarf

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Cambridge IGCSE™ (9–1)

PHYSICS (9–1)

0972/21

Paper 2 Multiple Choice (Extended)

May/June 2023

MARK SCHEME

Maximum Mark: 40

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

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Question	Answer	Marks
1	D	1
2	B	1
3	C	1
4	D	1
5	D	1
6	A	1
7	B	1
8	C	1
9	A	1
10	A	1
11	C	1
12	A	1
13	D	1
14	C	1
15	B	1
16	B	1
17	A	1
18	A	1
19	B	1
20	B	1
21	D	1
22	A	1
23	C	1
24	B	1
25	C	1
26	D	1
27	C	1
28	C	1

Question	Answer	Marks
29	B	1
30	C	1
31	C	1
32	D	1
33	C	1
34	B	1
35	A	1
36	C	1
37	A	1
38	D	1
39	D	1
40	D	1



Cambridge IGCSE™ (9–1)

PHYSICS

0972/22

Paper 2 Multiple Choice (Extended)

May/June 2023

45 minutes

You must answer on the multiple choice answer sheet.

You will need: Multiple choice answer sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)

INSTRUCTIONS

- There are **forty** questions on this paper. Answer **all** questions.
- For each question there are four possible answers **A**, **B**, **C** and **D**. Choose the **one** you consider correct and record your choice in soft pencil on the multiple choice answer sheet.
- Follow the instructions on the multiple choice answer sheet.
- Write in soft pencil.
- Write your name, centre number and candidate number on the multiple choice answer sheet in the spaces provided unless this has been done for you.
- Do **not** use correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.
- Take the weight of 1.0 kg to be 9.8 N (acceleration of free fall = 9.8 m/s^2).

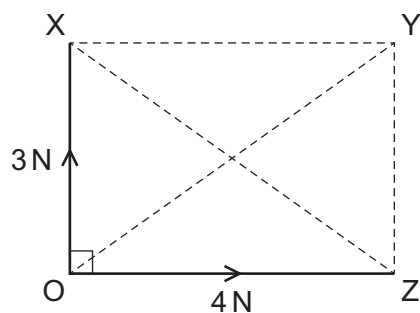
INFORMATION

- The total mark for this paper is 40.
- Each correct answer will score one mark.
- Any rough working should be done on this question paper.

This document has **16** pages. Any blank pages are indicated.



- 1 Forces of 3 N and 4 N act at right angles, as shown.



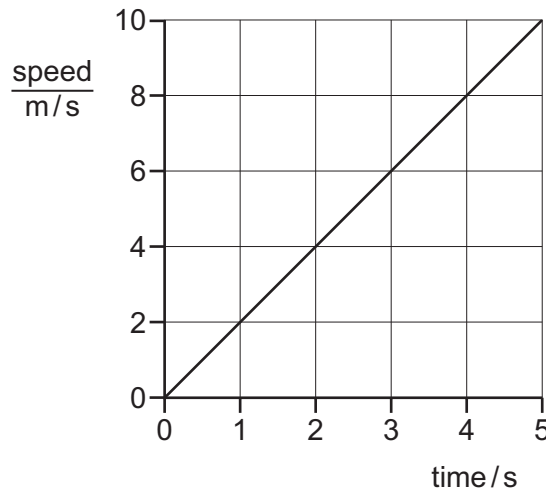
What is the resultant force?

- A** 1 N along XZ
B 5 N along XZ
C 5 N along OY
D 7 N along OY
- 2 A light ball is held at rest at the top of a tall cliff. It is released and falls through the air, eventually reaching its terminal velocity.

Which row describes the behaviour of the ball as it descends?

	the initial acceleration of the ball	the final acceleration of the ball
A	0	0
B	0	g
C	g	0
D	g	g

- 3 The graph represents the motion of a car.



How far has the car moved between 0 and 5 s?

- A** 2 m **B** 10 m **C** 25 m **D** 50 m

- 4 Which statement about mass or weight is **not** correct?

- A** Masses can be compared using a balance.
B Mass is a force.
C Weights can be compared using a balance.
D Weight is a force.

- 5 A sphere P, made of steel, has a weight of 10 N on Earth.

Another sphere Q, also made of steel, has a weight of 10 N on Mars.

The gravitational field strength on Earth is greater than the gravitational field strength on Mars.

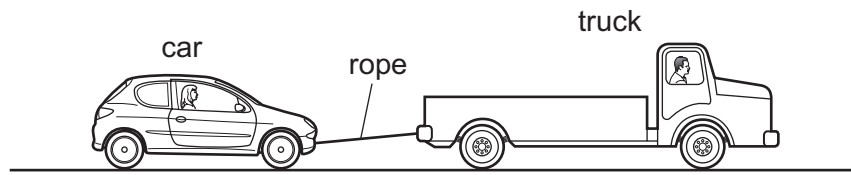
Which statement is correct?

- A** The mass of sphere P is the same as the mass of sphere Q.
B The mass of sphere P is less than the mass of sphere Q.
C On Mars, the weight of sphere P is the same as the weight of sphere Q.
D On Earth, the weight of sphere Q is less than 10 N.

- 6 Which two quantities must be known to determine the density of a material?

- A** mass and area
B mass and volume
C weight and area
D weight and volume

- 7 A truck is towing a car along a straight horizontal road at a constant speed.



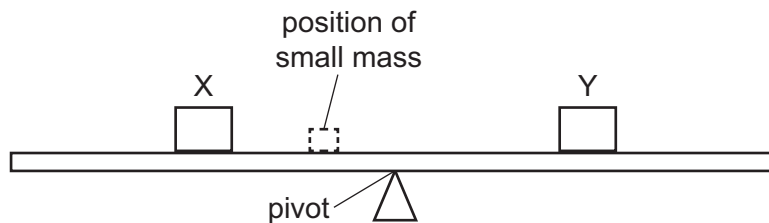
The rope breaks.

Which row gives the direction of the initial acceleration of the truck after the rope breaks and the reason for the acceleration?

	direction of acceleration of the truck	reason
A	left	the driving force is greater than the resistive forces on the truck
B	left	the driving force is smaller than the resistive forces on the truck
C	right	the driving force is greater than the resistive forces on the truck
D	right	the driving force is smaller than the resistive forces on the truck

- 8 A uniform beam is pivoted at the centre and two identical masses, X and Y, are placed so that the beam balances.

A smaller mass is then added at the position shown.



How can the masses be positioned so the beam balances again?

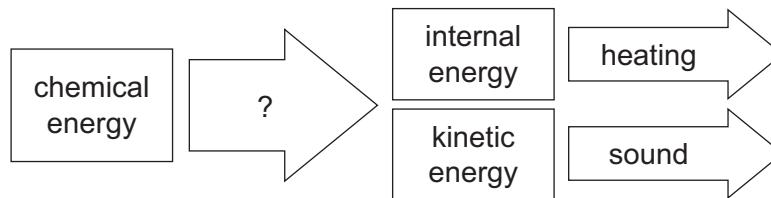
- A** Move X away from the pivot.
- B** Move X towards the pivot.
- C** Move Y towards the pivot.
- D** Move the small mass away from the pivot.

- 9 A resultant force of 2.0 N acts on an object of mass 3.0 kg for 6.0 s.

What is the change in velocity of the object?

- A 0.25 m/s B 1.0 m/s C 4.0 m/s D 36 m/s

- 10 The diagram shows the energy stores for a mobile (cell) phone and how the energy is transferred between stores.



What describes how the chemical energy is transferred?

- A electrical work done
 B mechanical work done
 C electromagnetic waves
 D sound waves
- 11 A wind turbine is 30% efficient and has an output of 2.5 MW of electrical power.
 What is the power input to the turbine?
- A 0.75 MW B 8.3 MW C 75 MW D 83 MW
- 12 Which two physical quantities must be used to calculate the power developed by a student running up a flight of steps?
- A force exerted and the vertical height of the steps only
 B force exerted and the time taken only
 C work done and the vertical height of the steps only
 D work done and the time taken only

- 13 The density of sea water is 1030 kg/m^3 .

The gravitational field strength on the Earth is 9.8 N/kg .

Atmospheric pressure is $101\,000 \text{ Pa}$.

At which depth in sea water is the total pressure due to the atmosphere and the water equal to $513\,000 \text{ Pa}$?

- A 40.8 m B 50.8 m C 400 m D 498 m

- 14** Four students describe the phrase ‘absolute zero’ during a lesson on the particle model.

Which student is correct?

- A** This is the lowest possible temperature.
- B** Particles in a solid start vibrating.
- C** Particles do not have any weight.
- D** Particles have the least gravitational potential energy.

- 15** Four students are asked to state and explain the relative magnitudes of the thermal expansion of solids and gases.

Which student is correct?

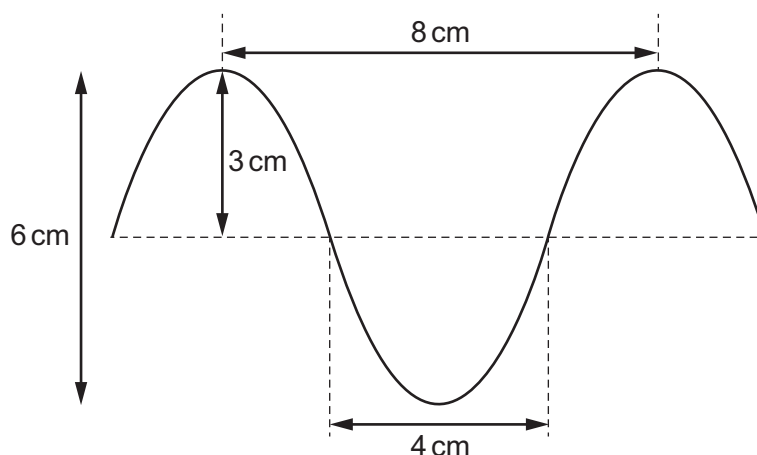
- A** Gases expand more than solids because the molecules in a gas are in random motion.
- B** Gases expand more than solids because the attractive forces between molecules are much weaker in gases.
- C** Solids expand more than gases because the molecules are closer together in solids.
- D** Solids expand more than gases because the molecules in a solid are in a regular pattern.

- 16** Four cups **A**, **B**, **C** and **D** contain hot coffee.

Which cup keeps the coffee warm the longest?

	the outside surface of the cup	the top of the cup
A	black	covered with a lid
B	black	no lid
C	white	covered with a lid
D	white	no lid

17 The diagram shows a wave.



What are the amplitude and the wavelength of this wave?

	amplitude / cm	wavelength / cm
A	3	4
B	3	8
C	6	4
D	6	8

18 Optical fibres are used to transmit digital signals using infrared radiation.

The average refractive index of the fibres is 1.50.

Which row describes a digital signal and gives the speed of infrared radiation in the fibres?

	digital signal	<u>speed of infrared radiation in the fibres</u> m/s
A	a signal that consists of only two values	2.0×10^8
B	a signal that consists of only two values	4.5×10^8
C	a signal that consists of a continuous range of values	2.0×10^8
D	a signal that consists of a continuous range of values	4.5×10^8

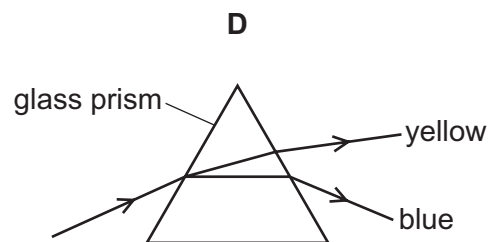
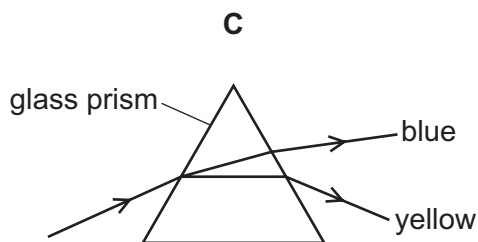
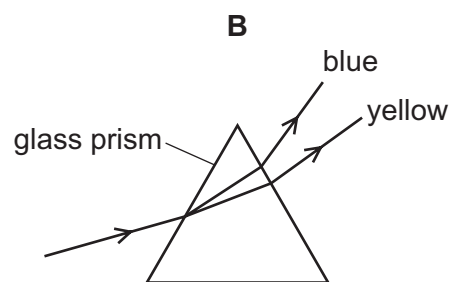
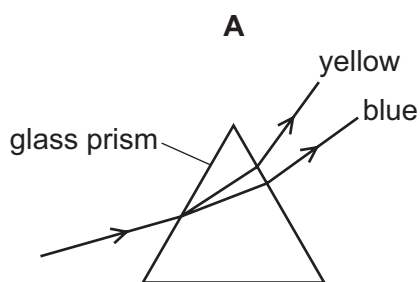
19 Which statement describes monochromatic light?

- A light of a single frequency
- B light transmitted by a transparent prism
- C visible light
- D white light

20 A beam of light consists of yellow and blue light.

The beam of light is incident on a glass prism.

Which diagram is correct?



21 A radio station broadcasts a signal with a frequency of 89 MHz.

What is the wavelength of this signal?

- A $3.7 \mu\text{m}$
- B 3.4 m
- C 3.7 km
- D 3.4 Mm

22 A boy shouts and hears the echo from a tall building 2.2 s later.

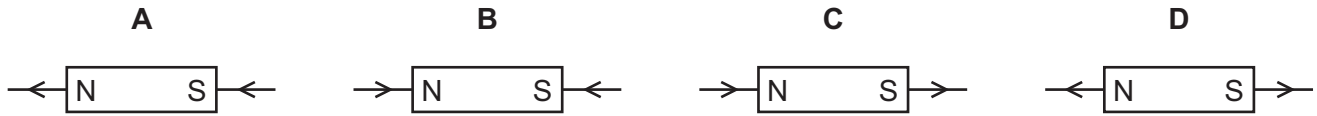
The speed of sound in air is 330 m/s.

How far away from the boy is the building?

- A 150 m
- B 300 m
- C 360 m
- D 730 m

23 The magnetic field of a bar magnet can be represented by magnetic field lines.

Which diagram shows two magnetic field lines correctly?



24 A plastic rod is rubbed with a dry cloth. The rod becomes positively charged.

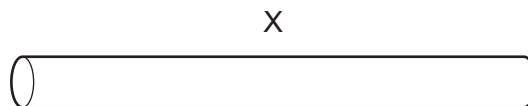
Why has the rod become positively charged?

- A** It has gained electrons.
- B** It has gained neutrons.
- C** It has lost electrons.
- D** It has lost neutrons.

25 Which statement about electric current in a conductor is correct?

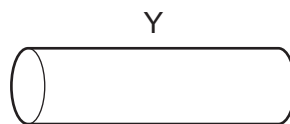
- A** In a d.c. circuit, the electric current gradually decreases along the conductor.
- B** In a d.c. circuit, the free electrons flow back and forth.
- C** In an a.c. circuit, the electric current remains exactly the same all the time.
- D** In an a.c. circuit, the flow of charge changes direction continually.

26 A piece of metal wire X with a uniform diameter has resistance R .



A second piece of wire Y is made of the same metal and has a uniform diameter.

Y has double the cross-sectional area of X and half the length of X.



What is the resistance of Y?

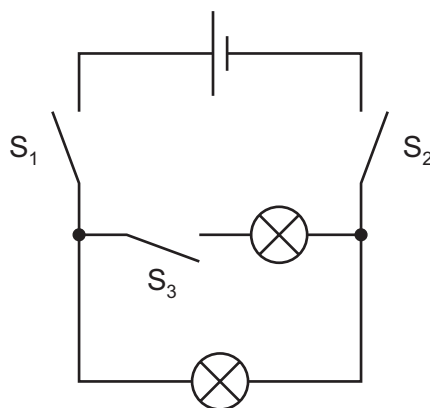
- A** $\frac{R}{4}$
- B** $\frac{R}{2}$
- C** R
- D** $4R$

- 27** An electric fire is connected to a 240 V supply and transfers energy at a rate of 1.0 kW.

How much charge passes through the fire in 1.0 h?

- A** 42 C **B** 250 C **C** 1.5×10^4 C **D** 2.4×10^5 C

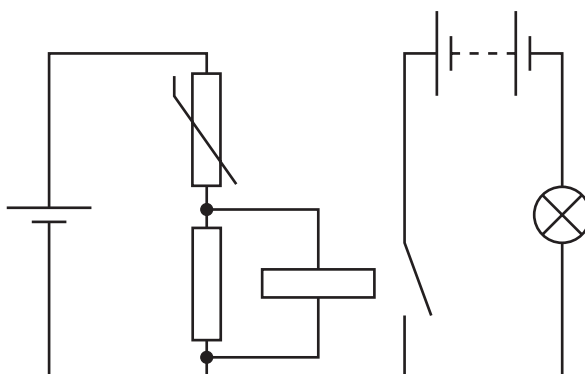
- 28** Two lamps are connected in parallel.



Which switches must be closed so that both lamps light?

- A** S_1 and S_2 only
B S_1 and S_3 only
C S_2 and S_3 only
D S_1 , S_2 and S_3

- 29** The diagram shows a circuit that switches on a lamp when there is a change in the environment.



Which change in the environment causes the lamp to be switched on?

- A** a decrease in light intensity
B a decrease in temperature
C an increase in light intensity
D an increase in temperature

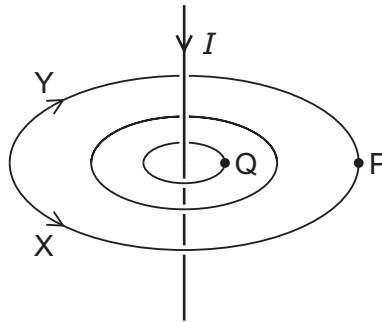
- 30 A step-down transformer is 100% efficient. It has an input voltage of 240 V a.c. and an output voltage of 60 V a.c.

The current in the primary coil is 0.50 A.

What is the current in the secondary coil?

- A** 0.13 A **B** 0.50 A **C** 2.0 A **D** 8.0 A

- 31 The diagram shows the pattern of the magnetic field due to the current I in a straight wire.



Which row is correct?

	direction of field	relative strength of field
A	X	greater at P than Q
B	Y	greater at P than Q
C	X	greater at Q than P
D	Y	greater at Q than P

- 32 In which device is the magnetic effect of a current **not** used?

- A** electromagnet
B loudspeaker
C potential divider
D relay

33 Fission and fusion are two types of nuclear process.

How does the total mass of the nuclides produced compare with the total mass of the original nuclide or nuclides in these nuclear processes?

	total mass of fission products compared to original nuclide	total mass of fusion products compared to original nuclides
A	same	same
B	more	less
C	less	more
D	less	less

34 The table shows the composition of three different nuclei.

nucleus	number of protons	number of neutrons
X	3	3
Y	3	4
Z	4	3

Which nuclei are isotopes of the same element?

A X, Y and Z **B** X and Y only **C** X and Z only **D** Y and Z only

35 Which change occurs in the nucleus of a radioactive atom during β -emission?

- A** A neutron transforms into a proton and an electron.
- B** A neutron transforms into a proton only.
- C** A proton transforms into a neutron and an electron.
- D** A proton transforms into a neutron only.

- 36 A radioactive isotope has a half-life of 8 days.

A detector close to a sample of this isotope gives a count rate of 200 counts per minute.

Without the source, the background count is 20 counts per minute.

What is the count rate due to the source after 8 days?

- A 80 counts per minute
 - B 90 counts per minute
 - C 100 counts per minute
 - D 110 counts per minute
- 37 Which statement about the Solar System is correct?
- A All the planets are rocky.
 - B Only the Earth has a moon.
 - C Pluto is a dwarf planet.
 - D There are many stars in the Solar System.

- 38 Comets are bodies which orbit the Sun in the Solar System.

What is the shape of the orbit and how is the Sun positioned within the orbit?

	shape of orbit	position of the Sun
A	circular	centre of orbit
B	circular	not at centre of orbit
C	elliptical	centre of orbit
D	elliptical	not at centre of orbit

- 39 Which nuclear reaction produces the release of energy to power a star?

- A nuclear fission of helium into hydrogen
- B nuclear fission of hydrogen into helium
- C nuclear fusion of helium into hydrogen
- D nuclear fusion of hydrogen into helium

40 What is the definition of the Hubble constant?

- A** the ratio of the speed at which a galaxy is receding from the Earth to its distance from the Earth
- B** the value of the change in wavelength of the galaxy's starlight due to redshift
- C** the constant used to represent the rate of expansion of the Universe in all directions
- D** the estimated constant equal to the age of the Universe

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Cambridge IGCSE™ (9–1)

PHYSICS (9–1)

0972/22

Paper 2 Multiple Choice (Extended)

May/June 2023

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4	B	1
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7	C	1
8	B	1
9	C	1
10	A	1
11	B	1
12	D	1
13	A	1
14	A	1
15	B	1
16	C	1
17	B	1
18	A	1
19	A	1
20	D	1
21	B	1
22	C	1
23	A	1
24	C	1
25	D	1
26	A	1
27	C	1
28	D	1

Question	Answer	Marks
29	D	1
30	C	1
31	D	1
32	C	1
33	D	1
34	B	1
35	A	1
36	B	1
37	C	1
38	D	1
39	D	1
40	A	1



Cambridge IGCSE™ (9–1)

PHYSICS

0972/21

Paper 2 Multiple Choice (Extended)

October/November 2023

45 minutes

You must answer on the multiple choice answer sheet.

You will need: Multiple choice answer sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)

INSTRUCTIONS

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- You may use a calculator.
- Take the weight of 1.0 kg to be 9.8 N (acceleration of free fall = 9.8 m/s^2).

INFORMATION

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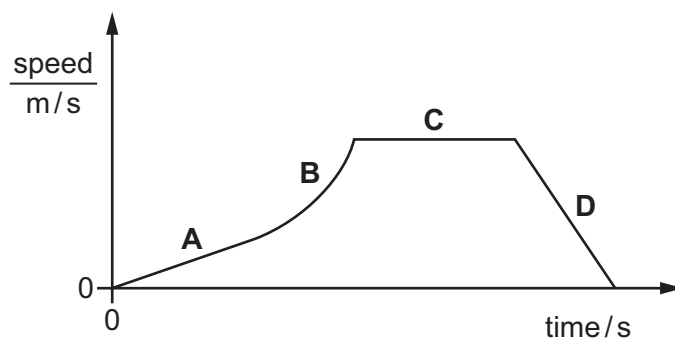


1 Which is a vector quantity?

- A density
- B mass
- C pressure
- D weight

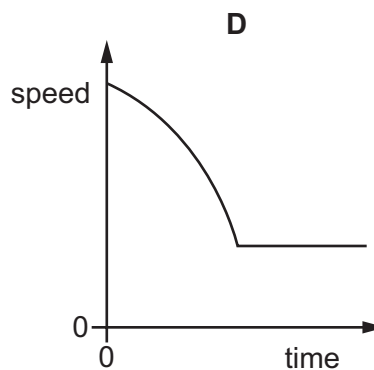
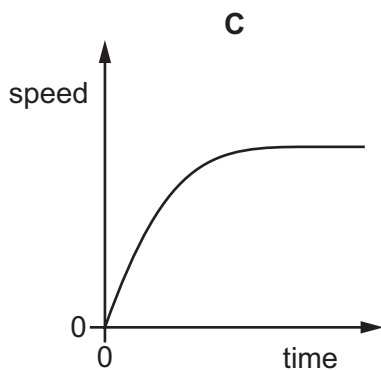
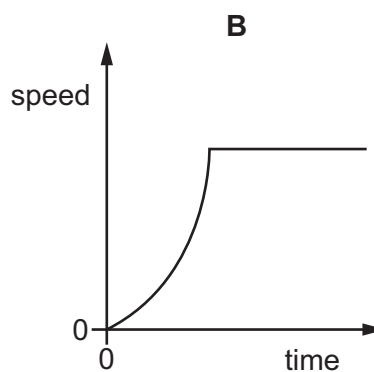
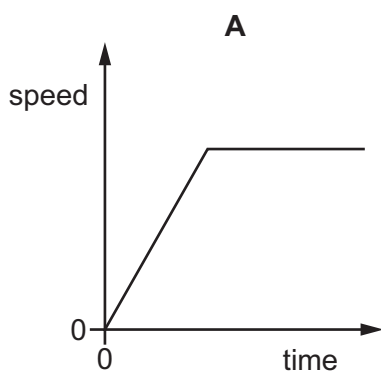
2 The graph shows the speed of a car travelling through a town.

Which section of the graph represents a period when the car is decelerating?



3 An object reaches terminal velocity after being dropped and falling through air.

Which graph shows how its speed varies with time?



- 4 The table shows the mass and volume of three different liquids, X, Y and Z.

liquid	mass / g	volume / cm ³
X	120	200
Y	80	67
Z	100	120

The liquids are placed in the same container. The liquids do not mix.

Which liquid is at the top of the container and which liquid is at the bottom?

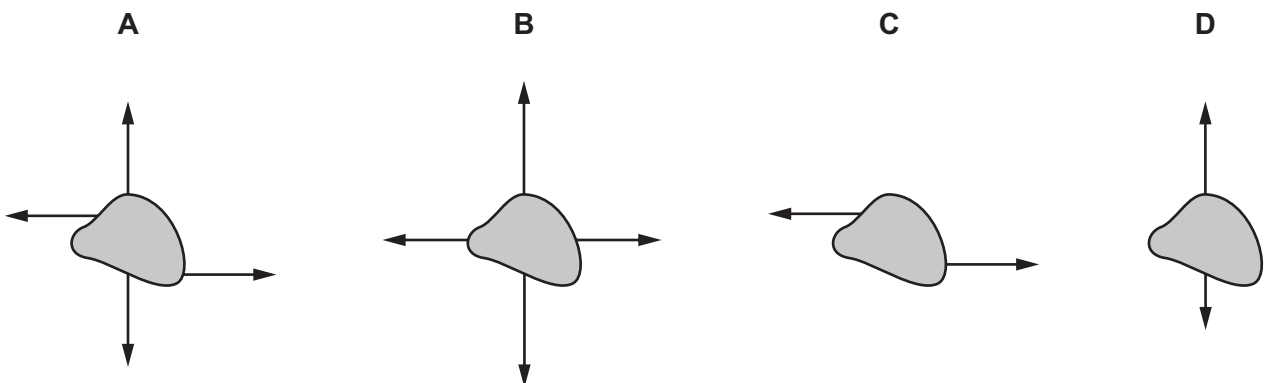
	liquid at top	liquid at bottom
A	X	Y
B	X	Z
C	Y	X
D	Y	Z

- 5 Which moving object has a resultant force acting on it?
- A** a diver rising vertically through water at constant speed
 - B** an aircraft circling an airport at constant speed
 - C** a train going up a straight incline at constant speed
 - D** a parachutist descending vertically at terminal velocity

- 6 Forces are applied to four identical objects.

The length of each arrow indicates the magnitude of the force.

Which object is in equilibrium?



- 7 A sphere X collides head on with a second identical sphere Y which is stationary.

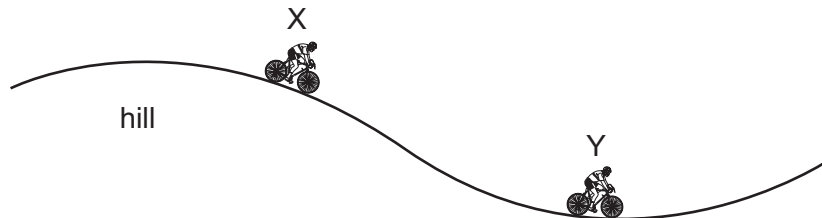
The mass of each sphere is 0.15 kg.

Sphere X is travelling at a velocity of 2.0 m/s before the collision and produces an impulse of 0.21 N s on sphere Y.

What is the velocity of sphere X after collision?

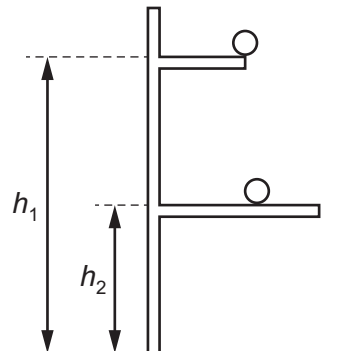
- A** 0.60 m/s in the opposite direction to Y
B 0.60 m/s in the same direction as Y
C 1.4 m/s in the opposite direction to Y
D 1.4 m/s in the same direction as Y
- 8 A cyclist travels down a hill from rest at point X without pedalling.

The cyclist applies his brakes and the cycle stops at point Y.



Which energy transfers have taken place between X and Y?

- A** gravitational potential → kinetic → internal (thermal)
B gravitational potential → internal (thermal) → kinetic
C kinetic → gravitational potential → internal (thermal)
D kinetic → internal (thermal) → gravitational potential
- 9 An object of mass m falls from a higher shelf to a lower shelf.



How much gravitational potential energy does the object lose?

- A** mgh_2 **B** $\left(\frac{m}{g}\right)h_2$ **C** $\left(\frac{m}{g}\right)(h_1 - h_2)$ **D** $mg(h_1 - h_2)$

- 10 A pump does 460 000 J of work to raise water to fill a tank. It takes 7 minutes to fill the tank.

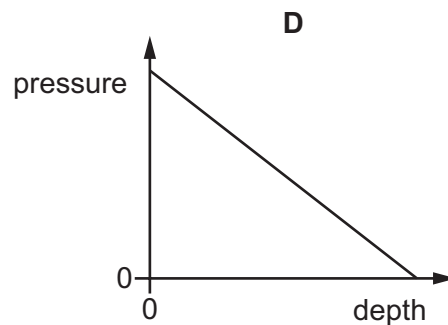
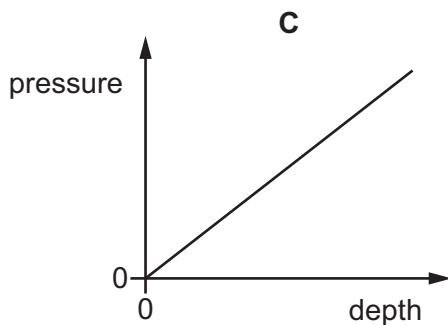
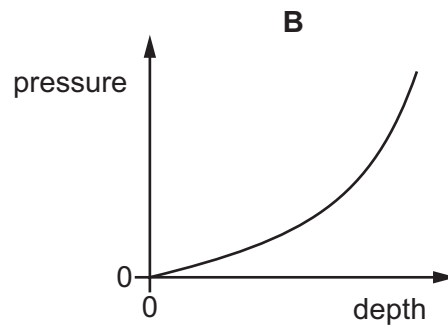
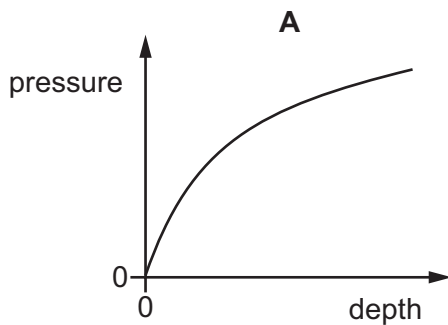
What is the power of the pump?

- A 1.1 kW B 66 kW C 3200 kW D 190 000 kW

- 11 The water in a swimming pool exerts a pressure at the bottom of the pool.

Which graph shows the relationship between the pressure exerted by the water and the depth of water in the pool?

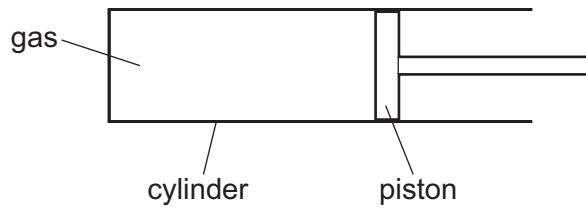
(Assume the density of water is constant.)



- 12 Why can a gas be compressed easily into a smaller volume?

- A The particles are far apart.
B The particles do not attract each other.
C The particles move randomly.
D The volume of each particle can be reduced.

- 13 A gas is contained in a cylinder by a movable piston.



The gas is heated so that it expands at **constant pressure**.

How is the force of each collision of a gas particle with the piston affected and how does the frequency of collisions between the gas particles and the piston change?

	force	frequency
A	increases	decreases
B	increases	increases
C	stays the same	decreases
D	stays the same	increases

- 14 On a warm day, a carton of fresh milk is covered with a wet cloth.

Why does this help to reduce the temperature of the milk?

- A** Some water evaporates from the cloth so the remaining water becomes cooler.
- B** The water has a very high specific heat capacity.
- C** The water insulates the milk from the warm air around it.
- D** Water is always colder than the air around it.

- 15 A chef heats some water in a pan on a hotplate.

The temperature of the water rises by 10°C in time t .

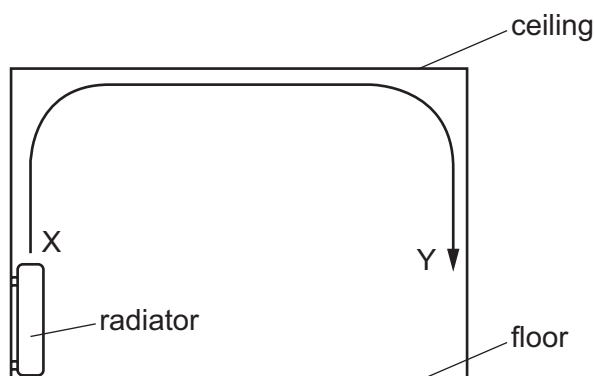
She then puts the same volume of oil in an identical pan on the same hotplate.

The specific heat capacity of water is 2.5 times that of oil and water is 1.1 times denser than oil.

What is the time for the temperature of the oil to rise by 10°C ?

- A** $0.36t$
- B** $0.44t$
- C** $2.3t$
- D** $2.8t$

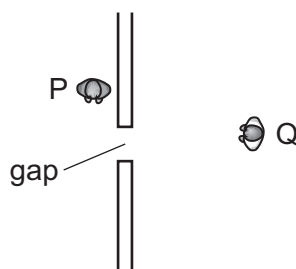
- 16 The diagram shows the view of a room heated by a radiator. The arrowed line from X to Y is the path of the convection current in the air.



Which row about the air temperature and the air density at X and at Y is correct?

	air temperature	air density
A	higher at X	higher at X
B	higher at X	higher at Y
C	higher at Y	higher at Y
D	higher at Y	higher at X

- 17 Two men, P and Q, stand close to a gap in a wall, as shown. Man P cannot see man Q but man P can hear man Q speaking.



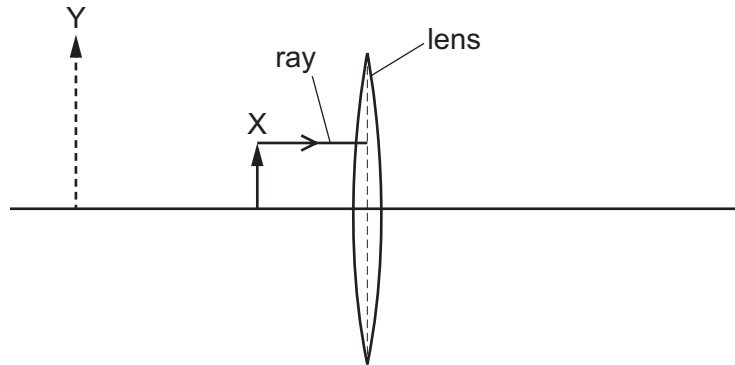
Which statement explains this?

- A** Light waves do not diffract at all because they are electromagnetic waves.
- B** Light waves have a range of frequencies but sound has just one frequency.
- C** Sound waves are of a higher frequency than light waves.
- D** Sound waves diffract a lot because their wavelength is a similar size to the width of the gap.

18 Which quantities relating to a wave on the surface of water can **both** be measured in metres?

- A amplitude and frequency
- B amplitude and wavelength
- C amplitude and wave speed
- D frequency and wavelength

19 The diagram shows part of a ray diagram that demonstrates the formation of a virtual image Y of object X by a converging lens.

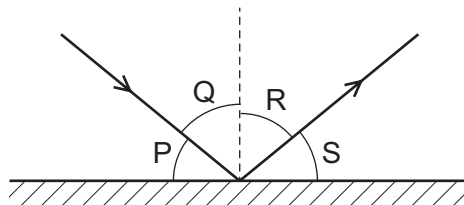


One ray of light from X is shown approaching the lens.

Which arrow shows the direction of this ray as it leaves the lens?



20 A ray of light is reflected by a plane mirror.



Which row shows the angle of incidence and the angle of reflection?

	angle of incidence	angle of reflection
A	P	Q
B	P	S
C	Q	R
D	R	S

- 21** A thin converging lens is used to produce a real image of an object.

Which statement about the real image is always correct?

- A** It is nearer to the lens than the object.
- B** It is on the opposite side of the lens to the object.
- C** It is the same size as the object.
- D** It is upright.

- 22** A radio transmitter broadcasts at a frequency of 200 kHz.

What is the wavelength of these radio waves?

- A** 6.7×10^{-4} m **B** 1.5 m **C** 1.5×10^3 m **D** 1.5×10^6 m

- 23** The element mercury exists as a solid, a liquid or a gas.

Which row gives a possible set of values of the speeds of sound through mercury?

	speed of sound in frozen mercury m/s	speed of sound in liquid mercury m/s	speed of sound in mercury vapour m/s
A	250	1500	2500
B	250	2500	1500
C	1500	250	2500
D	2500	1500	250

- 24** Which metal could be used for a permanent magnet and which metal could be used for the core of an electromagnet?

	permanent magnet	core of electromagnet
A	iron	copper
B	iron	steel
C	steel	copper
D	steel	iron

25 Which statement describes the direction of an electric field at a point?

- A the direction of the force on a negative charge
- B the direction of the force on an N pole
- C the direction of the force on a positive charge
- D the direction of the force on an S pole

26 Which statement about a voltmeter is correct?

- A It has a scale which is marked in amperes (A).
- B It must be connected in series in a circuit.
- C It measures potential difference (p.d.).
- D It must have three terminals.

27 A wire has a resistance of $8.0\ \Omega$.

A second wire of the same material has twice the length and twice the cross-sectional area.

What is the resistance of the second wire?

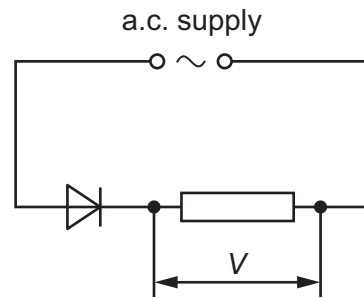
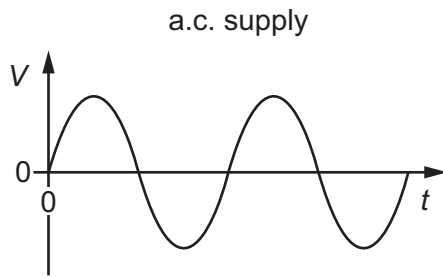
- A $4.0\ \Omega$ B $8.0\ \Omega$ C $16\ \Omega$ D $32\ \Omega$

28 A plastic rod is rubbed with a cloth. The rod becomes positively charged because of the movement of charged particles.

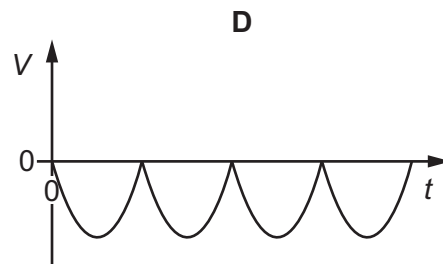
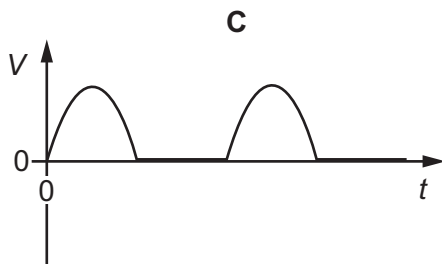
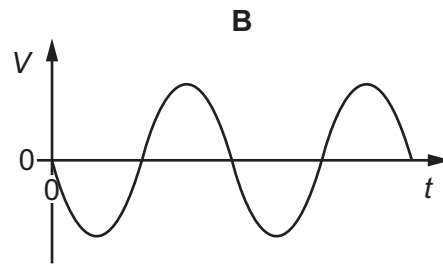
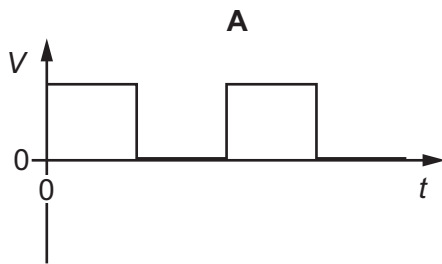
Which row gives the name of these charged particles and the direction in which they move?

	charged particles	direction of movement
A	electrons	from cloth to rod
B	electrons	from rod to cloth
C	protons	from cloth to rod
D	protons	from rod to cloth

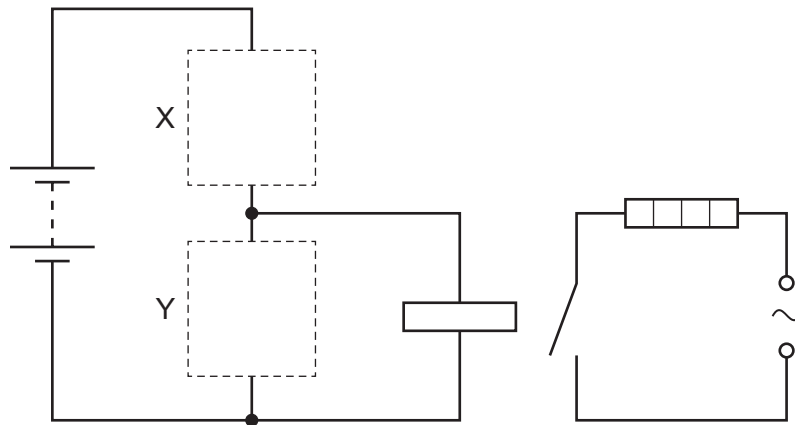
- 29 An alternating (a.c.) supply is connected to a diode and a resistor in series.



Which graph shows how the potential difference V across the resistor varies with time t ?



- 30 The diagram shows a circuit used to switch on a heater when the temperature drops below a certain value.



Which row shows the components that should be connected at X and at Y?

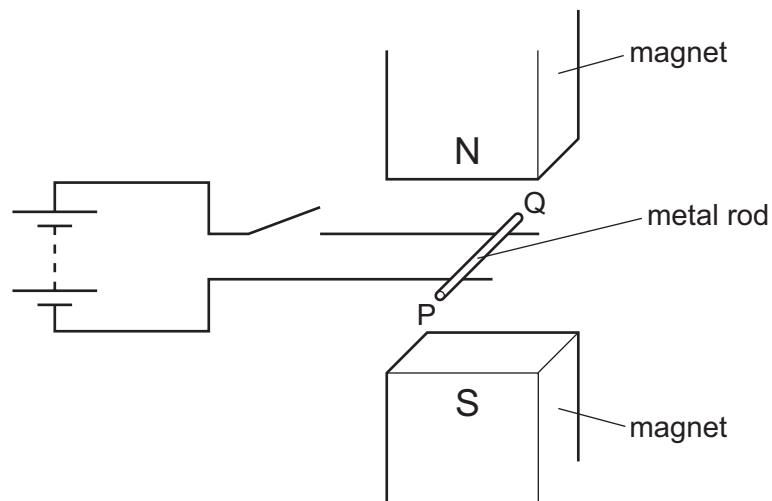
	X	Y
A		
B		
C		
D		

- 31 A current in a solenoid produces a uniform magnetic field inside the solenoid. The magnetic field direction is due east.

Which changes will produce a stronger magnetic field with a direction due west?

- A Use a smaller current and turn the solenoid through 180° .
- B Use a smaller current and reverse the current.
- C Use a larger current and turn the solenoid through 90° .
- D Use a larger current and reverse the current.

- 32** A metal rod PQ rests on two horizontal metal wires that are attached to a battery. The rod lies between the poles of a magnet.



When the switch is closed, the rod moves to the right.

What could be changed so that the rod moves to the left?

- A** Open the switch.
 - B** Reverse the battery terminals and exchange the poles of the magnet.
 - C** Reverse the battery terminals but without exchanging the poles of the magnet.
 - D** Turn the metal rod around (P and Q exchanged).
- 33** A transformer in a computer is used to transform the mains voltage of 240 V to 12 V.

The number of turns on the secondary coil is 2000.

Which statement about the transformer is correct?

- A** It is a step-down transformer and has 100 turns on its primary coil.
- B** It is a step-down transformer and has 40 000 turns on its primary coil.
- C** It is a step-up transformer and has 100 turns on its primary coil.
- D** It is a step-up transformer and has 40 000 turns on its primary coil.

34 In α -particle scattering, α -particles are incident on a thin metal foil.

Which row describes results from the experiment and a conclusion that the results lead to?

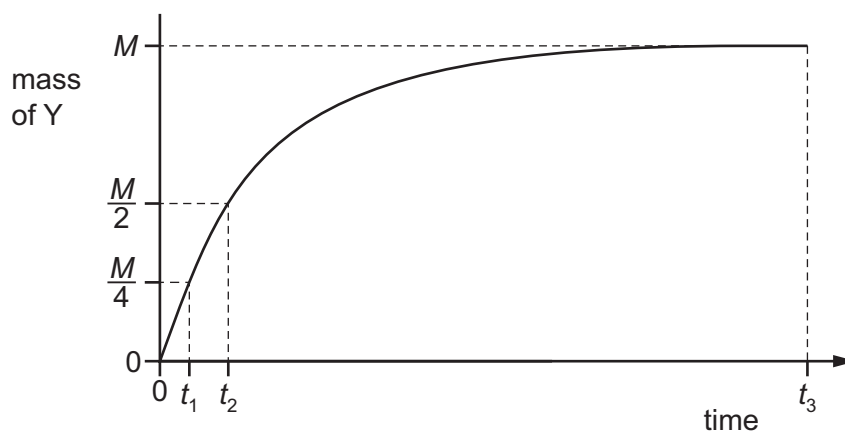
	results	conclusion
A	most of the α -particles pass straight through the foil	most of the atom is empty space
B	most of the α -particles pass straight through the foil	the nucleus is very large
C	very few of the α -particles pass straight through the foil	most of the atom is empty space
D	very few of the α -particles pass straight through the foil	the nucleus is very large

35 α -particles, β -particles and γ -rays are emitted by radioactive nuclei when they decay.

Which emissions can be deflected by an electric field?

- A** α -particles, β -particles and γ -rays
- B** α -particles and β -particles only
- C** β -particles and γ -rays only
- D** γ -rays and α -particles only

36 Radioisotope X decays to the stable isotope Y. The graph shows how the mass of Y present in a sample varies with time.



Which time interval gives the half-life of X?

- A** $t_2 - t_1$
- B** $t_3 - t_2$
- C** t_2
- D** $\frac{1}{2} t_3$

37 Which particle is absorbed by a nucleus to cause nuclear fission?

- A a neutron
- B a proton
- C an α -particle
- D a β -particle

38 It is summer in the northern hemisphere of the Earth in June.

Which statement explains why?

- A The Earth is closer to the Sun in June.
- B The Earth spins on its axis in the opposite direction to that in which it rotates around the Sun.
- C The Moon is full in June.
- D The north pole of the axis of the Earth's rotation is tilted towards the Sun in June.

39 Which statement about the Sun is correct?

- A The Sun is a dwarf star consisting mostly of hydrogen and oxygen.
- B The Sun is a giant star consisting mostly of helium and carbon dioxide.
- C The Sun is a medium-sized star consisting mostly of hydrogen and helium.
- D The Sun is a medium-sized star consisting mostly of nitrogen and oxygen.

40 The table lists some information about some stars.

Which star will eventually explode as a supernova?

	name of star	type of star	temperature / °C
A	Aldebaran	red giant	3 700
B	Betelgeuse	red supergiant	3 300
C	Geminga	neutron star	520 000
D	Sirius B	white dwarf	25 000

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Cambridge IGCSE™ (9–1)

PHYSICS

0972/21

Paper 2 Multiple Choice (Extended)

October/November 2023

MARK SCHEME

Maximum Mark: 40

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the October/November 2023 series for most Cambridge IGCSE, Cambridge International A and AS Level components, and some Cambridge O Level components.

Question	Answer	Marks
1	D	1
2	D	1
3	C	1
4	A	1
5	B	1
6	B	1
7	B	1
8	A	1
9	D	1
10	A	1
11	C	1
12	A	1
13	A	1
14	A	1
15	A	1
16	B	1
17	D	1
18	B	1
19	A	1
20	C	1
21	B	1
22	C	1
23	D	1
24	D	1
25	C	1
26	C	1
27	B	1
28	B	1

Question	Answer	Marks
29	C	1
30	B	1
31	D	1
32	C	1
33	B	1
34	A	1
35	B	1
36	C	1
37	A	1
38	D	1
39	C	1
40	B	1



Cambridge IGCSE™ (9–1)

PHYSICS

Paper 2 Multiple Choice (Extended)

0972/21

May/June 2024

45 minutes

You must answer on the multiple choice answer sheet.

You will need: Multiple choice answer sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)

INSTRUCTIONS

- There are **forty** questions on this paper. Answer **all** questions.
- For each question there are four possible answers **A**, **B**, **C** and **D**. Choose the **one** you consider correct and record your choice in soft pencil on the multiple choice answer sheet.
- Follow the instructions on the multiple choice answer sheet.
- Write in soft pencil.
- Write your name, centre number and candidate number on the multiple choice answer sheet in the spaces provided unless this has been done for you.
- Do **not** use correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.
- Take the weight of 1.0 kg to be 9.8 N (acceleration of free fall = 9.8 m/s^2).

INFORMATION

- The total mark for this paper is 40.
- Each correct answer will score one mark.
- Any rough working should be done on this question paper.

This document has **16** pages. Any blank pages are indicated.



- 1 In which row are quantities correctly categorised into scalar quantities and vector quantities?

	scalar quantities	vector quantities
A	mass and energy	weight and acceleration
B	gravitational field strength and time	force and electric field strength
C	speed and momentum	distance and force
D	distance and energy	velocity and temperature

- 2 Which name is given to the quantity of matter in an object?

- A** density
- B** mass
- C** volume
- D** weight

- 3 A body is moved from place X to place Y where the gravitational field strength is different.

What happens to its mass and to its weight due to the move?

	mass	weight
A	changes	changes
B	changes	stays the same
C	stays the same	changes
D	stays the same	stays the same

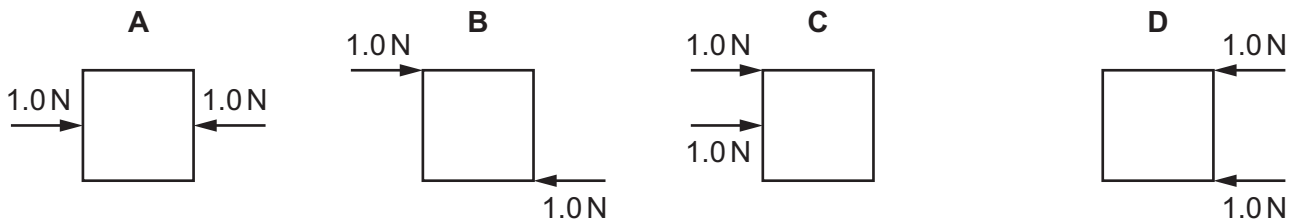
- 4 The mass of an object is 2.5 kg. The volume of the object is 480 cm³.

What is the density of the object?

- A** $5.2 \times 10^{-3} \text{ g/cm}^3$
- B** 5.2 g/cm^3
- C** 190 g/cm^3
- D** $1.2 \times 10^3 \text{ g/cm}^3$

- 5 Four objects each have two forces acting on them.

Which object is in equilibrium?



- 6 A ball of mass m falls vertically and hits a hard surface.

Its speed on hitting the surface is v_1 .

It rebounds vertically upwards with speed v_2 .

What is the change in momentum of the ball?

- A** mv_1 **B** mv_2 **C** $m(v_1 + v_2)$ **D** $m(v_2 - v_1)$

- 7 A stone is dropped from a tall tower and falls a distance of 50 m to the ground.

The stone has a mass of 3.0 kg.

At which speed does the stone hit the ground?

- A** 17 m/s **B** 31 m/s **C** 54 m/s **D** 150 m/s

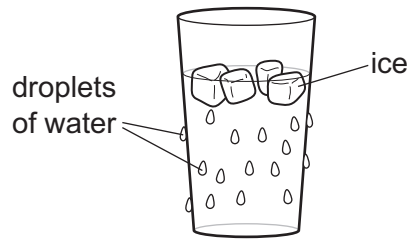
- 8 A coal-fired power station generates electricity. Coal is burned and the energy released is used to boil water. The steam from the water makes the generator move and this produces electricity.

Which row gives the name of the energy store in the coal and the energy store of the moving generator?

	coal	generator
A	chemical	hydroelectric
B	chemical	kinetic
C	geothermal	hydroelectric
D	geothermal	kinetic

- 9 A man puts some ice into a glass of water on a warm day.

After a short time, he notices that the ice disappears and that water droplets appear on the outside of the glass.

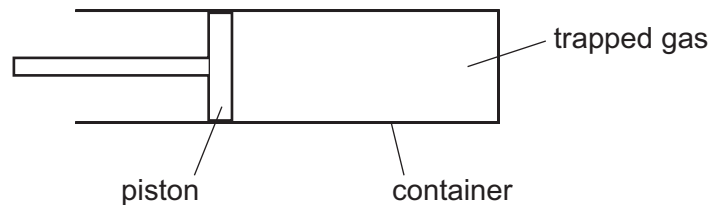


Which two changes of state are taking place?

- A condensation and freezing
 - B condensation and melting
 - C boiling and melting
 - D freezing and evaporation
- 10 Gas of volume 200 cm^3 is trapped inside a container by a piston.

The piston is pushed to the right and the volume of the gas decreases to 100 cm^3 .

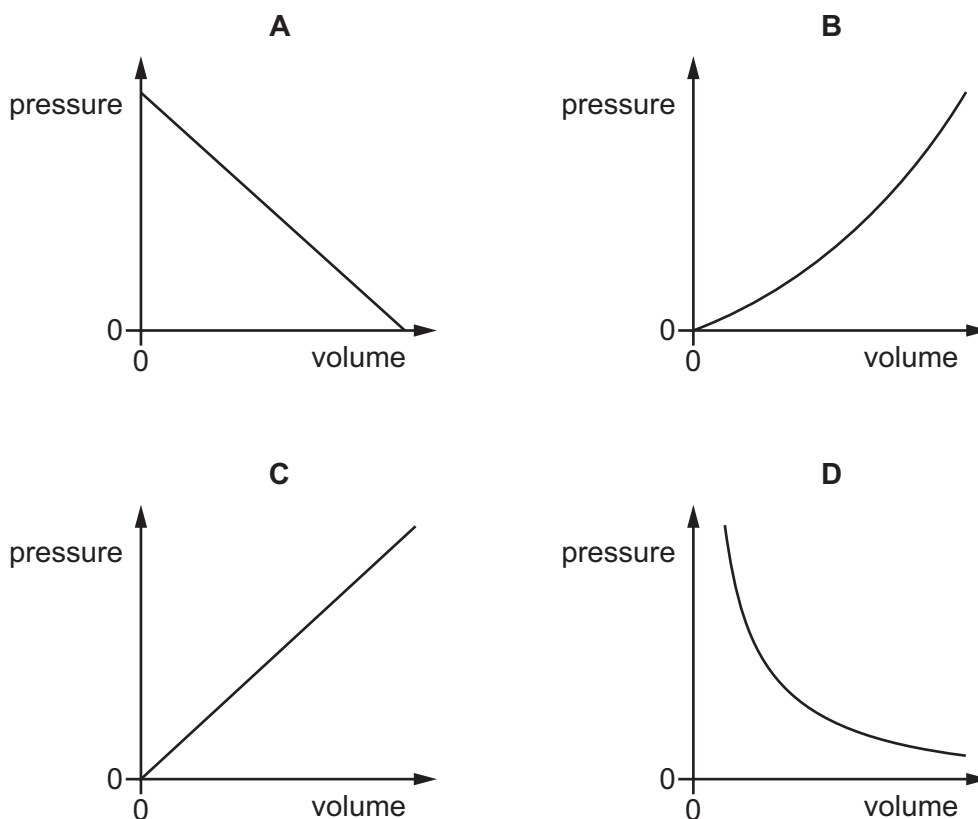
The temperature of the gas remains constant.



Which row states the effect that this has on the kinetic energy of the gas particles and the force per unit area exerted by the particles colliding with the inside walls of the container?

	kinetic energy	force per unit area
A	doubles	doubles
B	doubles	stays the same
C	stays the same	doubles
D	stays the same	stays the same

- 11 Which graph shows the relationship between the pressure and volume of a fixed mass of gas at constant temperature?



- 12 During evaporation of a liquid, the most energetic particles escape. The temperature of the remaining liquid changes.

Which row identifies where these particles escape from and describes the temperature change?

	place from which the particles escape	temperature of remaining liquid
A	body of the liquid	decreases
B	body of the liquid	increases
C	surface of the liquid	decreases
D	surface of the liquid	increases

- 13 Wet clothes are put out on a line to allow the water in the clothes to evaporate.

Which type of weather would cause the water to evaporate most quickly?

- A** a cold day with no wind
- B** a cold day with wind
- C** a hot day with no wind
- D** a hot day with wind

14 Due to an issue with question 14, the question has been removed from the question paper.

15 A solar water heater is designed to absorb energy from sunlight.

Which surface texture and colour would be best for the solar water heater?

- A** dull black
- B** shiny black
- C** dull white
- D** shiny white

16 Which statement about waves is correct?

- A** Waves do not transfer either energy or matter.
- B** Waves transfer both energy and matter.
- C** Waves transfer energy without transferring matter.
- D** Waves transfer matter without transferring energy.

17 The angle of incidence of a ray of light incident on a plane mirror is gradually increased.

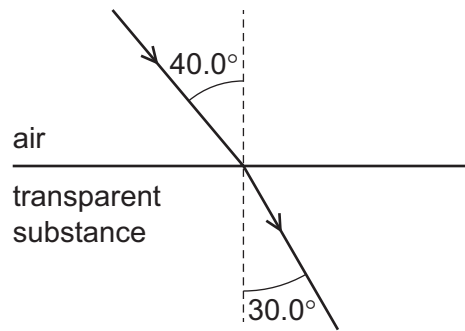
To the nearest degree, what is the maximum possible angle between the incident and reflected rays?

- A** 0°
- B** 45°
- C** 90°
- D** 180°

18 Which conditions are necessary for light to be totally internally reflected?

	incident light	angle of incidence
A	is in the less dense medium	less than the critical angle
B	is in the less dense medium	greater than the critical angle
C	is in the more dense medium	less than the critical angle
D	is in the more dense medium	greater than the critical angle

- 19 The diagram shows a ray of light passing from air into a transparent substance.



What is the refractive index of the transparent substance?

- A** 1.33 **B** 1.29 **C** 0.778 **D** 0.750
- 20 The diagram shows a ray of light in an optical fibre.
-
- The diagram shows a wavy line representing the optical fibre. A ray of light enters from the left and reflects off the boundaries of the fibre multiple times, staying within the fibre. An arrow at the end of the ray indicates the direction of light travel.
- Which statement correctly explains the condition for the maximum transmission of light by the optical fibre?
- A** The glass must slow the light as little as possible to make the critical angle for the fibre as large as possible.
- B** The glass must slow the light as little as possible to make the critical angle for the fibre as small as possible.
- C** The glass must slow the light as much as possible to make the critical angle for the fibre as large as possible.
- D** The glass must slow the light as much as possible to make the critical angle for the fibre as small as possible.
- 21 Which region of the electromagnetic spectrum is used for detecting fake bank notes?
- A** radio
- B** microwaves
- C** ultraviolet
- D** X-rays
- 22 A sound wave travels at 330 m/s . The distance between the centre of a compression and the centre of the nearest rarefaction in the sound wave is 2.5 cm .

What is the frequency of the sound wave?

- A** 66 Hz **B** 130 Hz **C** 6600 Hz **D** 13 000 Hz

23 Which waves are used in the medical scanning of soft tissue?

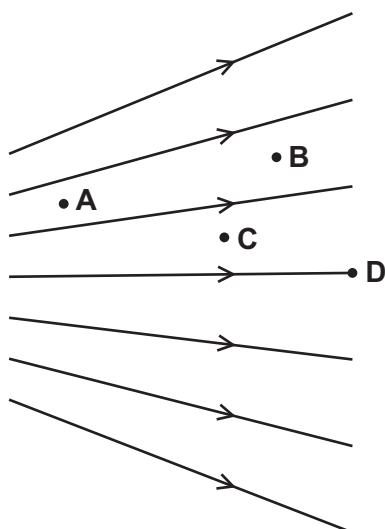
- A** gamma rays
- B** infrared
- C** microwaves
- D** ultrasound

24 Which row describes suitable materials for use in a temporary magnet and in a permanent magnet?

	temporary magnet	permanent magnet
A	soft iron	soft iron
B	soft iron	steel
C	steel	soft iron
D	steel	steel

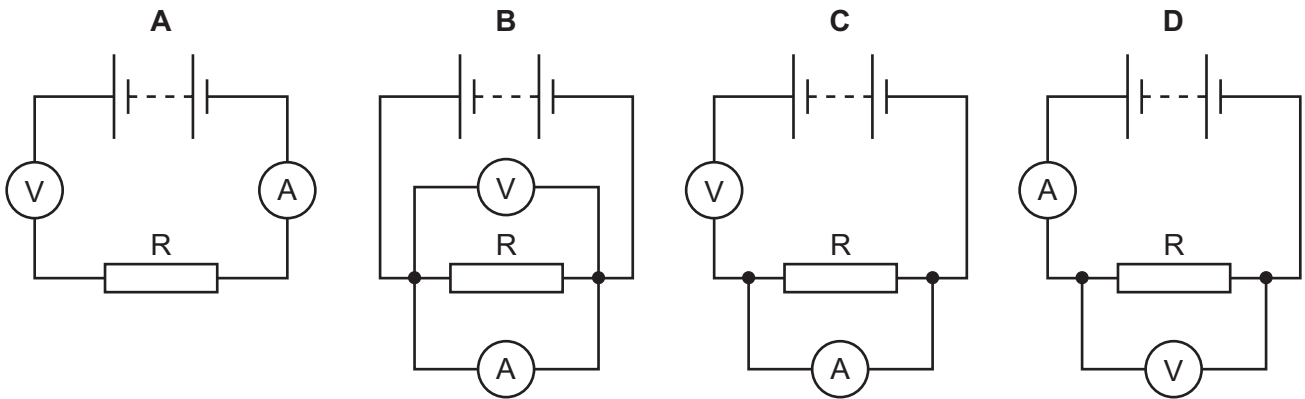
25 A magnetic field is represented in the diagram by magnetic field lines.

At which point is the magnetic field strongest?

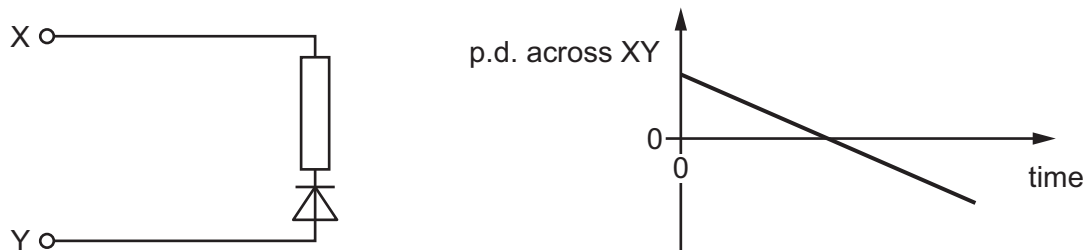


- 26** A student determines the resistance of resistor R . She uses a circuit including a voltmeter and an ammeter.

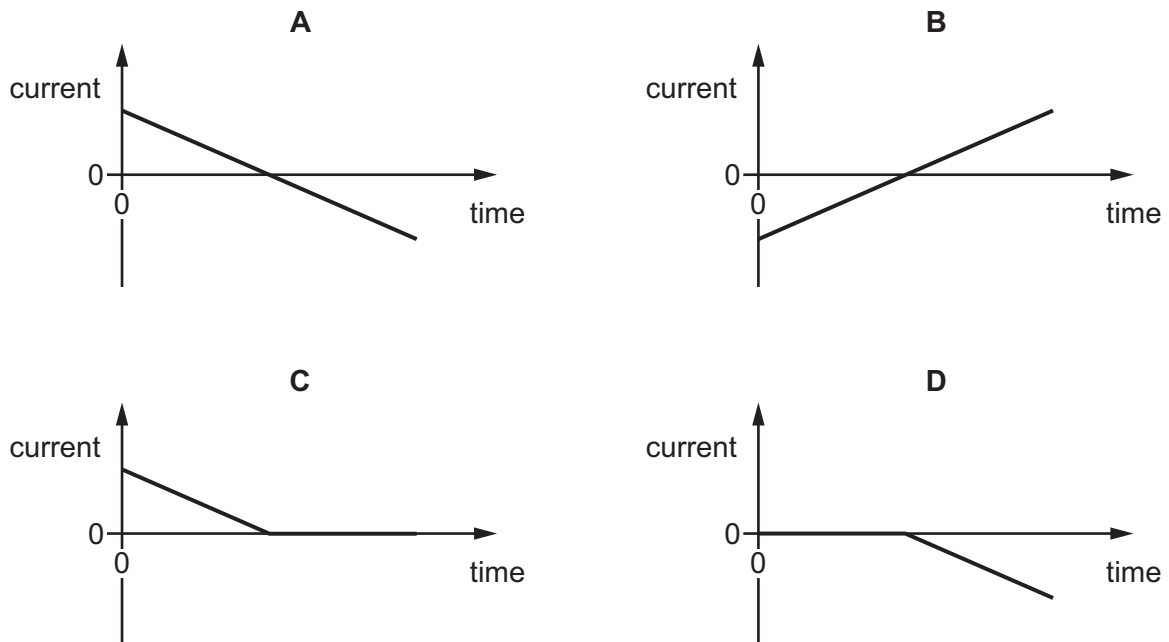
Which circuit does she use?



- 27** A diode and a resistor are connected across a variable d.c. supply. Terminal X is initially positive and at a maximum value. The potential difference (p.d.) across XY is adjusted so that it changes, as shown by the graph.



Which graph shows how the current in the circuit (from X to Y) varies during the same time interval?

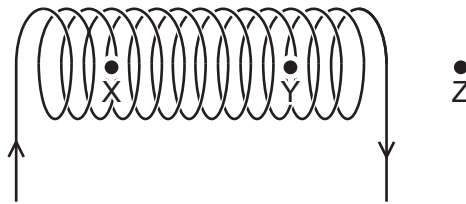


- 28** When a conductor in a complete circuit cuts across a magnetic field, a current is induced in the conductor.

Which statement about the induced current is correct?

- A** The induced current is in the same direction as the motion of the conductor.
- B** The induced current is in the opposite direction to the motion of the conductor.
- C** The direction of the induced current is in the same direction as the magnetic field.
- D** The direction of the induced current opposes the change causing it.

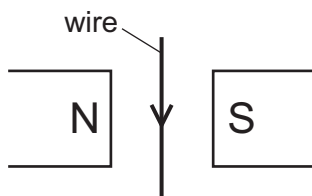
- 29** The diagram shows a solenoid carrying an electric current.



Which row compares the strength of the magnetic field due to the solenoid at points Y and Z with the strength of the magnetic field at point X?

	magnetic field strength at Y	magnetic field strength at Z
A	equal to X	less than X
B	equal to X	greater than X
C	greater than X	less than X
D	greater than X	greater than X

- 30** A conducting wire is placed between the poles of a magnet. When an electric current in the wire is in the direction shown, then the force on the wire acts out of the page.



Three statements of different conditions and how the wire is affected are given.

- 1 When the current is towards the top of the page and the direction of the magnetic field is unchanged, the force produced acts into the page.
- 2 When the current is towards the bottom of the page and the magnetic field is reversed, the force produced acts into the page.
- 3 When the current in the wire is alternating, the wire vibrates into and out of the page.

Which statements are correct?

- A** 1, 2 and 3 **B** 1 and 2 only **C** 1 and 3 only **D** 2 and 3 only

- 31** Which types of electric current are in the primary coil and the secondary coil of a step-up transformer?

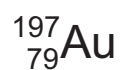
	primary coil	secondary coil
A	a.c.	a.c.
B	a.c.	d.c.
C	d.c.	a.c.
D	d.c.	d.c.

- 32** A scientist fires alpha-particles at a very thin sheet of gold and detects the particles that pass through.

Which statement about the results of the scattering experiment is correct?

- A** Alpha-particles are attracted to the nucleus of the atoms in the metal sheet.
- B** Half the mass of the atom is concentrated in the nucleus.
- C** Most of the alpha-particles are deflected, showing that the nucleus is very dense.
- D** Only a small number of alpha-particles are deflected, but some of these are deflected through large angles.

- 33 The notation represents the nucleus of a gold atom.



The relative charge on a proton is +1.

What is the relative charge on the gold nucleus?

- A** +79 **B** +118 **C** +197 **D** +276

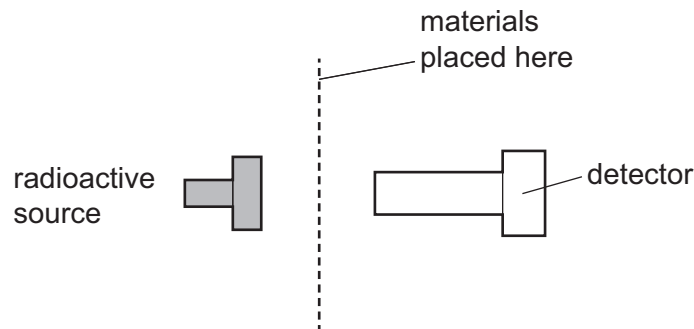
- 34 An isotope of hydrogen has the nuclide notation ^2_1H .

How many neutrons are in the nucleus of this isotope and what is the relative charge on the nucleus?

	number of neutrons	relative charge
A	1	+1
B	1	+2
C	2	+1
D	2	+2

- 35** A radioactive source is placed near a detector.

The radiation arriving at the detector from the source is measured for 10 minutes with different materials placed between the source and the detector.



material between source and detector	radiation detected / counts
none	5626
sheet of paper	5629
thick sheet of aluminium	2226
thick sheet of lead	255

Which types of radiation are emitted by the source?

- A** α -particles and γ -rays
 - B** α -particles only
 - C** β -particles and γ -rays
 - D** β -particles only
- 36** The reading on a detector placed near a radioactive material is 536 counts per second.

The background count rate is 44 counts per second.

The half-life of the radioactive material is 34 hours.

What is the reading on the detector after 68 hours?

- A** 44 counts per second
- B** 123 counts per second
- C** 134 counts per second
- D** 167 counts per second

37 A planet in the Solar System is at the point in its orbit where it is closest to the Sun.

Which row is correct?

	orbital speed	energy in gravitational potential store
A	at its maximum	at its maximum
B	at its maximum	at its minimum
C	at its minimum	at its maximum
D	at its minimum	at its minimum

38 What is the main process that powers the Sun?

- A** burning of helium and oxygen
- B** burning of hydrogen and oxygen
- C** nuclear fusion of hydrogen to form helium
- D** nuclear fission of helium to form hydrogen

39 A galaxy is 3.0×10^{20} km from the Earth.

At which speed is the galaxy moving away from the Earth?

- A** 660 km/s **B** 6600 km/s **C** 660 m/s **D** 6600 m/s

40 An astronomer observes a distant galaxy.

The table shows how the distance and the speed of recession of the galaxy are determined.

Which row is correct?

	distance	speed of recession
A	brightness of a supernova in the galaxy	change in wavelength of starlight from the galaxy
B	brightness of a supernova in the galaxy	brightness of the galaxy
C	brightness of the galaxy	change in wavelength of starlight from the galaxy
D	brightness of the galaxy	brightness of a supernova

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Cambridge IGCSE™ (9–1)

PHYSICS (9–1)**0972/21**

Paper 2 Multiple Choice (Extended)

May/June 2024**MARK SCHEME**Maximum Mark: 40

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the May/June 2024 series for most Cambridge IGCSE, Cambridge International A and AS Level and Cambridge Pre-U components, and some Cambridge O Level components.

This document consists of **3** printed pages.

Question	Answer	Marks
1	A	1
2	B	1
3	C	1
4	B	1
5	A	1
6	C	1
7	B	1
8	B	1
9	B	1
10	C	1
11	D	1
12	C	1
13	D	1
14	Question Discounted	1
15	A	1
16	C	1
17	D	1
18	D	1
19	B	1
20	D	1
21	C	1
22	C	1
23	D	1
24	B	1
25	A	1
26	D	1
27	D	1
28	D	1

Question	Answer	Marks
29	A	1
30	A	1
31	A	1
32	D	1
33	A	1
34	A	1
35	C	1
36	D	1
37	B	1
38	C	1
39	A	1
40	A	1



Cambridge IGCSE™ (9–1)

PHYSICS

0972/22

Paper 2 Multiple Choice (Extended)

May/June 2024

45 minutes

You must answer on the multiple choice answer sheet.

You will need: Multiple choice answer sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)

INSTRUCTIONS

- There are **forty** questions on this paper. Answer **all** questions.
- For each question there are four possible answers **A**, **B**, **C** and **D**. Choose the **one** you consider correct and record your choice in soft pencil on the multiple choice answer sheet.
- Follow the instructions on the multiple choice answer sheet.
- Write in soft pencil.
- Write your name, centre number and candidate number on the multiple choice answer sheet in the spaces provided unless this has been done for you.
- Do **not** use correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.
- Take the weight of 1.0 kg to be 9.8 N (acceleration of free fall = 9.8 m/s^2).

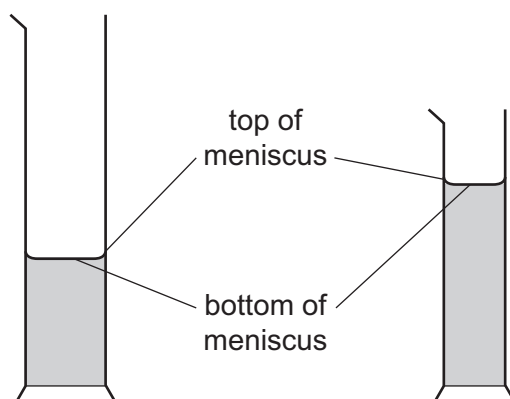
INFORMATION

- The total mark for this paper is 40.
- Each correct answer will score one mark.
- Any rough working should be done on this question paper.

This document has **16** pages. Any blank pages are indicated.



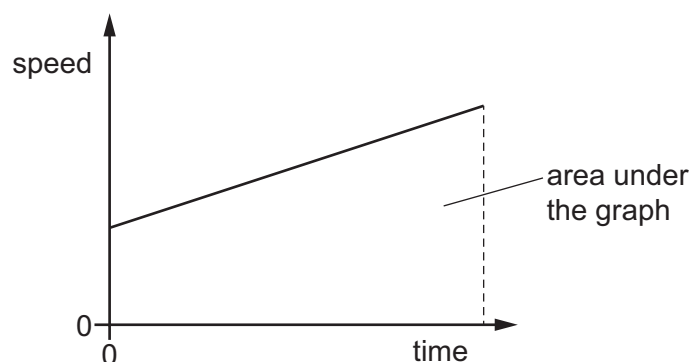
- 1 A student wishes to measure accurately the volume of approximately 40 cm^3 of water. She has two measuring cylinders, a larger one that can hold 100 cm^3 , and a smaller one that can hold 50 cm^3 . The water forms a meniscus where it touches the glass.



Which cylinder and which water level does the student use to ensure an accurate result?

	cylinder	water level
A	larger one	bottom of meniscus
B	larger one	top of meniscus
C	smaller one	bottom of meniscus
D	smaller one	top of meniscus

- 2 The motion of an object is represented by the speed–time graph shown.



Which quantity is equal to the area under the graph?

- A** acceleration
- B** average speed
- C** distance travelled
- D** kinetic energy

- 3 An astronaut of mass 80 kg is standing on a planet with gravitational field strength $g = 3.8 \text{ N/kg}$.

What is the weight of the astronaut on this planet?

- A 780 N B 300 N C 210 N D 21 N

- 4 A sealed container of volume 2000 cm^3 contains air at high pressure.

The container is placed on a top-pan balance.

The balance reads 200.00 g .

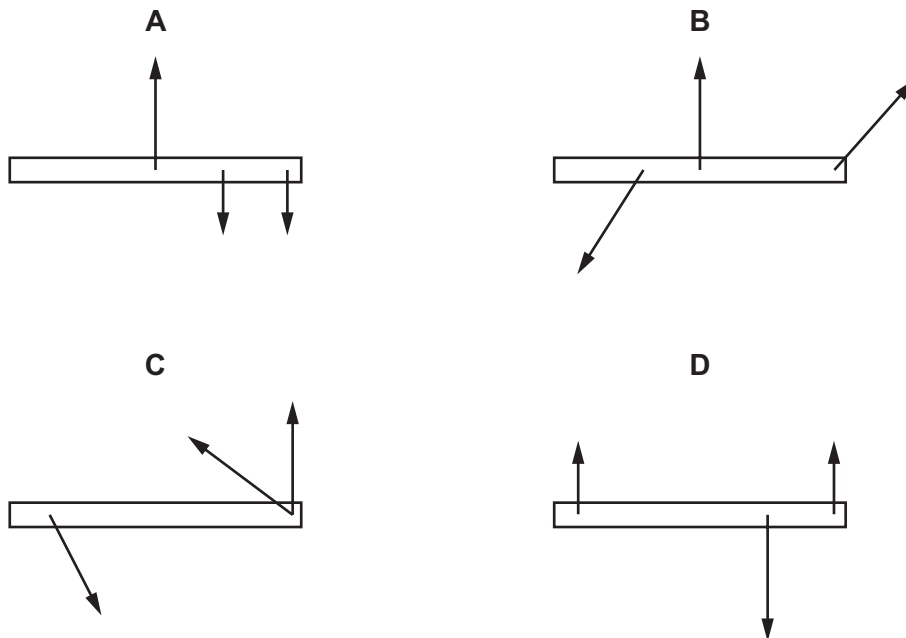
All the air is removed by a vacuum pump and the balance reading changes to 196.00 g .

What was the density of the pressurised air?

- A 0.00200 g/cm^3
 B 0.098 g/cm^3
 C 4.00 g/cm^3
 D 10.2 g/cm^3

- 5 Each diagram shows three forces on a beam.

In which situation is it possible for the three forces shown to be in equilibrium?



- 6 A gymnast jumps down from a high piece of apparatus.

She gradually bends her knees as she lands.

Which effect does this have?

- A She will have a smaller change of momentum as she lands.
- B She will lose less kinetic energy as she lands.
- C She will exert a smaller impulse on the ground as she lands.
- D She will experience a smaller force from the ground as she lands.

- 7 The table gives four energy resources and states whether the main source of energy for the resource is the Sun.

Which row is correct?

	energy resource	main source of energy is the Sun
A	geothermal	yes
B	oil	no
C	water held behind a dam	yes
D	wind	no

- 8 The equation $\Delta p = \rho g \Delta h$ can be used for a liquid.

What is the meaning of the term ρ ?

- A pressure due to the liquid
- B density of the liquid
- C total pressure due to the liquid and the air above the liquid
- D density of an object placed in the liquid

- 9 What are the correct terms for each change of state?

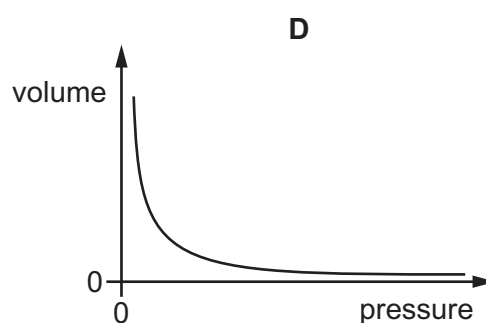
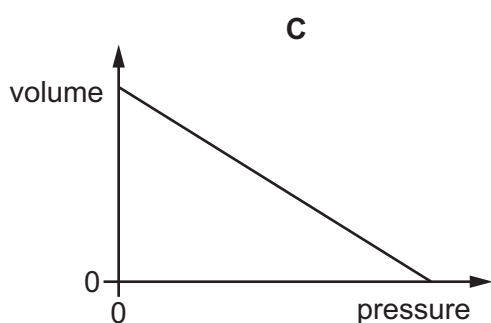
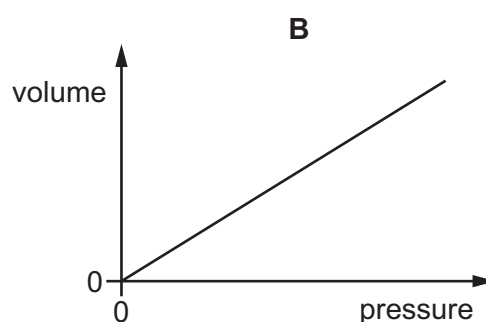
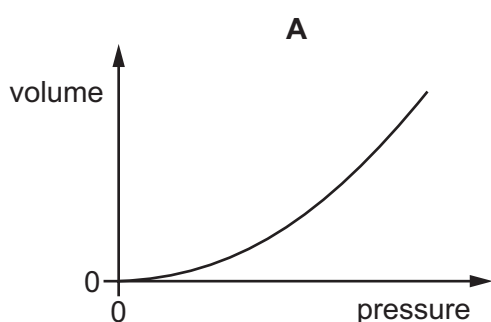
	liquid to solid	liquid to gas
A	solidification	condensation
B	melting	condensation
C	solidification	evaporation
D	melting	evaporation

- 10 Extremely small pollen grains in water are viewed through a microscope. The grains are seen to move continually and randomly.

What is the reason for this random movement?

- A The grains are moved by randomly moving water molecules.
- B The grains are moved by random convection currents in the water.
- C The grains are moved by random rays of light reflecting off them.
- D The grains are moved by the random motion of their own atoms.

- 11 Which graph shows how the volume of a fixed mass of gas at constant temperature varies with its pressure?



- 12 Which effect is caused by thermal expansion?

- A a metal surface heating up in direct sunlight
- B ice-cream melting on a hot day
- C a railway track buckling on a hot day
- D ice forming on a pond on a cold day

- 13** The temperature of the water at the bottom of a waterfall is greater than the temperature of the water at the top.

The energy in the gravitational potential store of the water at the top is transferred to the thermal store at the bottom.

The specific heat capacity of water is $4200 \text{ J/(kg } ^\circ\text{C)}$.

What is the temperature difference for a waterfall of height 21 m?

- A** 0.0050°C **B** 0.049°C **C** 20°C **D** 200°C

- 14** An ice cube is placed in a beaker and is heated.

The ice melts to form water, which evaporates at first and then boils.

The steam condenses on a cold window in the room.

Which process involves a transfer of energy from the ice, water or steam to the surroundings?

- A** melting
B evaporating
C boiling
D condensing

- 15** A teacher shows his class a polystyrene cup. The polystyrene is a thick plastic with lots of tiny air bubbles in it.

He asks the class why the cup is so good at keeping a hot drink warm. Three suggestions are made.

- 1 It contains air which is a poor thermal conductor.
- 2 The air is trapped in tiny bubbles so very little convection is possible.
- 3 The plastic is a poor thermal conductor.

Which suggestions are correct?

- A** 1, 2 and 3 **B** 1 and 2 only **C** 1 and 3 only **D** 2 and 3 only

- 16** Two samples of the same material have the same mass but different surface areas.

Each sample is heated to the same temperature and then left to cool to room temperature.

Each sample is allowed to cool to the same final temperature.



Which row correctly compares the decrease in internal energy and the initial rate of cooling for each sample?

	decrease in internal energy	initial rate of cooling
A	X loses more internal energy than Y	X cools down faster than Y
B	Y loses more internal energy than X	Y cools down faster than X
C	X and Y lose the same quantity of energy	X cools down faster than Y
D	X and Y lose the same quantity of energy	Y cools down faster than X

- 17** In which type of wave is the direction of vibration parallel to the direction of travel?

- A** electromagnetic waves
- B** seismic P-waves
- C** seismic S-waves
- D** water waves

- 18** Waves in a ripple tank are diffracted as they pass through a narrow gap.

What can be done to make the spreading due to diffraction greater?

- A** Decrease the frequency of the waves and keep the speed constant.
- B** Decrease the speed of the waves and keep the frequency constant.
- C** Increase the frequency of the waves and keep the speed constant.
- D** Increase the frequency of the waves and decrease the speed.

- 19 Diagram 1 shows the page of a book in front of a plane mirror.

An eye is looking at the image of the page.

Diagram 2 shows a large letter G on the page facing the mirror.

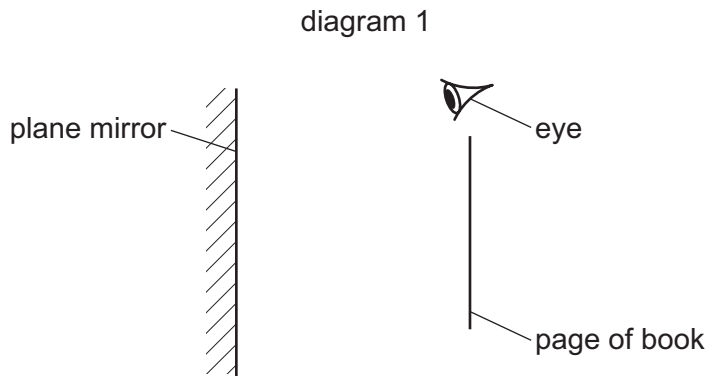


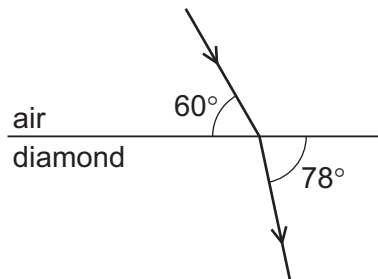
diagram 2



What is the appearance of the image of G seen by the eye?



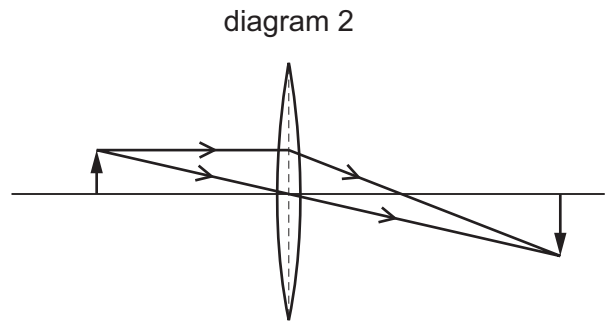
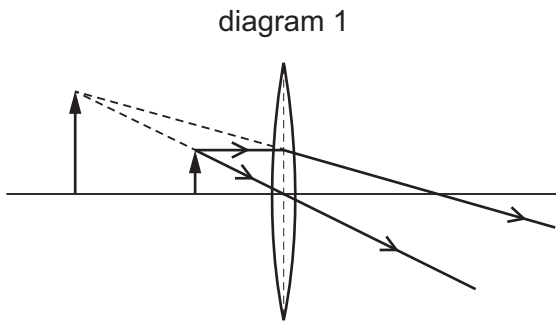
- 20 The diagram shows a ray of light passing from air into diamond.



What is the refractive index of the diamond?

- A 0.89 B 1.1 C 2.4 D 2.5

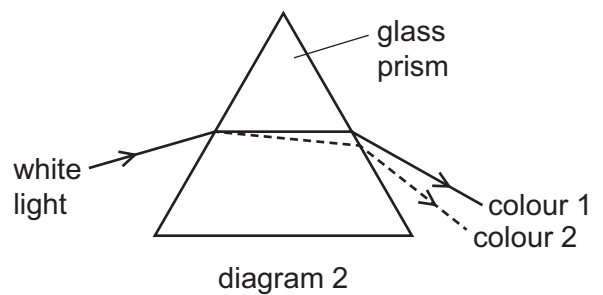
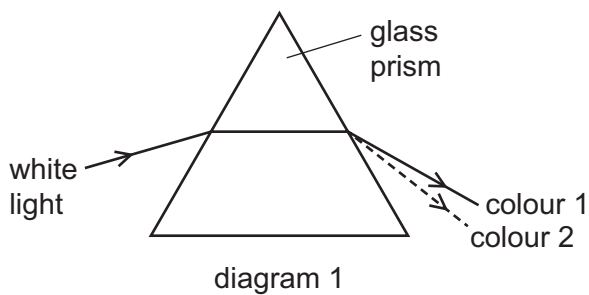
21 The ray diagrams show the formation of an image by two different converging lenses.



Which row describes the images produced?

	diagram 1	diagram 2
A	real	real
B	real	virtual
C	virtual	real
D	virtual	virtual

22 White light enters a glass prism. The light leaving the other side of the prism is separated into colours.



Which row correctly describes what happens?

	path taken by the light	colour 1	colour 2
A	diagram 1	red	violet
B	diagram 1	violet	red
C	diagram 2	red	violet
D	diagram 2	violet	red

- 23** Thermal radiation is part of the electromagnetic spectrum.

What is the name of this region of the spectrum?

- A** gamma rays
- B** infrared
- C** ultraviolet
- D** visible light

- 24** A remote-controlled vehicle is travelling on the surface of a planet. The vehicle senses an obstacle ahead. It sends a radio message to the control room from where it is being controlled. The control room is 2.4×10^6 km away from the vehicle. The control room sends a message back to the vehicle telling it to stop.

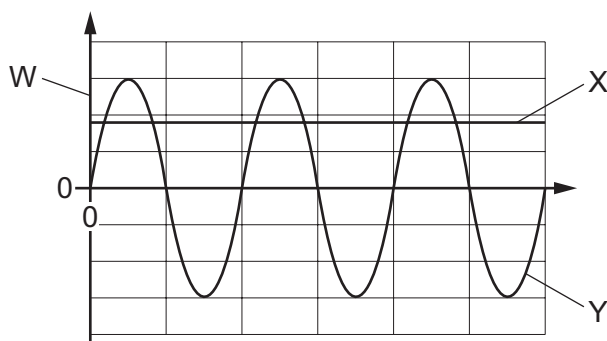
What is the minimum time that elapses between the vehicle sensing the obstacle and receiving the message back from the control room?

- A** 8.0 ms
- B** 16 ms
- C** 8.0 s
- D** 16 s

- 25** Which statement about the direction of a magnetic field at a point is correct?

- A** It is the direction of the force on an N pole placed at that point.
- B** It is the direction of the force on an S pole placed at that point.
- C** It is the direction of the force on a positive charge placed at that point.
- D** It is the direction of the force on a negative charge placed at that point.

- 26** A student draws graphs of an a.c. supply and a d.c. supply.



Which row shows the correct labels for W, X and Y?

	label of W	a.c. supply	d.c. supply
A	time	X	Y
B	time	Y	X
C	voltage	X	Y
D	voltage	Y	X

- 27 A resistor transfers 100 J of energy when a charge of 10 C flows through it.

What is the potential difference across the resistor?

- A 0.10 V B 10 V C 10 W D 1000 W

- 28 Wire X and wire Y are made from the same metal.

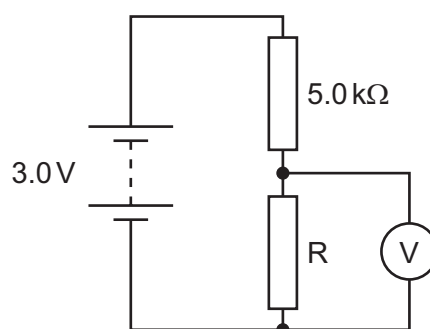
The table gives information about the two wires.

	length / m	cross-sectional area / mm ²	resistance / Ω
wire X	1.0	0.40	2.0
wire Y	1.3	0.30	

What is the resistance of wire Y?

- A 0.31 Ω B 0.78 Ω C 2.1 Ω D 3.5 Ω

- 29 The diagram shows two resistors used as a potential divider.



The reading on the voltmeter is 1.8 V.

What is the resistance of the resistor R?

- A 3.0 k Ω B 3.3 k Ω C 7.5 k Ω D 13 k Ω

- 30 Two electrical appliances are connected to the mains supply.

The cable connected to one appliance includes an earth wire.

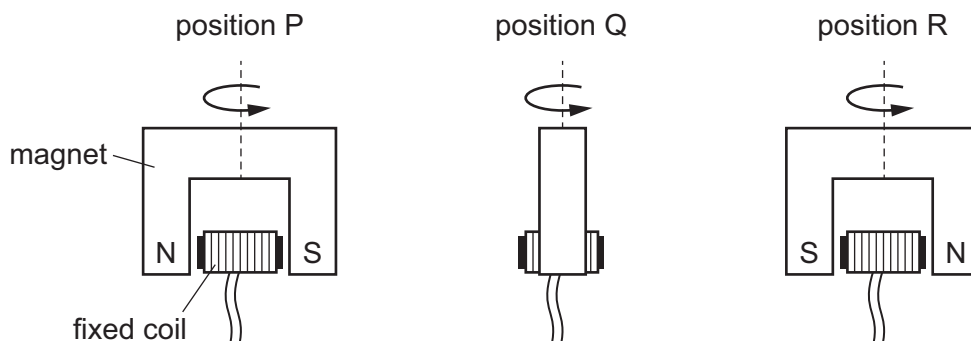
The cable connected to the second appliance does **not** need an earth wire.

What is a reason for this difference?

- A One appliance has a metal case, but the other appliance does not.
 B One appliance is fitted with a fuse, but the other appliance is not.
 C One appliance is fitted with a switch, but the other appliance is not.
 D One appliance needs more current than the other appliance.

- 31** The diagram shows a simple a.c. generator. The magnet rotates around a fixed coil wound round a fixed iron core. The electromotive force (e.m.f.) output is a sine wave, a series of crests, zeros and troughs.

The magnet is shown in three positions, P, Q and R, during its rotation. The magnetic field in the coil is zero at Q but changing most rapidly.



Which statement about the induced e.m.f. in the coil is correct?

- A** The magnitude of the e.m.f. at position Q is larger than that at position P.
B The magnitude of the e.m.f. at position Q is zero.
C The magnitude of the e.m.f. at position Q is smaller than at position R, but not zero.
D The magnitude of the e.m.f. at position R is larger than at position P.
- 32** A 24 V power supply and a 100% efficient transformer are used to light a 6.0 V, 12 W lamp.

What is the current in the power supply?

- A** 0.50 A **B** 2.0 A **C** 3.0 A **D** 8.0 A

- 33** The scattering of particles by a thin gold foil provided scientists with evidence for the nuclear atom.

Which particles were scattered by the gold nuclei in the thin foil?

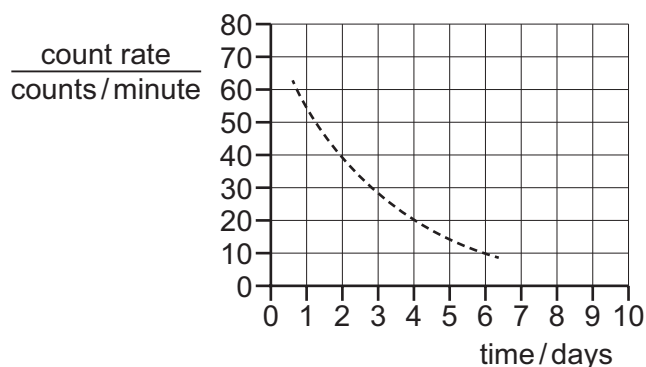
- A** α -particles
B β -particles
C neutrons
D protons

- 34** A nucleus X has 17 protons and 18 neutrons.

Which notation is correct for this nucleus?

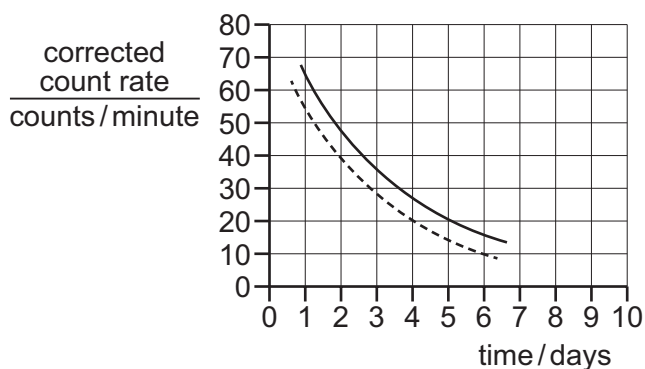
- A** ${}_{18}^{17}\text{X}$ **B** ${}_{35}^{17}\text{X}$ **C** ${}_{17}^{18}\text{X}$ **D** ${}_{17}^{35}\text{X}$

- 35 The dashed line on the graph shows the decay curve recorded from a sample of a particular radioactive isotope. The count rate includes background radiation.

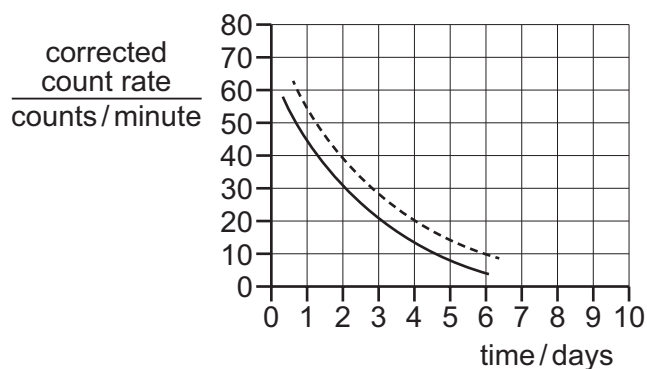


Which curve shows the corrected count rate for the decay?

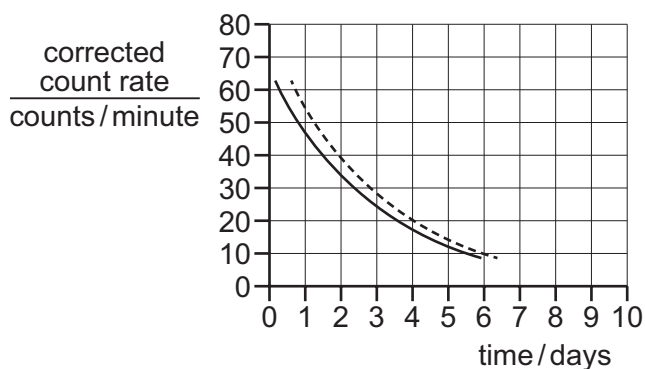
A



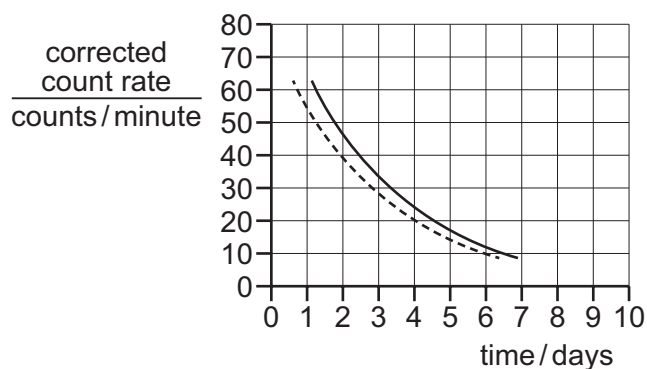
B



C



D

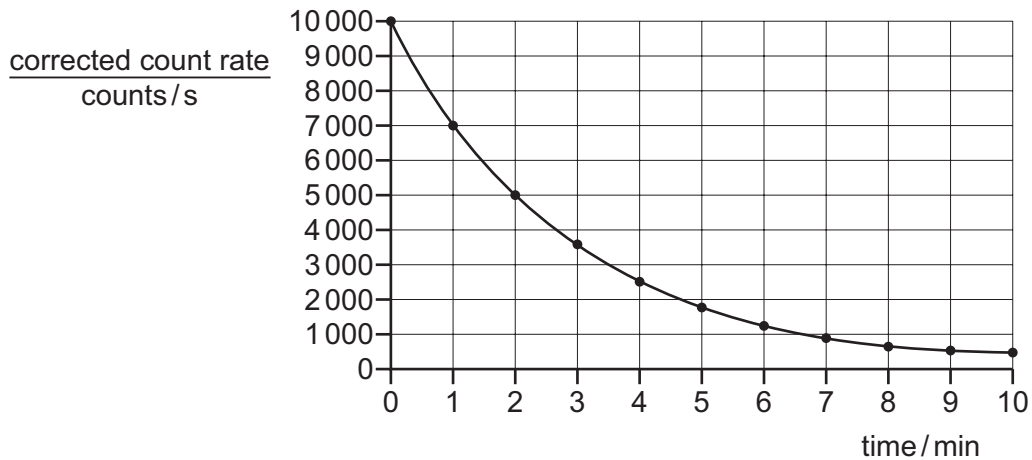


- 36 Which type of radioactive source is suitable for use in measuring and controlling the thickness of paper in a paper-manufacturing factory?

	type of emission	half-life
A	alpha	long
B	alpha	short
C	beta	long
D	beta	short

- 37 A medical tracer is used to investigate a tumour in a patient.

The graph shows the corrected count rate from the tracer against time.



What is the half-life of the source?

- A 2 minutes
 - B 10 minutes
 - C 5000 minutes
 - D 10000 minutes
- 38 The Earth orbits the Sun once in approximately 365 days.

The average radius of the orbit is 1.5×10^{11} m.

What is the average orbital speed of the Earth?

- A 3.0×10^4 m/s
 - B 1.1×10^8 m/s
 - C 2.6×10^9 m/s
 - D 1.3×10^{15} m/s
- 39 Which statement about the life cycle of a star is correct?
- A All stars eventually run out of hydrogen as fuel.
 - B A red giant forms a planetary nebula with a supernova at its centre.
 - C A white dwarf star eventually forms a black hole.
 - D Most stars expand and form protostars.

- 40** A galaxy at a distance of 3.1×10^{21} km is moving away from the Earth with a speed of 7000 km/s.

What is the Hubble constant H_0 calculated using this data and what is the speed of a galaxy at a distance of 6.2×10^{21} km from the Earth?

	H_0/s^{-1}	<u>speed of galaxy</u> km/s
A	2.3×10^{-18}	3 500
B	2.3×10^{-18}	14 000
C	4.4×10^{17}	3 500
D	4.4×10^{17}	14 000

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Cambridge IGCSE™ (9–1)

PHYSICS (9–1)**0972/22**

Paper 2 Multiple Choice (Extended)

May/June 2024**MARK SCHEME**Maximum Mark: 40

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This document consists of **3** printed pages.

Question	Answer	Marks
1	C	1
2	C	1
3	B	1
4	A	1
5	D	1
6	D	1
7	C	1
8	B	1
9	C	1
10	A	1
11	D	1
12	C	1
13	B	1
14	D	1
15	A	1
16	D	1
17	B	1
18	A	1
19	D	1
20	C	1
21	C	1
22	C	1
23	B	1
24	D	1
25	A	1
26	D	1
27	B	1
28	D	1

Question	Answer	Marks
29	C	1
30	A	1
31	A	1
32	A	1
33	A	1
34	D	1
35	B	1
36	C	1
37	A	1
38	A	1
39	A	1
40	B	1



Cambridge IGCSE™ (9–1)

PHYSICS

0972/21

Paper 2 Multiple Choice (Extended)

October/November 2024

45 minutes

You must answer on the multiple choice answer sheet.

You will need: Multiple choice answer sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)

INSTRUCTIONS

- There are **forty** questions on this paper. Answer **all** questions.
- For each question there are four possible answers **A**, **B**, **C** and **D**. Choose the **one** you consider correct and record your choice in soft pencil on the multiple choice answer sheet.
- Follow the instructions on the multiple choice answer sheet.
- Write in soft pencil.
- Write your name, centre number and candidate number on the multiple choice answer sheet in the spaces provided unless this has been done for you.
- Do **not** use correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.
- Take the weight of 1.0 kg to be 9.8 N (acceleration of free fall = 9.8 m/s^2).

INFORMATION

- The total mark for this paper is 40.
- Each correct answer will score one mark.
- Any rough working should be done on this question paper.

This document has **20** pages. Any blank pages are indicated.



- 1 A student uses a metre ruler to measure the length of a sheet of paper.

Which measurement is shown to the nearest millimetre?

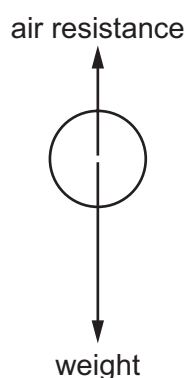
A 0.2932 m **B** 0.293 m **C** 0.29 m **D** 0.3 m

- 2 A rocket travels with an average speed of 6 km/s for 2 minutes.

What is the distance travelled by the rocket?

A 12 km **B** 50 km **C** 720 km **D** 12 000 km

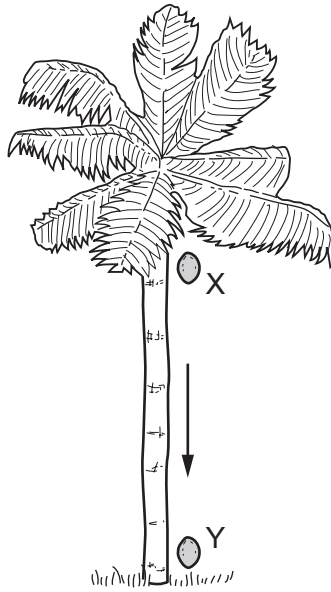
- 3 The diagram shows the vertical forces acting on a ball as it falls vertically through the air. The ball does not reach terminal velocity.



Which row describes what happens to the resultant force on the ball and what happens to the acceleration of the ball as it falls through the air?

	resultant force	acceleration
A	decreases	decreases
B	decreases	increases
C	increases	decreases
D	increases	increases

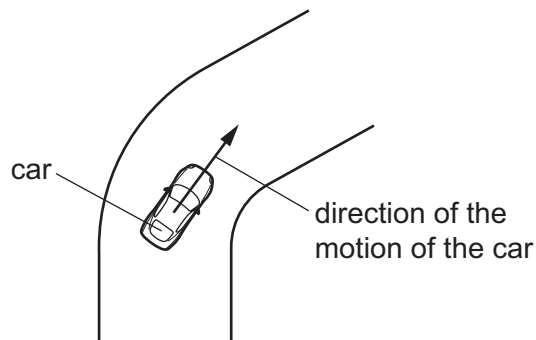
- 4 A coconut falls from a palm tree. At X, it has just started falling. Y is the point just before it hits the ground.



What is the acceleration of the coconut at X?

(Air resistance can be ignored.)

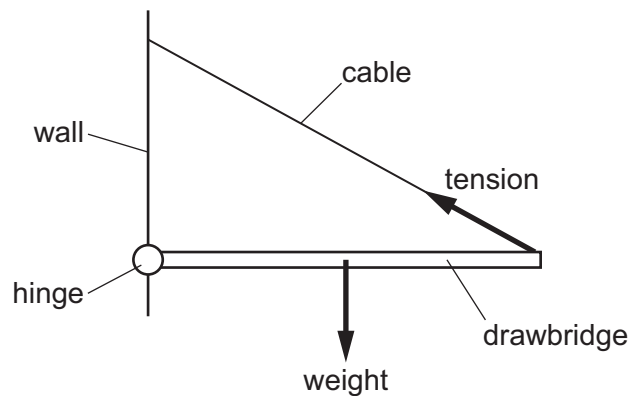
- A zero
 - B less than that at Y
 - C the same as that at Y
 - D more than that at Y
- 5 A car is driven round a bend in the road at a constant speed.



What is the direction of the resultant force on the car when it is going round the bend?

- A parallel to the motion and in the same direction as the motion
- B parallel to the motion and in the opposite direction to the motion
- C perpendicular to the motion and towards the inside of the bend
- D perpendicular to the motion and towards the outside of the bend

- 6 The diagram shows a drawbridge that is attached to a wall by a hinge at one end and a cable at the other.



The weight of the drawbridge and the tension in the cable are represented by the labelled arrows in the diagram. There is a third force at the hinge, which is not shown. The drawbridge is in equilibrium.

Which arrow shows a possible direction for the force at the hinge?



- 7 Which equation gives the momentum change of an object?

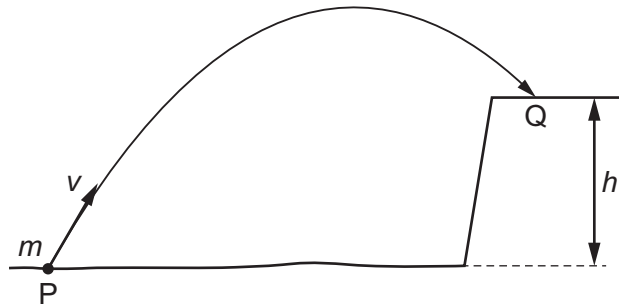
- A momentum change = $\frac{\text{force}}{\text{area}}$
- B momentum change = force $\times$ distance
- C momentum change = force $\times \frac{\text{distance}}{\text{time}}$
- D momentum change = force $\times$ time

- 8 Which statement about the use of nuclear fuel as an energy resource is correct?

- A It obtains its energy from the Sun.
- B It has **no** environmental impact.
- C Nuclear power stations do **not** need turbines.
- D The energy is released by nuclear fission.

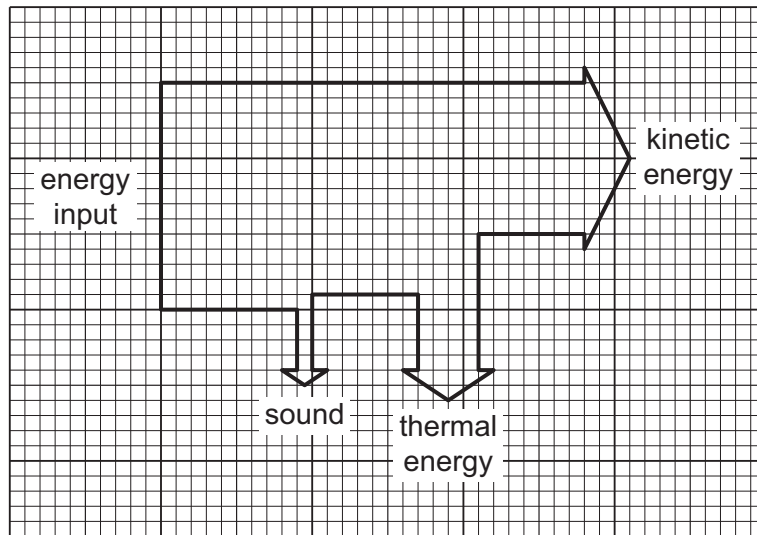
- 9 The diagram shows an object of mass m being projected with speed v from a point P to a point Q on top of a cliff height h .

The object does work W against air resistance as it travels from P to Q.



What is the kinetic energy of the object when it reaches point Q?

- A $\frac{1}{2}mv^2 + mgh + W$
- B $\frac{1}{2}mv^2 + mgh - W$
- C $\frac{1}{2}mv^2 - mgh + W$
- D $\frac{1}{2}mv^2 - mgh - W$
- 10 The diagram shows a Sankey diagram for an electric drill.



scale: distance between two lines = 40 J

What is the total wasted energy shown by the diagram?

- A 120 J B 200 J C 360 J D 560 J

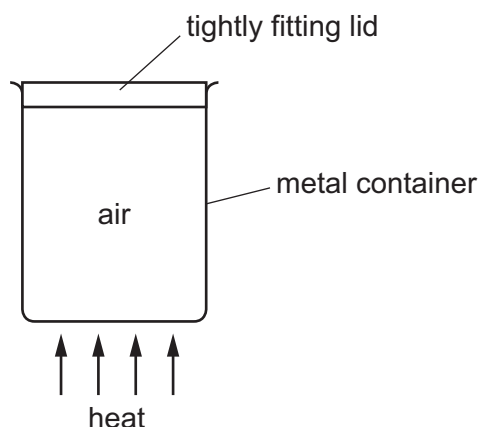
- 11 An elephant of weight 45 000 N stands with all four feet on the floor.

The average pressure on the floor due to the elephant foot in contact with it is 34 000 Pa.

What is the area of each foot of the elephant?

- A** 0.19 m^2 **B** 0.33 m^2 **C** 0.76 m^2 **D** 1.3 m^2

- 12 Some air is trapped inside a metal container with a tightly fitting lid.



The container is heated strongly behind a safety screen. The lid is blown off by the increased pressure of the air inside the container.

What causes the increase in pressure of the air inside the container?

- A** Each air particle expands and takes up more room.
B The air particles move more quickly.
C The number of particles inside the container increases.
D The volume occupied by the air decreases.
- 13 A very hot mug of coffee is near a large unheated swimming pool holding $2.0 \times 10^6 \text{ kg}$ of water.
- Which statement is correct?
- A** If the mug of coffee is tipped in the pool, the coffee will lose much more internal energy than the water in the pool gains.
B The internal energy of a substance depends only on its temperature.
C The internal energy of the water in the pool is lower than the internal energy of the mug of coffee.
D When the temperature of the mug of coffee falls, its internal energy will decrease.

- 14** A student measures the mass of water in an open container over two hours. The container is kept in a warm room. The results are in the table.

time / hours	mass of water / g
0.0	33.9
0.5	30.6
1.0	27.6
1.5	24.9
2.0	22.5

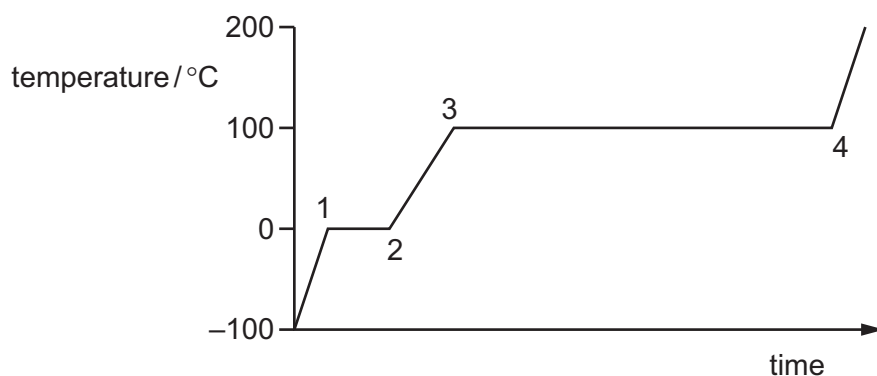
Why does the mass of the water change?

- A** The water evaporates.
 - B** The water freezes.
 - C** The water condenses.
 - D** The water boils.
- 15** A 1 kg block of aluminium requires more thermal energy to raise its temperature by 1 °C than a 1 kg block of copper requires.

Why is this?

- A** Aluminium is a better conductor of thermal energy than copper.
- B** Aluminium is a poorer conductor of thermal energy than copper.
- C** Aluminium has a higher specific heat capacity than copper.
- D** Aluminium has a lower specific heat capacity than copper.

- 16** A block of ice is at a temperature of -100°C . Energy is supplied at a constant rate. The graph shows how its temperature changes.



At which points have the ice completely changed state to water and all the water completely changed state to steam?

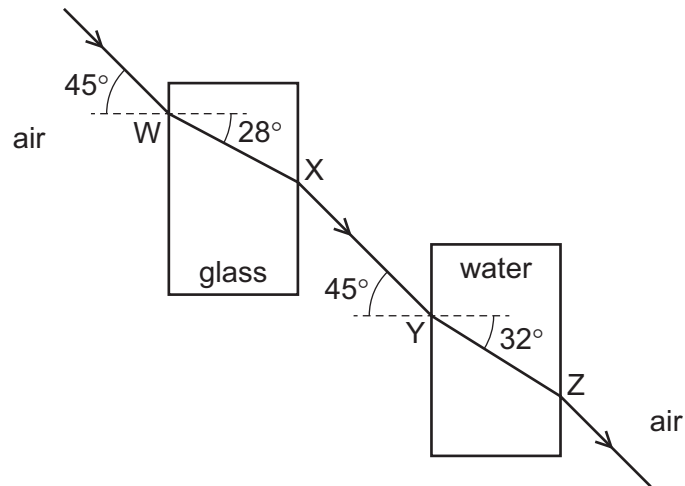
	completely changed to water	completely changed to steam
A	1	3
B	1	4
C	2	3
D	2	4

- 17** Water waves are reflected at a plane surface.

Which property of the waves is changed by the reflection?

- A** direction of propagation
- B** frequency
- C** speed
- D** wavelength

18 A light ray is passed through air, glass and water.

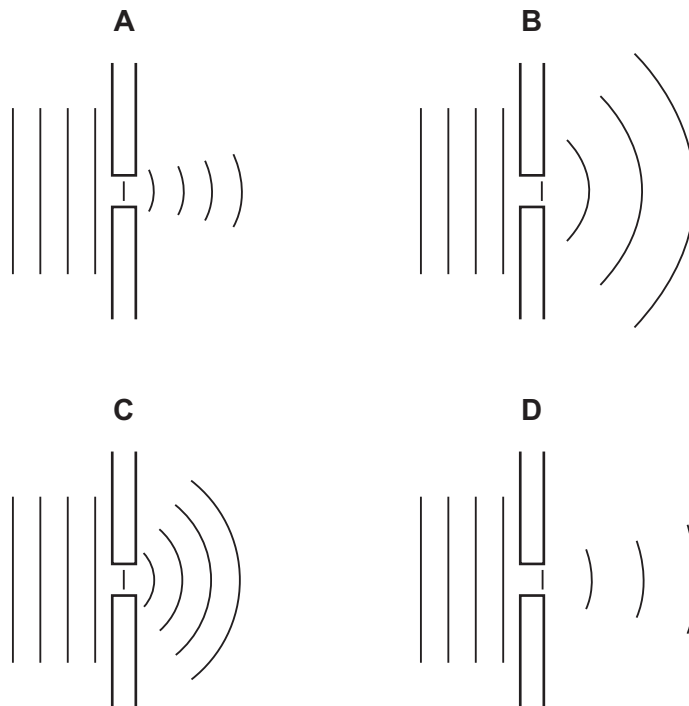


At which points does the light speed decrease?

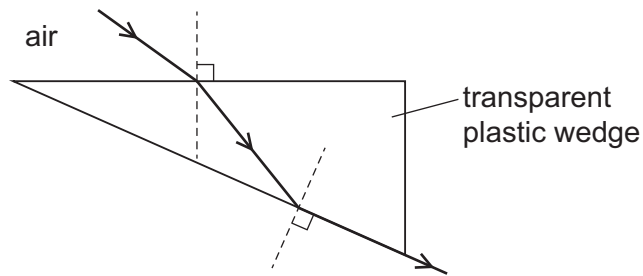
- A** W and X **B** W and Y **C** X and Z **D** Y and Z

19 Plane water waves approach a narrow gap in a barrier.

Which diagram shows the diffraction pattern that would occur?

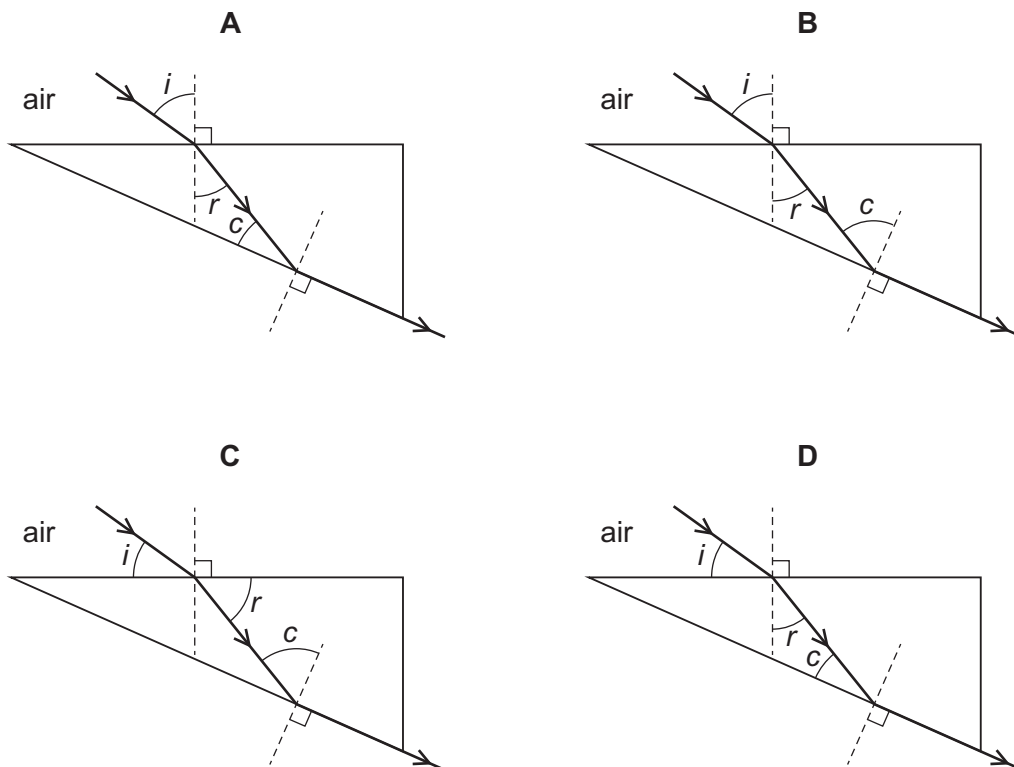


- 20 A ray of light enters a transparent plastic wedge from air and is incident on the lower surface at the critical angle, as shown.

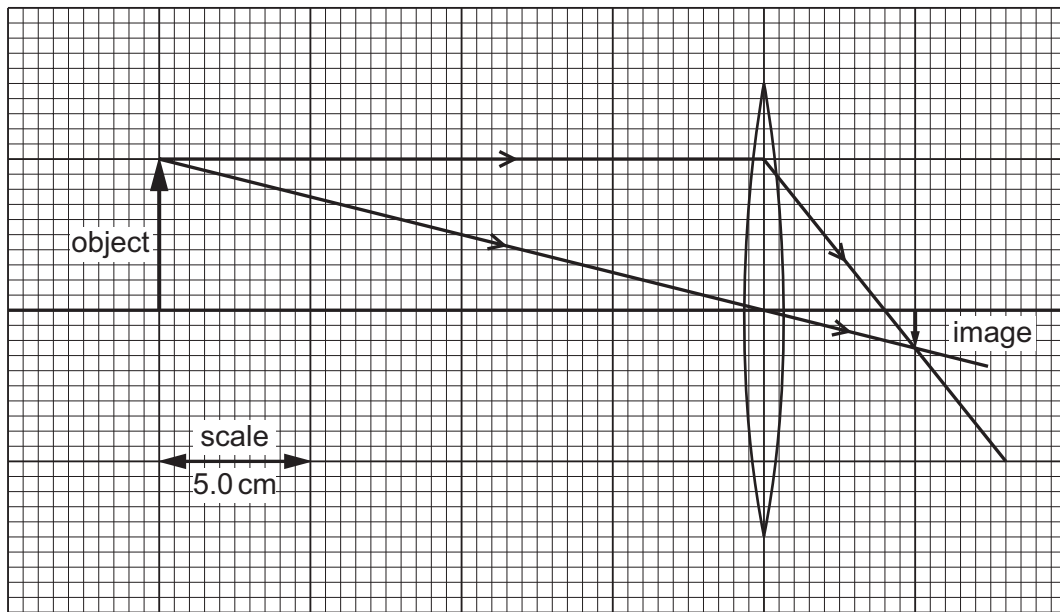


The angle of incidence is i , the angle of refraction is r , and the critical angle is c .

In which ray diagram are the angles labelled correctly?



- 21 An object is placed 20 cm in front of a thin converging lens. The scale diagram shows how the lens forms a real, inverted image.



Which row gives the focal length of the lens and the distance of the image from the lens?

	focal length of lens / cm	distance of image from lens / cm
A	4.0	5.0
B	5.0	4.0
C	8.0	10.0
D	10.0	8.0

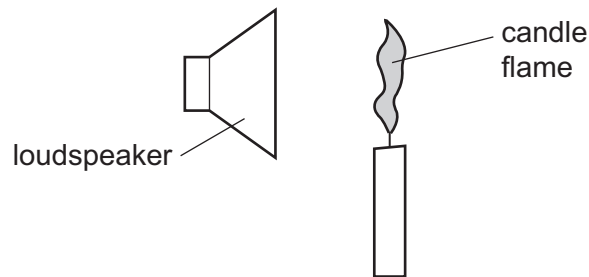
- 22 Radio waves are electromagnetic waves.

What is the wavelength of a radio wave of frequency 2.0×10^5 Hz?

- A** 0.00067 m **B** 0.0015 m **C** 670 m **D** 1500 m

- 23** A candle flame is placed in front of a loudspeaker.

The loudspeaker produces a sound wave that causes air particles to vibrate. The vibrating air particles make the candle flame vibrate in the same direction as the air particles.

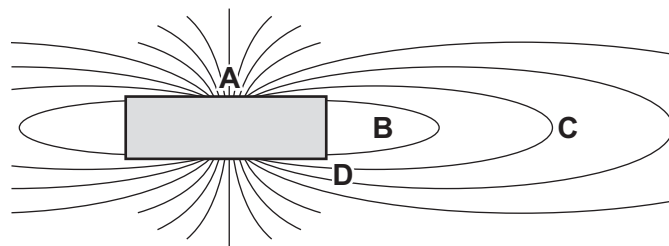


Which row shows the direction of vibration of the candle flame and the nature of sound waves?

	direction of vibration	nature of sound waves
A		longitudinal
B		transverse
C		longitudinal
D		transverse

- 24** The diagram shows part of the magnetic field around a strong magnet.

In which position does a magnetic pole experience the strongest force?



- 25** There is a current of 0.60 A in a closed circuit.

How much charge passes a point in the circuit in 2.5 min?

- A** 0.24 C **B** 1.5 C **C** 90 C **D** 5400 C

26 Which two physical quantities have the unit J/C?

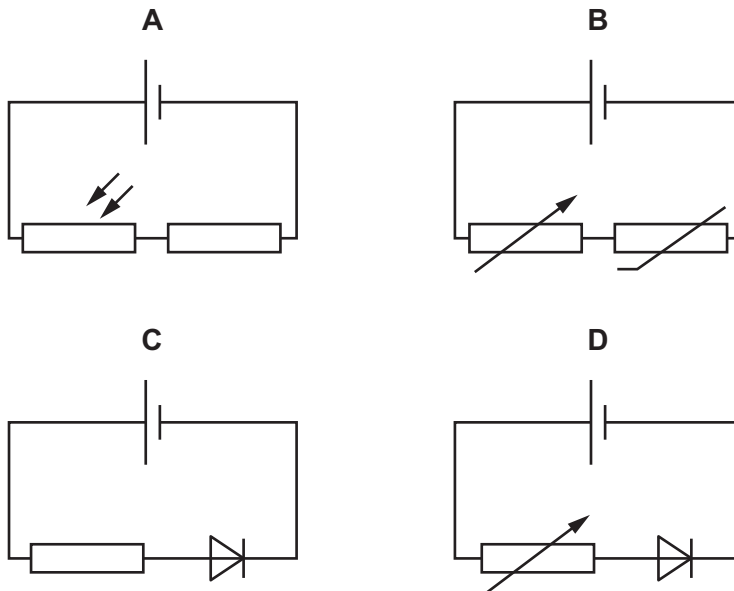
- A charge and energy
- B charge and potential difference (p.d.)
- C electromotive force (e.m.f.) and p.d.
- D e.m.f. and charge

27 A copper wire is placed in an electrical circuit and its resistance is measured. The wire is then replaced with a second copper wire with twice the length and a quarter of the diameter.

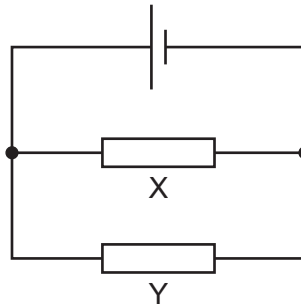
What is the ratio of the resistance for the two wires?

- A 1:2 B 1:4 C 1:8 D 1:32

28 Which circuit contains a diode connected in series with a fixed resistor?



- 29 In the circuit shown, resistor X has twice the resistance of resistor Y.



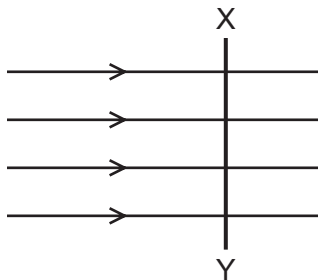
Which statement about the circuit is correct?

- A The current in the cell is greater than the current in X.
 - B The current in X is greater than the current in Y.
 - C The potential difference (p.d.) across the cell is greater than the p.d. across X.
 - D The p.d. across X is greater than the p.d. across Y.
- 30 A $4\ \Omega$ resistor and an $8\ \Omega$ resistor are connected in series with a power supply.

A voltmeter connected across the $8\ \Omega$ resistor reads $36\ \text{V}$.

What is the voltmeter reading when it is connected across the $4\ \Omega$ resistor?

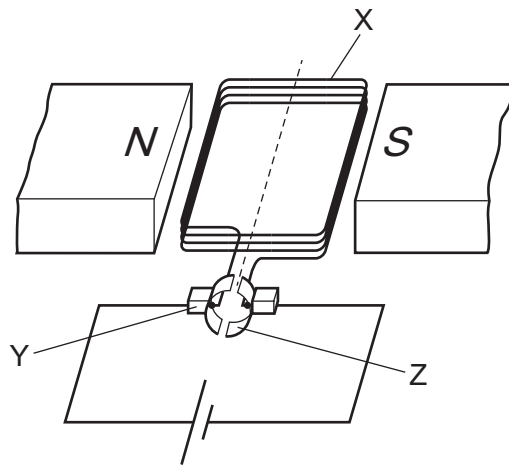
- A $12\ \text{V}$
 - B $18\ \text{V}$
 - C $72\ \text{V}$
 - D $108\ \text{V}$
- 31 A metal rod, XY, is placed in a magnetic field so that it is perpendicular to the field, as shown.



In which direction is the rod moved so that there is an induced current from Y to X?

- A left to right
- B right to left
- C into the paper, away from the observer
- D out of the paper, towards the observer

32 The diagram shows a basic electric motor.



What are the names of the features?

	X	Y	Z
A	axle	battery	brushes
B	coil	brushes	commutator
C	generator	wires	transformer
D	solenoid	connectors	switch

33 Two transformers, P and Q, each have 2000 turns on their primary coils and 1000 turns on their secondary coils. Both transformers are 100% efficient.

The secondary of transformer P is connected to the primary of transformer Q.

Q provides a motor with a current of 0.2 A.

What is the current being supplied to the primary of transformer P?

- A** 0.8 A **B** 0.4 A **C** 0.1 A **D** 0.05 A

- 34** α -particles are directed at a metal foil.

Most of the particles pass through the foil with little change in direction.

A small proportion of the particles are scattered back through large angles.

What does this evidence suggest about the structure of an atom?

- A** It consists of a charged centre much smaller than the size of the atom and with little of the mass of the atom.
 - B** It consists of a negative charge the size of the atom containing small positive charges scattered through it.
 - C** It consists of a charged centre much smaller than the size of the atom but with most of the mass of the atom.
 - D** It consists of a positive charge the size of the atom containing small negative charges scattered through it.
- 35** Which statement correctly compares the properties of alpha-particles and beta-particles?
- A** Alpha-particles are less penetrating than beta-particles because alpha-particles are less ionising.
 - B** Alpha-particles are less penetrating than beta-particles because alpha-particles are more ionising.
 - C** Alpha-particles are more penetrating than beta-particles because alpha-particles are less ionising.
 - D** Alpha-particles are more penetrating than beta-particles because alpha-particles are more ionising.
- 36** Which statement about alpha decay is correct?
- A** The nucleus loses electrons.
 - B** The nucleus changes to that of a different element.
 - C** The nucleus does not decay until after one half-life.
 - D** After two half-lives, alpha decay always stops.
- 37** An explosion in a nuclear reactor spreads the isotope caesium-137, $^{137}_{55}\text{Cs}$, across a large area.
- After 90.0 years have passed, the quantity of caesium-137 present is 12.5% of its original level.
- What is the half-life of caesium-137?
- A** 11.3 years **B** 22.5 years **C** 30.0 years **D** 45.0 years

38 Which type of object rotates around the Sun with an elliptical orbit?

- A planet
- B galaxy
- C red giant
- D moon

39 Which statement about the Sun is correct?

- A The nuclear reactions in the Sun create mainly radio waves and microwaves.
- B The Sun is powered by fission reactions in which hydrogen is converted into helium.
- C The Sun is powered by fission reactions in which lithium is converted into helium.
- D The Sun is powered by fusion reactions and electromagnetic radiation is produced.

40 To determine the value of the Hubble constant, an astronomer measures the distance of a galaxy from the Earth as 2.5×10^{21} km.

She also measures the speed of recession of the galaxy from the Earth as 5.2×10^3 km/s.

From these data, what is the value of the Hubble constant?

- A $7.7 \times 10^{-26} \text{ s}^{-1}$
- B $2.1 \times 10^{-18} \text{ s}^{-1}$
- C $4.8 \times 10^{17} \text{ s}^{-1}$
- D $1.3 \times 10^{25} \text{ s}^{-1}$

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Cambridge IGCSE™ (9–1)

PHYSICS**0972/21**

Paper 2 Multiple Choice (Extended)

October/November 2024

MARK SCHEME

Maximum Mark: 40

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

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1	B	1
2	C	1
3	A	1
4	C	1
5	C	1
6	D	1
7	D	1
8	D	1
9	D	1
10	B	1
11	B	1
12	B	1
13	D	1
14	A	1
15	C	1
16	D	1
17	A	1
18	B	1
19	C	1
20	B	1
21	A	1
22	D	1
23	C	1
24	A	1
25	C	1
26	C	1
27	D	1
28	C	1

Question	Answer	Marks
29	A	1
30	B	1
31	D	1
32	B	1
33	D	1
34	C	1
35	B	1
36	B	1
37	C	1
38	A	1
39	D	1
40	B	1



Cambridge IGCSE™ (9–1)

PHYSICS

0972/21

Paper 2 Multiple Choice (Extended)

May/June 2025

45 minutes

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- Do **not** write on any bar codes.
- You may use a calculator.
- Take the weight of 1.0 kg to be 9.8 N (acceleration of free fall = 9.8 m/s^2).

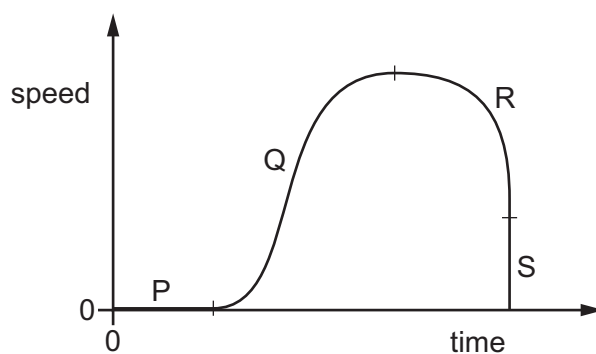
INFORMATION

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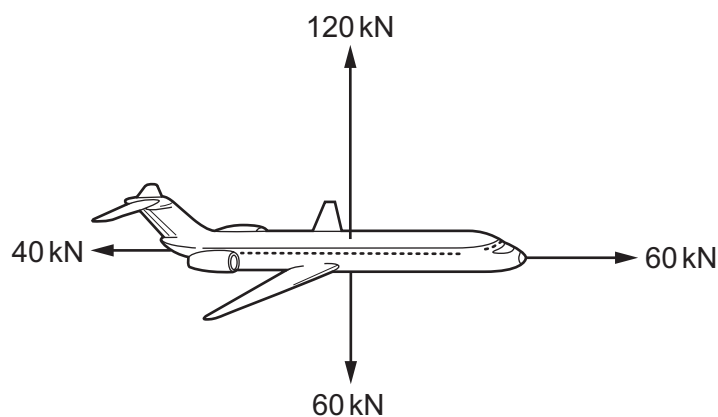


- 1 The speed–time graph for a racing car is divided into four sections, P, Q, R and S. The car starts the race but soon crashes into a wall of tyres.



Which sections of the graph show that there is an acceleration that is changing?

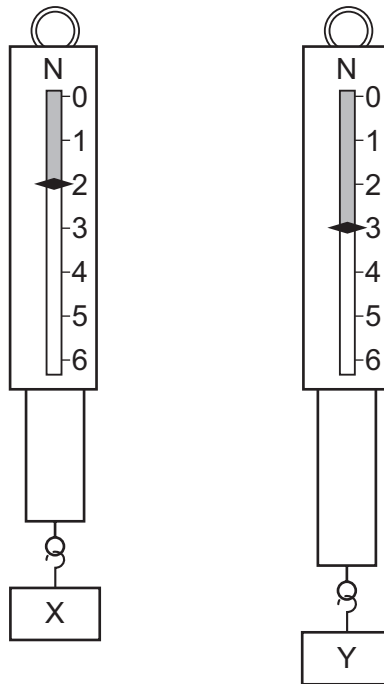
- A P, Q, R and S
 - B Q, R and S only
 - C Q and R only
 - D R and S only
- 2 The diagram shows the four forces acting on an aircraft in flight.



Which arrow shows the direction of the resultant of the four forces?



- 3 The diagrams show two metal blocks, X and Y, suspended from identical force meters. X and Y have identical dimensions.



Which statement about X and Y is correct?

- A They have different volumes and different weights.
- B They have different volumes and equal weights.
- C They have equal volumes and equal weights.
- D They have equal volumes and different weights.

- 4 A girl has a gold necklace.

Which apparatus does she need to find the density of the gold?

	balance	measuring cylinder	thermometer	ruler
A	✓	✓	✓	✓
B	✓	✓	✓	x
C	✓	✓	x	x
D	✓	x	x	x

key

✓ = needed

x = not needed

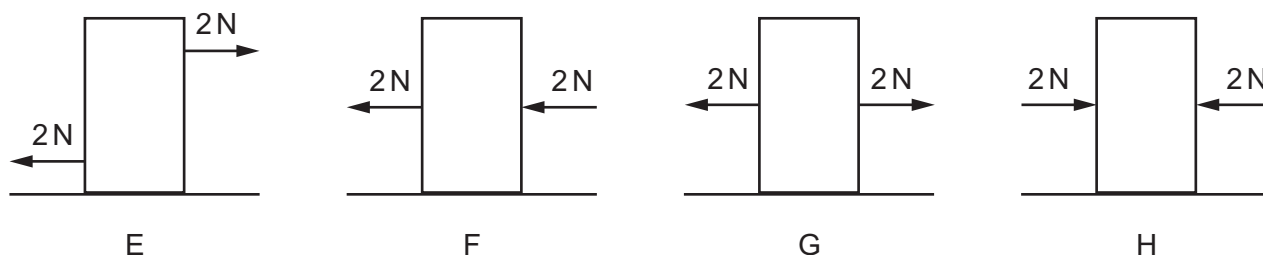
- 5 A spring has a length of 25 cm when a load of 2.0 N is suspended from it.

The spring has a length of 35 cm when a 7.0 N load is used.

What is the spring constant of the spring?

- A 0.20 N/cm
- B 0.50 N/cm
- C 2.0 N/cm
- D 5.0 N/cm

- 6 The diagrams E–H show a block of wood on a frictionless surface. In each diagram, the block has two forces acting on its sides.

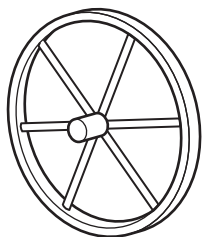


Which diagrams show the block in equilibrium?

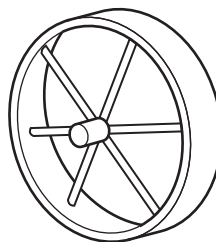
- A** E, F, G and H
 - B** E, G and H only
 - C** E and F only
 - D** G and H only
- 7 Which statement is **not** correct?
- A** the momentum of a system is conserved in all interactions
 - B** impulse = change of momentum
 - C** force acting = change of momentum $\times$ time
 - D** momentum = mass $\times$ velocity
- 8 A force acting on a moving ball causes its motion to change. This force stays constant.
- What makes the force produce a greater change in the motion of the ball?
- A** decreasing the total mass of the ball
 - B** increasing the temperature of the ball
 - C** using a ball with a hollow centre but the same mass
 - D** using a different material for the ball so that it has a lower density but the same mass

- 9 A farmer has two carts. The carts have the same weight, but one has four narrow wheels and the other has four wide wheels.

narrow wheel



wide wheel



Which cart wheels sink less into soft ground and what is the reason?

	cart wheels	reason
A	narrow	greater pressure on the ground
B	narrow	less pressure on the ground
C	wide	greater pressure on the ground
D	wide	less pressure on the ground

- 10 A hot gas and a cool liquid are at the same pressure.

Which row describes the separation and the motion of the particles of a hot gas compared to those of a cool liquid?

	separation of gas particles	motion of gas particles
A	greater than a cool liquid	faster than a cool liquid
B	greater than a cool liquid	slower than a cool liquid
C	less than a cool liquid	faster than a cool liquid
D	less than a cool liquid	slower than a cool liquid

- 11 A fixed mass of gas has a volume of 1.2 m^3 and is at a pressure of 200 kPa. The gas expands at constant temperature to a volume of 1.8 m^3 .

What is the new pressure of the gas?

- A** 120 kPa **B** 130 kPa **C** 300 kPa **D** 600 kPa

- 12 The specific heat capacity of solid P is greater than that of solid Q.

What does this statement mean?

- A Less energy is needed to raise the temperature by 1°C of unit mass of solid P than unit mass of solid Q.
- B Less energy is needed to melt unit mass of solid P than unit mass of solid Q.
- C More energy is needed to raise the temperature by 1°C of unit mass of solid P than unit mass of solid Q.
- D More energy is needed to melt unit mass of solid P than unit mass of solid Q.

- 13 When water evaporates, what escapes from the surface of the water?

- A individual atoms
- B individual molecules
- C individual protons
- D tiny drops of water

- 14 A sealed container of gas is heated. The pressure of the gas increases.

Which statement explains this increase in pressure?

- A The forces of the gas particles striking the walls of the container increase.
- B The forces of attraction between the gas particles and the walls of the container increase.
- C The gas particles collide with each other more frequently.
- D The gas particles lose more energy when they strike the walls of the container.

- 15 A student has a copper rod and a wooden rod. Both rods have the same dimensions.

The student puts one end of each rod into a beaker of crushed ice.

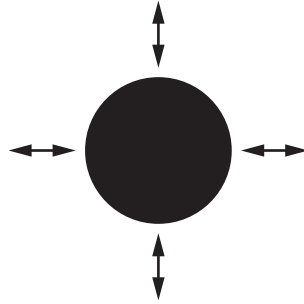
After a few minutes, the student picks up the rods by the ends **not** in the ice.

Which statement is correct?

- A The copper rod feels colder than the wooden rod because copper is a better conductor than wood.
- B The copper rod feels warmer than the wooden rod because copper is a better conductor than wood.
- C The wooden rod feels colder than the copper rod because wood is a better conductor than copper.
- D The wooden rod feels warmer than the copper rod because wood is a better conductor than copper.

- 16 The diagram shows a black object X at constant temperature.

The arrows represent the absorption and emission of thermal radiation at the surface.

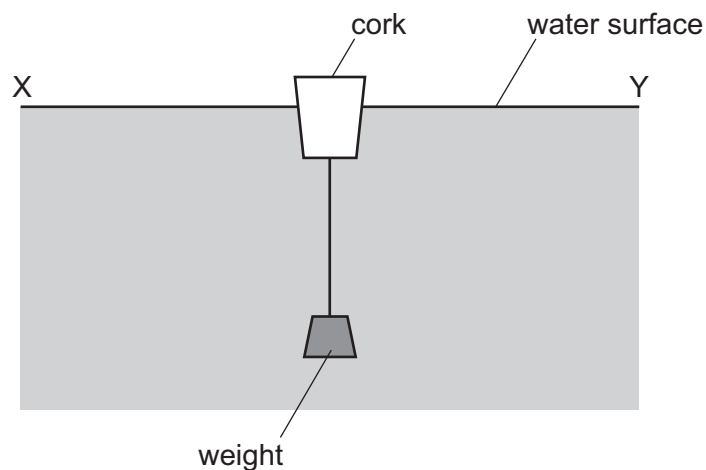


Object Y is at the same constant temperature as X and is identical except that it has a white surface.

Which row correctly compares object X with object Y?

	rate of absorption for X	rate of emission for X
A	higher	higher
B	lower	higher
C	higher	lower
D	lower	lower

- 17 The diagram shows a cork with a weight attached so that the cork floats upright in water.



Transverse waves travel across the water from X to Y.

In which direction do the waves make the cork move?

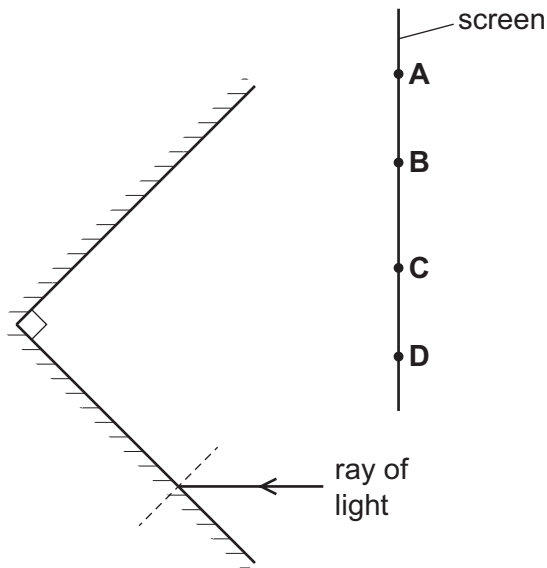
- A → ← right and left
 - B ↑↓ up and down
 - C → only to the right
 - D ← only to the left
- 18 A thin converging lens is used as a magnifying glass.

How does the image compare with the object?

- A real and inverted
- B real and upright
- C virtual and inverted
- D virtual and upright

- 19** The diagram shows two plane mirrors at 90° to each other. A ray of light is incident on one of the mirrors. The ray reflects off both mirrors before reaching a screen.

At which labelled point does the ray reach the screen?



- 20** The speed of light in a material is 50% of the speed of light in air.

What is the refractive index of the material?

- A** 0.5 **B** 2.0 **C** 6.0×10^8 **D** 1.5×10^8
- 21** A beam of white light is split into a spectrum of seven colours.

Which name is given to this process?

- A** diffraction
B dispersion
C reflection
D refraction
- 22** Satellites are sometimes used to receive and re-transmit television signals.

Which row shows the type of satellite and the electromagnetic waves used?

	type of satellite	electromagnetic waves used
A	low orbit	infrared
B	low orbit	microwaves
C	geostationary	infrared
D	geostationary	microwaves

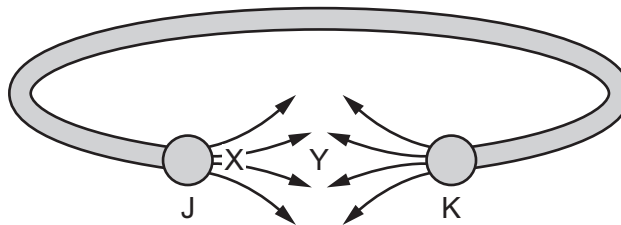
- 23** A sound wave travels through the air and strikes a solid block of steel.

Some of the sound is reflected from the steel, and some is transmitted through the steel.

How does the speed and the wavelength of the sound in steel compare to the speed and the wavelength of the sound in air?

	speed in steel	wavelength in steel
A	less than in air	less than in air
B	less than in air	greater than in air
C	greater than in air	less than in air
D	greater than in air	greater than in air

- 24** The diagram shows a bracelet containing two magnets. The arrows represent the pattern and the direction of the magnetic field due to the magnets.



Which row identifies the magnetic pole at J, the magnetic pole at K and where the magnetic field is the strongest?

	magnetic pole at J	magnetic pole at K	where the magnetic field is the strongest
A	N	N	X
B	N	N	Y
C	S	S	X
D	S	S	Y

25 Which row describes an electric field and its direction?

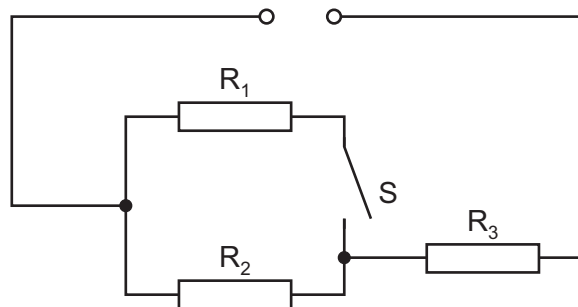
	description of electric field	direction of electric field
A	region in which an electric charge experiences a force	direction of the force on a positive charge
B	region in which an electric charge experiences a force	direction of the force on a negative charge
C	region in which an electric charge produces a current	direction of the force on a positive charge
D	region in which an electric charge produces a current	direction of the force on a negative charge

26 Copper wire is available in fixed lengths but in various diameters d . Each wire has a different resistance R .

Which relationship between R and d is correct?

- A** R is directly proportional to d .
- B** R is directly proportional to d^2 .
- C** R is inversely proportional to d .
- D** R is inversely proportional to d^2 .

27 The diagram shows three $20\ \Omega$ resistors, R_1 , R_2 and R_3 , and a switch S connected to a d.c. power supply.



Switch S is open.

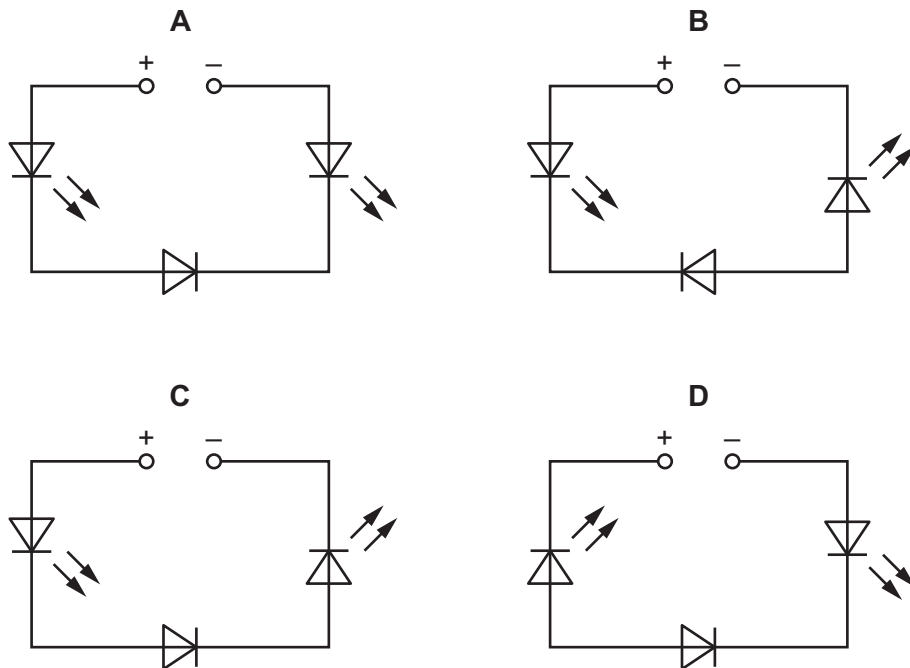
The potential difference (p.d.) across R_3 is 9.0 V .

What is the p.d. across R_3 when switch S is closed?

- A** 3.0 V
- B** 6.0 V
- C** 12 V
- D** 27 V

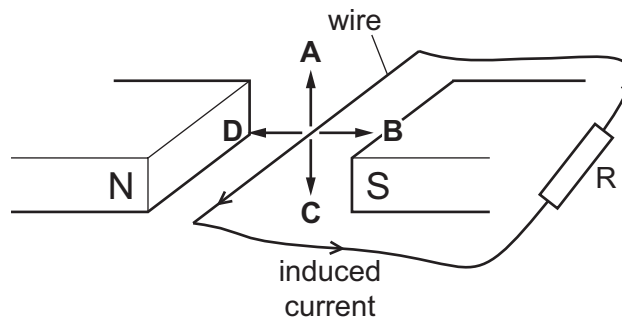
- 28 The circuits shown all include a d.c. power supply, one diode and two light-emitting diodes (LEDs).

In which circuit will both LEDs be turned on?



- 29 A wire connected to a resistor R is moved in a magnetic field. A current is induced in the direction shown.

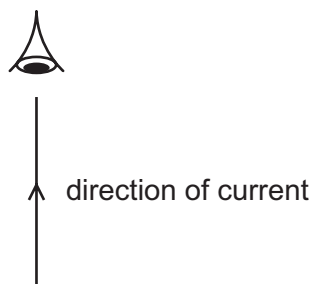
In which direction is the wire moved?



- 30 Which component is **not** found in an a.c. generator?

- A coil of wire
- B split-ring commutator
- C magnetic poles
- D slip rings

- 31 There is a current in a straight metal wire in the direction shown.



Which row describes the direction and the pattern of the magnetic field due to the wire when looking from vertically above the wire?

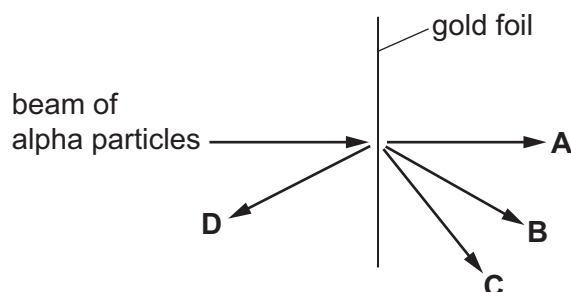
	direction	pattern
A	anticlockwise	circular
B	left to right	parallel lines
C	clockwise	circular
D	right to left	parallel lines

- 32 A simple transformer contains an iron core.

What is the function of the iron core in the transformer?

- A** to conduct current from the primary coil to the secondary coil
 - B** to enable the secondary coil to act as an electromagnet
 - C** to ensure there is a high voltage between the primary coil and the secondary coil
 - D** to link the magnetic field in the primary coil to the secondary coil
- 33 In an alpha particle scattering experiment, a beam of alpha particles is fired at a very thin piece of gold foil.

In which direction do most of the alpha particles travel after hitting the foil?



- 34** An atom of boron contains 5 protons and 5 electrons. The nucleon number of the atom is 11.

How many neutrons are there in the atom?

- A** 5 **B** 6 **C** 11 **D** 16

- 35** A detector placed near a radioactive isotope gives a measured count rate of 800 counts/minute. The half-life of the isotope is three hours.

The background count rate is 32 counts/minute.

What is the count rate on the detector after six hours?

- A** 192 counts/minute
B 200 counts/minute
C 224 counts/minute
D 232 counts/minute

- 36** Sodium-24 is a radioactive isotope that emits beta radiation.

Which nuclear equation shows how sodium-24 decays?

- A** ${}_{11}^{24}\text{Na} \rightarrow {}_{12}^{24}\text{Mg} + {}_{-1}^0\beta$
B ${}_{11}^{24}\text{Na} \rightarrow {}_{10}^{24}\text{Ne} + {}_{-1}^0\beta$
C ${}_{11}^{24}\text{Na} \rightarrow {}_9^{20}\text{F} + {}_2^4\beta$
D ${}_{11}^{24}\text{Na} \rightarrow {}_{13}^{20}\text{Al} + {}_2^4\beta$

- 37** Which safety precautions must be taken when using a source of gamma radiation?

- 1 Reduce the distance between the source and the person.
- 2 Reduce the time of exposure to the radiation.
- 3 Use a suitable shielding material between the source and the person.

- A** 1, 2 and 3 **B** 1 and 2 only **C** 1 and 3 only **D** 2 and 3 only

38 A student defines three periods of time:

- the time T_S taken for the Earth to orbit the Sun
- the time T_R taken for the Earth to rotate once on its axis
- the time T_M taken for the Moon to orbit the Earth.

Which row shows the shortest period of time and the longest period of time?

	shortest period of time	longest period of time
A	T_S	T_R
B	T_R	T_S
C	T_R	T_M
D	T_M	T_S

39 Which statement about the Universe is correct?

- A** The Universe consists of 100 000 galaxies.
- B** Redshift is a decrease in the observed wavelength of electromagnetic radiation emitted from receding galaxies.
- C** Light emitted from distant galaxies appears redshifted.
- D** Redshift of the light from distant galaxies gives evidence that the Universe is contracting.

40 Which statement about the cosmic microwave background radiation (CMBR) is correct?

- A** CMBR now has a frequency higher than that of infrared radiation.
- B** CMBR has always had exactly the same frequency.
- C** CMBR had a shorter wavelength in the past.
- D** CMBR was produced a long time after the Universe was formed.

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Cambridge IGCSE™ (9–1)

PHYSICS (9–1)**0972/21**

Paper 2 Multiple Choice (Extended)

May/June 2025**MARK SCHEME**Maximum Mark: 40

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the May/June 2025 series for most Cambridge IGCSE, Cambridge International A and AS Level components, and some Cambridge O Level components.

This document consists of **3** printed pages.

Question	Answer	Marks
1	C	1
2	B	1
3	D	1
4	C	1
5	B	1
6	D	1
7	C	1
8	A	1
9	D	1
10	A	1
11	B	1
12	C	1
13	B	1
14	A	1
15	A	1
16	A	1
17	B	1
18	D	1
19	B	1
20	B	1
21	B	1
22	D	1
23	D	1
24	A	1
25	A	1
26	D	1
27	C	1
28	C	1

Question	Answer	Marks
29	C	1
30	B	1
31	A	1
32	D	1
33	A	1
34	B	1
35	C	1
36	A	1
37	D	1
38	B	1
39	C	1
40	C	1



Cambridge IGCSE™ (9–1)

PHYSICS

0972/22

Paper 2 Multiple Choice (Extended)

May/June 2025

45 minutes

You must answer on the multiple choice answer sheet.

You will need: Multiple choice answer sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)

INSTRUCTIONS

- There are **forty** questions on this paper. Answer **all** questions.
- For each question there are four possible answers **A**, **B**, **C** and **D**. Choose the **one** you consider correct and record your choice in soft pencil on the multiple choice answer sheet.
- Follow the instructions on the multiple choice answer sheet.
- Write in soft pencil.
- Write your name, centre number and candidate number on the multiple choice answer sheet in the spaces provided unless this has been done for you.
- Do **not** use correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.
- Take the weight of 1.0 kg to be 9.8 N (acceleration of free fall = 9.8 m/s^2).

INFORMATION

- The total mark for this paper is 40.
- Each correct answer will score one mark.
- Any rough working should be done on this question paper.

This document has **16** pages.

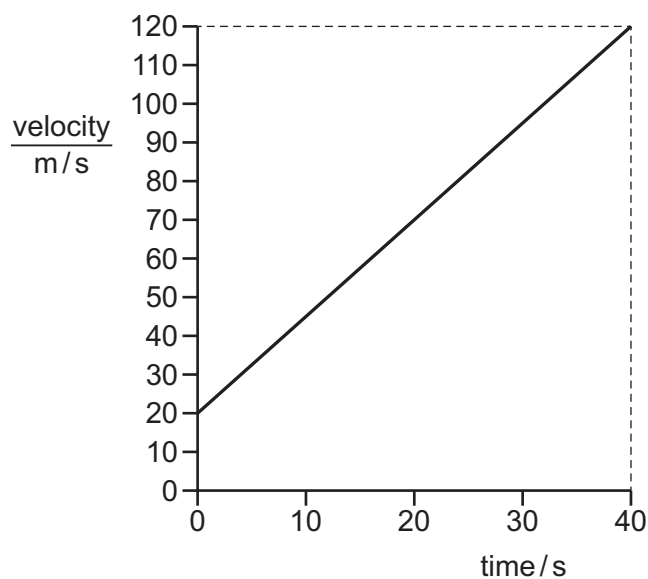


- 1 Which instrument is most suitable to determine the volume of a small irregularly shaped stone?
- A 30 cm ruler
 - B digital timer
 - C measuring cylinder
 - D tape measure

- 2 A steel ball is dropped from the top floor of a building. Air resistance can be ignored.

Which statement describes the motion of the ball?

- A The ball falls with constant acceleration.
 - B The ball falls with constant speed.
 - C The ball falls with decreasing speed.
 - D The ball falls with increasing acceleration.
- 3 The diagram shows a velocity–time graph for an object which is accelerating.



What is the acceleration of the object?

- A 0.40 m/s^2
- B 2.5 m/s^2
- C 3.0 m/s^2
- D 100 m/s^2

- 4 Diagram 1 shows a piece of flexible material that contains many pockets of air. Diagram 2 shows the same piece of flexible material after it has been compressed so that its volume decreases.

diagram 1
(before compression)

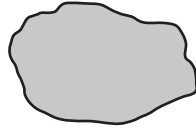
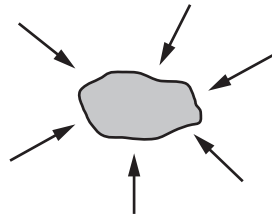


diagram 2
(after compression)



What happens to the mass and to the weight of the flexible material when it is compressed?

	mass	weight
A	increases	increases
B	increases	no change
C	no change	increases
D	no change	no change

- 5 A student determines the density of copper.

She has a rectangular block of copper.

Which other apparatus does she need?

	balance	ruler	stop-watch	thermometer
A	✓	✓	✓	✓
B	✓	✓	✓	✗
C	✓	✓	✗	✗
D	✓	✗	✗	✗

key

✓ = needed

✗ = **not** needed

- 6 A 0.15 kg mass is suspended from a vertical spring.

The mass is pulled downwards until the spring has extended by 12 cm from its unstretched length.

The spring constant of the spring is 30 N/m.

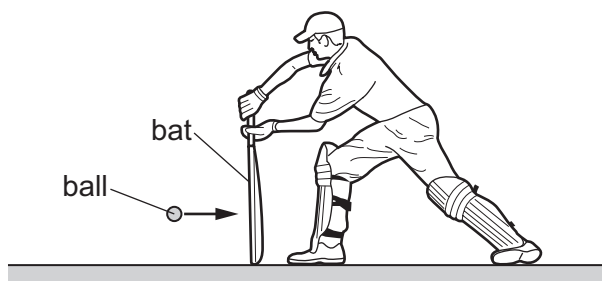
What is the magnitude of the initial acceleration of the spring when it is released?

- A** 14 m/s² **B** 24 m/s² **C** 28 m/s² **D** 34 m/s²

- 7 A cricket ball of mass 0.15 kg is moving at a speed of 30 m/s.

The diagram shows a batsman holding a bat stationary as the ball hits the bat and bounces back at the same speed in the opposite direction.

The ball is in contact with the bat for 1.0×10^{-3} s.



The table shows the change in momentum of the ball and the magnitude of the force applied to the ball.

Which row is correct?

	<u>change of momentum</u> kg m/s	force applied to ball / N
A	4.5	4500
B	4.5	9000
C	9.0	4500
D	9.0	9000

- 8 A stone is thrown vertically upwards at a speed of 5.0 m/s.

All of the energy initially in the kinetic store of the stone is transferred to its gravitational potential store when the stone reaches its maximum height.

Which equation gives the maximum height h reached by the stone?

A $h = \frac{9.8}{2 \times 5.0^2}$

B $h = \frac{5.0^2 \times 2}{9.8}$

C $h = \frac{5.0^2}{2 \times 9.8}$

D $h = \frac{2 \times 9.8}{5.0^2}$

9 Three energy resources are listed.

- 1 geothermal
- 2 wind
- 3 nuclear fuel

Which resources get their energy from the Sun?

- A** 1, 2 and 3 **B** 1 and 2 only **C** 1 only **D** 2 only

10 A sample of a substance has a shape that is determined by its container.

What are the possible states of this sample?

- A** gas, liquid and solid
- B** gas and liquid only
- C** gas and solid only
- D** solid and liquid only

11 A stationary smoke particle is hit by a fast-moving nitrogen molecule.

Which row describes the motion of the smoke particle and of the nitrogen molecule after the collision?

	smoke particle	nitrogen molecule
A	moves	rebounds
B	moves	stops
C	remains stationary	rebounds
D	remains stationary	stops

- 12 A bubble of gas rises from the bottom of a lake to the top.

The table shows readings of the gas pressure inside the bubble and the volume of the bubble.

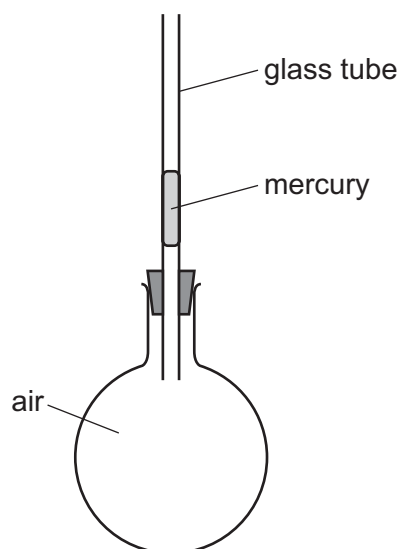
The temperature of the water in the lake is constant.

The values in three rows of the table are correct and one row is **not** correct.

Which row is **not** correct?

	pressure / kPa	volume / cm ³
A	360	2.4
B	270	3.2
C	160	4.6
D	120	7.2

- 13 The diagram shows a glass flask sealed with a small volume of mercury in a glass tube. When the flask is gently warmed, the mercury rises up the tube.

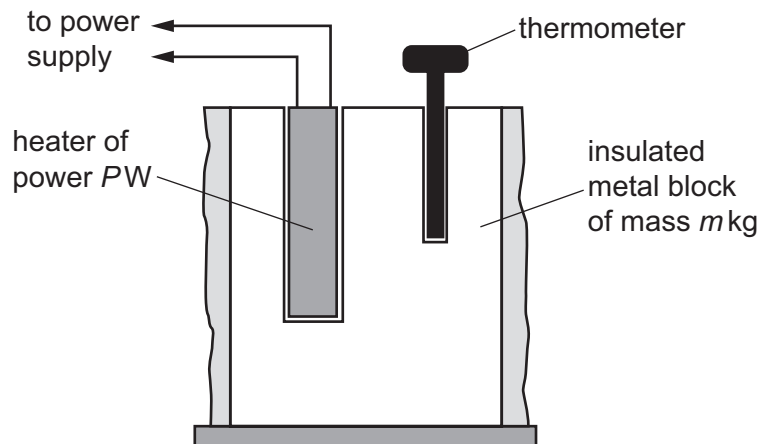


What is the main cause of the movement of the mercury?

- A** expansion of air in the flask
- B** expansion of the glass flask
- C** expansion of the glass tube
- D** expansion of the mercury

- 14** The diagram shows the equipment used in an experiment to determine the specific heat capacity of a metal.

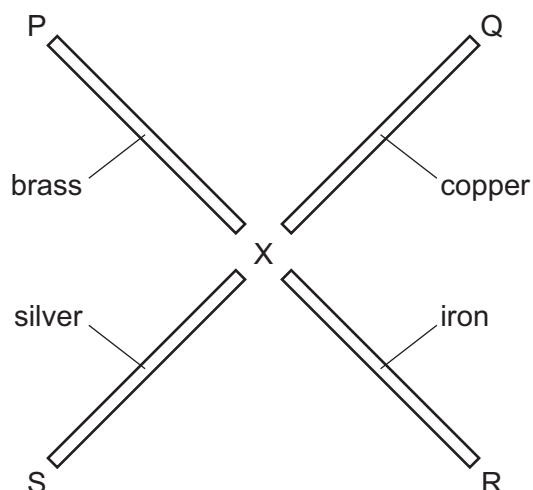
In the experiment, a heater is switched on for t s and the increase in temperature $\Delta\theta$ of a block of the metal is measured in $^{\circ}\text{C}$.



Which equation gives the specific heat capacity of the metal?

- A** $c = \frac{P}{tm\Delta\theta}$ **B** $c = \frac{Pt}{m\Delta\theta}$ **C** $c = \frac{tm\Delta\theta}{P}$ **D** $c = \frac{m\Delta\theta}{Pt}$

- 15 The diagram shows four rods. Each rod is made of a different metal.



Wax is used to attach small metal balls at the rod ends P, Q, R and S.

Each rod is the same size. They are heated uniformly by a Bunsen burner at point X.

As the rods warm up, the wax melts and the balls fall off.

Why does the ball on the silver rod fall first?

- A** Silver is the best conductor of heat.
- B** Silver is the worst conductor of heat.
- C** Silver is the best radiator of heat.
- D** Silver is the worst radiator of heat.

- 16 Three solid copper spheres, X, Y and Z, are heated to the same temperature.

sphere	diameter	colour of surface
X	$2d$	black
Y	d	white
Z	$2d$	white

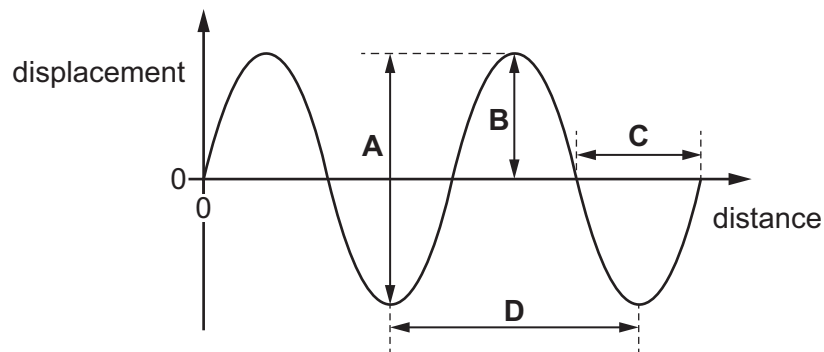
Which row gives the order for the rate of emission of radiation?

	highest rate of emission	second highest rate of emission	lowest rate of emission
A	X	Y	Z
B	X	Z	Y
C	Z	X	Y
D	Z	Y	X

17 Which wave will diffract the most when passing through a gap that is 0.85 m wide?

	wave speed m/s	wave frequency / Hz
A	300	280
B	300	400
C	350	280
D	350	400

18 Which arrow on the graph shows the amplitude of the wave?



19 Which row describes a seismic S-wave?

	type of wave	direction of vibration
A	longitudinal	parallel to the direction of travel of the wavefront
B	longitudinal	perpendicular to the direction of travel of the wavefront
C	transverse	parallel to the direction of travel of the wavefront
D	transverse	perpendicular to the direction of travel of the wavefront

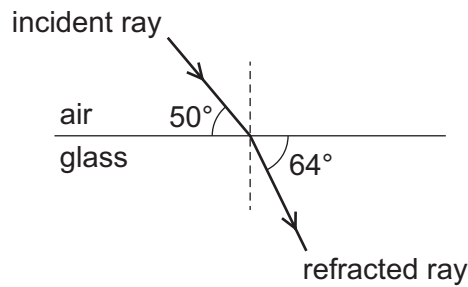
20 The image of an object is formed by a plane mirror.

Which row gives the correct properties of the image?

	real or virtual	distance of the image from the mirror
A	real	closer to the mirror than the object
B	real	the same distance from the mirror as the object
C	virtual	closer to the mirror than the object
D	virtual	the same distance from the mirror as the object

- 21** A ray of light, travelling in air, is incident on an air–glass boundary.

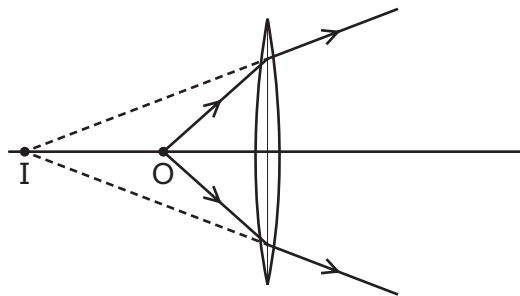
The angles between the incident ray, the refracted ray and the boundary are shown.



Which equation is used to calculate the refractive index n of the glass?

- A** $n = \frac{\sin 40^\circ}{\sin 26^\circ}$ **B** $n = \frac{\sin 40^\circ}{\sin 64^\circ}$ **C** $n = \frac{\sin 50^\circ}{\sin 26^\circ}$ **D** $n = \frac{\sin 50^\circ}{\sin 64^\circ}$

- 22** A small object O is placed near a converging lens, as shown. The lens forms an image I.



Which statement is correct?

- A** The image I is diminished.
B The image I is inverted.
C The image I is real.
D The object O is closer to the lens than its principal focus.
- 23** What is the approximate speed of light in air?
- A** 300 m/s
B 300 km/s
C 3.0×10^8 m/s
D 3.0×10^8 km/s

- 24 A ship sends an ultrasound pulse to the seabed to measure the depth of the sea.

The speed of ultrasound waves in sea water is 1500 m/s.

The pulse takes 4.6 seconds to travel to the seabed and back to the ship.

What is the depth of the sea?

- A 330 m B 650 m C 3500 m D 6900 m

- 25 A dolphin has a range of audible frequencies of 150 Hz–150 kHz.

Which range of frequencies can be heard both by humans with good hearing and by dolphins?

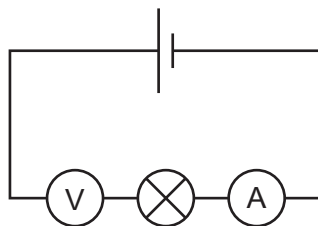
- A 20 Hz–150 Hz
B 20 Hz–150 kHz
C 20 kHz–150 kHz
D 150 Hz–20 kHz

- 26 A positively charged balloon is placed against a wall. The charge makes the balloon stick to the wall.

Which statement explains this?

- A Positive charges on the balloon attract electrons in the wall.
B Positive charges on the balloon attract protons in the wall.
C Positive charges on the balloon repel electrons in the wall.
D Positive charges on the balloon repel protons in the wall.

- 27 A student is trying to determine the resistance of a lamp. He sets up the **incorrect** circuit shown.



Which change would allow him to take the measurements needed to calculate the resistance of the lamp?

- A Disconnect the ammeter and move it to the left of the lamp.
B Swap the positions of the ammeter and the voltmeter in the circuit.
C Disconnect the ammeter and connect it in parallel with the lamp.
D Disconnect the voltmeter and connect it in parallel with the lamp.

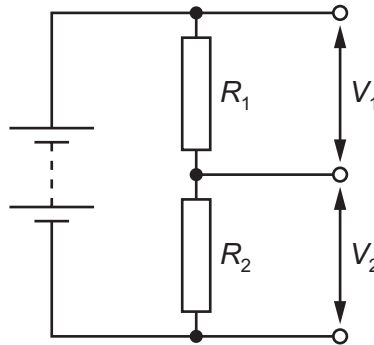
28 A metallic wire X has a resistance.

What is the difference in resistance for a wire of the same material as X that is:

- shorter than X but the same diameter
- the same length as X but with a smaller diameter?

	resistance of shorter wire	resistance of wire with smaller diameter
A	greater	greater
B	greater	smaller
C	smaller	greater
D	smaller	smaller

29 The diagram shows the circuit for a potential divider.



Which equation is used to calculate V_1 ?

A $V_1 = \frac{R_1}{V_2 \times (R_1 + R_2)}$

B $V_1 = \frac{V_2 \times R_1}{R_2}$

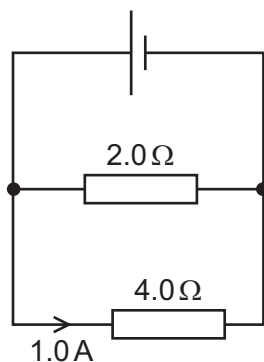
C $V_1 = \frac{V_2 \times R_1}{R_1 + R_2}$

D $V_1 = \frac{R_1}{V_2 \times R_2}$

30 Which electrical device is used to convert a.c. to d.c.?

- A diode
- B relay
- C thermistor
- D transformer

31 A cell is connected to a parallel combination of a 2.0Ω resistor and a 4.0Ω resistor. The current in the 4.0Ω resistor is 1.0 A .

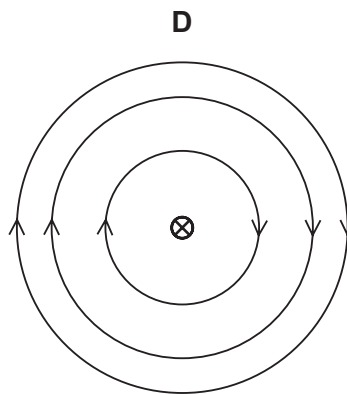
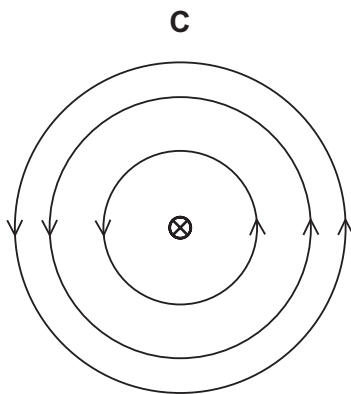
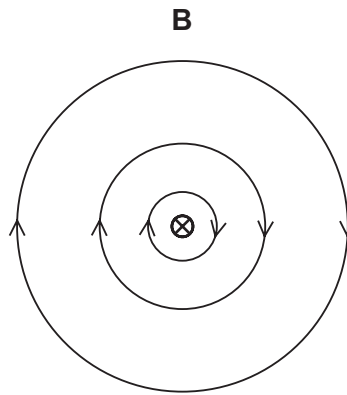
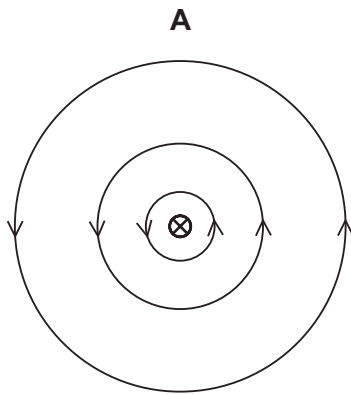


What is the current in the cell?

- A 1.0 A
- B 1.5 A
- C 2.0 A
- D 3.0 A

- 32** When there is an electric current in a long straight wire, a magnetic field is created around the wire.

Which diagram shows the correct pattern and direction of magnetic field lines around a long straight wire carrying current into the page?



key

⊗ wire carrying current into page

- 33** The primary voltage supplied to a transformer is 3.0 V. The secondary voltage is 5.0 V and the secondary current is 2.0 A. The transformer is 100% efficient.

What is the primary current in the transformer?

- A** 1.2 A **B** 2.0 A **C** 3.3 A **D** 7.5 A

- 34** When a thin gold foil is bombarded with alpha particles, some alpha particles are deflected through large angles.

Which statement explains this deflection?

- A** Most of the atom consists of empty space.
B All of the positive charge and most of the mass of the gold atom are concentrated in a small volume.
C Positive charge in the gold atom is spread evenly throughout the atom.
D All of the negative charge is concentrated at its centre.

- 35 A detector near a radioactive source shows a reading of 2000 counts / minute.

How is the corrected count rate determined?

- A by moving the detector closer to the source
 - B by placing an absorber between the detector and the source
 - C by repeating and averaging the measurements
 - D by taking away the background radiation count rate
- 36 Which statement about alpha, beta and gamma radiation is **not** correct?
- A Alpha particles are deflected most in an electric field.
 - B Beta particles are deflected most in a magnetic field.
 - C Gamma radiation is **not** deflected by a magnetic field.
 - D Gamma radiation is least ionising.

- 37 A sample of a radioactive material has a half-life of 20 minutes.

Which statement is correct?

- A After 30 minutes, less than half of the material has decayed.
 - B After 40 minutes, all of the radioactive material has decayed.
 - C After 60 minutes, an eighth of the radioactive material remains.
 - D After 120 minutes, a sixth of the radioactive material remains.
- 38 Which statement is correct?
- A The Earth orbits the Sun once in approximately 24 hours.
 - B The Earth orbits the Moon once in approximately 1 week.
 - C The Moon orbits the Earth once in approximately 1 month.
 - D The Sun orbits the Earth once in approximately 365 days.

- 39 Stars are powered by nuclear reactions that release energy.

Which statement describes the nuclear reaction in stable stars?

- A Nuclear fission converts helium into hydrogen.
- B Nuclear fission converts hydrogen into helium.
- C Nuclear fusion converts helium into hydrogen.
- D Nuclear fusion converts hydrogen into helium.

40 A galaxy is moving away from the Earth at a velocity of 3000 km/s.

What is the distance between the Earth and the galaxy?

- A** 7.3×10^{-25} m
- B** 7.3×10^{-22} m
- C** 1.4×10^{21} m
- D** 1.4×10^{24} m

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Cambridge IGCSE™ (9–1)

PHYSICS

0972/21

Paper 2 Multiple Choice (Extended)

October/November 2025

45 minutes

You must answer on the multiple choice answer sheet.

You will need: Multiple choice answer sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)

INSTRUCTIONS

- There are **forty** questions on this paper. Answer **all** questions.
- For each question there are four possible answers **A**, **B**, **C** and **D**. Choose the **one** you consider correct and record your choice in soft pencil on the multiple choice answer sheet.
- Follow the instructions on the multiple choice answer sheet.
- Write in soft pencil.
- Write your name, centre number and candidate number on the multiple choice answer sheet in the spaces provided unless this has been done for you.
- Do **not** use correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.
- Take the weight of 1.0 kg to be 9.8 N (acceleration of free fall = 9.8 m/s^2).

INFORMATION

- The total mark for this paper is 40.
- Each correct answer will score one mark.
- Any rough working should be done on this question paper.

This document has **16** pages. Any blank pages are indicated.



- 1 A student determines the diameter of a metal ball.

Which apparatus is needed to get an accurate measurement?

- A a ruler and a protractor
- B a ruler only
- C a protractor and a set square
- D a ruler and two rectangular blocks

- 2 An athlete runs at a speed of 8 m/s for 10 s, and then at a speed of 6 m/s for 12 s.

Which calculation gives the average speed of the athlete in m/s?

- A $\frac{8+6}{2}$
- B $\frac{(8 \times 10) + (6 \times 12)}{22}$
- C $\frac{(8 \div 10) + (6 \div 12)}{22}$
- D $\frac{(10 \div 8) + (12 \div 6)}{22}$

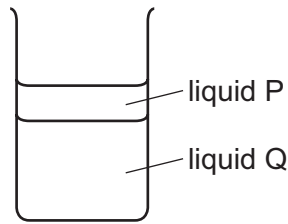
- 3 A piece of paper torn out of an exercise book is shown.

	describes how a massive object affects
	the space around it, including the size and
	direction of the force it would exert on other
	objects placed in that space.

What is being described?

- A gravitational charge
- B gravitational field
- C gravitational mass
- D gravitational potential energy

- 4 The diagram shows one liquid floating on top of another liquid.



Which statement explains why liquid P floats on top of liquid Q?

- A The mass of liquid Q is greater than the mass of liquid P.
 - B The mass of 1 cm^3 of liquid Q is greater than the mass of 1 cm^3 of liquid P.
 - C The volume of liquid Q is greater than the volume of liquid P.
 - D The volume of 1 g of liquid Q is greater than the volume of 1 g of liquid P.
- 5 A spring has a spring constant k .
- What is a possible unit for k ?
- A metre per newton
 - B newton metre
 - C newton per metre
 - D newton per metre squared
- 6 Which statement about the moment of a force is **not** correct?
- A If an object is balanced about a pivot, the resultant moment on the object must be zero.
 - B The moment of a force is a measure of its turning effect.
 - C The moment of a force about a point is equal to: force $\times$ perpendicular distance from the point.
 - D The moment of a force about a point increases when the perpendicular distance of the force from the point decreases.
- 7 The momentum of a car changes from $10\,000 \text{ kg m/s}$ to $15\,000 \text{ kg m/s}$ in a time of 8.0 s .

What is the resultant force acting on the car?

- A 630 N B 3100 N C 40 000 N D 200 000 N

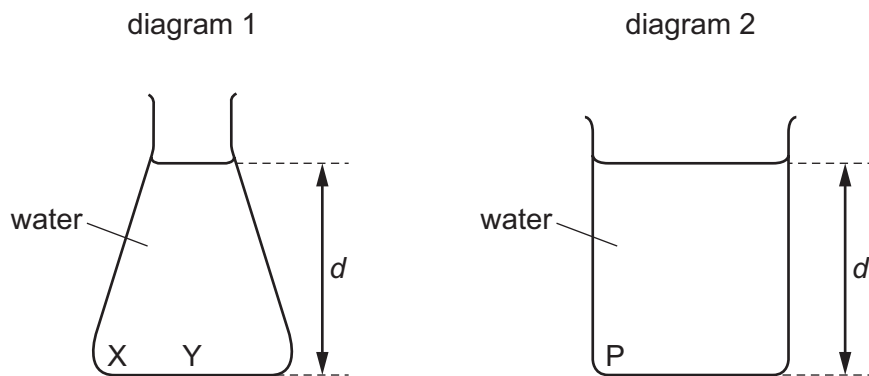
- 8 A machine is very efficient.

What does this mean?

- A It produces a large amount of power.
- B It uses very little energy.
- C It wastes very little energy.
- D It works very quickly.

- 9 Diagram 1 shows a conical flask containing water.

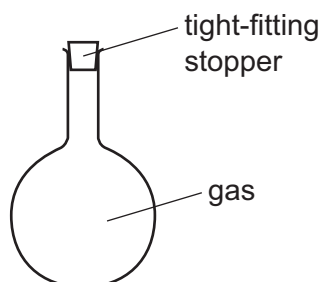
Diagram 2 shows a beaker with the same base area as the flask and containing water of the same depth d .



Which statement is correct?

- A The pressure at P is equal to the pressure at X.
- B The pressure at P is greater than the pressure at Y.
- C The pressure at P is greater than the pressure at X.
- D The pressure at Y is greater than the pressure at X.

- 10 A flask with a tight-fitting stopper contains a gas. The gas and flask are initially at room temperature.



The gas and flask are cooled.

Which row describes the average speed of the gas particles and the pressure of the gas as the temperature decreases?

	average speed of the gas particles	pressure of the gas
A	decreases	decreases
B	decreases	stays the same
C	increases	increases
D	increases	stays the same

- 11 Brownian motion is the random movement of microscopic particles in a liquid or gas.

Which statement about Brownian motion is correct?

- A** Brownian motion only occurs when the liquid or gas is heated.
 - B** Much heavier fast-moving particles that **cannot** be seen collide with the microscopic particles.
 - C** Much lighter fast-moving particles that **cannot** be seen collide with the microscopic particles.
 - D** The microscopic particles move to avoid the fast-moving liquid or gas molecules.
- 12 The distance between two electricity pylons is 60 m. The cable that connects the two pylons is 62 m in length.

Why is the cable longer than the distance between the two pylons?

- A** to allow for contraction of the cable in cold weather
- B** to create a slope in the cable for electrons to flow down
- C** to keep the current low and the voltage high
- D** to reduce magnetic fields around the cable

- 13 A glass beaker contains a mass of 0.20 kg of water at a temperature of 20 °C.

The specific heat capacity of water is 4200 J/(kg °C).

A brass block of mass 0.15 kg is at a temperature of 170 °C. The block is carefully lowered into the beaker of water.

The final temperature of both the water and brass is 30 °C.

What is the specific heat capacity of the brass?

(Assume that no water is lost and no thermal energy has been transferred to the beaker or the surroundings.)

- A 400 J/(kg °C)
 - B 660 J/(kg °C)
 - C 1200 J/(kg °C)
 - D 5600 J/(kg °C)
- 14 Which statement about convection is **not** correct?
- A It enables water in a pan on a cooker to get evenly heated.
 - B It happens in liquids and gases.
 - C It occurs because cooler liquids rise above warmer liquids.
 - D It occurs because the density of a liquid decreases when it is heated.
- 15 Iron is a better conductor of thermal energy than rubber.

Which statement explains this?

- A Electrons in iron can move from one part of the iron to another. Electrons in rubber **cannot** move in this way.
- B Electrons in iron vibrate with more energy than those in rubber.
- C Positive particles in iron can move from one part of the iron to another. Positive particles in rubber **cannot** move in this way.
- D Positive particles in iron vibrate with more energy than those in rubber.

- 16** Two metal cans are identical, except that one has a shiny silver outer surface and the other has a dull black outer surface. They each have the same mass of hot water at the same temperature sealed inside them. They are both in a vacuum in the darkness of outer space.

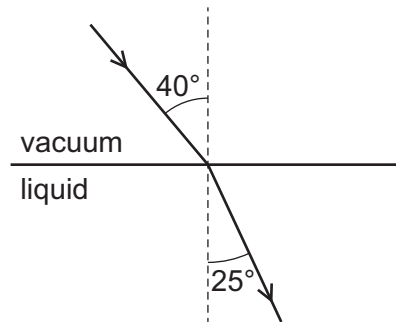
How does the temperature of the water in each one change?

- A** The temperature of the water does **not** change.
 - B** The water in the black can cools more slowly than that in the silver can.
 - C** The water in the silver can cools more slowly than that in the black can.
 - D** They both cool down at the same rate.
- 17** A ray of light reflects from a plane mirror.
- The angle between the incident ray and the surface of the mirror is 35° .
- Which statement is correct?
- A** The angle of incidence is 35° .
 - B** The angle of incidence is 55° .
 - C** The angle of reflection is 70° .
 - D** The angle of reflection is 110° .
- 18** Parallel waves in a ripple tank are diffracted as they pass through a gap in a barrier.
- Three changes that can be made to this arrangement are listed.
- 1 Decrease the wavelength of the waves.
 - 2 Increase the amplitude of the waves.
 - 3 Decrease the size of the gap.

Which changes will cause the shape of the diffracted waves to be less curved?

- A** 1, 2 and 3 **B** 1 and 3 only **C** 1 only **D** 2 and 3 only

- 19 A beam of light passes through a vacuum and then enters a transparent liquid. The diagram shows the path it takes.

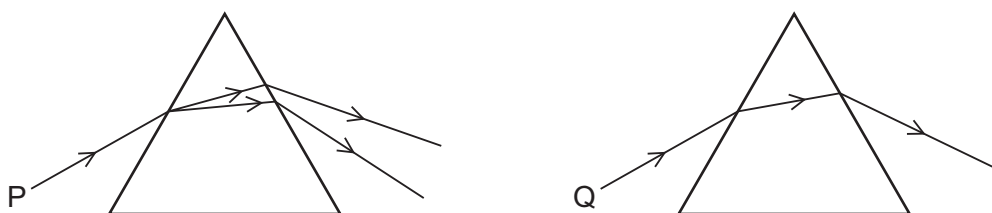


The light travels through the vacuum at a speed of $3.0 \times 10^8 \text{ m/s}$.

What is the speed of light in the liquid?

- A $1.9 \times 10^8 \text{ m/s}$
 - B $2.0 \times 10^8 \text{ m/s}$
 - C $4.6 \times 10^8 \text{ m/s}$
 - D $4.8 \times 10^8 \text{ m/s}$
- 20 Which statement about a thin converging lens is correct?
- A All rays of light refracted by the lens pass through the principal focus.
 - B All rays initially parallel to the principal axis of the lens are refracted through the principal focus.
 - C The focal length of the lens is the distance between the image and the principal focus.
 - D The focal length of the lens is the distance between the object and the image.

- 21 The diagram shows two narrow beams of light, P and then Q, passing through the same prism. Beams P and Q are both incident on the same point on the prism and at the same angle.



Which row correctly compares beams P and Q?

	dispersion	nature of light
A	only Q is dispersed	P is monochromatic and Q is white light
B	only Q is dispersed	P is white light and Q is monochromatic
C	only P is dispersed	P is monochromatic and Q is white light
D	only P is dispersed	P is white light and Q is monochromatic

- 22 Ultraviolet radiation and infrared radiation are both parts of the electromagnetic spectrum.

However, they differ in wavelength and frequency.

Which row describes how they differ?

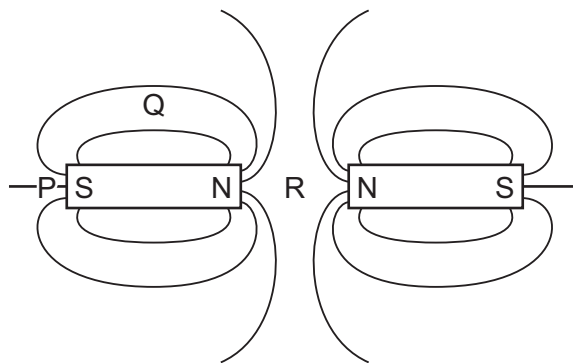
	ultraviolet	infrared
A	longer wavelength than infrared	higher frequency than ultraviolet
B	longer wavelength than infrared	lower frequency than ultraviolet
C	shorter wavelength than infrared	higher frequency than ultraviolet
D	shorter wavelength than infrared	lower frequency than ultraviolet

- 23** A police car with its siren sounding is stationary in heavy traffic. A pedestrian notices that, although the loudness of the sound produced does **not** change, the pitch varies.

Which row describes the amplitude and the frequency of the sound?

	amplitude	frequency
A	constant	constant
B	constant	varying
C	varying	constant
D	varying	varying

- 24** The diagram shows part of the magnetic field around two magnets.



Which list gives the order of magnetic field strengths at the positions P, Q and R, from weakest to strongest?

- A** P → Q → R
 - B** P → R → Q
 - C** Q → P → R
 - D** R → Q → P
- 25** A fully charged battery stores a charge of 18 000 C. The battery is charged with a current of 830 mA.

How long does it take to fully charge a completely discharged battery?

- A** about 10 minutes
- B** about 3.5 hours
- C** about 4 hours
- D** about 6 hours

26 What is the definition of electromotive force?

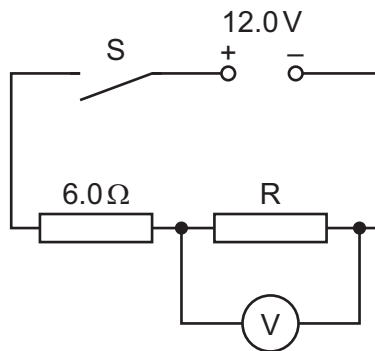
- A the electrical work done by a source in moving a unit charge around a complete circuit
- B the electrical work done by a unit charge passing through a component
- C the mechanical work done by a source in moving a unit charge around a complete circuit
- D the mechanical work done by a unit charge passing through a component

27 A cell passes a current of 2.0 A in a circuit for 30 s. In this time, the cell transfers 120 J of energy.

What is the electromotive force (e.m.f.) of the cell?

- A 0.50 V B 1.5 V C 2.0 V D 8.0 V

28 The diagram shows two resistors in series connected to a power supply.



The reading on the voltmeter is 3.0 V when the switch is closed.

What is the resistance of resistor R?

- A 0.67 Ω B 1.2 Ω C 1.5 Ω D 2.0 Ω

29 Lamps can be connected in series or in parallel to a power supply.

What is an advantage of connecting the lamps in parallel rather than in series?

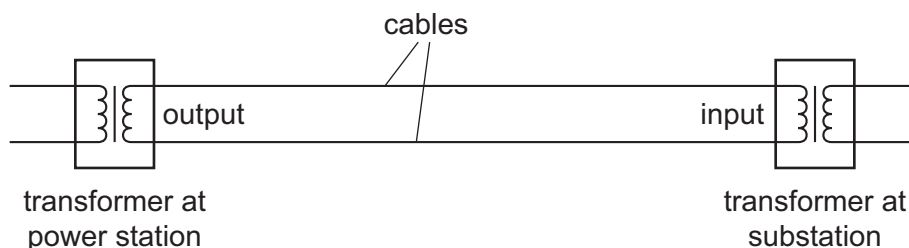
- A All of the lamps can be switched on and off using the same switch.
- B Fewer connecting leads are needed to make the circuit.
- C When one lamp stops working, the others may continue to work.
- D Less power is transferred from the supply.

- 30** The lamps on a bicycle are powered by a simple electrical generator.

A permanent magnet rotates around a stationary coil of wire wound on an iron core.

Which statement is correct?

- A** The current in the lamps decreases as the speed of rotation increases.
- B** The current in the lamps decreases when the magnet is replaced by a stronger magnet.
- C** The output electromotive force (e.m.f.) will be higher if the gap between the magnet and the coil is bigger.
- D** The output e.m.f. will be higher if there are more turns on the coil.
- 31** A current-carrying coil in the magnetic field between the poles of two magnets experiences a turning effect.
- Which change would decrease the turning effect?
- A** decreasing the distance between the magnets
- B** increasing the current in the coil
- C** using stronger magnets
- D** decreasing the number of turns on the coil
- 32** Cables transmit electrical power from the output of the transformer at a power station to the input of another transformer at a substation.



The power at the output of the transformer at the power station is 400 MW.

Which situation delivers the most power to the input of the transformer at the substation?

	potential difference at power station transformer output / kV	diameter of cables
A	200	large
B	200	small
C	400	large
D	400	small

- 33** $^{239}_{94}\text{Pu}$ is an isotope of plutonium.

The nucleus can absorb a neutron and then split to form two other nuclei.

Two nuclei that form are gold, $^{207}_{79}\text{Au}$, and phosphorus, $^{31}_{15}\text{P}$.

In this reaction, neutrons are also produced.

What is the number of neutrons produced when the nucleus splits?

- A** 1 **B** 2 **C** 4 **D** 5

- 34** Three statements about alpha (α) particles and beta (β) particles are listed.

- 1 The magnitude of the charge on an α -particle is twice the charge on a β -particle.
- 2 The mass of an α -particle is large compared with the mass of a β -particle.
- 3 The kinetic energy of an α -particle is small compared to the kinetic energy of a β -particle.

Which statements help to explain the different ionising effects of α -particles and β -particles?

- A** 1, 2 and 3 **B** 1 and 2 only **C** 1 and 3 only **D** 2 and 3 only

- 35** A beta-particle is a fast-moving electron.

Which statement explains how beta-particles are emitted from an atom?

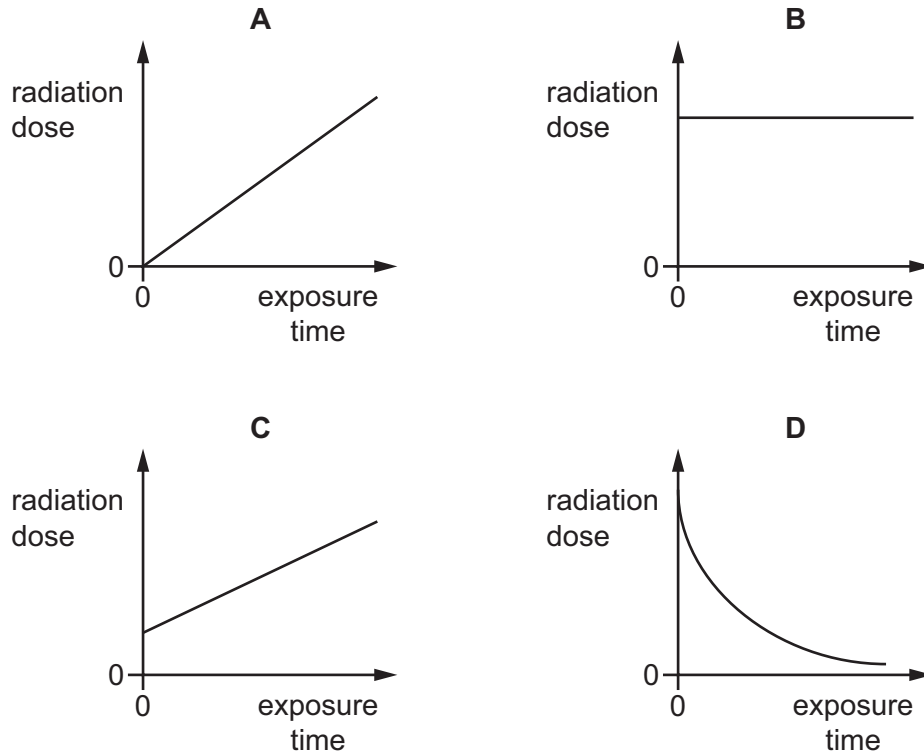
- A** An electron is emitted as a beta-particle from an inner electron shell of the atom.
B An electron is emitted as a beta-particle from an outer electron shell of the atom.
C A neutron changes into a proton and a beta-particle is emitted from the nucleus.
D A proton changes into a neutron and a beta-particle is emitted from the nucleus.

- 36** Which properties must a radioisotope have for its radiation to make it suitable to kill bacteria in food?

	half-life	type of radiation emitted
A	less than one minute	γ only
B	several hours	α only
C	several hours	γ only
D	several thousand years	α only

- 37** A student tries to predict the effect of exposure time on the radiation dose received from a source of ionising radiation. Radiation dose measures the amount of ionising radiation received by a living organism.

Which graph shows the correct trend between the variables?



- 38** A satellite is in orbit around the Earth. The orbit has radius r and period T .

Which expression gives the orbital speed of the satellite?

- A** $\frac{r}{2\pi T}$ **B** $\frac{T}{2\pi r}$ **C** $\frac{2\pi r}{T}$ **D** $\frac{2\pi T}{r}$

- 39** Comets move round the Sun.

Which statement describes the orbit of a comet?

- A** The orbit is circular with the Sun at the centre of the circle.
- B** The orbit is circular with the Sun **not** at the centre of the circle.
- C** The orbit is elliptical with the Sun at the centre of the ellipse.
- D** The orbit is elliptical with the Sun **not** at the centre of the ellipse.

- 40** A galaxy is at a distance of 60 million light-years from the Earth. The galaxy is moving away at a speed of 1300 km/s.

One light-year is about 9.5×10^{12} km.

There are about 3.2×10^7 seconds in one year.

Which expression can be used to estimate the age of the Universe in years?

- A** $\frac{60 \times 10^6 \times 3.2 \times 10^7}{1300 \times 9.5 \times 10^{12}}$
- B** $\frac{60 \times 10^6 \times 9.5 \times 10^{12}}{1300 \times 3.2 \times 10^7}$
- C** $\frac{1300 \times 9.5 \times 10^{12}}{60 \times 10^6 \times 3.2 \times 10^7}$
- D** $\frac{1300 \times 3.2 \times 10^7}{60 \times 10^6 \times 9.5 \times 10^{12}}$

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Cambridge IGCSE™ (9–1)

PHYSICS**0972/21**

Paper 2 Multiple Choice (Extended)

October/November 2025**MARK SCHEME**Maximum Mark: 40

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

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This document consists of **3** printed pages.

Question	Answer	Marks
1	D	1
2	B	1
3	B	1
4	B	1
5	C	1
6	D	1
7	A	1
8	C	1
9	A	1
10	A	1
11	C	1
12	A	1
13	A	1
14	C	1
15	A	1
16	C	1
17	B	1
18	C	1
19	B	1
20	B	1
21	D	1
22	D	1
23	B	1
24	D	1
25	D	1
26	A	1
27	C	1
28	D	1

Question	Answer	Marks
29	C	1
30	D	1
31	D	1
32	C	1
33	B	1
34	B	1
35	C	1
36	C	1
37	A	1
38	C	1
39	D	1
40	B	1